Reaction Intermediates — JEE Advanced Organic Chemistry

Carbocations (in depth), carbanions, free radicals, carbenes, nitrenes, benzyne and radical ions: theory, generation, stability, arrow-validated mechanisms, and worked solutions to the exercises you supplied. Status: Parts 0–4, 5A–5H, 7.0–7.1 and 8–14 are written; the 50 original stability questions (Part 5), the migratory-aptitude chapter (Part 6) and the rest of the rearrangement series (Part 7) are not yet written.

Contents

Part 0 Reaction Intermediates: Bond Fission, Intermediates, Electrophiles and Nucleophiles 4
- 0.1 Homolytic and heterolytic bond fission 4
- 0.2 The reaction intermediates at a glance 4
- 0.3 Electrophiles and nucleophiles (definitions) 5

Part 1 Carbocations: Definitions, Classification, Geometry and Bonding 5
- 1.1 Definitions: carbocation, carbenium ion, carbonium ion 5
- 1.2 Geometry, hybridisation and the empty p orbital 6
- 1.3 Orbital overlap requirements and delocalisation of charge 7
- 1.4 Hypercoordinate (carbonium) ions 7
- 1.5 Ion pairs, solvent and counterion 7

Part 2 Generation of Carbocations 8
- 2.A Carbocations from alcohols 8
- 2.B Carbocations from alkyl halides and sulfonates (ionisation) 10
- 2.C Carbocations from alkenes, dienes and alkynes (electrophilic addition) 13
- 2.D Solvolysis and S_N1 substitution 15
- 2.E Other routes: diazonium ions, Lewis acids, hydride abstraction, acid-mediated ionisations 17
- 2.F Carbocations from amines: diazotisation with nitrous acid 19
- 2.G Carbocations from aromatic compounds: arenium (σ-complex, Wheland) ions 21
- 2.H Lewis-acid-assisted ionisation, ether cleavage, and aromatic carbocations 22

Part 3 Leaving-Group Ability 25
- 3.1 What a leaving group is, and what 'good' means 25
- 3.2 Factors that determine leaving ability 25
- 3.3 Why there is no single universal order 26
- 3.4 Fifteen leaving-group comparison problems 26

Part 4 Carbocation Stability: Complete Advanced Treatment 29
- 4.1 Inductive effect 29
- 4.2 Mesomeric (resonance) effects 29
- 4.3 Hyperconjugation 30
- 4.4 Aromaticity and antiaromaticity 31
- 4.5 Cyclopropylcarbinyl stabilisation and 'dancing resonance' 31
- 4.6 Neighbouring-group participation and bridged ions 32
- 4.7 d-Orbital resonance claims 32
- 4.8 Steric inhibition of resonance (SIR) 32
- 4.9 Nonclassical carbocations 33
- 4.10 Carbonium ions (hypercoordinate carbon) 33
- 4.11 Additional effects 33

Part 5A Worked Solutions: the Supplied 'Exercise 2 – Carbocation Stability Order' (redrawn) 34
- Items from your exercise that need confirmation 41

Part 5B Worked Solutions: the Supplied 'Electronic Displacement Effects' Carbocation Problems (redrawn) 41

Part 5C Worked Solutions: Supplied 'How Many' Counting Problems (redrawn) 45

Part 5D Worked Solutions: More Supplied Counting Problems (heterolysis, statements, rearrangement, $AgNO_3$) 48

Part 5E Your 'Stability of Carbocation' Notes (cation.pdf): everything added, redrawn and checked 50
- 5E.1 The factor list 51
- 5E.2 Worked orders 1–12 (redrawn) 51
- 5E.3 'Important order': the pK_{R^+} ladder 52
- 5E.4 Gas-phase hydride affinities (kcal mol^{-1}, as in your note) 53
- 5E.5 Tabulated pK_{R^+} values from the note 54

5E.6 Examples from the note (March / Carey sets): where they appear in this book 54
5E.7 More pK_{R^+} structures from the note (pages 5 and 6) 55
5E.8 Pentavalent carbonium ions and σ-bond resonance 56
5E.9 Non-classical cations from the note 57
5E.10 Aryl-cation substituent series (pages 10 and 11) 58
5E.11 Carbanion stability (the note ends here) 59
5E.12 Effect on the earlier answer D-27 59

Part 5F Complete Worked Answers: the 'Electronic Displacement Effects' Exercise Book and the Allen Q48–51 Set 59

5F.0 Allen Q48–51 (stability orders, from the first image set) 60
5F.1 Exercise O-I (Q1–Q51) 60
5F.2 Exercise O-II (Q1–Q40) 63
5F.3 Exercise S-I (Q1–Q25) 66
5F.4 Exercise J-Main (Q1–Q31) 66
5F.5 Exercise J-Advanced (Q1–Q18) 67

Part 5G Complete Worked Answers: the Nucleus 'Carbocation and Free Radical' Exercise 68

5G.0 Cations drawn for reference 68
5G.1 Exercise I – mains oriented (Q1–Q53) 68
5G.2 Exercise II-A (multiple correct, Q1–Q18) 70
5G.3 Exercise II-B (Q1–Q13) 70
5G.4 JEE-Main and JEE-Advanced (objective) 71

Part 5H Orders and Rules from Your Lecture Notes (Lessons 1–5, Doubt-Clearing Sessions) 71

5H.1 Master stability order (Doubt-clearing session) 72
5H.2 Rearrangement rules from Carbocation-V 72
5H.3 Leaving-group and ionisation rules from Carbocation-IV 72
5H.4 Reactions of carbocations (Lessons 5 and FR-II) 73
5H.5 Dehydration of alcohols (Free-Radical III/IV lessons) 73
5H.6 Pinacol rearrangement rules (Free-Radical IV) 73
5H.7 Electrophiles and nucleophiles (Lesson 2) – the part of your topic list 73
5H.8 Carbocation structure (Doubt-clearing session) 73

Part 7.0 Carbocation Rearrangements: Thermodynamic and Kinetic Control, Mechanisms, and Named Reactions 74

7.0.1 Orders you must know 74
7.0.2 Step-validated 1,2-shifts and named rearrangements 75

Part 7 Carbocation Rearrangements — Counting Problems (opening section) 79

7.1 How many rearrange? How many ring-expand? How many shifts are possible? 80

Part 8 Carbanions: Structure, Generation, Stability and Reactions 81

8.1 Structure and shape 82
8.2 Generation of carbanions 82
8.3 Acidity of C–H bonds and the stability of the conjugate carbanion 82
8.4 Factors that stabilise a carbanion 83
8.5 Mechanisms in which a carbanion is the key intermediate (arrow-validated) 83
8.6 Kinetic and thermodynamic enolates; why optical activity is lost 85
8.7 Reactions of carbanions (summary) 86
8.9 The carbon–metal bond, ion pairs and aggregates: what 'R⁻' really is in the flask 86
8.10 Quantitative acidity of C–H: scales, equilibria, and the subtle cases 87
8.11 Enolates: formation, regiochemistry, ambident reactivity 88
8.12 Carbanion-based carbon–carbon bond-forming reactions (named reactions) 89
8.13 Organometallic reagents as carbanion sources: what they do and do not do 90
8.14 Stereochemistry and reaction pathways of carbanions (Cram's experiments) 90
8.15 Ylides: carbanions stabilised by an adjacent onium centre 90
8.16 Choosing a base: which proton is removed first? 91
8.20 Problems on carbanions 96

Part 9 Free Radicals: Structure, Generation, Stability and Reactions 105

9.0 Homolysis and bond dissociation energy 105
9.1 Structure and geometry 105
9.2 Stability of free radicals 105
9.3 Generation of free radicals 106
9.4 Reactions of free radicals 106
9.5 Radical halogenation of alkanes: thermochemistry and selectivity 106
9.6 Counting problems in radical halogenation 107
9.7 Allylic and benzylic halogenation; the HBr peroxide effect; polymerisation 107
9.8 Mechanisms (fish-hook arrows; each step electron-count validated) 107
9.9 Radical substitution of aromatics and related reactions 109
9.11 Electronic structure of carbon radicals: SOMO, spin density, EPR 109
9.12 Stability of radicals: bond dissociation energies and radical stabilisation energy 110
9.13 Reactivity and selectivity: the Hammond postulate for hydrogen abstraction 111

9.14 Radical addition to C=C; regiochemistry, polarity and Baldwin–Beckwith rules 111
9.15 Radical chain reactions: initiation, propagation, termination, inhibition 112
9.16 Radical polymerisation 112
9.17 Fragmentation and rearrangement of radicals 113
9.18 Other named radical reactions and their key features 113
9.19 Stereochemistry of radical reactions 114
9.20 Additional validated mechanisms 115
9.21 Problems on free radicals 116

Part 10 Carbenes: Structure, Generation, Singlet and Triplet States, Reactions 124
10.1 Singlet and triplet carbenes 124
10.2 Generation of carbenes 124
10.3 Reactions of carbenes 125
10.4 Mechanisms (arrow-validated) 125
10.5 The Reimer–Tiemann and carbylamine reactions in detail 126
10.7 Electronic structure in depth: why CH_2 is a triplet and CF_2 a singlet 127
10.8 Generation of carbenes: a complete catalogue 128
10.9 Reactions of carbenes: a systematic survey 128
10.10 Carbenoids: the ‘free carbene’ is the exception 129
10.11 Stable carbenes: from curiosity to catalysts 130
10.12 Evidence for carbenes and experimental techniques 130
10.13 Additional validated mechanisms 131
10.14 Problems on carbenes 132

Part 11 Nitrenes: Structure, Generation, Reactions and the Rearrangements that Mimic Them 138
11.1 Singlet and triplet nitrenes 138
11.2 Generation of nitrenes 139
11.3 Rearrangements that proceed through nitrene-like (nitrenoid) steps 139
11.4 Reactions of nitrenes: summary 140
11.6 Electronic structure of nitrenes, nitrenoids and nitrenium ions 140
11.7 Generation of nitrenes: a complete catalogue 141
11.8 Reactions of nitrenes in detail 141
11.9 The nitrogen-migration family: scope, mechanism, stereochemistry, applications 142
11.10 Choosing the right rearrangement: a decision table 143
11.11 Additional validated mechanisms 143
11.12 Problems on nitrenes 144

Part 12 Benzyne (Arynes): Structure, Generation, Evidence and Reactions 150
12.1 Structure and bonding 150
12.2 Generation of benzyne 150
12.3 Mechanisms (arrow-validated) 151
12.4 Evidence for the benzyne mechanism 152
12.5 Regiochemistry of nucleophilic addition to substituted benzynes 152
12.6 Reactions of benzyne 152
12.8 Electronic structure of benzyne in depth 153
12.9 Generation of benzyne: complete catalogue 153
12.10 Evidence for benzyne as an intermediate: Roberts, trapping, isotope effects 154
12.11 Regiochemistry of nucleophilic addition to substituted benzynes: Roberts and Gilchrist rules 154
12.12 Reactions of benzyne: a systematic classification 154
12.13 Heteroarynes and cycloalkynes 155
12.14 Three pathways to nucleophilic aromatic substitution compared 155
12.15 Additional validated mechanisms 156
12.16 Problems on benzyne 156

Part 13 Radical Ions: Radical Anions and Radical Cations 163
13.2 Dissolving-metal reductions: regiochemistry and scope 164
13.3 Radical cations: mass spectrometry, anodic chemistry, photoredox 164
13.4 Radical anions in substitution: $S_{RN}1$ and electron-transfer mechanisms 165
13.5 Additional validated mechanisms 165
13.1 Problems on radical ions 166

Part 14 Reaction Intermediates Compared, and Integrated Problems 168
14.1 The six intermediates at a glance 168
14.2 Decision guide: which intermediate does this reaction go through? 168
14.3 Integrated problems 169

Answers Part 3 — Leaving-Group Problems: Answers and Explanations 170

Answers Part 7.1 — Counting Problems: Answers and Explanations 173

Part 0 Reaction Intermediates: Bond Fission, Intermediates, Electrophiles and Nucleophiles

0.1 Homolytic and heterolytic bond fission

A covalent A–B bond can break in two ways. In **homolysis** the bonding pair is shared equally: each fragment keeps one electron, giving two radicals. Single-electron movement is shown with **fishhook** (half-headed) arrows. In **heterolysis** the whole pair goes to one fragment, giving a cation and an anion; the electron pair movement is shown with a **full-headed** curved arrow that starts at the bond and ends at the atom that keeps the pair. Reaction arrows (→, ⇌) are never used for electron flow.

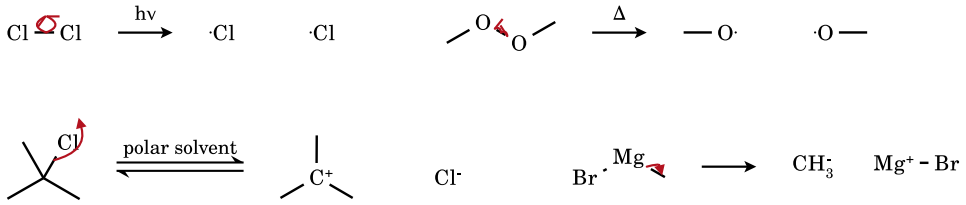


Figure 0.1 Homolysis (fishhook arrows; Cl_2 with light, a peroxide with heat) and heterolysis (full-headed arrows; an alkyl halide in a polar solvent, a C–metal bond).

- **Homolysis** is favoured by non-polar bonds (Cl–Cl, O–O, C–C, N=N of azo compounds), non-polar solvents or the gas phase, heat or UV light, and by radical-stabilising substituents on the fragments. Typical bond dissociation enthalpies (kcal mol^{-1}): Cl–Cl 58, Br–Br 46, I–I 36, HO–OH ≈ 51, RO–OR ≈ 38, CH_3–CH_3 90, H–CH_3 105, H–$C(CH_3)_3$ ≈ 96. Weaker bond ⇒ lower temperature needed.
- **Heterolysis** is favoured by polar bonds (large $\Delta\chi$), polar protic solvents that solvate both ions (hydrogen bonding to the anion, dipole/lone-pair interaction with the cation), a stable cation and a stable anion, and Lewis-acid assistance. In the gas phase heterolysis costs far more energy than homolysis; the solvation energy is what makes it feasible in solution.
- **Direction of heterolysis**: the electron pair stays with the more electronegative atom / the better leaving group (R–X → $R^+ + X^-$; R–MgBr → $R^- + MgBr^+$ with carbon taking the pair because C is more electronegative than Mg).
- **Typical initiators of homolysis**: peroxides (benzoyl peroxide, di-tert-butyl peroxide), azobisisobutyronitrile (AIBN, loses N_2 at 60–80 °C), halogens with light, N-bromosuccinimide with light/peroxide.

0.2 The reaction intermediates at a glance

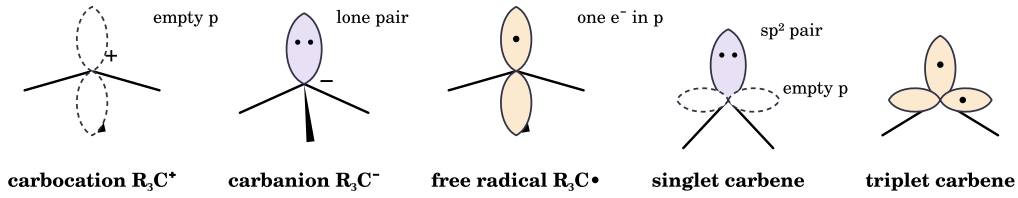


Figure 0.2 Orbital cartoons of the intermediates treated in this book (R = substituent). Solid lobes = occupied; dashed lobes = empty.

Intermediate	Valence e^- at the centre	Shape / hybridisation	Typical generation	Characteristic fate
Carbocation (carbenium)	6 (sextet)	planar, sp^2; linear sp for vinyl, acylium	heterolysis of R–X, ROH_2^+; H^+ + alkene; diazonium N_2 loss	nucleophile capture, E1 ($-H^+$), rearrangement, addition to π bonds
Carbanion	8 (3 bonds + lone pair)	pyramidal sp^3 (alkyl); planar when delocalised; sp (acetylide)	deprotonation; metal–halogen exchange; decarboxylation; Grignard/organolithium C–M	protonation, SN2, addition to C=O, E1cB, elimination
Free radical	7	planar (or shallow pyramid), sp^2; SOMO is a p orbital	homolysis (heat/light/initiator); single-electron transfer	chain propagation (abstraction/addition), recombination, disproportionation
Carbene	6 (2 bonds + 2 nonbonding e^-)	singlet: bent sp^2, pair + empty p; triplet: bent/linear, two unpaired e^-	α-elimination ($CHCl_3$/KOH); diazo or diazirine loss of N_2; carbenoid (Zn/Cu, CH_2I_2)	cyclopropanation, C–H insertion, ylide formation, Wolff rearrangement

Nitrene	6 (1 bond + 2 lone-pair-type e^- sets)	R–N: singlet or triplet (triplet usually ground state)	azide photolysis/pyrolysis; α-elimination from N-haloamides (concerted in Hofmann/Curtius)	insertion, aziridination, rearrangement, dimerisation
Benzyne (aryne)	ring C with in-plane 'third' bond	strained, planar; extra bond from sp^2 overlap in plane	ArX + $NaNH_2$/KNH_2; anthranilic acid + RONO; 2-(Me_3Si)C_6H_4OTf + F^-	nucleophilic addition, Diels–Alder, [2+2]
Radical ion	odd electron + charge	delocalised SOMO in π^* (anion) or π (cation)	Na/NH_3 (Birch), Na + ketone/nitrile, electrolysis, photo-induced ET	protonation, dimerisation, fragmentation, further e^- transfer

0.3 Electrophiles and nucleophiles (definitions)

A **nucleophile** is a species that donates an electron pair to form a new bond to an electron-poor atom (Lewis base character in a bond-forming step). An **electrophile** is a species that accepts an electron pair to form a new bond (Lewis acid character). In curved-arrow language, the arrow always leaves a nucleophile and arrives at an electrophile. Basicity and acidity refer to proton transfer (equilibrium); nucleophilicity and electrophilicity refer to bond formation to atoms other than H (rate). Only the *definitions and classification* are treated here; no ranking of nucleophilicity is attempted.

Class	Nucleophiles	Electrophiles
Anions	HO^-, RO^-, X^-, CN^-, RS^-, N_3^-, R^- (carbanions, enolates), H^- (hydride donors)	—
Neutral with lone pair	H_2O, ROH, NH_3, RNH_2, R_3N, R_2S, R_3P, ROR	—
π systems	alkenes, alkynes, arenes, enols/enol ethers/enamines (C-nucleophiles)	C=C bearing EWG (Michael acceptors), C ≡ C–EWG, arenes with NO_2 (SNAr)
σ bonds	C–M (M = Li, Mg, Zn), B–H, Al–H, Si–C (allyl-/aryl-silanes)	polarised C–X (X = halogen, OTs) – carbon is electrophilic
Cations	—	H^+, R^+, ArN_2^+, NO_2^+, NO^+, X^+ (from X_2), Hg^{2+}, Ag^+, acylium RCO^+, iminium, oxocarbenium
Neutral, electron-deficient	—	BF_3, $AlCl_3$, SO_3, $FeBr_3$, $ZnCl_2$ (Lewis acids); carbenes (singlet); R–C=O, C=N, C ≡ N carbon; CO_2; epoxides; α,β-unsaturated carbonyls
Radicals	•CR_2OH, •$CH_2CH(OH)$… (nucleophilic, SOMO high)	•CF_3, •CCl_3, Br•, RC(=O)O•, •$CH(CO_2R)_2$ (electrophilic, SOMO low)

AMBIDENT NUCLEOPHILES (TWO REACTIVE SITES)

- CN^- (C or N), NO_2^- (N or O), SCN^- (S or N), enolates (C or O), phenoxide (O or ring C), thiourea/thioamide (S or N). The site of attack depends on the electrophile and conditions (hard/soft, ion pairing, solvent); e.g. R–Br + KCN gives mainly R–CN, R–Br + AgCN gives mainly R–NC (isonitrile) because Ag^+ ties up the carbon end and pushes S_N1-like attack through N; enolate + R–X (soft) gives C-alkylation; enolate + R_3SiCl or RCOCl (hard) favours O.
- Ambident *electrophiles*: α,β-unsaturated carbonyls (C=O carbon, 1,2-attack; β-carbon, 1,4-attack), allyl halides (S_N2 vs S_N2'), epoxides, aldehyde/imine carbons.

Identification routine. (1) Draw lone pairs and π bonds; (2) mark atoms with δ^+ (attached to electronegative atoms) or an empty orbital; (3) join the most electron-rich site to the most electron-poor site; (4) check that every atom keeps a legal valence and the charges add up. Species such as carbenes are *ambiphilic* (singlet carbenes have a lone pair and an empty p orbital) and radicals can behave as either, according to the energy of the SOMO.

Part 1 Carbocations: Definitions, Classification, Geometry and Bonding

1.1 Definitions: carbocation, carbenium ion, carbonium ion

Carbocation is the general term for any species with a positively charged carbon atom. Two families are distinguished by the *coordination number* of that carbon:

- **Carbenium ions** (classical, tricoordinate): the positive carbon has **three** substituents and an empty p orbital, e.g. CH_3^+, $(CH_3)_3C^+$, $C_6H_5CH_2^+$. Formally they are derived from a neutral carbene or an alkane by removal of H^-. These are the ions of ordinary organic mechanisms (S_N1, E1, electrophilic addition).

- **Carbonium ions** (non-classical, hypercoordinate): the positive carbon is **five- (or higher)** coordinate through a three-centre two-electron (3c–2e) bond, e.g. CH_5^+, $C_2H_7^+$; formally derived by protonation (or alkylation) of an alkane σ bond. The 2-norbornyl cation is a bridged (σ-delocalised) species of this bonding type.
- The older usage 'carbonium ion' for *every* carbocation (including t-Bu^+) is obsolete; the two names must not be interchanged. In this book 'carbocation' is the generic term, 'carbenium' is used when precision is needed, and 'carbonium' always means a hypercoordinate (3c–2e) ion.

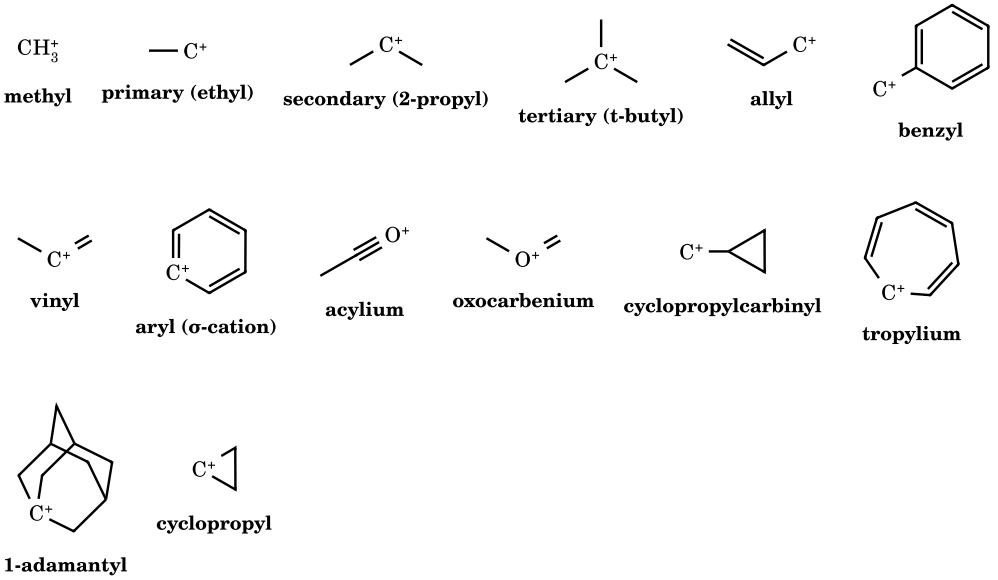


Figure 1.1 Gallery of carbocation types. The skeletal convention is used: hydrogens on carbon are not drawn; a symbol 'C⁺' is a cationic carbon with its implicit hydrogens (e.g. 'C⁺' in the ethyl cation is CH_2^+).

1.2 Geometry, hybridisation and the empty p orbital

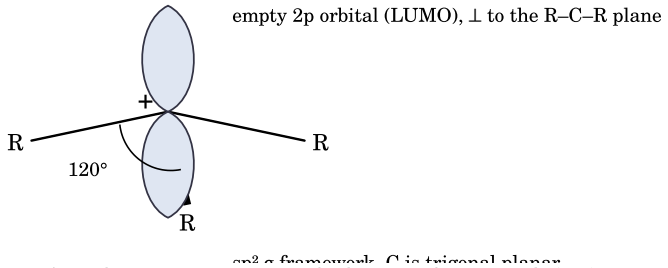


Figure 1.2 A carbenium ion: trigonal-planar σ framework (sp^2), empty 2p orbital perpendicular to the plane.

- **Planarity.** The three σ bonds use sp^2 hybrids (120°); the remaining p orbital is empty and is the LUMO. Planarity places the empty orbital in a position to be attacked from *either* face – the basis of racemisation in S_N1.
- **Bond lengths.** C^+–C bonds are shorter than normal $C(sp^3)$–$C(sp^3)$ bonds because of greater s character and hyperconjugative double-bond character.
- **Linear cations.** Vinyl ($R–C^+=CR_2$) and acylium ($R–C\equiv O^+$) cations have an sp carbon (180°); the vinyl cation's empty p orbital lies *in* the plane of the double bond, orthogonal to the π bond, so the π system cannot stabilise it by resonance. The phenyl (aryl) cation has its empty orbital in the plane of the ring, orthogonal to the aromatic π system – hence it is exceptionally unstable.
- **Pyramidalisation is penalised.** If the carbon is forced out of planarity (bridgehead of a small bicycle), overlap with substituent σ orbitals is poor and the p character is lost: 1-norbornyl (bicyclo[2.2.1]hept-1-yl) cations are very hard to form, whereas the larger, more flexible 1-adamantyl and bicyclo[2.2.2]oct-1-yl ions form far more easily.
- **Electrophilicity.** The empty orbital makes a carbocation a strong electrophile and Lewis acid: gas-phase hydride affinities (recalled, kcal mol^{-1}, approximate): CH_3^+ ≈ 312, i-Pr^+ ≈ 249, t-Bu^+ ≈ 231, benzyl ≈ 238, allyl ≈ 256 – the smaller the hydride affinity, the more stable the cation relative to its hydrocarbon. In solution Mayr's E parameter ranks benzhydrylium ions from E ≈ −7 (strongly donor-substituted) to E ≈ +6 (Ph_2CH^+).

1.3 Orbital overlap requirements and delocalisation of charge

A cation is stabilised to the extent that electron density from neighbouring orbitals can flow into its empty p orbital. Three kinds of donor exist and each has a geometric requirement: (i) **lone pairs** (n → p) on an attached heteroatom (OR, NR_2, SR, halogen): the lone-pair orbital must be parallel to the empty p orbital; (ii) **π bonds** (π → p) in an allyl, benzyl or vinyl system: the π orbitals must be parallel (coplanar σ frameworks); (iii) **σ bonds** ($\sigma_{C–H}$, $\sigma_{C–C}$ → p) – hyperconjugation: the donor σ bond must be close to parallel to the empty p orbital, i.e. roughly perpendicular to the plane of the cation (dihedral angle near 0°; at 90° the overlap is zero).

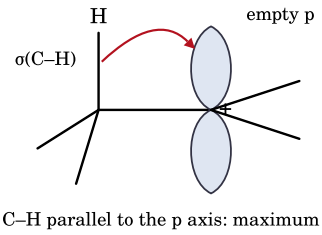
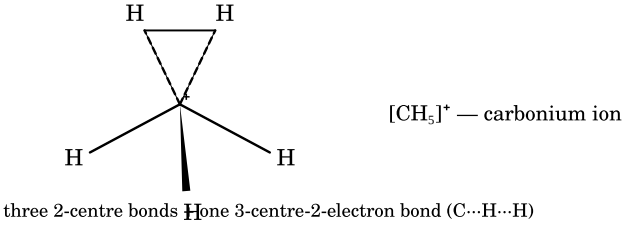


Figure 1.3 Hyperconjugation: the neighbouring σ(C–H) bond donates into the empty p orbital when the two are parallel.

Localised versus delocalised charge. In t-Bu^+ the charge is formally on one carbon but is shared with nine hydrogens by hyperconjugation; in allyl and benzyl cations resonance places partial positive charge on two or more carbons (shown with double-headed arrows between contributors, never with reaction arrows); in σ-bridged non-classical ions (2-norbornyl, cyclopropylcarbinyl, phenonium) the charge is spread over three or more carbons through *partial bonds* and no single Lewis structure is adequate. Resonance contributors differ only in electron positions; the nuclei do not move. If atoms move, the species are different (isomers, rapidly equilibrating or bridged).

1.4 Hypercoordinate (carbonium) ions



$[CH_5]^+$ — carbonium ion

Figure 1.4 CH_5^+: three ordinary C–H bonds and a three-centre two-electron C···H_2 unit (H_2 bound side-on to a CH_3^+ fragment).

Protonation of methane in a superacid or in a mass spectrometer ion source forms CH_5^+. High-level calculations and gas-phase infrared spectra show a structure with a CH_3^+ unit and a bound H_2, but with extremely facile exchange of all five hydrogens: it is a fluxional, hypercoordinate species with only *eight* valence electrons for five bonds. A carbonium ion is therefore **not** simply a 'more substituted' or 'more stable' carbenium ion; its 3c–2e bonding is a different bonding type. Their stability cannot be placed on the same 3° > 2° > 1° scale. Their usual fate is loss of H_2 (giving a carbenium ion; Part 2.E6), loss of a hydrocarbon fragment, or rapid proton scrambling.

1.5 Ion pairs, solvent and counterion

Ionisation passes through **contact (intimate) ion pairs** R^+X^-, then **solvent-separated ion pairs** $R^+\|X^-$, then free ions (Winstein). Nucleophile capture from the contact pair is stereo-selective (the counterion shields one face), from the solvent-separated pair largely statistical. The solvent ionises the substrate by (a) hydrogen bonding to the leaving group (protic solvents, TFE, HFIP are best) and (b) a high dielectric constant that screens charge; the nucleophilicity of the solvent matters much less in S_N1 than in S_N2. A **non-nucleophilic, weakly coordinating counterion** (SbF_6^-, $B(C_6F_5)_4^-$, carboranes, $Al(OC(CF_3)_3)_4^-$) and a non-basic solvent (SO_2ClF, SO_2, CH_2Cl_2) allow even tertiary cations to be observed (Olah, 1960s onward).

> **Scope and honesty note.** This book's definitions follow current IUPAC practice and standard physical-organic-chemistry usage. The IUPAC Gold Book, the textbooks listed in Part 17 and the archive.org page requested were not accessible in this authoring environment; where a numerical value is quoted from memory it is marked 'recalled' or given as an order only. See the Source Map for what was and was not consulted.

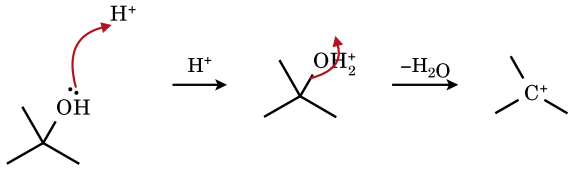
Part 2 Generation of Carbocations

Each of the following mechanisms was checked by computer: the curved arrows were applied to the drawn reactant graph and the result compared with the drawn product (valence, charge, atom conservation). Examples are grouped by generation route: 2.A alcohols, 2.B alkyl halides and sulfonates, 2.C alkenes/dienes/alkynes, 2.D solvolysis, 2.E other routes, 2.F amines with nitrous acid, 2.G aromatic compounds (arenium ions), 2.H Lewis-acid-assisted ionisation, ether cleavage and aromatic cations.

2.A Carbocations from alcohols

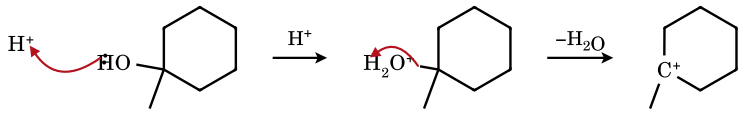
Alcohols do not ionise directly because HO^- is a poor leaving group (conjugate acid H_2O, $pK_a \approx 15.7$). Protonation, or coordination to a Lewis acid, converts it into H_2O (pK_a of $H_3O^+ \approx -1.7$) and ionisation then becomes possible *only if* the cation is stabilised enough: tertiary, benzylic, allylic or α-heteroatom-substituted. Primary and unactivated secondary alcohols generally react by concerted routes (E2, S_N2, or ionisation with simultaneous 1,2-shift) and are drawn that way below. Every arrow starts at an electron pair (lone pair or bond) and ends where that pair finishes.

2.A1 tert-Butyl alcohol, dilute strong acid



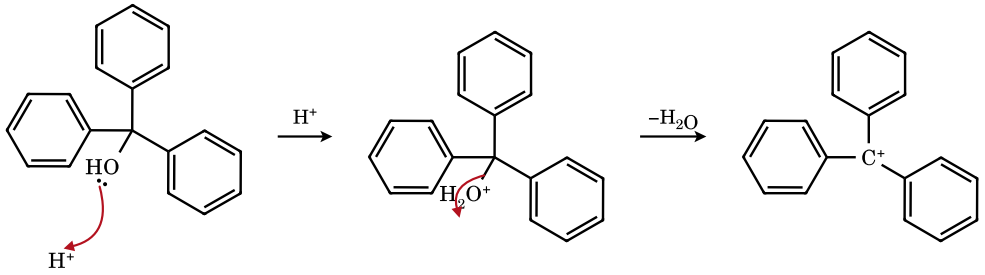
Rate-limiting loss of water from the oxonium ion gives the planar tert-butyl cation (nine hyperconjugating C–H bonds). In water the cation is captured faster than it eliminates unless the mixture is heated.

2.A2 1-Methylcyclohexanol, H_2SO_4



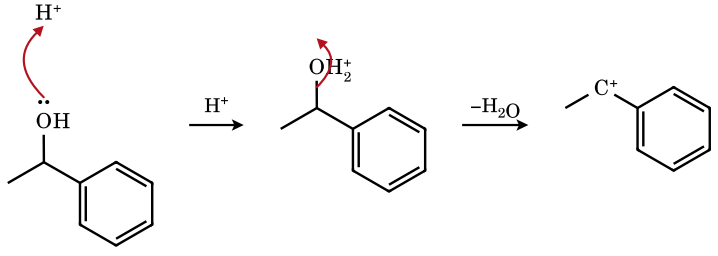
A tertiary ring cation: the cationic carbon is sp^2. E1 loss of H^+ competes and gives 1-methylcyclohexene (the Zaitsev alkene) on warming.

2.A3 Triphenylmethanol, conc. H_2SO_4



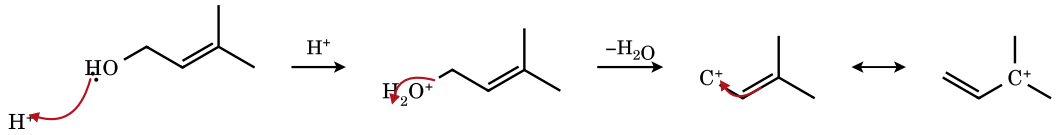
The yellow trityl cation is delocalised into three rings, which twist about 30° out of the central plane (propeller) because ortho-H···ortho-H contacts forbid full planarity. Isolable as BF_4^- or ClO_4^- salts.

2.A4 1-Phenylethanol, HBr (conc.)



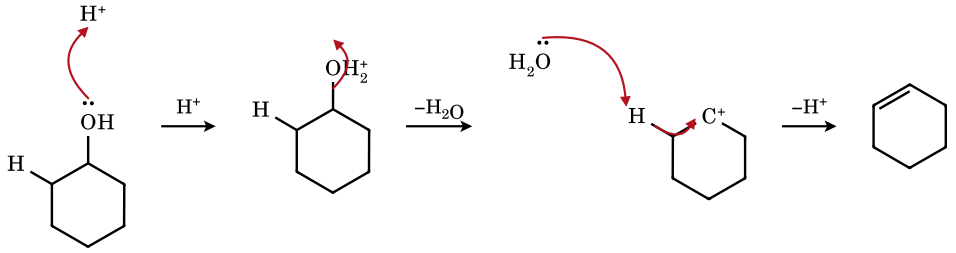
A secondary benzylic cation, stabilised by +M delocalisation into the ring (charge at o- and p-carbons). Bromide then captures it (S_N1).

2.A5 Prenol (3-methylbut-2-en-1-ol), H^+



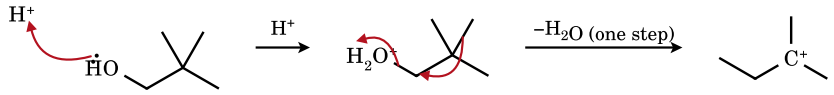
Ionisation puts the charge at C-1 (primary) *and*, by the double-headed resonance arrow, at C-3 (tertiary): one allyl cation, two contributors, with most of the positive charge on the tertiary terminus. Nucleophiles can attack at either end (S_N1 and S_N1').

2.A6 Cyclohexanol, 85 % H_3PO_4, heat (E1)



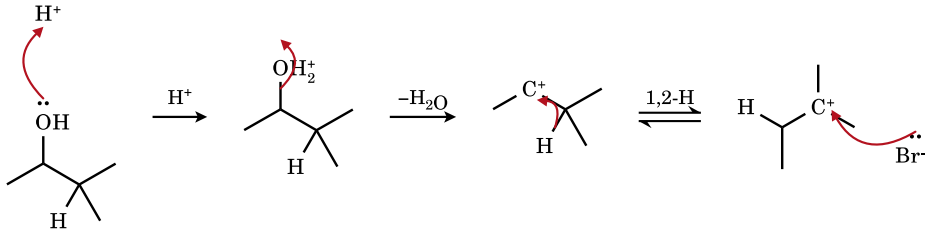
A secondary cation is formed only at elevated temperature and in poorly nucleophilic acid (H_3PO_4, HSO_4^-). The base removes a β-H whose C–H bond overlaps the empty p orbital; the C–H pair becomes the new π bond (E1). H_3O^+ regenerates the catalyst.

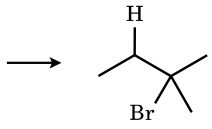
2.A7 Neopentyl alcohol, HBr, heat: ionisation with concerted CH_3 shift



A free primary cation $(CH_3)_3CCH_2^+$ is not an intermediate: the C–CH_3 σ bond anti to the departing water pushes in as the leaving group departs, so the tertiary tert-amyl cation is reached directly (anchimeric σ-assistance). The same applies to neopentyl halides and tosylates.

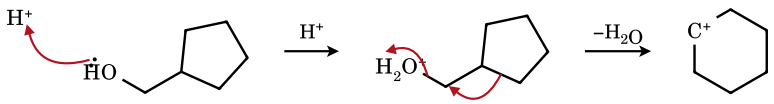
2.A8 3-Methylbutan-2-ol, conc. HBr





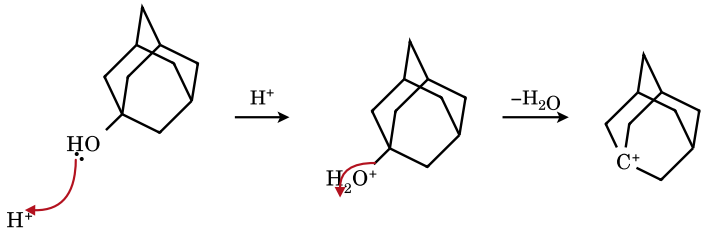
The secondary cation is formed with (or just before) the hydride shift: C–H (σ) → empty p. The tertiary cation is lower in energy and is captured by Br^-: the major product is 2-bromo-2-methylbutane, not 2-bromo-3-methylbutane.

2.A9 Cyclopentylmethanol, H^+: ionisation with ring expansion



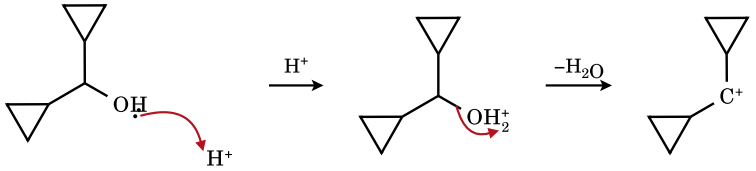
The ring C–C bond migrates to the departing CH_2 as water leaves, so the ring grows from five to six atoms and a secondary cation results; no primary cation intervenes. Ring strain relief is small (cyclopentane → cyclohexane) – the driving force is mainly 1° → 2°.

2.A10 1-Adamantanol, H_2SO_4 (bridgehead cation)



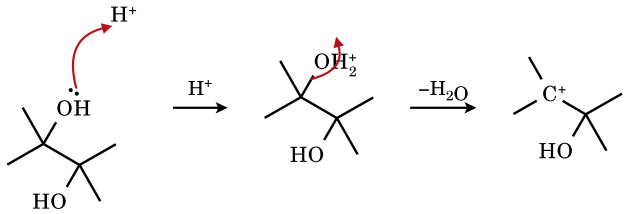
The 1-adamantyl cation is a tertiary bridgehead cation that forms readily: the cage is rigid, strain-free in the cation and the three C–C bonds at C-1 are held in good overlap with the empty orbital ($\sigma_{C–C}$ hyperconjugation). Contrast with bicyclo[2.2.1]hept-1-yl, where bridgehead cations are strongly disfavoured.

2.A11 Dicyclopropylmethanol, H^+



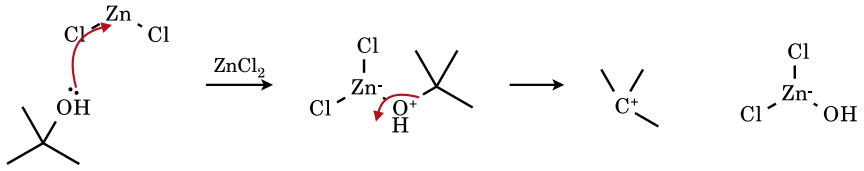
Each cyclopropyl ring donates its bent (Walsh-type) C–C σ bonds into the empty p orbital, which must lie in the *bisected* geometry. Dicyclopropylcarbinyl cations are far more stable than isopropyl- or even benzyl-type cations of the same substitution (see 5.5).

2.A12 Pinacol, 2,3-dimethylbutane-2,3-diol, H_2SO_4 (first step of the pinacol rearrangement)



The β-hydroxy tertiary cation is formed; its OH group is only inductively withdrawing and cannot stabilise it by resonance. A 1,2-methyl shift (Part 7) converts it into the oxygen-stabilised cation, which is the driving force.

2.A13 tert-Butyl alcohol, $ZnCl_2$/HCl (Lucas reagent), Lewis-acid ionisation

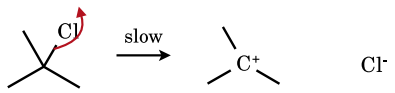


Zn(II) binds the oxygen lone pair; the C–O bond then ionises to the cation and the zincate $[HOZnCl_2]^-$. Tertiary alcohols give turbidity in seconds, secondary in minutes, primary not at all – the Lucas test is an S_N1 test.

2.B Carbocations from alkyl halides and sulfonates (ionisation)

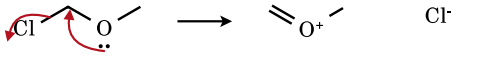
Heterolysis of a C–X bond gives R^+ X^-; in solution the first species is an *ion pair* (contact, then solvent-separated) and only later free ions. Ionisation is helped by (i) a good leaving group ($I^- \approx Br^- > Cl^- \gg F^-$; sulfonates better still), (ii) a strongly ionising, hydrogen-bond-donating solvent, (iii) a halophilic Lewis acid (Ag^+, $AlCl_3$, SbF_5) and (iv) a cation stabilised by hyperconjugation, resonance or neighbouring-group bridging. Each example below is checked atom-by-atom: arrows start at the electron pair that moves and end where it comes to rest.

2.B1 tert-Butyl chloride in water/ethanol: ion pair



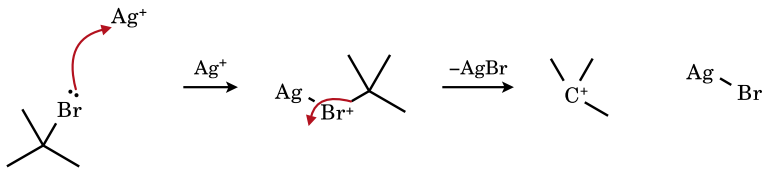
The rate-determining step: the σ(C–Cl) pair leaves with chlorine. In a solvent of high dielectric constant and H-bond donor ability (water, TFE, HCO_2H) the ion pair is stabilised and ionisation is faster by roughly five orders of magnitude than in ethanol (the Grunwald–Winstein Y scale is defined on exactly this substrate).

2.B9 Chloromethyl methyl ether (MOM-Cl)



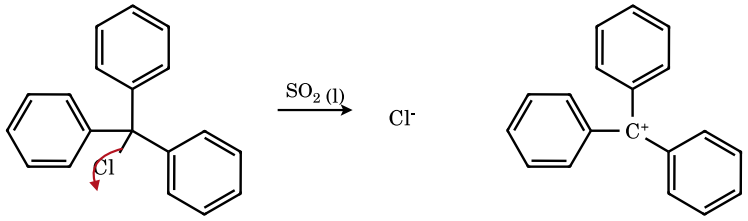
An α-heteroatom lone pair pushes out the halide: the 'carbocation' is really an oxocarbenium ion $CH_2{=}O^+CH_3$ in which every atom has an octet. α-Chloroethers ionise many orders of magnitude faster than simple alkyl chlorides of the same substitution.

2.B2 tert-Butyl bromide + $AgNO_3$ (Ag^+ assistance)



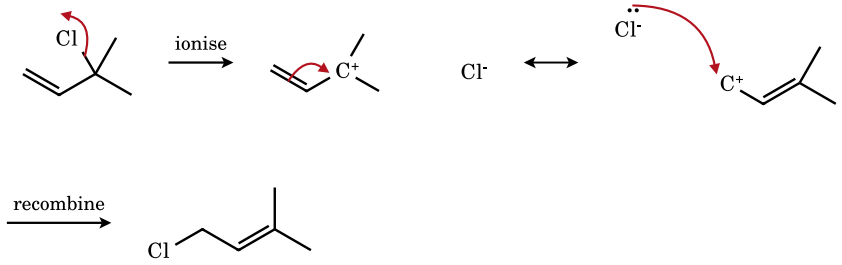
Ag^+ accepts a bromine lone pair (σ-complex), so the leaving group is effectively AgBr, which precipitates. This is why silver ion makes even primary neopentyl and secondary halides ionise – and rearrange.

2.B3 Trityl chloride in liquid SO_2



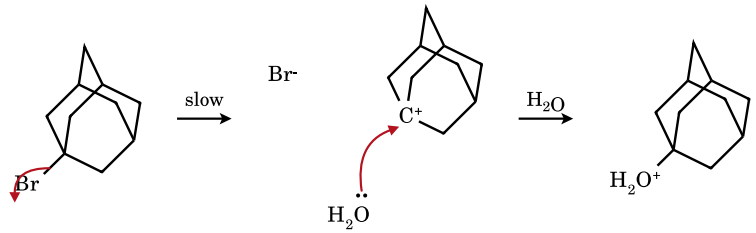
Ph_3CCl is a covalent solid but ionises in SO_2 (conducting yellow solution). The cation is stabilised by three aryl groups; compare Ph_2CHCl (ionises in the same solvents much more slowly) and $PhCH_2Cl$ (barely).

2.B4 3-Chloro-3-methylbut-1-ene ⇌ prenyl chloride: ion pair and allylic transposition



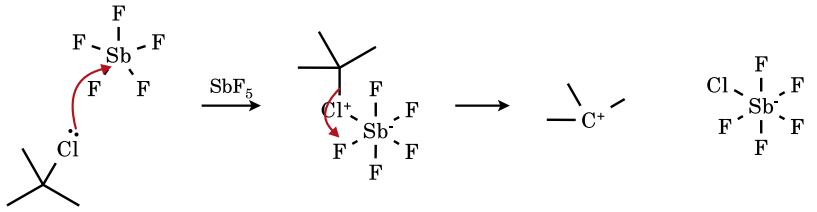
The same allyl cation is formed from either isomer. Chloride (kept nearby as an ion pair) returns to the less hindered primary terminus, giving the more substituted, thermodynamically more stable alkene (1,3-shift, S_Ni' / S_N1').

2.B5 1-Bromoadamantane in 80 % aqueous ethanol (bridgehead S_N1)



S_N2 is impossible at a bridgehead (no backside access) yet 1-adamantyl bromide solvolyses readily, because the adamantyl cage is nearly strain-free in the planar-ish cation. 1-Bromobicyclo[2.2.1]heptane, in contrast, is essentially inert (see 5.11).

2.B6 tert-Butyl chloride + SbF_5 in SO_2ClF, –78 °C (Olah)



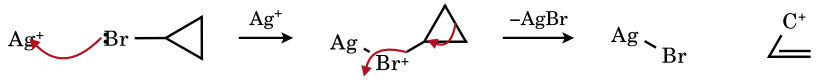
With a very strong halide acceptor and a non-nucleophilic medium the ion pair persists indefinitely: $(CH_3)_3C^+\ SbF_5Cl^-$ was observed by NMR (a single ^1H signal, the nine hydrogens equivalent by rapid methyl rotation), proving that a tertiary alkyl cation can be a real, long-lived species.

2.B7 Neopentyl bromide + $AgNO_3$ (concerted ionisation–methyl shift)



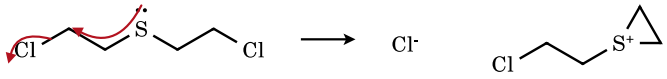
Neopentyl halides cannot undergo S_N2 (β-branching blocks backside attack) nor form a primary cation: when ionisation is forced, the anti-periplanar C–CH_3 bond migrates as bromide leaves.

2.B8 Cyclopropyl bromide + Ag^+: ionisation with disrotatory ring opening



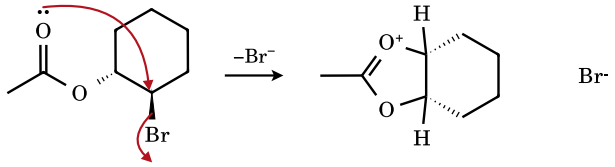
A cyclopropyl cation (sp^2 carbon in a 60° ring) is so strained that it is never formed as such: the C–C bond opposite the leaving group opens as the halide departs, giving the allyl cation (disrotatory, Woodward–Hoffmann, 2π electrons). Cyclopropyl halides solvolyse far more slowly than comparable secondary halides, and only with the assistance of the ring-opening.

2.B10 Sulfur mustard, $ClCH_2CH_2SCH_2CH_2Cl$: neighbouring-group participation



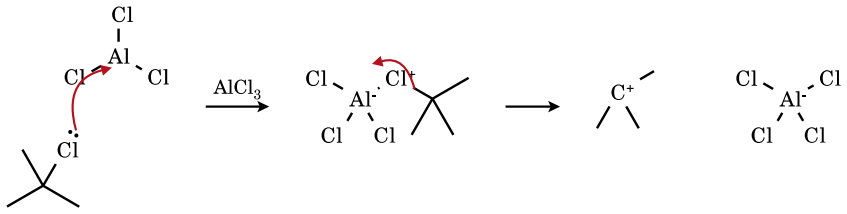
The β-sulfur displaces chloride intramolecularly (anchimeric assistance) to give a cyclic episulfonium (thiiranium) ion – a bridged cation in which charge sits mainly on sulfur and no open primary cation exists. It is a potent alkylating agent.

2.B11 trans-2-Bromocyclohexyl acetate: acetoxonium ion



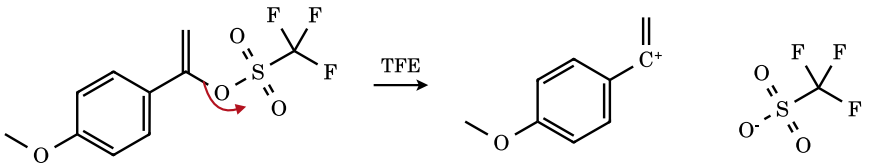
Only the *trans* (diaxial-capable) isomer reacts fast: the carbonyl oxygen attacks C-2 from the side opposite to Br, giving a 1,3-dioxolan-2-ylium (acetoxonium) ion; positive charge is shared by both oxygens and the carbon. Opening by a nucleophile at C-2 gives *net retention* (two inversions).

2.B12 tert-Butyl chloride + $AlCl_3$ (Friedel–Crafts alkylation)



$AlCl_3$ abstracts chloride to give t-$Bu^+\ AlCl_4^-$. With primary halides the complex R–Cl→$AlCl_3$ is itself the electrophile and may rearrange (see 2.E4).

2.B13 1-(4-Methoxyphenyl)vinyl triflate in CF_3CH_2OH: vinyl cation

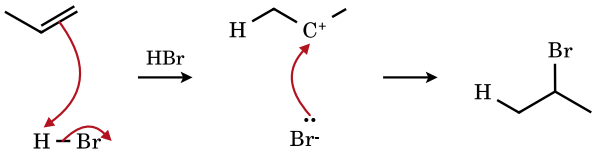


Vinyl (alkenyl) cations are real, linear sp-hybridised species but are high in energy; they form only from excellent leaving groups (triflate) and require stabilisation by an α-aryl (here a p-anisyl +M donor) or an α-alkyl group. The empty p orbital lies in the plane of the alkene, orthogonal to the π system.

2.C Carbocations from alkenes, dienes and alkynes (electrophilic addition)

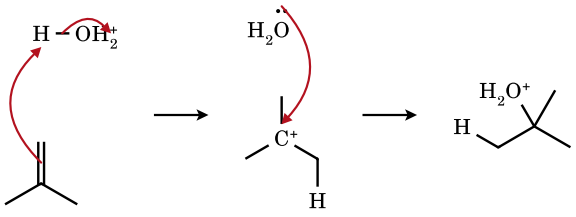
A π bond is a nucleophile. Its electron pair attacks the electrophile (H^+, a carbocation, X^+, Hg^{2+}) and forms a σ bond to *one* carbon; the other carbon is left electron-deficient. The proton adds so that the **more stable** cation (Markovnikov) is produced, and the arrow always runs from the π bond to the electrophile. When the electrophile has a lone pair that can reach back (Br, S, Hg), a **bridged onium ion** – not an open carbocation – is the intermediate, which is why no rearrangement and *anti* stereochemistry result.

2.C1 Propene + HBr



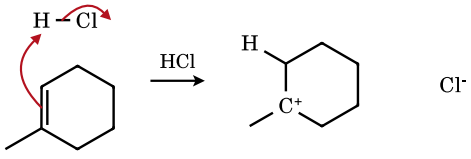
Protonation at C-3 puts the cation on the secondary C-2 (Markovnikov). The rate-limiting step is H^+ transfer from HBr to the alkene.

2.C2 2-Methylpropene + H_3O^+ (acid hydration)



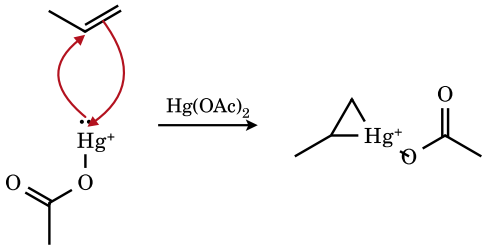
Only the tertiary cation is formed (the primary isobutyl cation would be ≈ 15 kcal/mol higher in energy). Capture by water and loss of H^+ gives tert-butyl alcohol; hydration is reversible (E1 reverse).

2.C3 1-Methylcyclohexene + HCl



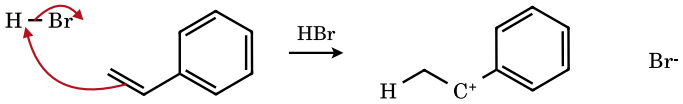
The proton adds to the less substituted alkene carbon, giving the tertiary ring cation. Capture by chloride from the open face, from either side, gives 1-chloro-1-methylcyclohexane (no stereocentre is created).

2.C7 Propene + $Hg(OAc)^+$ (oxymercuration)



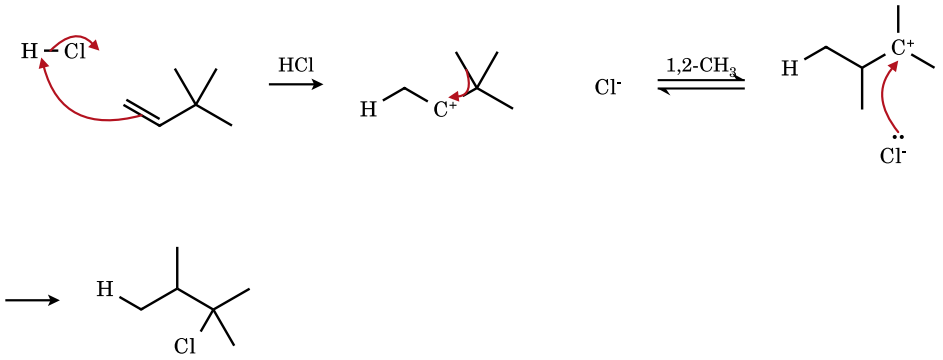
The mercurinium ion is bridged and delocalised; the nucleophile (H_2O) attacks the more substituted carbon (greater positive charge) with *anti* stereochemistry and, because no free cation forms, **without rearrangement**. $NaBH_4$ then replaces Hg by H.

2.C4 Styrene + HBr



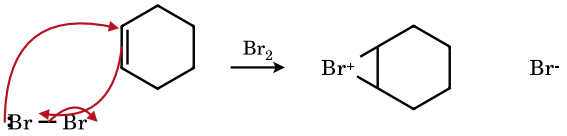
H^+ adds to the terminal CH_2 so that the cation is benzylic. The rate of addition to ring-substituted styrenes follows σ^+ ($\rho \approx -3$): a +M group in the para position speeds up the reaction.

2.C5 3,3-Dimethylbut-1-ene + HCl: addition with rearrangement



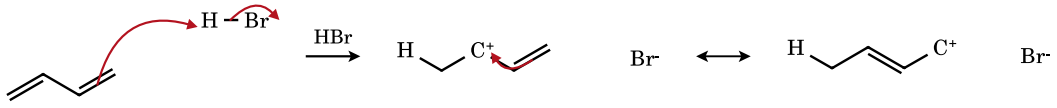
The secondary cation (neopentyl-type, with an adjacent quaternary carbon) is trapped in part by Cl^- (2-chloro-3,3-dimethylbutane, minor) but mostly undergoes a methyl shift to the tertiary cation: 2-chloro-2,3-dimethylbutane is the major product. A classic proof that alkene hydrohalogenation involves a cation.

2.C6 Cyclohexene + Br_2: bromonium ion



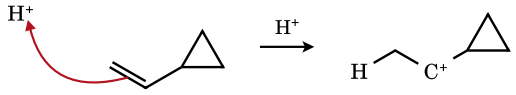
A bromine lone pair closes onto the carbon that lost the π electrons: the intermediate is a three-membered bromonium ion, not an open carbocation. Bromide then opens it from the face opposite to the ring (anti addition, trans-1,2-dibromocyclohexane).

2.C8 Buta-1,3-diene + HBr: allylic cation



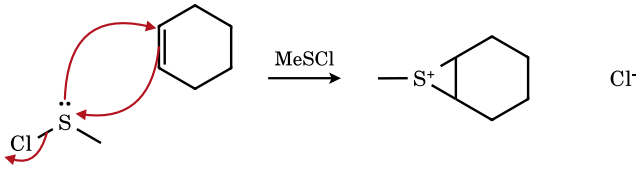
Protonation at the terminal carbon is preferred because the resulting cation is allylic (two contributors). Br^- can attack C-2 (1,2-adduct, formed faster at 0 °C – kinetic) or C-4 (1,4-adduct, more stable – thermodynamic); see Part 9.

2.C9 Vinylcyclopropane + H^+



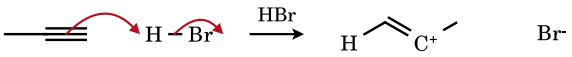
A cyclopropylcarbinyl cation results, strongly stabilised by the neighbouring C–C σ bonds of the ring (bisected conformation, Part 5.5): vinylcyclopropanes protonate faster than simple 1,1-disubstituted alkenes.

2.C10 Cyclohexene + MeSCl: thiiranium ion



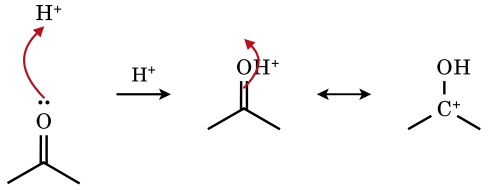
Sulfur's lone pair bridges the two carbons exactly as bromine does. Episulfonium ions give *anti*-addition products and can open at either carbon (Markovnikov or anti-Markovnikov, depending on substitution and solvent).

2.C11 Propyne + HBr: vinyl cation



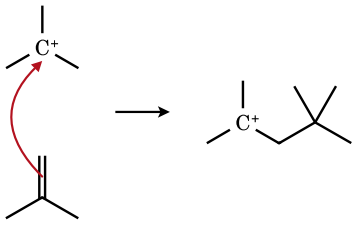
A secondary vinyl cation (linear, empty p in the alkene plane) is much less stable than a secondary alkyl cation, so alkynes are protonated more slowly than alkenes; with HX the addition is often termolecular (AdE3), avoiding a free vinyl cation.

2.C12 Acetone + H^+: oxocarbenium (protonated carbonyl)



A protonated carbonyl is a hydroxy-carbenium ion – a tertiary cation stabilised by the +M effect of OH. Resonance structure (right) shows why the carbon becomes electrophilic; acid-catalysed additions to C=O all start here.

2.C13 2-Methylpropene + t-Bu^+: C–C bond formation (cationic dimerisation)

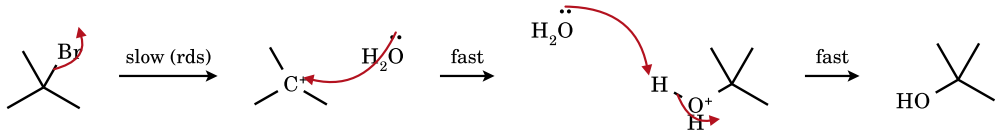


A carbocation is itself an electrophile for alkenes. The new tertiary cation can lose H^+ to give the diisobutylene isomers (2,4,4-trimethylpent-1-ene and -2-ene) or add a further isobutene (oligomerisation, polyisobutylene).

2.D Solvolysis and S_N1 substitution

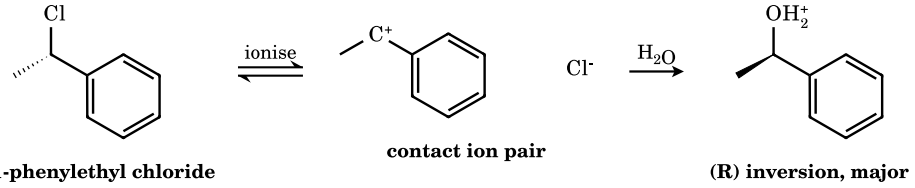
In solvolysis the solvent is both the ionising medium and the nucleophile. The kinetics are first order in substrate (rate = k[RX], independent of added nucleophile), and the stereochemical and product evidence is the fingerprint of a cationic intermediate: partial racemisation, rearranged skeletons, mixtures of S_N1 and E1 products, common-ion rate depression and σ^+ correlations. A rate law alone cannot prove a *free* cation: ion pairs and neighbouring groups often intervene, so the rate and the product-forming steps must be distinguished.

2.D1 tert-Butyl bromide in water: the full S_N1 sequence

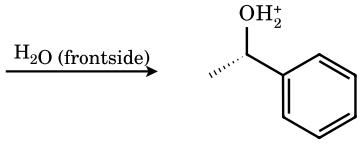


Rate = k[t-BuBr]. The cation is captured by the solvent (rate of capture ⩾ rate of formation, so capture is not kinetically visible) and a second water removes H^+. Competing E1 gives 2-methylpropene (≈ 10 % in water, more in ethanol or at higher temperature).

2.D2 Stereochemistry: (S)-1-phenylethyl chloride in aqueous acetone



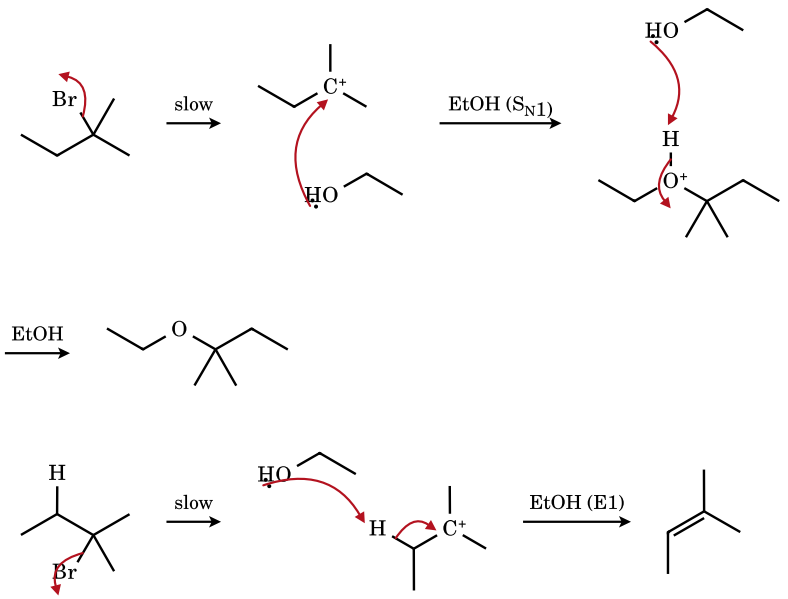
(S)-1-phenylethyl chloride — **contact ion pair** — **(R) inversion, major**



(S) retention, minor

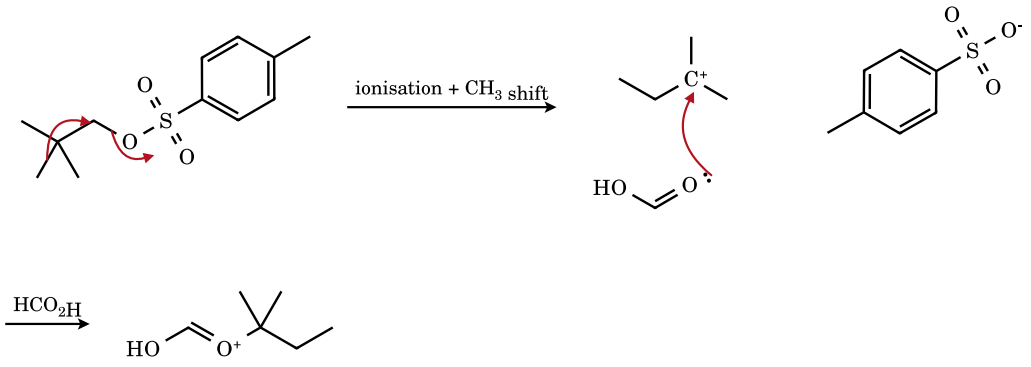
The planar cation can be attacked from either face, so racemisation is expected; observation gives *partial* racemisation with net inversion. The leaving chloride, still sitting on the front face in the contact ion pair, shields that face; the solvent attacks preferentially from the back. Free (solvent-separated) cations would give 50:50. The ratio is an experimental number, not a prediction.

2.D3 2-Bromo-2-methylbutane in ethanol: S_N1 and E1 from the same cation



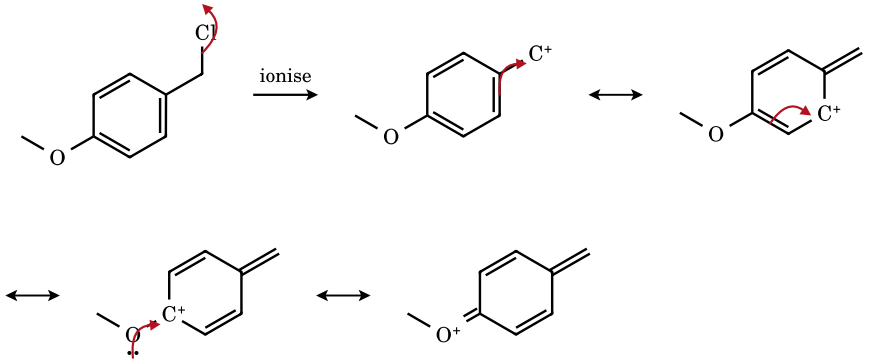
One cation, two destinations: nucleophilic capture at carbon (ether) or loss of a β-hydrogen (alkene). Higher temperature favours E1 (greater entropy gain). The more substituted alkene (2-methylbut-2-ene) dominates; the C–H bond that is lost must overlap the empty p orbital.

2.D4 Neopentyl tosylate in formic acid (formolysis): S_N1 with rearrangement



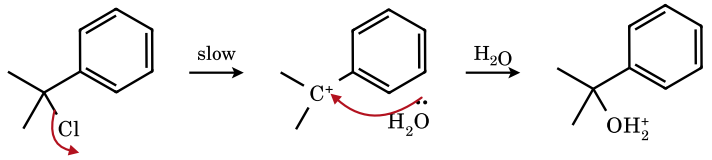
The only substitution product is tert-amyl formate (2-methylbut-2-yl formate); no neopentyl formate. A neopentyl system cannot do S_N2, so the ionising solvent pushes it through the σ-assisted pathway. The mixture also gives 2-methylbut-2-ene (E1).

2.D5 4-Methoxybenzyl chloride in water: why +M groups accelerate S_N1



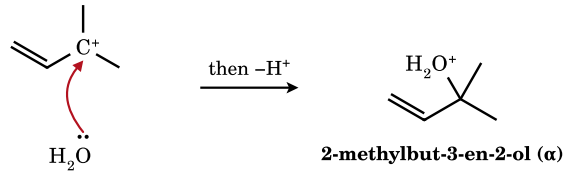
Ionisation puts + on the benzylic CH_2; three successive π-shifts carry it to C-4, where the OMe lone pair completes the octet: a p-quinone-methide oxocarbenium ion. The methoxy group lowers the barrier by tens of kJ mol^{-1} relative to H ($\sigma^+_{p\text{-}OMe} = -0.78$).

2.D6 2-Chloro-2-phenylpropane (cumyl chloride) in 90 % aqueous acetone



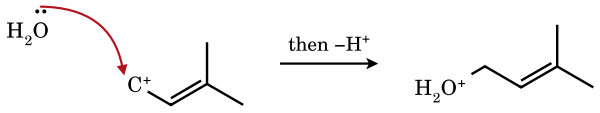
The tertiary benzylic cation is the standard system used to define σ^+ substituent constants (Brown–Okamoto, $\rho \approx -4.5$): the very large negative ρ shows that substantial positive charge is delocalised into the ring in the transition state.

2.D7 Prenyl chloride in water: S_N1 and S_N1' products from one allyl cation



2-methylbut-3-en-2-ol (α)

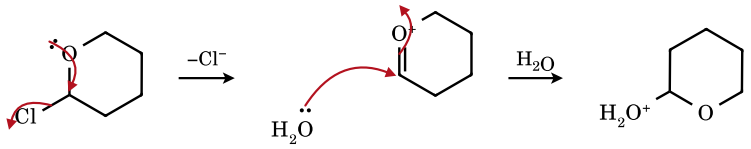
attack at C-2 (tertiary)



attack at C-4 (primary) **prenol (γ)**

The delocalised cation is attacked at either terminus, and both isomeric chlorides give the same two alcohols. Under kinetic control the tertiary alcohol is the major product (most of the positive charge resides on the tertiary carbon); prenol, which has the more substituted alkene, accumulates only if the tertiary alcohol can re-ionise (acidic, longer reaction times) – kinetic versus thermodynamic control (Part 9).

2.D8 2-Chlorotetrahydropyran: α-oxygen stabilisation

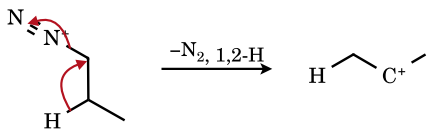


Glycosyl-type α-halo ethers ionise with ring-oxygen assistance (the axial lone pair overlaps $\sigma^*_{C–Cl}$, the anomeric effect in action). The cyclic oxocarbenium ion is attacked from either face: this is the basis of glycoside synthesis (Koenigs–Knorr, with Ag^+ activation).

2.E Other routes: diazonium ions, Lewis acids, hydride abstraction, acid-mediated ionisations

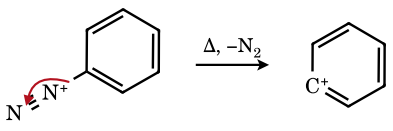
Some cations are produced neither from an alcohol/halide nor from an alkene. The common feature is a very good leaving group (N_2, H_2, H_2O, ROH) or a very strong electrophile (H^+ in superacid, Ph_3C^+, $AlCl_3$). Radical and concerted pathways are *not* counted here: e.g. the Hofmann–Löffler or Kolbe reactions are radical, and Tiffeneau–Demjanov, Curtius and Hofmann reactions proceed with migration concerted with loss of N_2 (no free cation).

2.E1 n-Propylamine + HNO_2 (via n-propyldiazonium)



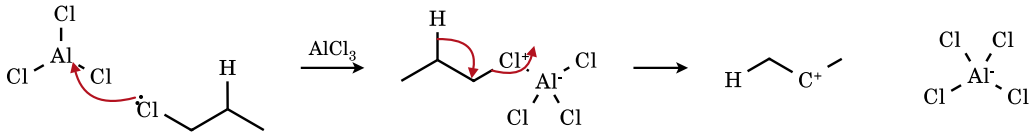
Alkyldiazonium ions are the most reactive leaving-group systems known (N_2, a thermodynamically very stable molecule). Even a primary one gives a mixture in which the 2-propyl derivatives (2-propanol, 2-chloropropane) outweigh the 1-propyl ones, together with propene, because the departing N_2 is pushed out by a neighbouring migrating H or alkyl group. The result is a classic reason why nitrous acid deamination of primary amines is synthetically useless.

2.E2 Benzenediazonium ion in hot water: aryl cation



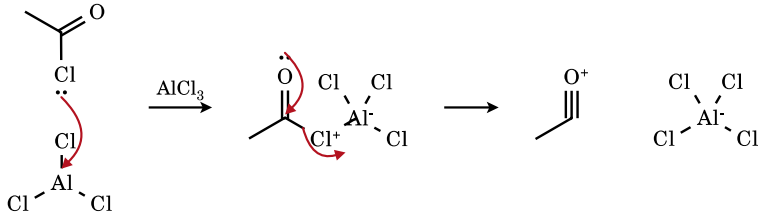
The aryl cation has its empty orbital in the plane of the ring (an sp^2 σ orbital orthogonal to the π system), so it gets no resonance help and is among the least stable carbocations; it is formed only because N_2 is such a good leaving group. It reacts with the nearest nucleophile (H_2O → phenol), even F^- in the Balz–Schiemann reaction.

2.E3 n-Propyl chloride + $AlCl_3$: 1,2-hydride shift during ionisation



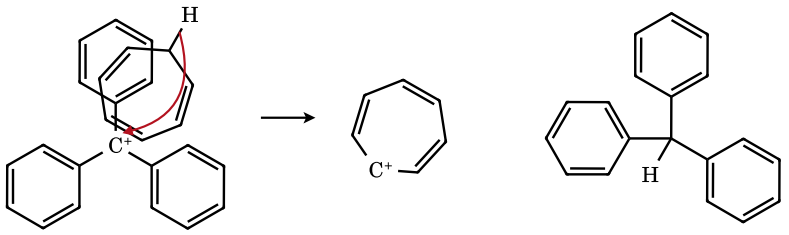
The polarised complex R–Cl→$AlCl_3$ transfers a hydride to the carbon that is losing Cl^-: isopropyl cation (as an ion pair with $AlCl_4^-$). This is why benzene + n-propyl chloride/$AlCl_3$ gives cumene, not n-propylbenzene.

2.E4 Acetyl chloride + $AlCl_3$: acylium ion



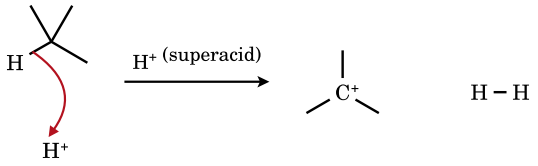
The oxygen lone pair drives out chloride and forms a third C–O bond; every atom in $CH_3–C\equiv O^+$ has an octet. Acylium ions are linear (sp-hybridised C) and do **not** rearrange, so Friedel–Crafts acylation gives unrearranged ketones; the ketone product deactivates the ring, which is why acylation (unlike alkylation) stops at one substitution.

2.E5 Hydride abstraction: cycloheptatriene + $Ph_3C^+\ BF_4^-$



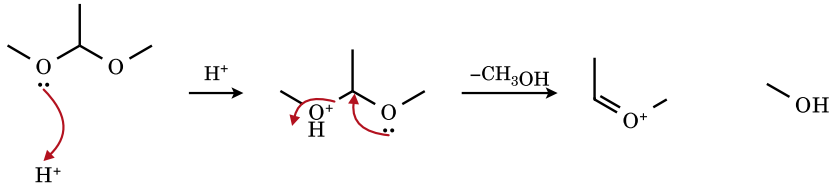
The C–H σ pair of the sp^3 CH_2 transfers as H^- to the trityl cation (an equilibrium that lies far to the right because the product is the aromatic 6π tropylium ion, pK_{R+} ≈ +4.7, against −6.6 for trityl). The seven-membered ring is planar, all seven C–C bonds are equal, and the positive charge is shared equally.

2.E6 Isobutane in $FSO_3H{\cdot}SbF_5$ (magic acid): σ-protonation then loss of H_2



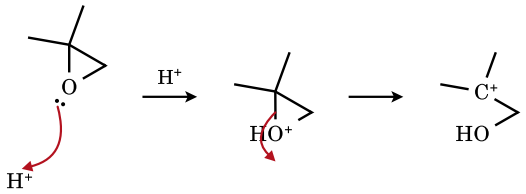
A C–H σ bond is a very weak nucleophile, but the proton of a superacid attacks it, forming a three-centre two-electron bond (a *carbonium* ion, $R_3C(H_2)^+$, see 1.4). Loss of H_2 leaves the carbenium ion. This is the basis of H/D exchange of alkanes in superacid and of alkane isomerisation (Olah).

2.E7 Acid hydrolysis of acetaldehyde dimethyl acetal



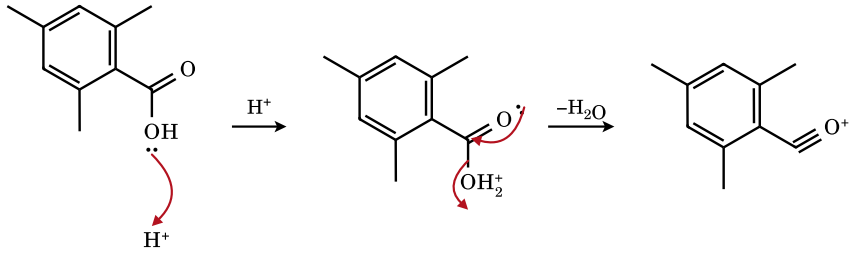
Protonation of one OMe turns it into a good leaving group (CH_3OH); the lone pair of the other oxygen pushes it out. The oxocarbenium ion is the intermediate that water attacks in the rate-limiting or following step: acetals are stable to base but labile to dilute acid.

2.E8 Acid-catalysed opening of 2,2-dimethyloxirane



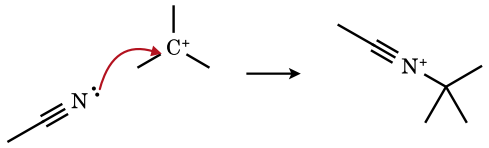
Protonation makes the oxirane a good electrophile; the C–O bond to the *more substituted* carbon is longer and weaker and breaks first, so the nucleophile (H_2O, ROH, X^-) ends up on the tertiary carbon (contrast base-promoted opening, S_N2 at the less hindered CH_2). The true transition state is a loose S_N2 with S_N1 character ('borderline'); a full tertiary cation is shown to indicate the charge distribution.

2.E9 2,4,6-Trimethylbenzoic acid in conc. H_2SO_4: $A_{AC}1$ acylium



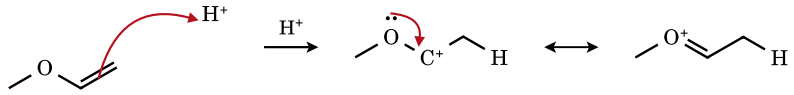
Ordinarily esterification/hydrolysis of benzoic acids goes by addition–elimination ($A_{AC}2$) through a tetrahedral intermediate. The two ortho methyl groups block nucleophilic addition to C=O, so the reaction switches to ionisation to a mesitoyl acylium ion, which is linear and so relieves the steric clash. Dissolving in H_2SO_4 and pouring into alcohol gives the ester in high yield (Newman).

2.E10 Ritter reaction: nitrile + tert-butyl cation



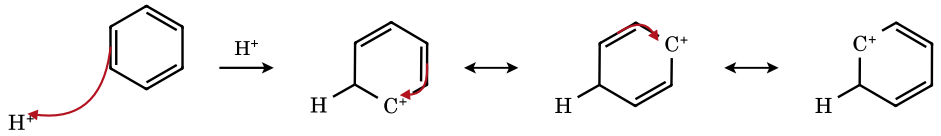
A nitrile nitrogen lone pair (sp-hybridised, weakly basic but unhindered) captures the cation: the nitrilium ion R–C ≡ N^+–R′ is attacked by water at carbon to give an N-tert-butylamide. Only cations that form readily (tertiary, benzylic) work.

2.E11 Methyl vinyl ether + H^+: protonation of an enol ether



Protonation of the β-carbon gives a cation that is immediately an oxocarbenium ion because the α-oxygen lone pair donates into the empty p orbital. Enol ethers are hydrolysed orders of magnitude faster than simple alkenes are hydrated for this reason (the rate-limiting step is protonation at carbon).

2.E12 Benzene + H^+ (superacid): the arenium (Wheland) ion



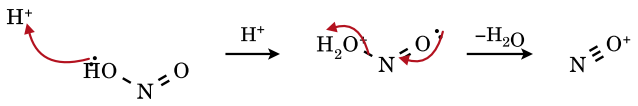
$C_6H_7^+$ is the intermediate in every electrophilic aromatic substitution. The positive charge is delocalised over the ortho and para carbons only (three contributors; the meta carbons carry none); the sp^3 CH_2 breaks the aromatic ring, which is why this step costs so much energy and is rate-determining.

2.F Carbocations from amines: diazotisation with nitrous acid

Nitrous acid (from $NaNO_2$ + HX, never stored) is the reagent that turns a **primary** amino group (–NH_2), the worst of leaving groups, into the best known one (N_2). The active electrophile in dilute mineral acid is the **nitrosonium ion NO^+**. What happens next depends on the type of amine:

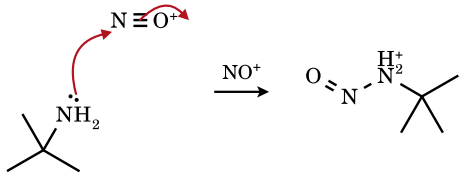
Amine	$NaNO_2$ / HCl, 0–5 °C	Carbocation?
Primary aliphatic RCH_2NH_2, R_3CNH_2	$R–N_2^+$ (not isolable) → loses N_2 at once	**Yes** (or a concerted shift/substitution with N_2 loss); mixture of alcohols, alkenes, halides; rearranged products common
Primary aromatic $ArNH_2$	$Ar–N_2^+$ $X^–$: stable in cold solution (resonance with the ring)	**No** at 0–5 °C. On warming in water the diazonium ion loses N_2 to give the aryl cation (phenol)
Primary benzylic $ArCH_2NH_2$	$ArCH_2N_2^+$ (not stable)	**Yes**: benzylic cation $ArCH_2^+$
Secondary R_2NH (and ArNHR)	N-nitrosamine $R_2N–N=O$ (yellow oil)	**No** (no N–H left to lose from N after nitrosation; the N–N bond is not an N_2-releasing diazonium)
Tertiary R_3N	R_3NH^+ $NO_2^–$ (salt); no stable product	**No** (tertiary aryl amines such as $PhNMe_2$ undergo C-nitrosation at the para carbon)

2.F1 Nitrosonium ion from nitrous acid



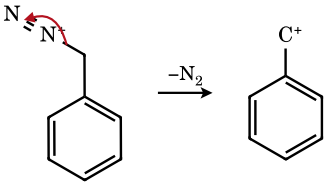
Protonation of the hydroxyl oxygen turns it into water; the oxygen lone pair of the N=O then forms a third N–O bond as water leaves, giving $N \equiv O^+$ (isoelectronic with CO and N_2). NO^+ is the electrophile that attacks amines (N-nitrosation).

2.F2 tert-Butylamine + NO^+: N-nitrosation (first step)



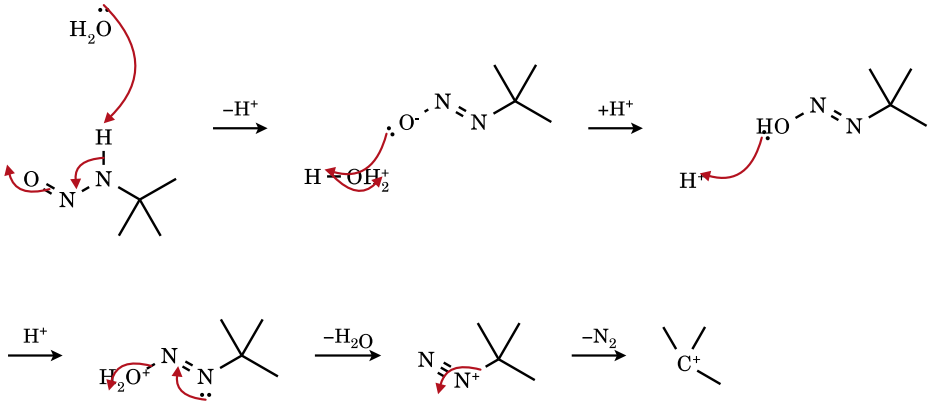
The amine nitrogen lone pair attacks nitrogen of NO^+; one $N \equiv O$ π bond moves onto oxygen. The N-nitrosoammonium ion then loses H^+ and tautomerises, as drawn in 2.F3.

2.F4 Benzylamine + HNO_2: benzylic cation



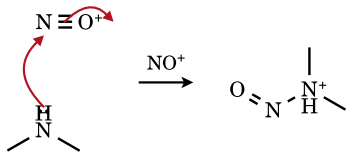
Benzyldiazonium ions lose N_2 at once, even at 0 °C, because the benzylic cation is resonance-stabilised. The cation is captured by water (benzyl alcohol); compare ArN_2^+, where the diazonium group is attached to an sp^2 carbon and is stable in the cold.

2.F3 …from N-nitroso-tert-butylammonium to the cation (R–NH–N=O → R–N=N–OH → $R–N_2^+$ → R^+)



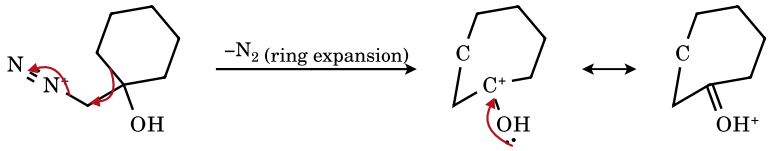
Loss of N–H proton and shift of the π bond gives the diazotate; protonation on oxygen (diazohydroxide R–N=N–OH), then a second protonation and loss of water gives the alkyldiazonium ion $R–N \equiv N^+$; its C–N σ pair leaves with N_2. For a tertiary alkyl group the product is the tert-butyl cation (captured by water or losing H^+). With a primary alkyl group the same sequence is followed by a concerted shift (2.E1).

2.F5 Secondary amine + NO^+: N-nitrosamine (no carbocation)



The nitrosoammonium ion can only lose its N–H proton to give $R_2N–N=O$; there is no hydrogen to move to oxygen to form a diazonium, so the sequence stops at the nitrosamine (a carcinogen class). No carbocation forms.

2.F6 Tiffeneau–Demjanov: 1-(aminomethyl)cyclohexanol + HNO_2 (diazonium, then ring expansion)

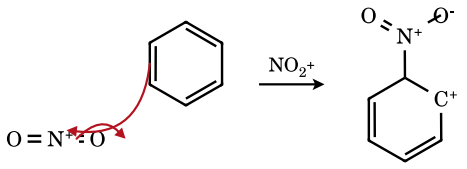


The ring C–C bond anti to the departing N_2 migrates to the CH_2 as N_2 leaves, so the six-membered ring becomes seven-membered; the cation produced is the protonated ketone (oxygen lone pair donates), which loses H^+ to give cycloheptanone. A free primary cation is never formed.

2.G Carbocations from aromatic compounds: arenium (σ-complex, Wheland) ions

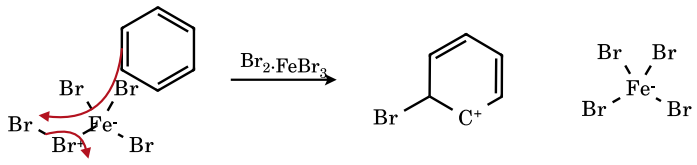
An aromatic ring is a poor nucleophile because attack on an electrophile E^+ destroys aromaticity. The product is a **cyclohexadienyl (arenium) cation**: one sp^3 carbon carrying E and H, and the other five carbons conjugated over **three contributors** with the charge at the two *ortho* and the *para* positions relative to the sp^3 carbon (never the meta carbons). Loss of H^+ from the sp^3 carbon restores the aromatic ring (substitution). Formation of the arenium ion is the slow step in every electrophilic aromatic substitution.

2.G1 Nitration: benzene + NO_2^+ (HNO_3/H_2SO_4)



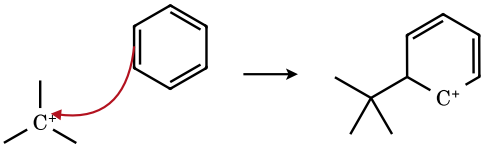
The nitronium ion (O=N⁺=O, linear) is generated by H_2SO_4 protonating HNO_3 ($HNO_3 + 2\ H_2SO_4 \rightarrow NO_2^+ + H_3O^+ + 2\ HSO_4^-$). Its attack on benzene gives the nitro-substituted arenium ion; HSO_4^- removes the ring proton.

2.G2 Bromination: benzene + $Br_2/FeBr_3$



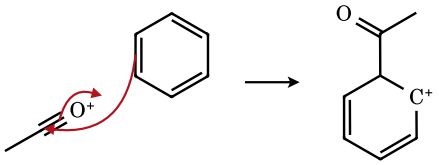
$FeBr_3$ polarises Br–Br ($Br^{\delta+}\cdots Br–FeBr_3^{\delta-}$); the π electrons attack the electrophilic bromine and displace $[FeBr_4]^-$. The same arenium ion forms regardless of which Br leaves with the Lewis acid.

2.G3 Friedel–Crafts alkylation: benzene + t-Bu^+ (t-BuCl/$AlCl_3$)



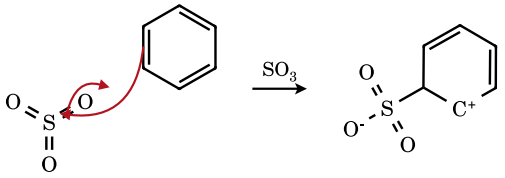
A carbocation (from R–Cl + $AlCl_3$, from an alkene + H^+, or from an alcohol + acid) is attacked by the arene. Cations that can rearrange (n-propyl, n-butyl, neopentyl…) give rearranged alkylbenzenes: cumene, not n-propylbenzene, from n-PrCl/$AlCl_3$. The product ring is more reactive than benzene, so polyalkylation is a problem.

2.G4 Friedel–Crafts acylation: benzene + CH_3CO^+ (acylium ion)



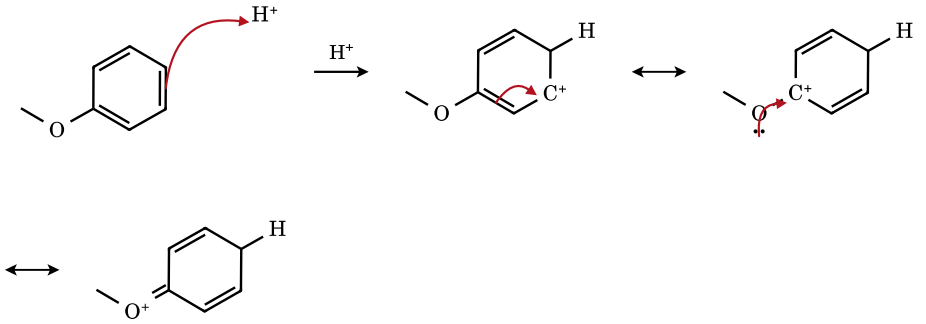
The acylium ion adds to the ring through its electrophilic carbon (one C ≡ O π bond becomes the C=O of the ketone). The acyl group deactivates the ring, so one substitution only, and acylium ions do not rearrange.

2.G5 Sulfonation: benzene + SO_3 (oleum)



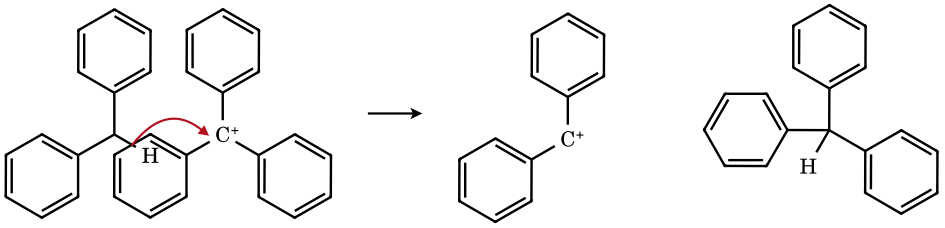
SO_3 (a neutral, very electrophilic sulfur) is attacked by the π bond; the S=O π electrons move onto oxygen, giving a zwitterionic arenium ion. Sulfonation is reversible (dilute hot acid removes SO_3H), unlike nitration or halogenation.

2.G6 Protonation of anisole at the para position (ipso to H)



Protonating C-1 (para to OMe) puts positive charge at the ortho and para carbons, and one contributor places it on C-4, which carries OMe: the oxygen lone pair then delocalises it (an oxocarbenium contributor). Protonation meta to OMe lacks this contributor, so ortho/para attack is faster – the origin of ortho/para direction by +M groups.

2.G7 Hydride abstraction from an alkylarene: diphenylmethane + Ph_3C^+



Aryl groups on a C–H carbon make it a hydride donor: loss of H^- leaves the diarylmethyl (benzhydryl) cation, stabilised by two rings. Ph_3C^+ is a hydride acceptor only for strongly donating hydrocarbons (cycloheptatriene, 2.E5); with Ph_2CH_2 the equilibrium lies far on the left (pK_{R+} −6.6 versus −13.3).

2.H Lewis-acid-assisted ionisation, ether cleavage, and aromatic carbocations

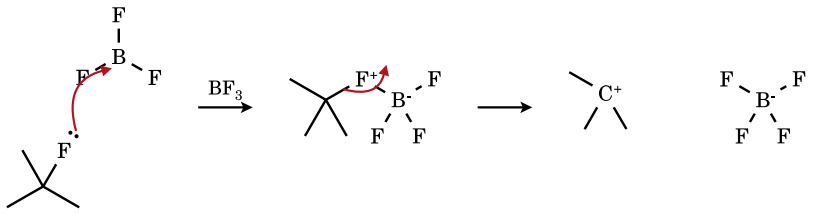
Why a Lewis acid works. Ionisation R–X → R^+ + X^- is the rate-determining, *endothermic* step (heterolysis); the capture of R^+ by a nucleophile is fast and exothermic. A halide, ether or alcohol that does not ionise alone can be pushed over the barrier by an electrophile that binds the leaving group: H^+ (ROH → ROH_2^+), Ag^+ (→ AgX), or a Lewis acid. Neutral alcohols do not react with NaI (HO^- is a poor leaving group and I^- is no electrophile) but do react with HI, because H^+ first converts OH into the good leaving group H_2O. The rate of R–OH + HX follows the cation stability, 3° > 2° > 1°.

Class of leaving group	Examples	Behaviour
Best (leave spontaneously)	R–N_2^+ (N_2), R–OH_2^+ (H_2O), R–O^+R_2, R–S^+R_2, triflate	ionise at room temperature or below when the cation is reasonably stable
Moderate (need help from a Lewis acid/electrophile, or a polar H-bonding solvent)	R–X (X = Cl, Br, I), R–OTs, R–OCOR, ethers R–OR′ after protonation	Ag^+, BF_3, BCl_3, ZnX_2, AlX_3, $SbCl_5/SbF_5$, H^+: the electrophile binds the leaving group, the nucleophile (if any) then attacks the cation
Poor (do not leave)	R–H, R–R, R–OH, R–OR, R–NH_2	cannot be ionised without prior conversion (protonation, sulfonylation, diazotisation)

Lewis acid / electrophile	Reaction with R–X	Why it works
Ag^+	R–X + Ag^+ → R^+ + AgX↓	very insoluble AgX drives the ionisation (halophilic)

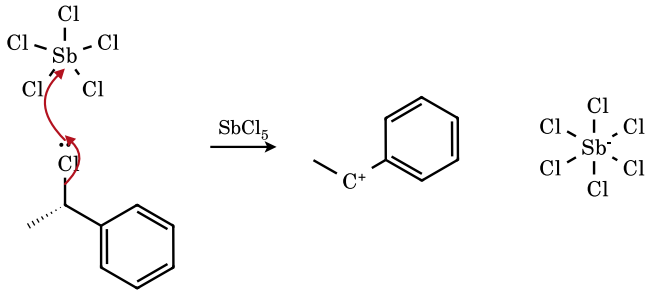
BF_3, BCl_3, BeX_2, ZnX_2, AlX_3	$R–X + MX_n \rightarrow R^+ + [X–MX_n]^-$	electron-deficient (hypovalent) centre accepts a halide lone pair into an empty orbital
$SbCl_5$, SbF_5	$R–X + SbCl_5 \rightarrow R^+ + SbCl_6^-$ (SbF_6^-)	coordination number expands from 5 to 6: a strong Lewis acid giving a very weakly nucleophilic counterion

2.H1 tert-Butyl fluoride + BF_3: even the strongest C–X bond is ionised



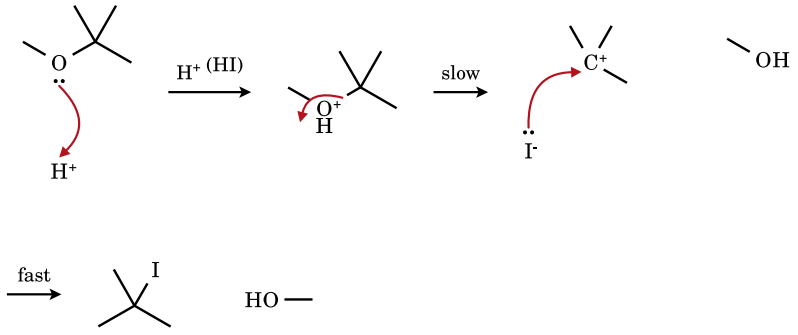
The fluorine lone pair fills the empty p orbital of boron; the C–F bond then ionises to the tert-butyl cation and tetrafluoroborate. Alkyl fluorides, inert to solvolysis, are therefore the most Lewis-acid-sensitive halides because of the strong B–F bond that is formed.

2.H2 (S)-1-Phenylethyl chloride + $SbCl_5$: planar cation, loss of optical activity



The cationic carbon is sp^2 and planar (empty p orbital), so the single stereocentre of the reactant is lost: the intermediate is achiral and attack by a nucleophile from either face gives racemic product – the same reasoning that explains racemisation in S_N1. Chloride is held by $SbCl_5$ as $SbCl_6^-$ and cannot return.

2.H3 Methyl tert-butyl ether + HI: S_N1 cleavage of an ether



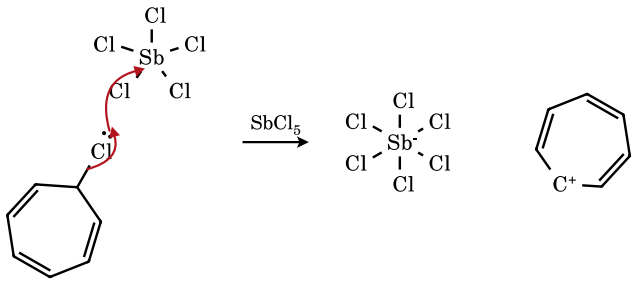
Protonation makes the ether oxygen a good leaving group (CH_3OH), and the C–O bond that breaks is the one that gives the *more stable cation* (tertiary, not methyl): methanol and t-BuI, not CH_3I and t-BuOH. With a primary alkyl and a methyl group (e.g. $CH_3OCH_2CH_3$) cleavage goes by S_N2 at the less hindered carbon instead (CH_3I + EtOH).

2.H4 3-Chlorocyclopropene + $SbCl_5$: the cyclopropenium cation (aromatic, 2π)



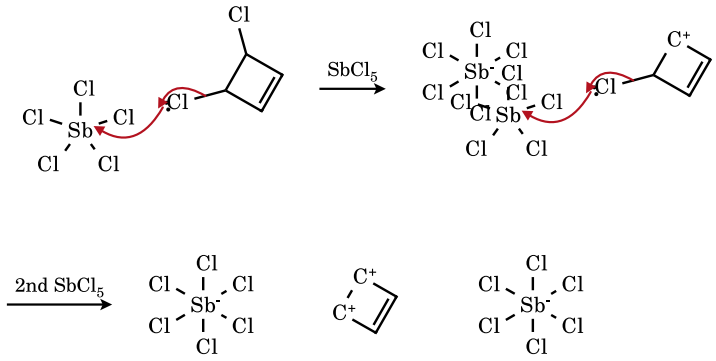
Ionisation gives $C_3H_3^+$, a planar ring of three p orbitals holding two π electrons (4n+2, n = 0): aromatic, and therefore formed from an otherwise hopelessly strained *cyclopropyl* cation. The positive charge is shared equally by the three carbons (D_{3h}).

2.H5 7-Chlorocycloheptatriene + $SbCl_5$: the tropylium cation (aromatic, 6π)



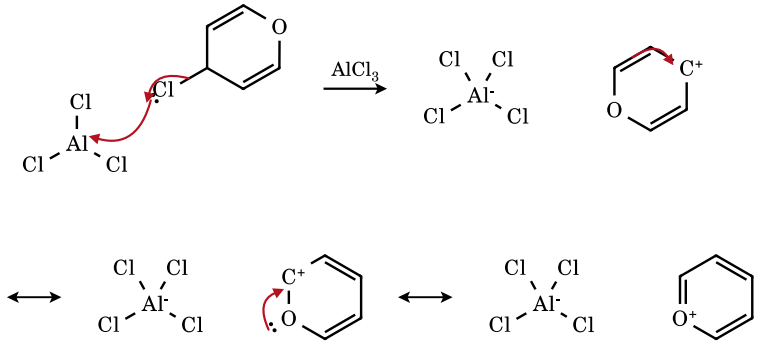
The sp^3 carbon becomes sp^2 and completes a cycle of seven p orbitals with six π electrons. Tropylium chloride and bromide are in fact ionic salts (water-soluble solids, not covalent alkyl halides), the clearest example that aromaticity can make a 'halide' ionic.

2.H6 3,4-Dichlorocyclobutene + 2 $SbCl_5$: the cyclobutadiene dication (aromatic, 2π)



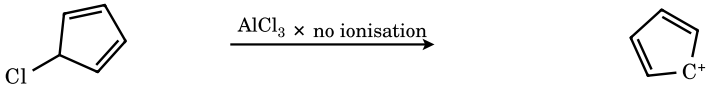
Removal of two chloride ions leaves $C_4H_4^{2+}$, a planar ring of four p orbitals with only **two** π electrons (4n+2, n = 0): the dication is aromatic, whereas neutral cyclobutadiene (4π) is antiaromatic. This is why two equivalents of $SbCl_5$ ionise both C–Cl bonds. The drawing localises the π bond and the two cationic carbons; the real ion is delocalised over all four carbons.

2.H7 4-Chloro-4H-pyran + $AlCl_3$: the pyrylium cation (aromatic, 6π, O^+)



Loss of chloride from C-4 gives a cation conjugated with the C=C–O unit; two further electron shifts bring the charge onto oxygen, whose lone pair completes the six-π-electron ring: pyrylium ($C_5H_5O^+$), an aromatic oxonium ion. The oxygen carries the formal + charge but the carbons 2, 4 and 6 carry most of the electrophilicity.

2.H8 …and where it fails: 5-chlorocyclopentadiene + $AlCl_3$



5-chlorocyclopenta-1,3-diene **cyclopentadienyl cation (4π, antiaromatic)**

Ionisation would give the cyclopentadienyl cation, a planar cycle of five p orbitals with four π electrons (4n): antiaromatic and so unstable (computed ground state: triplet) that 5-chlorocyclopentadiene does *not* ionise with Lewis acids, even though a simple secondary allylic cation of similar substitution would. 3,4-Dichlorocyclobutene (2.H6) and 3-chlorocyclopropene (2.H4) ionise *because* the cations are (4n+2) aromatic. **Rule: electron count decides.**

Part 3 Leaving-Group Ability

3.1 What a leaving group is, and what 'good' means

A **leaving group** (nucleofuge) is the fragment that departs with the bonding electron pair in a heterolytic substitution or elimination: R–LG → R^+ + LG^- (or neutral LG from a cationic substrate). 'Good' is a statement about a *rate* (kinetic, a lowered barrier for the C–LG cleavage step) that correlates with – but is not identical to – the *thermodynamic* stability of the departed LG, i.e. the acidity of its conjugate acid H–LG. The correlation exists because in the transition state the LG carries a partial negative charge; the better it can stabilise that charge, the lower the barrier (Brønsted-type relationship, typically β_{lg} between −0.3 and −1.0 for substitution).

Leaving group (LG)	Conjugate acid	pK_a (aq., approx.)	Remarks
I^-, Br^-, Cl^-	HI, HBr, HCl	−10, −9, −7	large, polarisable; weak C–X bonds (I < Br < Cl)
F^-	HF	3.2	very poor in S_N1/S_N2 (strong C–F bond); good in SNAr (see Q5)
$CF_3SO_3^-$ (triflate)	CF_3SO_3H	≈ −14	charge delocalised over 3 O, CF_3 –I effect; a 'super' LG
p-$O_2NC_6H_4SO_3^-$ (nosylate)	$ArSO_3H$	≈ −6.5 (est.)	EWG on the aryl ring improves LG
p-$CH_3C_6H_4SO_3^-$ (tosylate)	TsOH	≈ −2.8	standard crystalline sulfonate
$CH_3SO_3^-$ (mesylate)	MsOH	≈ −1.9	similar to tosylate in rate
H_2O (from ROH_2^+)	H_3O^+	−1.7	neutral LG; HO^- of ROH is not a leaving group
$CF_3CO_2^-$, $CH_3CO_2^-$	CF_3CO_2H, AcOH	0.5, 4.76	carboxylates: moderate; acetates ionise only for cations of tertiary/benzylic stability
N_2 (from RN_2^+)	—	—	best known LG: very stable neutral molecule, huge entropy gain
NH_3 (from RNH_3^+)	NH_4^+	9.25	poor: Hofmann elimination needs heat; NH_2^- (pK_a 38) never leaves
HO^-, RO^-, H^-, R^-	H_2O, ROH, H_2, RH	15.7, 16–19, 35, ≈ 50	do not leave in S_N/E1 unless activated (E1cB, intramolecular, ring strain)

3.2 Factors that determine leaving ability

- **Basicity / conjugate-acid strength.** Weaker bases are better leaving groups: the LG^- is more stable. Within a period this is the dominant trend ($I^- \gg F^-$ in protic solvent; $TfO^- \gg AcO^- \gg HO^-$).
- **Bond strength and polarisability.** Down a group the C–X bond weakens (C–F ≈ 110, C–Cl ≈ 84, C–Br ≈ 70, C–I ≈ 57 kcal mol^{-1}) and the large, soft anion is stabilised by polarisation of the TS; the pK_a trend (HF 3.2 → HI −10) reinforces this. In the gas phase F^- is a worse LG than I^- for the same reason.
- **Resonance and charge delocalisation.** Sulfonates (charge over 3 O), carboxylates (2 O), triflate (3 O + CF_3) are good LGs because the charge is spread; alkoxides and amides are not.
- **Induction.** EWGs on the LG (CF_3, NO_2, halogens on an aryl ring) stabilise the anion: $TfO^- \gg MsO^-$; p-O_2N–$C_6H_4SO_3^-$ > TsO^-.
- **Solvation.** Protic solvents hydrogen-bond to small, charge-dense anions (F^-, Cl^-, RO^-), stabilising them in the TS and promoting ionisation; the effect on the *ordering* of halides is small for S_N1 but large for nucleophiles in S_N2. Aprotic dipolar solvents (DMSO, DMF) favour S_N2 by leaving anions bare.
- **Neutral versus charged substrates.** Protonation, alkylation or diazotisation converts a bad LG (HO^-, NH_2^-) into a good neutral one (H_2O, NH_3, N_2) – the underlying reason for acid catalysis and for conversion of alcohols to sulfonates.
- **Neighbouring-group assistance.** A lone pair or π/σ bond anti-periplanar to the C–LG bond can displace the LG intramolecularly (anchimeric assistance; sulfur > nitrogen-free base > bromine > oxygen in typical hydrolyses): the C–LG bond breaks faster than the intrinsic LG ability would suggest.
- **Stereoelectronic and steric factors.** An axial LG on a cyclohexane ring is anti-periplanar to ring C–C/C–H bonds and relieves 1,3-diaxial strain on ionisation; an anti-periplanar C–Si bond (β-silicon effect) accelerates departure by orders of magnitude; large groups on the substrate relieve strain (B-strain) on ionisation.
- **Ion-pair behaviour.** The LG may remain close to the cation (internal return). Poor ion-pair return from large, delocalised LGs (OTs, OTf) increases the net rate relative to Cl^-.

3.3 Why there is no single universal order

The order $I^- > Br^- > Cl^- \gg F^-$ holds for S_N1, S_N2 and E2 at sp^3 carbon, but is reversed ($F \gg Cl \approx Br > I$) in nucleophilic aromatic substitution, where C–LG cleavage is *not* rate-limiting: the first step (attack of the nucleophile to give the Meisenheimer complex) is slower, and the very electronegative fluorine stabilises that TS by induction. Likewise in acyl substitution the order of acyl derivatives ($RCOCl > (RCO)_2O > RCO_2R' > RCONR_2$) reflects both LG ability *and* how strongly the LG donates into C=O. In E2 the LG changes the *regiochemistry*: poorer LGs (F^-, NR_3) make the TS carbanion-like and favour the Hofmann (less substituted) alkene. And in transition-metal oxidative addition the order is Ar–I > Ar–Br ≈ Ar–OTf > Ar–Cl, controlled by bond strength and metal–LG interaction, not by pK_a.

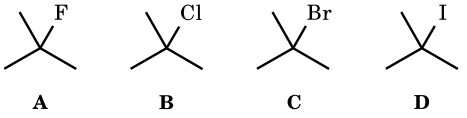
The following comparisons are made at stated conditions. A leaving-group ranking is meaningful only for a **fixed substrate, mechanism, solvent and temperature**.

Process	Rate-limiting step	Leaving-group dependence
S_N1 / E1	C–LG heterolysis	Strong ($\beta_{lg} \approx -0.5$ to -1); solvent H-bond donation important
S_N2	concerted attack + C–LG cleavage	Moderate ($\beta_{lg} \approx -0.3$ to -0.6); solvent-nucleophile interactions also matter
E2	concerted; anti-periplanar H–C–C–LG	Moderate; also changes Zaitsev/Hofmann ratio and TS looseness
E1cB	carbanion formation or LG expulsion	LG may be as poor as HO^- or RO^- if carbanion is stabilised (enolate)
SNAr	nucleophilic addition	Weak, 'element effect' $F \gg Cl \approx Br > I$
Acyl substitution	addition or collapse of tetrahedral intermediate	Strong, but coupled with resonance donation

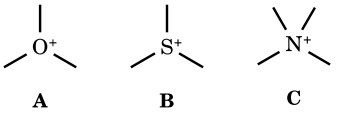
3.4 Fifteen leaving-group comparison problems

Rank **A–D** as instructed. Every statement of order refers to the stated conditions only. Rate orders are established qualitative trends in the literature; no numerical ratios are claimed unless stated.

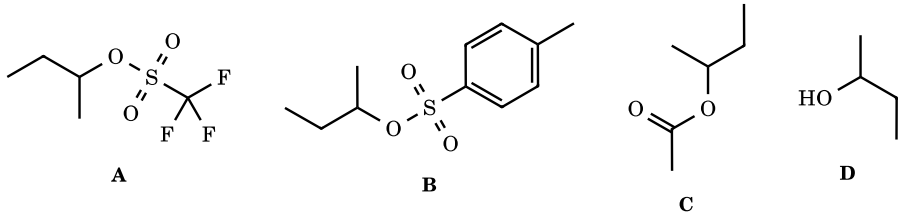
LG1. Arrange A–D in **decreasing** first-order solvolysis rate in 80 % aq. ethanol at 25 °C (same alkyl group; the only variable is the halogen).



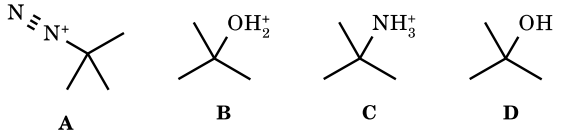
LG10. Methyl-transfer (S_N2) from the onium salts A–C to a thiolate in CH_3CN at 25 °C. Arrange in **decreasing** reactivity.



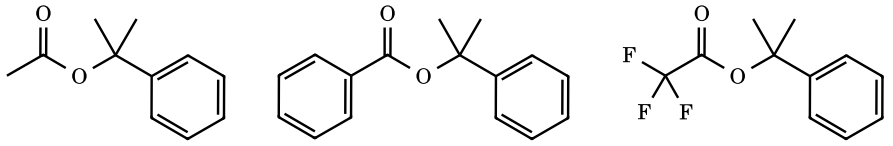
LG2. R = 2-butyl in all four. Arrange the derivatives in **decreasing** rate of ionisation in warm (50 °C) neutral ethanol (no acid added).



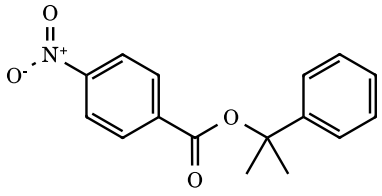
LG3. For the tert-butyl derivatives A–D in acidic aqueous solution at 25 °C, arrange in **decreasing** ease of C–LG heterolysis (LG = the neutral molecule that leaves, or HO^- for D).



LG4. The four tert-cumyl esters A–D are solvolysed in 70 % aq. acetone at 25 °C. Arrange in **decreasing** rate of ionisation.

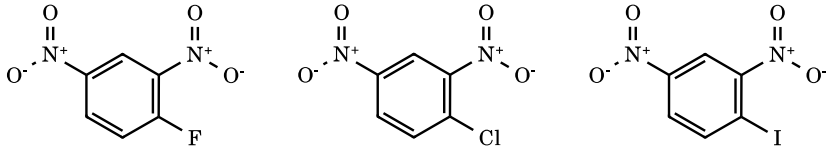


A **B** **C**



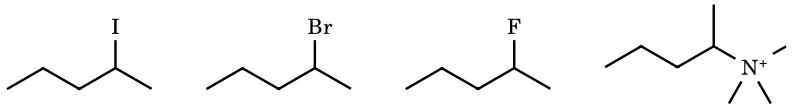
D

LG5. Arrange the 1-halo-2,4-dinitrobenzenes A–C in **decreasing** rate of reaction with piperidine (or methoxide) in ethanol at 25 °C.



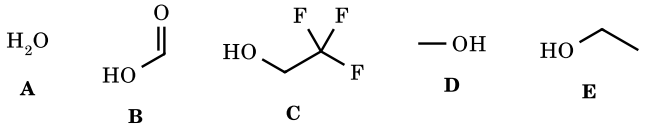
A **B** **C**

LG6. Each 2-pentyl derivative A–D is treated with NaOEt/EtOH (E2). Arrange in **decreasing** percentage of the *less substituted* alkene, pent-1-ene (Hofmann product), in the product mixture.



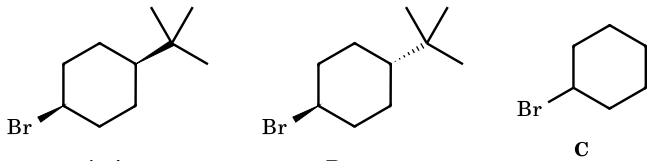
A **B** **C** **D**

LG7. Arrange the solvents A–E in **decreasing** ionising power for the solvolysis of tert-butyl chloride at 25 °C (Grunwald–Winstein Y scale). Solvents: A H_2O, B HCO_2H, C CF_3CH_2OH, D CH_3OH, E CH_3CH_2OH.



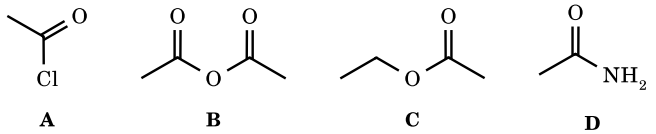
A **B** **C** **D** **E**

LG8. Bromides A–C undergo E2 with NaOEt in EtOH at 50 °C. Arrange in **decreasing** rate. (A cis-4-tert-butylcyclohexyl bromide; B trans-4-tert-butylcyclohexyl bromide; C bromocyclohexane. The t-Bu group is locked equatorial.)



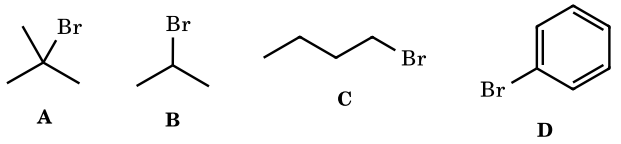
A cis **B trans** **C**

LG9. Arrange the acetyl derivatives A–D in **decreasing** rate of nucleophilic acyl substitution by water (neutral, 25 °C).



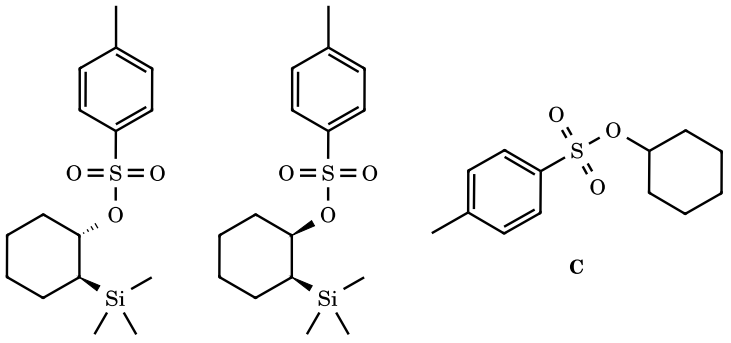
A **B** **C** **D**

LG11. Treat each compound A–D with ethanolic $AgNO_3$ at 25 °C. Arrange in **decreasing** rate of AgBr precipitation (no precipitate = last).



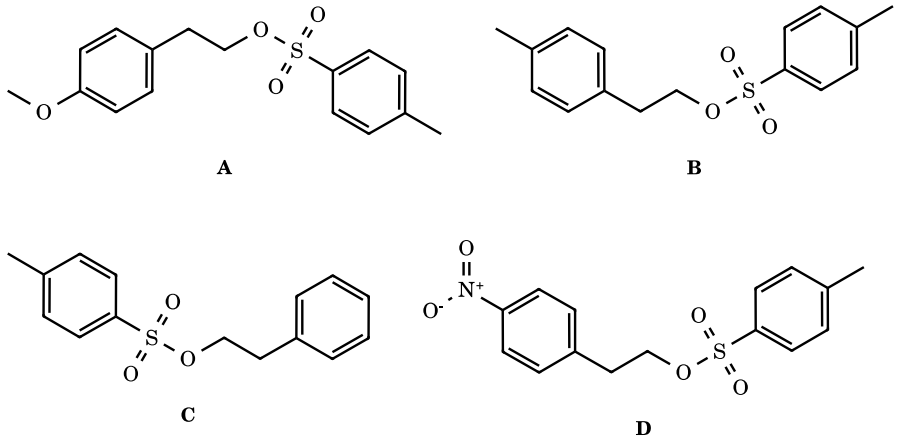
A **B** **C** **D**

LG12. Tosylates A–C are acetolysed at 50 °C. Arrange in **decreasing** rate. (A trans-2-(trimethylsilyl)cyclohexyl tosylate; B cis-2-(trimethylsilyl)cyclohexyl tosylate; C cyclohexyl tosylate.)



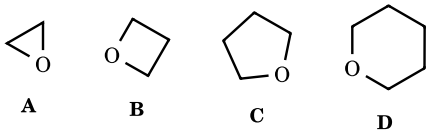
A trans **B cis** **C**

LG13. Arrange the 2-arylethyl tosylates A–D in **decreasing** rate of acetolysis at 75 °C (neighbouring-group participation by the aryl ring is the major pathway).



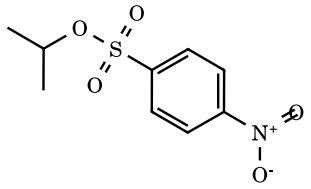
A **B** **C** **D**

LG14. Arrange the cyclic ethers A–D in **decreasing** ease of acid-catalysed ring opening by aqueous HBr at 25 °C.

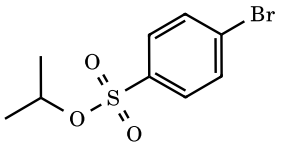


A **B** **C** **D**

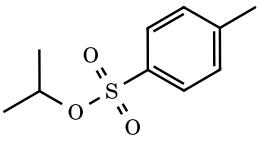
LG15. Arrange the 2-propyl arenesulfonates A–D in **decreasing** rate of solvolysis in aq. ethanol at 25 °C.



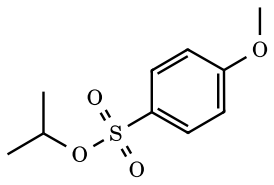
A p-NO_2



B p-Br



C p-CH_3



D p-OCH_3

Part 4 Carbocation Stability: Complete Advanced Treatment

What 'stability' means. Four different observables are routinely mixed up and must be kept apart: (1) *gas-phase thermodynamic stability*, measured by the hydride ion affinity (HIA) $R^+ + H^- \rightarrow R–H$ (smaller HIA = more stable cation) or the corresponding calculated energies – this reflects the *intrinsic electronic* stabilisation, free of solvent; (2) *stability in solution toward water*, pK_{R+} = pH at which $[R^+] = [ROH]$, applicable only to very stable cations; (3) *substituent stabilisation* expressed by σ^+ constants (Brown–Okamoto), derived from the solvolysis of $ArC(CH_3)_2Cl$; (4) *ionisation rates* (solvolysis), which depend on the stability of the cation *and* on the leaving group, solvent, ion-pair return, nucleophilic solvent assistance (k_s), neighbouring-group participation (k_Δ) and strain relief, and therefore track the cation's stability only when all of these are held constant and the transition state is late. Every problem in Part 5 states which observable is meant.

> Orders obtained in the gas phase can differ from, or even invert, orders in solution (alkyl-substituted benzyl cations: gas phase t-Bu > i-Pr > Et > Me, solution Me ≳ Et > i-Pr > t-Bu – the Baker–Nathan order – because larger alkyl groups are solvated less effectively). Where an order below depends on the medium, the medium is stated.

4.1 Inductive effect

A σ-framework effect: electronegativity differences polarise σ bonds. **+I** (electron-donating) groups – alkyl groups, and especially more polarisable and more electropositive groups such as $SiMe_3$, $SnMe_3$ – push electron density toward the cationic carbon and lower its energy; **–I** groups (F, Cl, OR, NR_3^+, NO_2, CN, CF_3, C=O) pull density away and raise it. The effect falls off roughly by a factor of 2–3 per intervening bond, so it is dominant for α and β substituents, small for γ. The field (through-space) component is important in rigid frameworks (4-substituted bicyclo[2.2.2]octyl systems). **Important caveat:** –I groups attached by a lone-pair atom (OR, SR, NR_2, Hal) at the α position are overall *stabilising* because the +M effect outweighs –I ($CH_3OCH_2^+$ is far more stable than $CH_3CH_2^+$), while the same groups at β position are destabilising unless they can bridge. Strongly –I groups on the cationic carbon are dramatically destabilising: α-CF_3 and α-carbonyl groups reduce solvolysis rates of tertiary systems by many orders of magnitude (Tidwell).

4.2 Mesomeric (resonance) effects

A π-bond or lone pair attached to the cationic carbon delocalises the charge (π→p or n→p overlap). **+M** groups (NR_2 > NHR > OH ≈ OR > SR > aryl ≈ vinyl > halogen) stabilise; **–M** groups (C=O, CN, NO_2, SO_2R, CF_3 by negative hyperconjugation/induction) destabilise by competing for the same orbital or by withdrawing π density. Resonance changes *only electron positions* and is shown with the double-headed arrow ⇔; curved arrows show how to go from one contributor to the next.

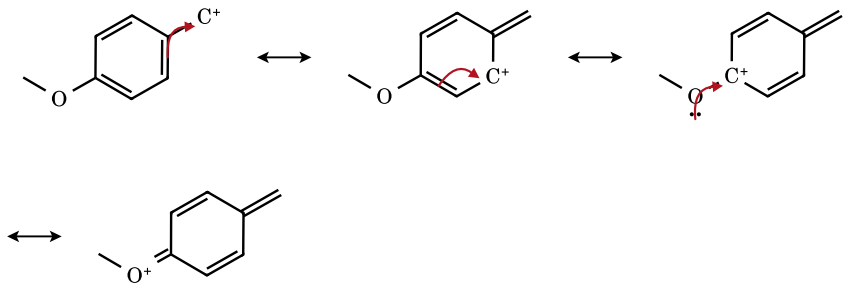
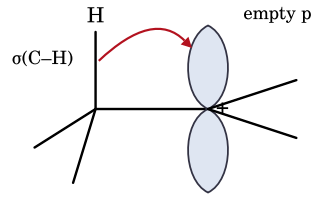


Figure 4.1 p-Methoxybenzyl cation: charge delocalised from CH_2 to the ortho/para carbons, and finally onto oxygen (an oxocarbenium contributor; every atom has an octet).

- **Position matters.** Charge in a benzyl cation sits on the ortho and para carbons, not meta. A +M group at p (or o) conjugates directly with the charge (σ^+(p-OMe) = −0.78) while at m only its −I effect operates (σ^+(m-OMe) = +0.05). A −M group at p (σ^+(p-NO_2) = +0.79) is worse than at m (+0.67), because at p it is conjugated with the charge-bearing carbon.
- **Halogens** have strong −I but a real +M effect (lone-pair donation); at p-F the two nearly cancel (σ^+ = −0.07), at p-Cl −I wins slightly (+0.11), at m-Cl only −I operates (+0.40).
- **Resonance requires coplanarity.** A π donor can only help if its p orbitals can align with the empty p orbital: see 4.8 (steric inhibition of resonance) and the vinyl/aryl cations (empty orbital in the nodal plane of the π system: no resonance).
- **Resonance energy** is large for the first donor and levels off for successive ones: allyl ≈ 20+ kcal mol^{-1} better than a localised secondary cation of the same substitution in the gas phase; each further vinyl unit adds less (pentadienyl, heptatrienyl).
- **The ranking of mesomeric donors** depends on orbital energy match and overlap: N (2p) > O (2p) > S (3p, poorer overlap but more polarisable) for the same carbon framework; in solution oxocarbenium ions are further stabilised by solvation.

4.3 Hyperconjugation

Donation of σ-bond electrons into the empty p orbital: **σ(C–H) → p** and **σ(C–C) → p** (and σ(C–M) → p for M = Si, Ge, Sn, which are much better donors because σ_{C-M} lies higher in energy). The stabilisation scales with (i) the donor ability of the σ bond (C–C ≳ C–H in the gas phase; C–Sn > C–Ge > C–Si ⋙ C–C), (ii) the number of aligned bonds and (iii) the alignment angle θ between the σ bond and the empty p orbital (overlap ∝ $\cos^2\theta$; zero at 90°). In a freely rotating alkyl group on average ≈ 2 of the 3 C–H bonds are well aligned at any time; **simply counting α-hydrogens is not a valid ranking device** – t-Bu^+ (9 α-H) vs 2-adamantyl, bridgehead or conformationally locked cations show that orbital alignment, strain and polarisability also matter. Computations show that C–C hyperconjugation is as important as C–H; polarisability (stabilisation of the charge by larger alkyl groups) makes larger alkyl groups better in the gas phase (recalled HIA, kcal mol^{-1}: t-Bu^+ ≈ 231 → t-amyl$^+$ ≈ 225 → Et_3C^+ lower still).



C–H parallel to the p axis: maximum overlap

Figure 4.2 σ(C–H) → p hyperconjugation (the C–H bond must lie roughly parallel to the p-orbital axis).

Heteroatom hyperconjugation. Lone pairs and σ bonds of heteroatoms attached β to the cation can also donate: β-silicon (C–Si anti-periplanar to the leaving group) stabilises a β-carbocation by ≈ 38 kcal mol^{-1} relative to hydrogen in some computations (≈ 10^{12} rate acceleration in rigid systems), β-Ge and β-Sn are even stronger; β-halogens at 3° carbons may bridge instead (4.6). Negative hyperconjugation (n→σ*, e.g. in α-sulfonyl carbanions) is the anionic analogue and does not apply to cations.

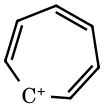
4.4 Aromaticity and antiaromaticity



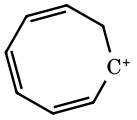
cyclopropenium $C_3H_3^+$ 2π, aromatic



cyclopentadienyl $C_5H_5^+$ 4π, antiaromatic



tropylium $C_7H_7^+$ 6π, aromatic



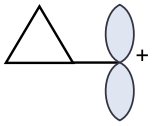
homotropylium $C_8H_9^+$ homoaromatic

Figure 4.3 Cyclic cations with a planar array of p orbitals: (4n+2) π electrons stabilise (cyclopropenium, tropylium); 4n π electrons destabilise (cyclopentadienyl cation, a triplet-ground-state species that is far less stable than the allyl cation). Homotropylium is stabilised by through-space (homo)conjugation across one sp^3 carbon.

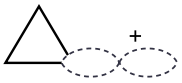
- **Aromatic cations:** cyclopropenium (pK_{R+} of $C_3Ph_3^+$ +3.1; tris(dialkylamino)- and tricyclopropyl derivatives far higher), tropylium (pK_{R+} +4.7: isolable as stable Br^-, BF_4^- salts, planar, all C–C equal), cyclooctatetraene dications.
- **Antiaromatic cations:** cyclopentadienyl cation and the 9-fluorenyl cation destabilised by 4π delocalisation. The 9-phenylfluorenyl cation ($pK_{R+} \approx -10.8$) is some 10^4 times less stable than trityl (–6.6) although the fluorenyl ring is planar – a clean demonstration of antiaromatic destabilisation. By contrast the 9-phenylxanthylium ion ($pK_{R+} \approx +1.1$) is strongly stabilised: the ring is planar and the oxygen lone pair donates without producing a 4π cycle.
- **Always check geometry and electron count** before declaring a cyclic conjugated cation aromatic: planarity may be unattainable (e.g. cyclooctatetraenyl cation puckers), and an sp^3 centre interrupts the cycle.

4.5 Cyclopropylcarbinyl stabilisation and 'dancing resonance'

A cyclopropane C–C bond is a bent ('banana') σ bond with high p character and high-lying HOMO; the cyclopropyl group therefore behaves like a π donor towards an adjacent empty p orbital – **provided the cation adopts the *bisected* conformation**, in which the empty p orbital is parallel to the plane of the ring and overlaps the two vicinal ring bonds. In the perpendicular conformation the overlap is nil. As a donor the cyclopropyl group is better than phenyl (σ^+: c-C_3H_5 –0.41, C_6H_5 –0.18) and, in tertiary systems, better than methyl: tricyclopropylcarbinyl cation (pK_{R+} –2.3) is more stable than trityl (–6.6).



bisected: p ∥ ring plane — good overlap



perpendicular: poor overlap

Figure 4.4 Cyclopropylcarbinyl cation: bisected (left, strongly stabilised) and perpendicular (right, no stabilisation) conformations. Orbital view; rings viewed edge-on.



cyclopropylcarbinyl **cyclobutyl** **homoallyl (contributor)**

*Figure 4.5 The $C_4H_7^+$ manifold. The solvolysis of cyclopropylcarbinyl and of cyclobutyl derivatives gives, in water, cyclopropylcarbinol : cyclobutanol : homoallyl alcohol ≈ 48 : 47 : 5 (Roberts, Mazur). These are **product ratios**, not cation populations; the cation is better described as a single σ-delocalised 'bicyclobutonium'-like structure than as a mixture of three distinct classical ions.*

'Dancing resonance' describes how, in this ion, the three CH_2 carbons exchange identity by rapid, degenerate rearrangement through the delocalised structure, so that the carbons become equivalent on the NMR timescale. It is not ordinary resonance, because the nuclei move. The textbook drawing of three localised cations connected by ⇌ arrows is therefore a *caricature*: calculations and superacid NMR place the bicyclobutonium-like (σ-bridged, C_s) and the bisected cyclopropylcarbinyl structures within about 1 kcal mol^{-1} of each other and the barrier between them is minute; the open homoallyl and the classical puckered cyclobutyl ions are not distinct, deep minima.

4.6 Neighbouring-group participation and bridged ions

A genuine bridged intermediate forms when a group with a lone pair, π bond or σ bond **displaces the leaving group from the rear** (anti-periplanar) and ends up bonded to *both* carbons. It differs from ordinary resonance in three ways: (i) the neighbour **moves** toward the cationic carbon and a new σ bond (3-membered or 5-membered ring) is formed; (ii) the stereochemical outcome is dictated by two successive inversions (net retention); (iii) the rate is higher than the intrinsic leaving-group ability suggests (anchimeric assistance).

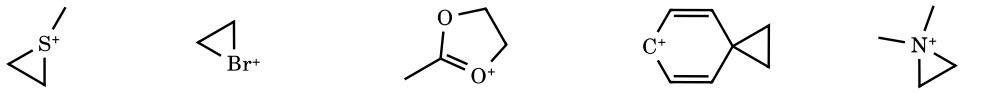


Figure 4.6 Bridged onium and phenonium ions. In each, positive charge is carried largely by the bridging heteroatom (or spread over the cyclohexadienyl ring in phenonium) and no open 'primary' cation exists.

- **Order of anchimeric power** for a given leaving group: roughly S ≈ N (free lone pair) > I > Br > Cl ≈ aryl > O (ether) – a lone pair donates best when its atom is large/polarisable and not strongly bound to its own nucleus; the actual rate enhancement depends on solvent, ring size formed and conformation.
- **π participation:** an aryl ring attacks C from the rear through its π face to give phenonium (Cram, Winstein); a C=C does the same in anti-7-norbornenyl tosylate (≈ 10^{11} faster than 7-norbornyl) and in homoallyl → cyclopropylcarbinyl systems.
- **σ participation:** a C–C or C–H σ bond (e.g. C1–C6 in exo-2-norbornyl; Wagner–Meerwein) acts as the nucleophile – see 4.9.
- **Test to separate bridging from resonance:** if two structures differ in the *position of atoms* (a ring closes), bridging is involved; if only electrons move, it is resonance.

4.7 d-Orbital resonance claims

Older textbooks explain the stabilisation of α-sulfonyl and α-phosphoryl carbanions, the planarity of silylamines, or the shortness of S–O bonds in sulfones/phosphates by **d_π–p_π bonding** (3d orbitals on S, P, Si accept electron density). Modern calculations show that 3d orbitals are too high in energy and too diffuse to participate significantly; d-functions in basis sets act as polarisation functions only. The effects are explained instead by: (i) **negative hyperconjugation** – donation of a lone pair into an antibonding σ*(S–O), σ*(S–C) or σ*(P–C) orbital; (ii) **polarisation** of the large, polarisable atom; (iii) strongly polarised 'ylidic' bonds (S^+–O^-, P^+–C^-) without a true double bond and with the octet intact. For **carbocations** the practical consequences are: α-silyl and α-sulfonyl substituents do *not* stabilise a cation (α-silyl cations are destabilised relative to alkyl; a sulfone is a strong –I group), whereas **β-silyl** stabilisation is σ-hyperconjugative and strong; and a sulfur lone pair (R–S–C^+) donates through *p–p* overlap (thionium ions) without d orbitals. Hypervalent descriptions (sulfonium ylide S=C) are therefore teaching shorthand, not electronic reality.

4.8 Steric inhibition of resonance (SIR)

If bulky ortho (or flanking) groups force the π donor out of coplanarity with the cationic centre, overlap falls as cos θ (energy as $\cos^2\theta$) and the resonance stabilisation is partly lost. The classic quantitative case: the dimethylamino group is an extreme π donor ($\sigma^+ = -1.7$) in 4-(dimethylamino)benzyl and trityl-type cations; but when two methyl groups flank the NMe_2 (3,5-dimethyl-4-dimethylamino) the amine is twisted ≈ 60–90° out of the ring plane, the n→π donation is largely switched off and the group behaves close to a simple inductive substituent (σ^+ much less negative; the nitrogen lone pair cannot reach the charge).

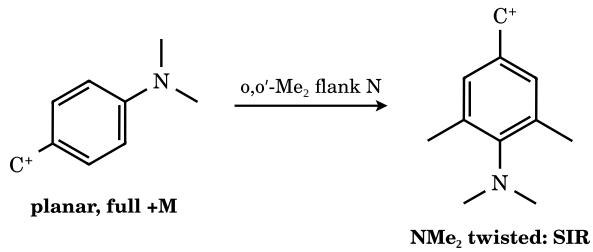
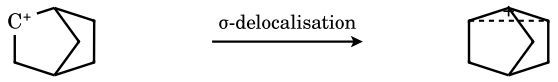


Figure 4.7 Steric inhibition of resonance: the flanking methyls (3,5 to the CH_2^+, ortho to N) twist the amine; the nitrogen lone pair then lies in the ring plane and cannot overlap with the π system.

> **No ranking by sterics alone.** SIR changes the *resonance* contribution only. Methyl groups at the same time exert +I/hyperconjugation and (in planar systems) can stabilise. Comparisons in Part 5 therefore always fix the other effects or state the observable.

4.9 Nonclassical carbocations

In a **classical** carbocation the charge is localised on one carbon (or delocalised by resonance only) and every bond is a 2c–2e bond. In a **nonclassical** ion σ electrons are delocalised over three or more nuclei by a 3c–2e bond, with *partial* bonds drawn as dotted lines. The best-established examples: 2-norbornyl, cyclopropylcarbinyl/bicyclobutonium, phenonium (aryl-bridged), protonated cyclopropane ($C_3H_7^+$ corner-protonated form), halonium ions, and the carbonium ions CH_5^+, $C_2H_7^+$.



classical (open) 2-norbornyl **σ-bridged (non-classical)**

*Figure 4.8 2-Norbornyl cation: classical structure (left, positive charge on C-2) and the σ-bridged non-classical structure (right, schematic: the C1–C6 σ bond delocalises toward C-2 – dotted line – and the positive charge is shared by C1, C2 and C6 inside the three-centre triangle). The two are **not** resonance forms (atoms move); the bridged structure is the better description of the real ion.*

- **Kinetic evidence** (Winstein, 1950s): exo-2-norbornyl brosylate solvolyses ≈ 350 times faster than the endo isomer; both give exo product only and the product from optically active exo is racemic, as expected if σ(C1–C6) assists ionisation anti-periplanar to the exo leaving group, producing a symmetric (C_s) bridged ion that is attacked at either of two equivalent carbons.
- **Spectroscopic evidence** (Olah, 1960s–): in superacid at low temperature the ^{13}C NMR shows rapid 6,2- and 3,2-hydride shifts and Wagner–Meerwein exchange that average carbon signals; at very low temperature (–150 °C, solid-state studies) the signals match a bridged structure with a strongly deshielded pentacoordinate-type carbon environment.
- **Structural evidence:** a crystal structure of the 2-norbornyl cation with an $[Al_2Br_7]^-$ counterion (Scholz, Kramer, Sieber et al., *Science* 2013) shows the bridged geometry, with C–C distances between the three bridging carbons intermediate between single and bridged-cyclopropane values. (Recalled from the literature; the source was not accessible here.)
- **The classical side (H. C. Brown):** the high exo/endo ratio can be explained by steric hindrance to ionisation of the endo isomer and by rapid equilibration of classical ions. The consensus today is that the **parent 2-norbornyl cation is bridged**, whereas tertiary derivatives (2-methyl-2-norbornyl, 2-aryl-2-norbornyl) are **classical** because the substituent already supplies enough stabilisation.
- **General rule:** bridging is a 'last resort' stabilisation. It occurs when the classical ion would be primary or secondary and unstabilised and the geometry offers an anti-periplanar σ (or π, or lone-pair) donor; it disappears when an aryl, cyclopropyl or tertiary substitution stabilises the cation already.

4.10 Carbonium ions (hypercoordinate carbon)

CH_5^+ (Figure 1.4) is the prototype. It has **eight** valence electrons for **five** C–H connections, so at least one bond is 3c–2e; the structure is fluxional (all five H exchange) and has an H_2 unit loosely bound to CH_3^+. $C_2H_7^+$ (protonated ethane) has a C–C or C–H σ bond protonated. In superacid, alkanes are protonated to carbonium ions, which lose H_2 (→ carbenium) or CH_4/alkanes (→ smaller carbenium ions: protolytic cracking, isomerisation). **Carbonium and carbenium ions are different bonding types and cannot be put on one stability scale by counting substituents.** Their relation is chemical, not energetic ranking: $R_3C–H + H^+ \rightleftharpoons [R_3C(H_2)]^+ \rightarrow R_3C^+ + H_2$.

4.11 Additional effects

- **Ring strain and I-strain.** Ionisation changes carbon from sp^3 to sp^2. H. C. Brown: 1-methylcyclopentyl chloride solvolyses ≈ 40× faster than t-BuCl, 1-methylcyclohexyl ≈ 3× slower (the six-membered ring loses a stable staggered arrangement and gains eclipsing strain), 1-methylcycloheptyl faster again; cyclopropyl and cyclobutyl cations are strained (bond angle 60°/90° versus 120°) and ionise only with ring-opening/participation.
- **Bridgehead constraint.** The cation must be able to flatten. Rates of solvolysis (80 % EtOH, 25 °C, relative to t-BuBr): 1-adamantyl ≈ 10^{-3}, bicyclo[2.2.2]oct-1-yl smaller, bicyclo[2.2.1]hept-1-yl (norbornyl) ≈ 10^{-13} (orders of magnitude recalled from Bingham–Schleyer; the sequence is secure) – a measure of the strain, not just of the cation's intrinsic stability (Bartlett, Schleyer).
- **Geometry/alignment.** Anti-periplanar σ bonds (C–Si, C–C) and gauche vs anti hyperconjugation; bisected vs perpendicular cyclopropylcarbinyl; rigid frameworks fix the alignment.
- **Field effects** of remote polar groups; **counterion** and ion pairing (a tight, nucleophilic counterion such as Cl^- can covalently collapse the cation; SbF_6^- cannot); **solvation** (H-bond donors stabilise the incipient anion;

ion-dipole solvation favours small, localised cations, which is why alkyl order inverts in solution relative to the gas phase).

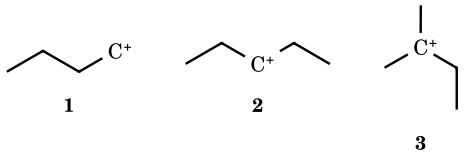
- **Isotope effects.** A CD_3 group is a poorer hyperconjugative donor than CH_3 (zero-point-energy difference): β-d_3 substitution slows ionisation by a factor ≈ 1.3–2.4 per CD_3, a probe of hyperconjugation in the transition state.

Effect	Acts through	Strength guide	Common error to avoid
Resonance (+M)	p(π)/n → p	NR_2 > OR > SR > aryl > vinyl; strongest, needs coplanarity	forgetting SIR; counting meta as conjugated
Hyperconjugation	σ → p	C–Sn > C–Ge > C–Si > C–C ≳ C–H; angle-dependent	counting α-H blindly
Induction/field	σ framework, space	–I destabilises; +I stabilises; falls ≈ 2–3× per bond	overlooking +M of α-heteroatoms
Aromaticity	cyclic (4n+2) π	very large (tropylium, cyclopropenium)	calling 4n cations 'stable'
Ring strain	rehybridisation	I-strain (5 > 6), bridgehead penalties	ignoring conformational change on ionisation
Bridging (σ/π/n)	3c–2e or 3-ring onium	large for secondary/primary systems only	drawing both bridged and open ions as separate intermediates
Solvent/ion pair	electrostatics, H-bonding	can invert gas-phase orders	using solution rates as intrinsic stability

Part 5A Worked Solutions: the Supplied 'Exercise 2 – Carbocation Stability Order' (redrawn)

These are the stability-order problems you supplied (Exercise 2, Q1–Q35, and two multiple-choice questions). The structures are **redrawn here from the images** and the reasoning is new. Each answer is the order of **decreasing** stability, using the standard JEE-Advanced conventions: stability rises with substitution, resonance donors (+M), hyperconjugation (number and quality of C–H/C–C bonds in the correct position), aromaticity; falls with –I/–M groups, antiaromaticity and bridgehead (non-planar) geometry. For isotopes: C–H hyperconjugation > C–D > C–T when the isotopic group is on the carbon *next to* the cationic centre, but only the +I order (T > D > H) operates when the group is further away. Items I could not read with certainty (Q4, Q18, Q19, Q22, Q33, Q34) are listed at the end for you to confirm.

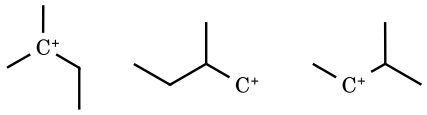
E2-1.
Arrange 1–3 in decreasing stability.



1 2 3

Answer: 3 > 2 > 1
Tertiary > secondary > primary: the number of alkyl groups on the cationic carbon (hyperconjugation and +I).

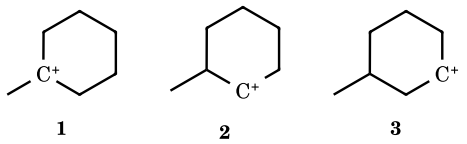
E2-2.
Arrange 1–3 in decreasing stability.



1 2 3

Answer: 1 > 3 > 2
1 is tertiary, 3 secondary, 2 primary (the 2-methylbutyl cation would rearrange, but as drawn it is primary).

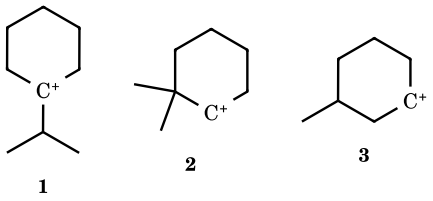
E2-3.
Arrange 1–3 in decreasing stability.



1 **2** **3**

Answer: 1 > 2 > 3
1 is tertiary. 2 and 3 are secondary; the methyl group is adjacent to the cationic carbon in 2 (its C–H and C–C bonds hyperconjugate/β-branching), but only γ in 3 (+I only).

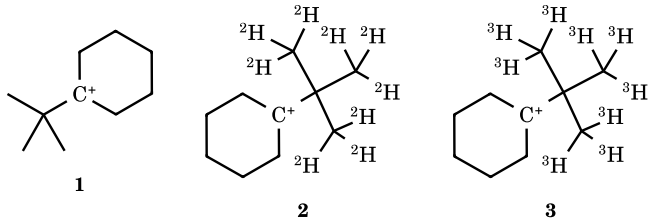
E2-5.
Arrange 1–3 in decreasing stability.



1 **2** **3**

Answer: 1 > 2 > 3
1 is tertiary. 2 and 3 are secondary; 2 has the more alkyl-branched neighbourhood (a quaternary carbon next to C^+), 3 only a remote methyl.

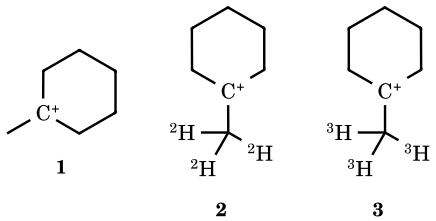
E2-6.
Arrange 1–3 in decreasing stability.



1 **2** **3**

Answer: 3 > 2 > 1
The labelled methyl groups are on the carbon *next to* the quaternary carbon, i.e. γ to C^+: no C–H/C–D/C–T bond is hyperconjugating. Only the inductive effect operates, and D and T are slightly more electron-releasing than H (T > D > H).

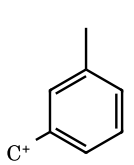
E2-7.
Arrange 1–3 in decreasing stability.



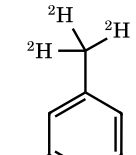
1 **2** **3**

Answer: 1 > 2 > 3
Now the labelled group is directly on C^+: hyperconjugation σ(C–X) → p operates, and C–H is a better donor than C–D, which is better than C–T (lower zero-point energy, shorter amplitude for the heavier isotope). β-Deuterium slows ionisation by about 1.3–2 per CD_3.

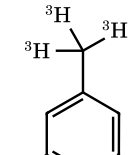
E2-8.
Arrange 1–3 in decreasing stability.



1



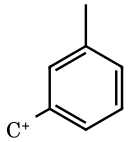
2



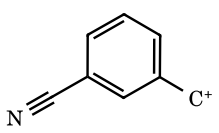
3

Answer: 3 > 2 > 1
A meta substituent cannot hyperconjugate with the benzylic cation (no resonance contributor puts charge on the meta carbon), so only the +I effect counts: $CT_3 > CD_3 > CH_3$.

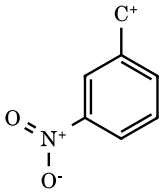
E2-9.
Arrange 1–3 in decreasing stability.



1



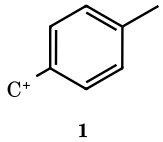
2



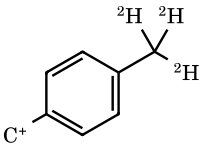
3

Answer: 1 > 2 > 3
m-CH_3 is weakly +I, m-CN and m-NO_2 are –I (σ_m 0.56 and 0.71): the stronger the –I, the less stable the cation.

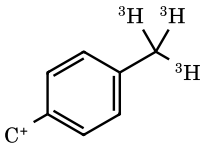
E2-10.
Arrange 1–3 in decreasing stability.



1



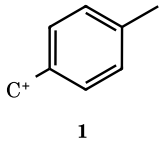
2



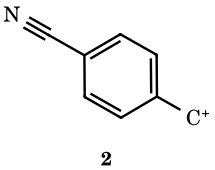
3

Answer: 1 > 2 > 3
p-Alkyl groups stabilise by hyperconjugation (charge is on the para carbon): C–H > C–D > C–T.

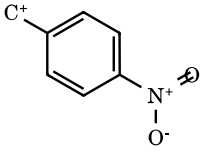
E2-11.
Arrange 1–3 in decreasing stability.



1



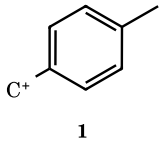
2



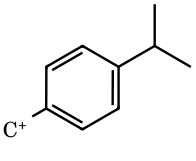
3

Answer: 1 > 2 > 3
p-CH_3 donates; p-CN and p-NO_2 withdraw by –I and –M (σ^+ 0.66 and 0.79), NO_2 more strongly.

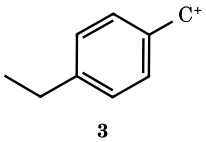
E2-12.
Arrange 1–3 in decreasing stability.



1



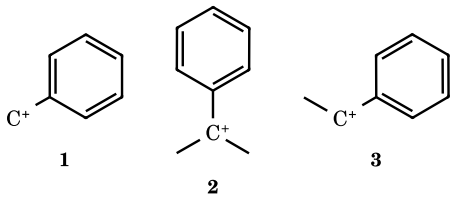
2



3

Answer: 1 > 3 > 2
Baker–Nathan order in solution: the number of hyperconjugating α-C–H bonds on the ring carbon: CH_3 (3) > CH_2R (2) > CHR_2 (1). (In the gas phase the order reverses, because polarisability then dominates.)

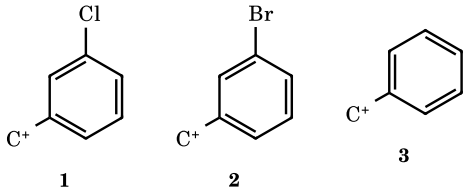
E2-13.
Arrange 1–3 in decreasing stability.



1 **2** **3**

Answer: 2 > 3 > 1
All are benzylic; each α-methyl group adds hyperconjugation/+I: tertiary (cumyl) > secondary (1-phenylethyl) > primary (benzyl).

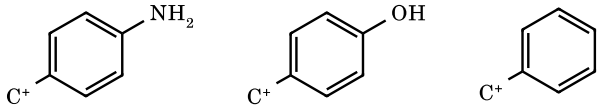
E2-14.
Arrange 1–3 in decreasing stability.



1 **2** **3**

Answer: 3 > 2 > 1
No substituent is best. A meta halogen is only –I (no +M can reach the benzylic cation from the meta position), and Cl is more electronegative than Br (σ_m^+ 0.40 vs 0.41, nearly equal; the conventional answer is Cl more destabilising).

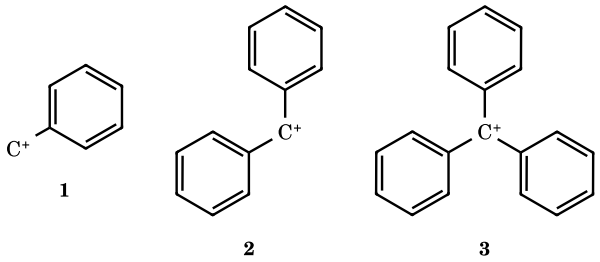
E2-15.
Arrange 1–3 in decreasing stability.



1 **2** **3**

Answer: 1 > 2 > 3
p-NH_2 and p-OH are +M donors (the quinoid iminium/oxocarbenium contributor); nitrogen is the better donor (σ^+ –1.3 vs –0.92).

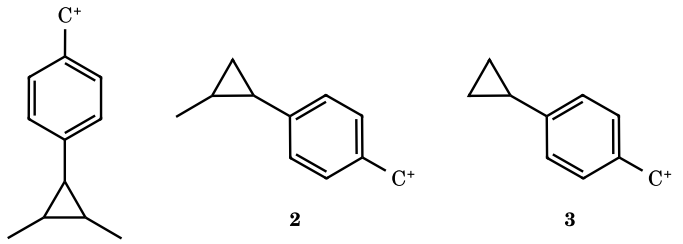
E2-16.
Arrange 1–3 in decreasing stability.



1 **2** **3**

Answer: 3 > 2 > 1
Each additional phenyl delocalises the charge further: trityl (pK_{R+} –6.6) > diphenylmethyl (–13.3) > benzyl.

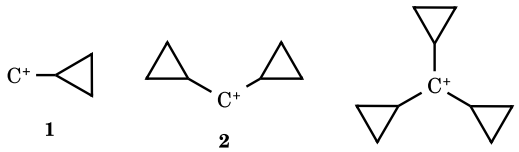
E2-17.
Arrange 1–3 in decreasing stability.



1 2 3

Answer: 1 > 2 > 3
A cyclopropyl ring on the benzene ring donates through its bent bonds (σ^+ −0.41); each extra methyl on the ring adds +I/hyperconjugation to the donor.

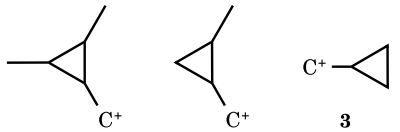
E2-20.
Arrange 1–3 in decreasing stability.



1 2 3

Answer: 3 > 2 > 1
Each cyclopropyl group donates its bent σ bonds in the bisected conformation: tricyclopropylmethyl (pK_{R+} −2.3, more stable than trityl) > dicyclopropyl > cyclopropylmethyl.

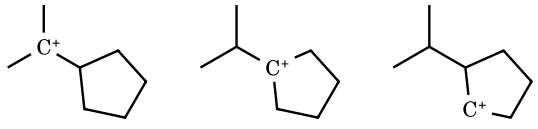
E2-21.
Arrange 1–3 in decreasing stability.



1 2 3

Answer: 1 > 2 > 3
Methyl groups on the cyclopropane ring increase its donor strength (+I and hyperconjugation).

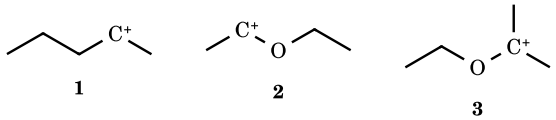
E2-23.
Arrange 1–3 in decreasing stability.



1 2 3

Answer: 2 > 1 > 3
2 and 1 are tertiary, 3 secondary. Ring cation 2 beats 1 because a five-membered ring prefers sp^2 carbon (I-strain: 1-methylcyclopentyl solvolyses ≈ 40× faster than t-butyl).

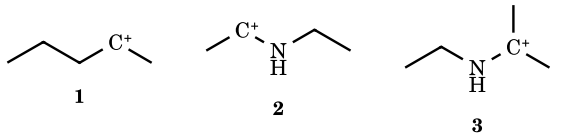
E2-24.
Arrange 1–3 in decreasing stability.



1 2 3

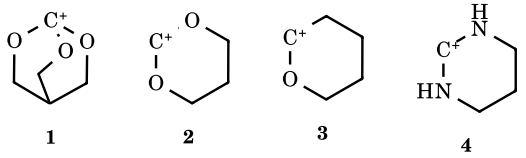
Answer: 3 > 2 > 1
An α-oxygen lone pair gives an oxocarbenium ion (+M ≫ hyperconjugation); the tertiary α-alkoxy ion also has two methyl groups.

E2-25.
Arrange 1–3 in decreasing stability.



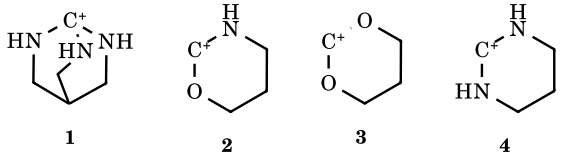
Answer: 3 > 2 > 1
An α-nitrogen lone pair gives an iminium ion; the more alkylated carbon (3) is better still. All are far more stable than the simple secondary alkyl cation 1.

E2-26.
Arrange 1–4 in decreasing stability.



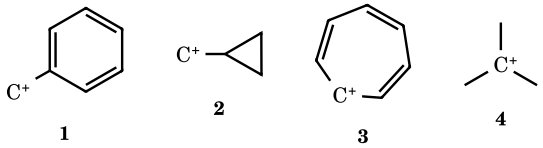
Answer: 4 > 2 > 3 > 1
Two nitrogen lone pairs (an amidinium-type ion, 4) > two oxygens (2) > one oxygen (3). Ion 1 would be best electronically (three oxygens) but it sits at a bridgehead of a rigid cage and cannot become planar, so the empty p orbital cannot overlap with the oxygen lone pairs: it is the least stable.

E2-27.
Arrange 1–4 in decreasing stability.



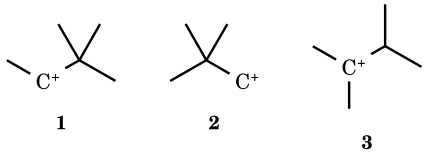
Answer: 4 > 2 > 3 > 1
N is a better π donor than O: N,N (4) > N,O (2) > O,O (3); the cage cation 1 is non-planar at the bridgehead and cannot be stabilised by resonance.

E2-28.
Arrange 1–4 in decreasing stability.



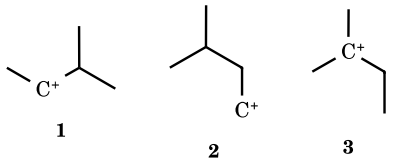
Answer: 3 > 2 > 1 > 4
Tropylium (aromatic 6π) is the most stable; cyclopropylmethyl (bisected σ-donation, 'dancing resonance') > benzyl (resonance) > tert-butyl (hyperconjugation only) *by the usual JEE convention that resonance ≫ hyperconjugation*. Note: measured gas-phase hydride affinities put t-Bu$^+$ (≈ 231 kcal mol^{-1}) slightly *below* benzyl (≈ 238), i.e. t-Bu$^+$ is the more stable in the gas phase; in solution the conventional order is the experimentally expected one for solvolysis.

E2-29.
Arrange 1–3 in decreasing stability.



Answer: 3 > 1 > 2
3 is tertiary; 1 is secondary (and will rearrange by a methyl shift to a tertiary ion); 2 is the primary neopentyl cation.

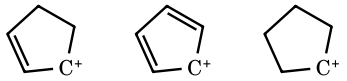
E2-30.
Arrange 1–3 in decreasing stability.



1 2 3

Answer: 3 > 1 > 2
Tertiary > secondary > primary.

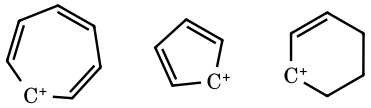
E2-31.
Arrange 1–3 in decreasing stability.



1 2 3

Answer: 1 > 3 > 2
1 is allylic (resonance); 3 is a simple secondary cation; 2, the cyclopentadienyl cation, is antiaromatic (4π) and the least stable.

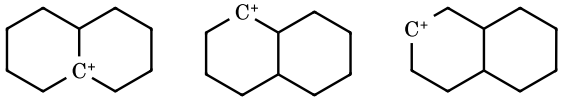
E2-32.
Arrange 1–3 in decreasing stability.



1 2 3

Answer: 1 > 3 > 2
Tropylium (aromatic, 6π) > cyclohex-2-enyl (allylic) > cyclopentadienyl (antiaromatic).

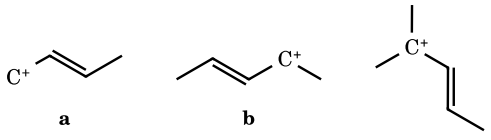
E2-35.
Arrange 1–3 in decreasing stability.



1 2 3

Answer: 1 > 2 > 3
1 is tertiary (ring fusion); 2 and 3 are secondary, and 2 has the branched ring-fusion carbon adjacent to C^+ (more hyperconjugation) while 3 does not.

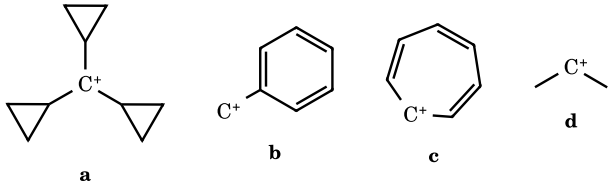
E2-91.
Which allylic carbocation is the most stable? (a) $CH_3–CH=CH–CH_2^+$, (b) $CH_3–CH=CH–CH^+–CH_3$, (c) $CH_3–CH=CH–C^+(CH_3)_2$, (d) all have the same stability.



a b c

Answer: (c)
Each is an allyl cation delocalised over two carbons, but the alkyl groups at the termini differ: (a) has a primary terminus, (b) a secondary, (c) a tertiary terminus. Charge is shared by both termini, so the more substituted ion (c) is the most stable. (d) is wrong: the alkyl substitution differs.

E2-93.
Which of the following carbocations is the most stable? (a) tricyclopropylmethyl, (b) benzyl, (c) tropylium (cycloheptatrienyl), (d) isopropyl.



a b c d

Answer: (c)
Tropylium is aromatic (6π, planar, charge shared by seven carbons; pK_{R+} +4.7): the order is tropylium > tricyclopropylmethyl > benzyl > isopropyl.

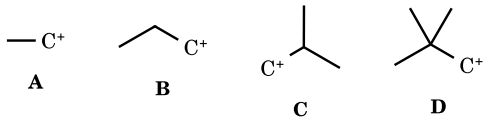
Items from your exercise that need confirmation

I could not read these structures reliably from the images, so I have **not** answered them. Please tell me (or send a clearer image of) the exact substituents: **Q4** (positions of the two methyl groups on each cyclohexyl cation), **Q18** and **Q19** (the ring substituents on the benzyl cations are not visible), **Q22** (the polyspiro cyclopropane/cyclobutane cations: the geometry of the spiro rings relative to the cationic centre decides whether cyclopropyl donation can operate), **Q33** (ring positions of the methyl groups) and **Q34** (which carbons of the oxygen heterocycles carry the double bonds).

Part 5B Worked Solutions: the Supplied 'Electronic Displacement Effects' Carbocation Problems (redrawn)

These are the carbocation-stability items from the 'Exercise O-II' set you supplied (Q8–Q16 of that exercise). Structures are redrawn and the reasoning is new. Orders are **decreasing** stability unless the question says otherwise, using JEE-Advanced conventions; where the textbook key conflicts with measured data it is flagged. The carbanion and radical items from the same pages (Q17–Q20) are answered in the carbanion and free-radical chapters, and the aromaticity, bond-length and resonance-energy items belong with those topics.

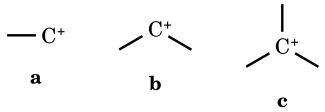
E3-8.
Select the least stable cation.



A B C D

Answer: (D) neopentyl
All four are primary, so count the hyperconjugating C–H bonds on the carbon next to C^+: ethyl 3, n-propyl 2, isobutyl 1, neopentyl 0 (its α-carbon is quaternary). Stability ethyl > propyl > isobutyl > neopentyl, so neopentyl is the least stable (the printed key in your PDF agrees). **Correction:** an earlier build of this book answered (A) using a +I/polarisability argument; that is the gas-phase trend, but the exam answer follows the α-H count.

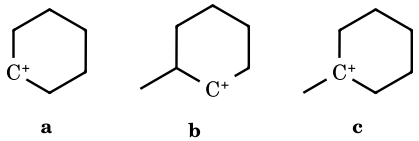
E3-9(i).
Arrange in decreasing stability.



a b c

Answer: c > b > a
Tertiary > secondary > primary.

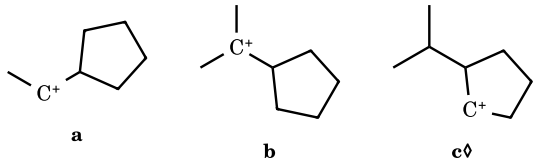
E3-9(ii).
Arrange in decreasing stability.



a b c

Answer: c > b > a
c is tertiary; a and b are secondary, and b carries an extra adjacent methyl-bearing carbon (more α-C–C/C–H hyperconjugation).

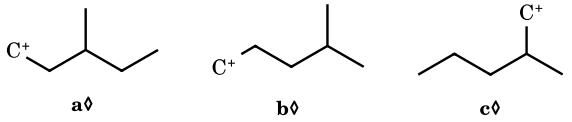
E3-9(iii).
Arrange in decreasing stability (cyclopentyl-substituted cations).



a b c◊

Answer: b > c > a (key)
b is tertiary. a and c are secondary; the key puts the ring cation c above the exocyclic cation a (cyclopentyl cations are favoured by relief of I-strain, and in c the isopropyl group adds hyperconjugating C–C/C–H bonds next to C^+). ◊ The position of the isopropyl group in c is my reading of the drawing.

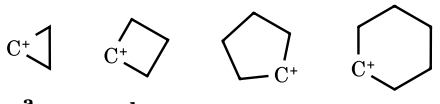
E3-9(vi).
Arrange in decreasing stability (three alkyl cations from the chain drawings).



a◊ b◊ c◊

Answer: key: b > c > a; your Allen solution page: a > b > c
Conflict. The printed answer key of your PDF gives b > c > a for 9(vi), but the worked solution on the 'Electronic displacement effects' page you also sent gives a > b > c (α-H = 2, 2, 1, then +I). I cannot tell from the drawings which chain is which, so I do not choose between them. The rule they use is: count α C–H bonds first (more is better), and if equal, the stronger +I (larger or more branched alkyl) wins.

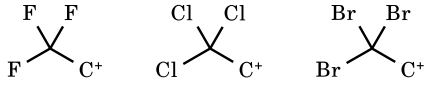
E3-9(iv).
Arrange the cycloalkyl cations in decreasing stability.



a b c d

Answer: d > c > b > a (textbook key)
Textbook key: angle strain at the sp^2 centre (60° in cyclopropyl, 90° in cyclobutyl) destabilises the small rings. **Conflict with data:** solvolysis rates (I-strain) make cyclopentyl faster than cyclohexyl, and cyclobutyl is stabilised by non-classical bridging, so the experimental order is closer to c > d > b > a. If a question forces one answer, use the key; know the data.

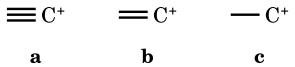
E3-9(v).
Arrange in decreasing stability.



a b c

Answer: c > b > a
All three groups are –I; the strength of the effect follows electronegativity, F ⩾ Cl > Br, so CBr_3 destabilises least.

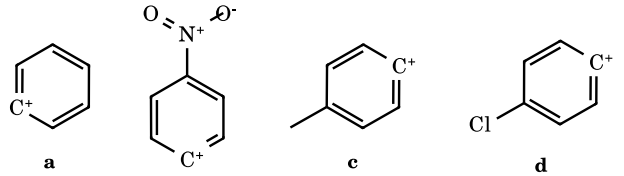
E3-9(vii).
Arrange in decreasing stability.



a b c

Answer: c > b > a
The empty orbital sits on sp (50% s) or sp^2 (33% s) carbon: the more s-character, the more tightly held the electrons and the less stable the cation. Ethyl uses a p orbital.

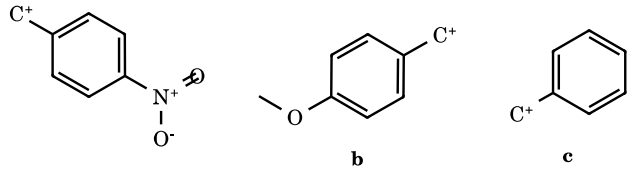
E3-9(viii).
Arrange the aryl cations in decreasing stability.



a b c d

Answer: c > a > d > b
The empty sp^2 orbital lies in the ring plane, orthogonal to the π system, so there is no resonance: only the inductive effect operates. CH_3 is +I; Cl is –I and NO_2 is more strongly –I.

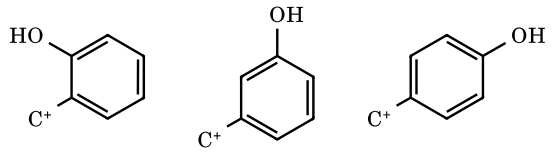
E3-9(ix).
Arrange the benzyl cations in decreasing stability.



a b c

Answer: b > c > a
p-OMe donates by +M (oxocarbenium contributor); NO_2 withdraws by –M and –I.

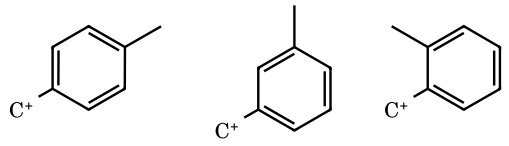
E3-9(x).
Arrange the hydroxybenzyl cations in decreasing stability.



a b c

Answer: c > a > b
Ortho and para OH can place the positive charge on oxygen (+M); meta cannot, and there OH is only –I. Para is better than ortho by the usual key (ortho: the –I effect is stronger at short distance and the oxygen lone pair is nearer the charge).

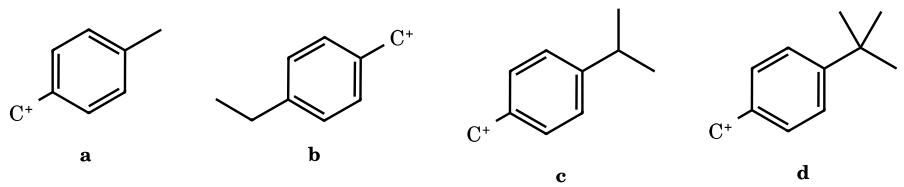
E3-9(xi).
Arrange the methylbenzyl cations in decreasing stability.



a b c

Answer: c > a > b
Ortho and para CH_3 hyperconjugate with the delocalised charge; meta cannot. Textbook key puts ortho first (closer +I and hyperconjugation); steric effects can reverse this in real systems.

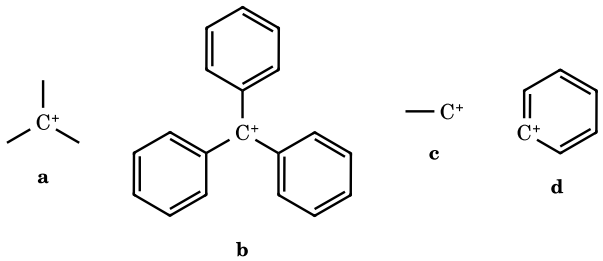
E3-9(xii).
Arrange the p-alkylbenzyl cations in decreasing stability.



a b c d

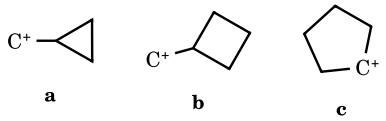
Answer: a > b > c > d
Baker–Nathan order: the number of hyperconjugating α-C–H bonds is 3, 2, 1, 0. (Pure inductive donation would give the opposite order.)

E3-9(xiii).
Arrange in decreasing stability.



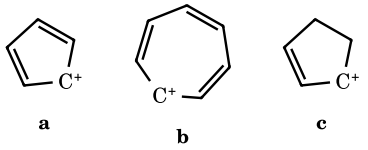
Answer: b > a > c > d
Trityl (three-ring delocalisation) > tertiary alkyl > primary alkyl > phenyl cation (no resonance, charge in an sp^2 orbital).

E3-9(xv).
Arrange in decreasing stability.



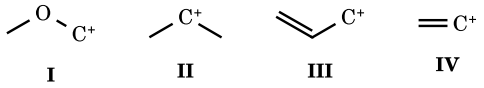
Answer: a > c > b
Cyclopropylmethyl is stabilised by bent-bond donation (σ → p overlap in the bisected conformation); cyclopentyl is secondary; cyclobutylmethyl is an ordinary primary cation.

E3-9(xvi).
Arrange in decreasing stability.



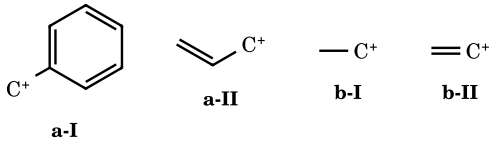
Answer: b > c > a
Tropylium is aromatic (6π); cyclopent-2-enyl is allylic; cyclopentadienyl is antiaromatic (4π) and the least stable of all.

E3-10.
Which statements are correct? (I) $CH_3OCH_2^+$ is more stable than $CH_3CH_2^+$. (II) $(CH_3)_2CH^+$ is more stable than $CH_3CH_2CH_2^+$. (III) $CH_2{=}CH{-}CH_2^+$ is more stable than $CH_3CH_2CH_2^+$. (IV) $CH_2{=}CH^+$ is more stable than $CH_3CH_2^+$. (A) I, II (B) III, IV (C) I, II, III (D) II, III, IV



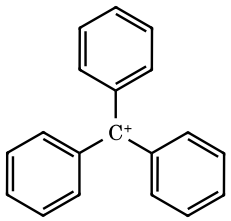
Answer: (C)
I: α-oxygen lone pair gives an oxocarbenium ion. II: secondary > primary. III: allyl resonance. IV is false: the vinyl cation has an sp-hybridised cationic carbon, which is less stable than ethyl.

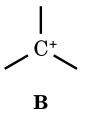
E3-11a/b.
Which ion is more stable? (a) $C_6H_5CH_2^+$ or $CH_2{=}CH{-}CH_2^+$; (b) $CH_3CH_2^+$ or $CH_2{=}CH^+$.



Answer: (a) benzyl; (b) ethyl
(a) Benzyl delocalises the charge over four carbons (seven contributors incl. ring), allyl over two. (b) The vinyl cation is sp-hybridised and the empty p orbital has no hyperconjugating partners except one C–H.

E3-13.
Which is the most stable? (A) Ph_3C^+ (B) $(CH_3)_3C^+$ (C) $(CH_3)_2CH^+$ (D) $CH_2{=}CH{-}CH_2^+$


A

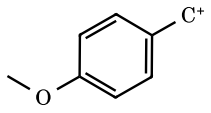

B

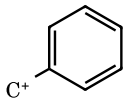

C

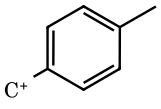

D

Answer: (A)
Trityl: charge is delocalised over three rings (ten carbons' worth of contributors); pK_{R+} = −6.6 compared with about −15 for t-butyl.

E3-14.
Increasing order of stability of (a) p-$CH_3O{-}C_6H_4CH_2^+$ (b) $C_6H_5CH_2^+$ (c) p-$CH_3{-}C_6H_4CH_2^+$ (d) $CH_3CH_2^+$.


a

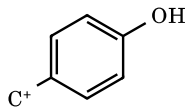

b

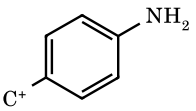

c

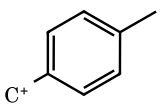

d

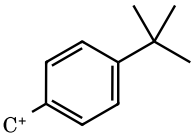
Answer: d < b < c < a → (A)
Ethyl (no resonance) < benzyl < p-methylbenzyl (hyperconjugation) < p-methoxybenzyl (+M).

E3-15.
Correct order of stability: (1) p-HO– (2) p-H_2N– (3) p-CH_3– (4) p-$(CH_3)_3C$– benzyl cation.


1

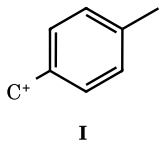

2

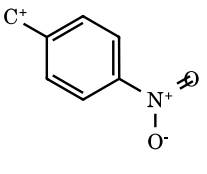

3

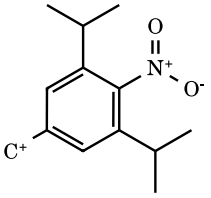

4

Answer: 2 > 1 > 3 > 4 → (D)
NH_2 is a stronger +M donor than OH (σ^+ −1.3 vs −0.92); CH_3 has three α-C–H bonds, t-Bu none.

E3-16.
Increasing order of stability: (I) p-methylbenzyl, (II) p-nitrobenzyl, (III) 3,5-diisopropyl-4-nitrobenzyl cation.


I


II

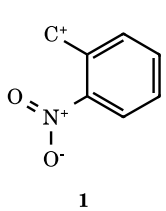

III

Answer: II < III < I → (B)
Steric inhibition of resonance: the two flanking isopropyl groups twist the nitro group out of the ring plane, so it loses its –M effect and only –I remains. III is therefore less destabilised than II (full –M and –I), but still less stable than the methyl-substituted I.

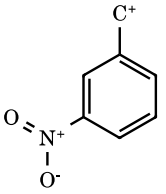
Part 5C Worked Solutions: Supplied 'How Many' Counting Problems (redrawn)

Counting problems need every item judged correctly, so the reasoning for each item is given. The structures are **redrawn from your images**; where a drawing was ambiguous I state the reading I used, because a different reading can change the count. Items marked ◊ are the ones to double-check against your original.

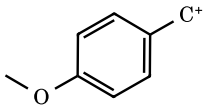
C-7.
How many of these cations are more stable than the benzyl cation?



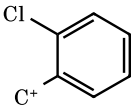
1



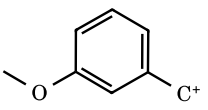
2



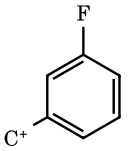
3



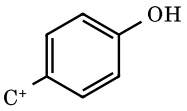
4



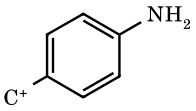
5



6



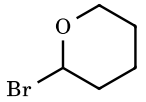
7



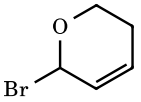
8

Answer: 3 (p-OCH_3, p-OH, p-NH_2)
Only +M donors in the ortho/para position beat hydrogen: p-OMe, p-OH and p-NH_2 (σ^+ −0.78, −0.92, −1.3). o-NO_2 and m-NO_2 are strongly –I (–M too in the ortho case). m-OMe and m-F act only by –I, because no resonance contributor puts charge on the meta carbon. o-Cl is –I and its +M is weak (σ^+_p Cl = +0.11, i.e. slightly destabilising), so it is below benzyl.

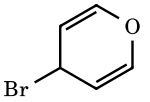
C-11a.
Among the pairs, how many have (i) more reactive than (ii) towards $AgNO_3$? (Reactivity = ease of ionising C–X, i.e. stability of the carbocation formed; SN1 path.)



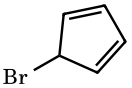
A-i



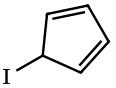
A-ii◊



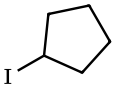
B-i



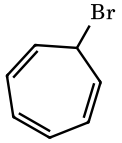
B-ii



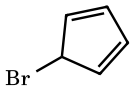
C-i◊



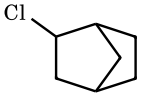
C-ii



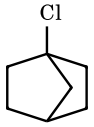
D-i



D-ii



E-i

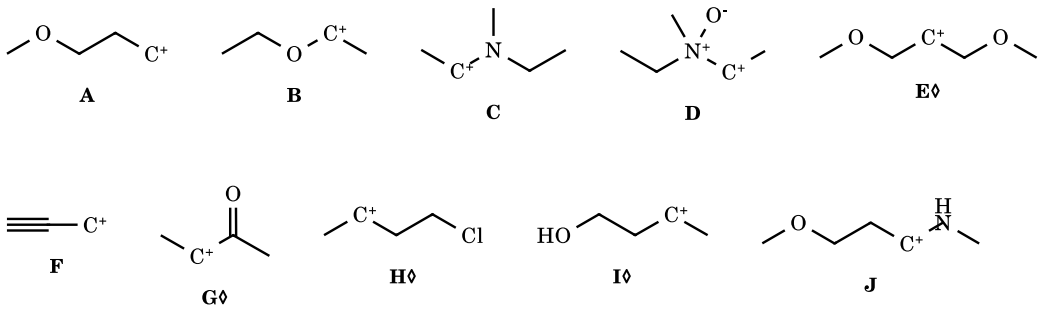


E-ii

Answer: 3 (B, D and E)
A: both give an oxocarbenium ion; in my reading (ii) is also allylic, so (ii) is more reactive (not counted). If (ii) is an isolated or enol-ether double bond the pair would count, so check the position of the double bond. **B**: (i) loses Br^- to give the *pyrylium* ion, aromatic (6π, O lone pair + two C=C); (ii) would give the antiaromatic cyclopentadienyl cation. (i) ≫ (ii): counts. **C**: reading (i) as 5-iodocyclopenta-1,3-diene, ionisation gives the antiaromatic cyclopentadienyl cation, much worse than a plain secondary cyclopentyl cation, so (i) is *less* reactive (not counted). If (i) were 3-iodocyclopentene (allylic) it would count. **D**: (i) gives the aromatic tropylium ion; (ii) the antiaromatic cyclopentadienyl cation: counts. **E**: (i) is a secondary 2-chloronorbornane (ionises, with σ-participation); (ii) is a bridgehead chloride, whose cation cannot become planar: counts.

C-12a.

How many of the following are more stable than the isopropyl cation, owing to +R together with –I effects?

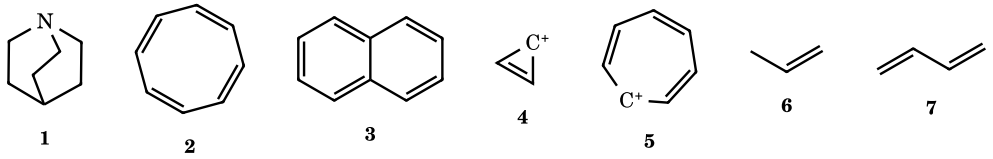


Answer: 3 (B, C and J)

A +R group must be bonded *directly* to C^+ with a free lone pair. B is an oxocarbenium ion and C and J are iminium ions: all three are far more stable than a secondary alkyl cation. D: the nitrogen lone pair is tied up in the N→O bond, so there is no +R and a strong –I (N^+–O^-). A, E, H and I have the heteroatom or halogen one or more carbons away from C^+: only –I operates, which destabilises. F: propargyl is resonance-stabilised but the sp carbon is –I, and it is less stable than secondary alkyl (gas-phase hydride affinity higher than i-Pr^+). G: α-carbonyl cation, destabilised by –I and –M.

C-11b.

Species: quinuclidine, cyclooctatetraene, naphthalene, cyclopropenyl cation, tropylium ion, propene, 1,3-butadiene. x = number in which resonance takes place; y = number in which every C–C bond length is equal. Find $(x^2 + y^2 + 1)/(x + y + 1)$.

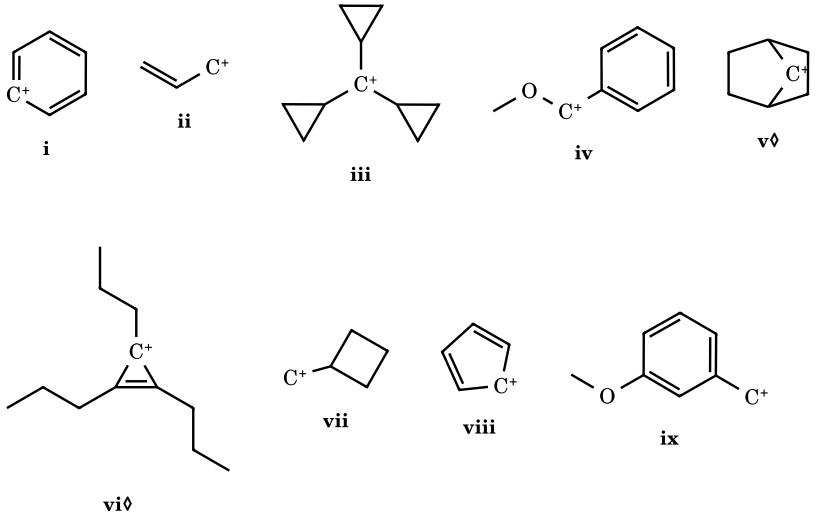


Answer: x = 4, y = 2, value = 21/7 = 3

Resonance (π delocalisation) occurs in naphthalene, the cyclopropenyl cation, tropylium and 1,3-butadiene: **x = 4**. Quinuclidine has no π system; cyclooctatetraene is tub-shaped, so its double bonds do not conjugate (alternating bond lengths); propene only has hyperconjugation. All C–C bonds are equal only in the cyclopropenyl and tropylium cations (complete delocalisation round a symmetric ring): **y = 2**. Naphthalene (1.36/1.42 Å) and butadiene have unequal bonds; in quinuclidine C2–C3 and C3–C4 are not symmetry-equivalent. (16 + 4 + 1)/(4 + 2 + 1) = 3.

C-12b.

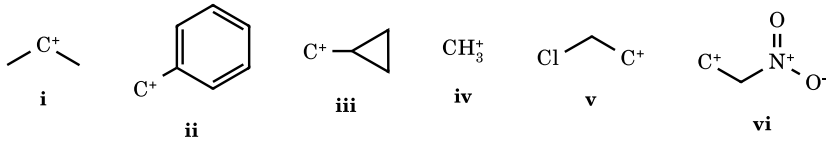
How many of the following cations are more stable than the benzyl cation?



Answer: 3 (iii, iv, vi)

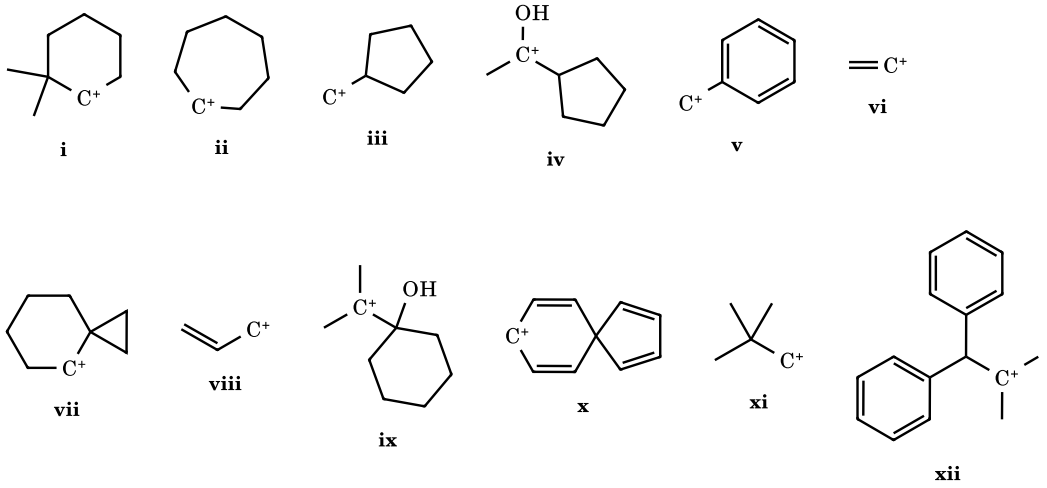
iii: tricyclopropylmethyl (pK_{R+} –2.3) is more stable than benzyl. iv: α-methoxybenzyl is both benzylic and an oxocarbenium ion. vi: an alkyl-substituted cyclopropenyl cation is aromatic (2π, pK_{R+} about +10). i: phenyl cation has its empty orbital in the ring plane, no resonance. ii: allyl is less stabilised than benzyl. v: the norbornyl cation at C7 (or at a bridgehead) is held non-planar and cannot delocalise. vii: ordinary primary cation. viii: antiaromatic. ix: m-OMe is only –I.

C-15.
How many of the given cations are more stable than the ethyl cation?



Answer: 3 (i, ii, iii)
i: secondary > primary. ii: resonance. iii: bent-bond donation from the cyclopropane ring. iv: methyl cation is less stable than ethyl (no hyperconjugation). v and vi: the –I effect of Cl and NO_2 destabilises the classical ion. (Real β-chloroalkyl cations are in fact bridged chloronium ions, which are lower in energy than a classical ethyl cation; the exam-level answer treats v as destabilised.)

C-23.
How many carbocations undergo rearrangement?



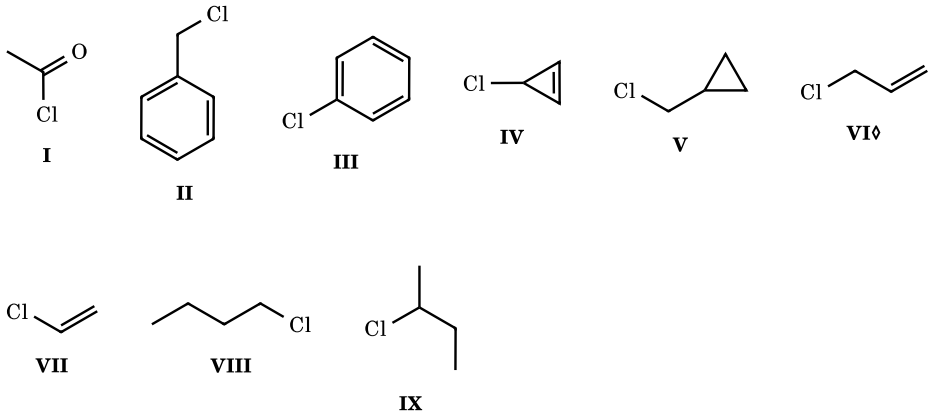
Answer: 7 (i, iii, vii, ix, x, xi, xii)
A cation rearranges only if a 1,2-shift gives a more stable ion. **i** 2,2-dimethylcyclohexyl (2°) → methyl shift → tertiary. **iii** cyclopentylmethyl (1°) → hydride shift to 1-methylcyclopentyl (3°), or ring expansion to cyclohexyl. **vii** cyclopropylcarbinyl-type: ring expansion of the cyclopropane to a cyclobutyl ring gives a tertiary ion. **ix** pinacol-type: ring-carbon migration to the cationic carbon expands the ring and gives a protonated ketone (2,2-dimethylcycloheptanone). **x** this is an ipso arenium ion with a spiro carbon: a 1,2 alkyl shift restores the aromatic ring (dienone–phenol type). **xi** neopentyl → methyl shift → tert-amyl. **xii** $Ph_2CH–C^+Me_2$ → hydride shift → $Ph_2C^+–CHMe_2$ (diphenylmethyl-type, more stable). No rearrangement: **ii** cycloheptyl (a hydride shift gives the same type of ion), **iv** an α-hydroxy carbenium ion is already stabilised by oxygen, and the shift would lose that, **v** benzyl, **vi** vinyl, **viii** allyl (resonance, not rearrangement).

Part 5D Worked Solutions: More Supplied Counting Problems (heterolysis, statements, rearrangement, $AgNO_3$)

Structures are redrawn from your images. Items marked ◊ were hard to read; the reading I used is stated, and a different reading can change the count. Conventions as before: the JEE key ranks resonance above hyperconjugation, so benzyl is taken as more stable than t-butyl even though gas-phase hydride affinities say otherwise.

D-27.

For how many compounds is the heterolytic C–Cl bond energy less than that of $(CH_3)_3C–Cl$? (A lower heterolysis energy means a more stable cation is formed than t-Bu^+.)

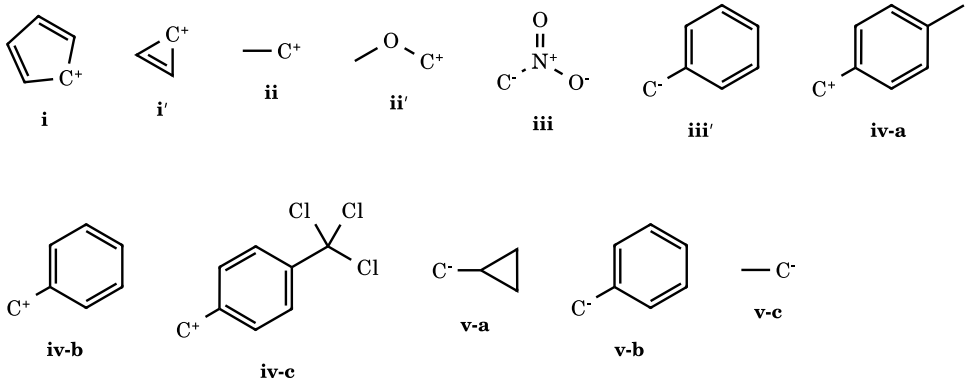


Answer: 4 (I, II, IV, V)

I gives the acylium ion $CH_3C \equiv O^+$, stabilised by oxygen lone-pair donation: counts (conventional key). **II** benzyl (resonance over the ring): counts. **IV** 3-chlorocyclopropene gives the aromatic cyclopropenyl cation: counts. **V** cyclopropylmethyl cation (bent-bond donation, more stable than benzyl): counts. **III** and **VII**: aryl and vinyl cations (sp^2/sp, no resonance, C–Cl also strengthened by p–π overlap): no. **VIII** primary and **IX** secondary: less stable than tertiary. **VI** allyl: a primary cation with two resonance forms, ranked below the tertiary cation by the usual key (gas-phase hydride affinity agrees: allyl is less stable than t-Bu^+). If your source ranks allyl above tertiary the count would be 5.

D-38.

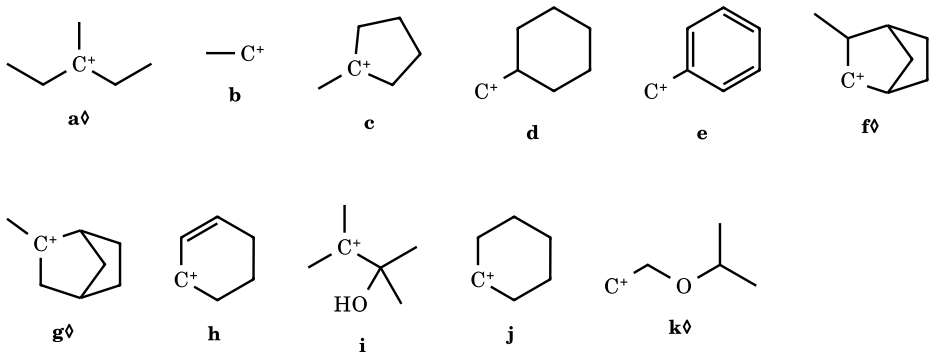
Find the number of correct statements.



Answer: 5 (i, ii, iii, iv, vi correct; v is wrong)

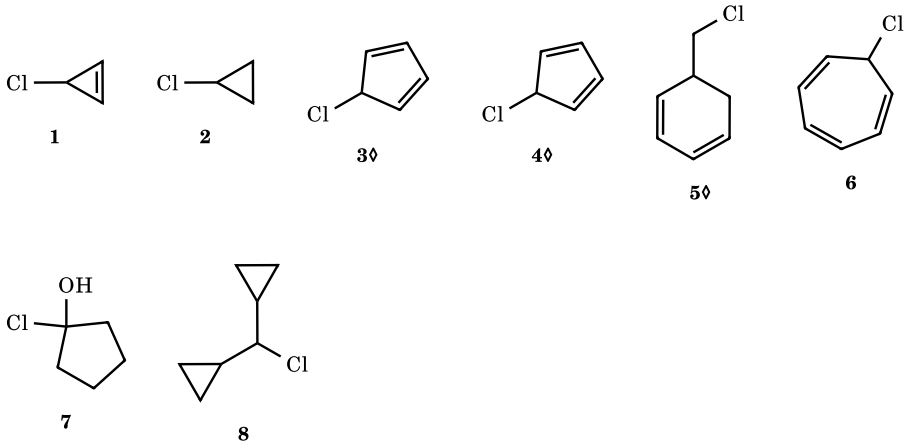
(i) Cyclopentadienyl cation is antiaromatic (4π); cyclopropenyl is aromatic (2π): the statement is **correct**. (ii) Ethyl is less stable than methoxymethyl (oxocarbenium, +M): **correct**. (iii) $^-CH_2NO_2$ is delocalised onto oxygen (nitronate), much better than benzyl anion: **correct**. (iv) p-CH_3 donates, CCl_3 withdraws (–I): p-methylbenzyl > benzyl > p-CCl_3-benzyl: **correct**. (v) Benzyl anion is delocalised into the ring and is much more stable than cyclopropylmethyl anion, which only has a weak s-character effect: the correct order is benzyl > cyclopropylmethyl > ethyl, so the statement is **wrong**. (vi) Singlet carbene stability follows π-donation from halogen lone pairs into the empty p orbital, best for 2p–2p overlap with F, then Cl, Br, I: $CF_2 > CCl_2 > CBr_2 > CI_2$: **correct**.

D-19 / D-21b.
How many of these carbocations can undergo rearrangement? (The same set appears twice in your pages.)



Answer: 4 (d, f, i, k)
d cyclohexylmethyl (1°): hydride shift to 1-methylcyclohexyl (3°) or ring expansion to cycloheptyl. **f** 3-methyl-2-norbornyl (2°): hydride shift from the methyl-bearing carbon gives the tertiary 2-methyl-2-norbornyl ion. **i** pinacol-type $(CH_3)_2C^+$–C(OH)(CH_3)–: a methyl shift gives a protonated ketone. **k** the β-alkoxy primary cation $^+CH_2CH_2OR$: a 1,2-hydride shift gives the oxocarbenium ion CH_3CH^+OR, far more stable. No rearrangement: **a** tertiary (3-methyl-3-pentyl, as I read it); **b** ethyl (the only shift is degenerate); **c** tertiary; **e** benzyl; **g** tertiary 2-methylnorbornyl (a shift would only give another tertiary or a less stable ion); **h** allylic; **j** cyclohexyl (the 1,2-H shift is degenerate; ring contraction to methylcyclopentyl is only seen in superacid, so it is not counted at this level). ◊ If **a** is really a secondary cation next to a quaternary carbon, it would rearrange and the count becomes 5.

D-21a.
How many of the following give a white precipitate with aqueous $AgNO_3$? (AgCl forms only if the C–Cl bond ionises readily, i.e. the cation is very stable.)



Answer: 4 (1, 6, 7, 8)
1 3-chlorocyclopropene → aromatic cyclopropenyl cation. **6** 7-chlorocycloheptatriene → tropylium chloride (an ionic salt, immediate AgCl). **7** 1-chlorocyclopentanol: an α-chloro alcohol loses Cl^- to give the protonated ketone (oxocarbenium), then cyclopentanone. **8** dicyclopropylmethyl chloride: the cation is stabilised by two cyclopropyl groups. **2** cyclopropyl chloride: the cyclopropyl cation is strained and unstable (ionisation and ring-opening are concerted and slow). **3, 4** I read these as 5-chlorocyclopenta-1,3-dienes: loss of Cl^- would give the antiaromatic cyclopentadienyl cation, so no precipitate. **5** I read this as a chloromethyl group on a saturated ring position, an ordinary primary chloride: no precipitate. ◊ Please check items 3–5 against your page: if any of them is allylic (for example 3-chlorocyclopentene) it would give a precipitate.

Part 5E Your 'Stability of Carbocation' Notes (cation.pdf): everything added, redrawn and checked

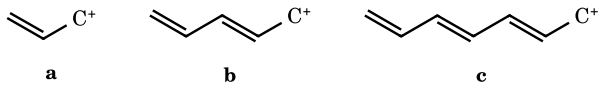
This part reproduces the content of the 11-page handwritten note you supplied: the factor list, the worked orders 1–12, the pK_{R^+} ladder, the gas-phase hydride-affinity table, the tabulated pK values, σ-bond resonance and the non-classical ions, and the aryl-cation substituent series. Structures are redrawn; numerical values are **transcribed from your pages** (they are literature values quoted in the note, not independently re-measured here). Where something in the note disagrees with another part of the same note, or with the data it quotes, the disagreement is marked ⚑ **Check** and the reasoning is given, so you can decide which to use in an exam.

5E.1 The factor list

Stabilising factor	Rule from the note	Comment
+R / +M	stability ∝ +R/+M	lone-pair or π donation into the empty p orbital; the strongest effect (oxocarbenium, iminium, arylmethyl with p-OMe/NR_2)
+H (hyperconjugation)	∝ number of α C–H (and C–C) bonds	counted on the carbon next to C^+; 9α > 6α > 4α in example 3
+I	∝ electron release by alkyl	matters when hyperconjugation is absent or equal
–I, –H, –M	stability ∝ 1/(–I), 1/(–H), 1/(–M)	withdrawing groups, loss of hyperconjugating bonds, and acceptors destabilise

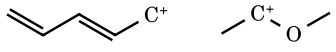
The note adds two measured quantities: the gas-phase **hydride affinity** (HIA; lower = more stable cation) and the solution $\mathbf{pK_{R^+}}$ (higher = more stable cation). They are separate observables and, as the note's own data show, do not always give the same order.

5E.2 Worked orders 1–12 (redrawn)



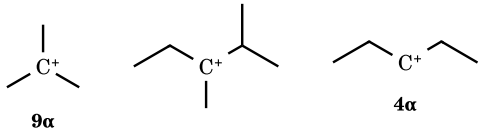
a **b** **c**

1. Longer conjugation: a < b < c (allyl < pentadienyl < heptatrienyl); more resonance contributors spread the charge over more carbons.



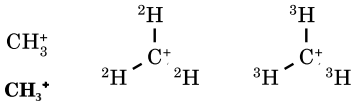
a **b**

2. a < b. The pentadienyl cation a has a sextet on carbon; in b oxygen gives a lone pair so every atom has an octet (oxocarbenium).



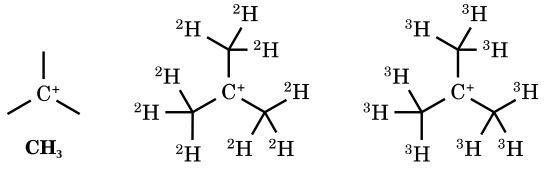
9α **6α** **4α**

3. 9α > 6α > 4α, i.e. hyperconjugation by α C–H bond count (H > I). ⚑ Check: as drawn (second and third), the counts follow the note; the middle cation is tertiary and the last secondary, so the order is also tertiary > … secondary, which agrees.



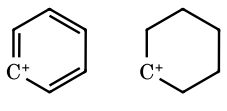
CH_3^+ **CD_3^+** **CT_3^+**

4. CH_3^+ < CD_3^+ < CT_3^+. The isotopes sit on C^+, so no C–X bond is hyperconjugating: only the +I order T > D > H is left.



CH_3 **CD_3** **CT_3**

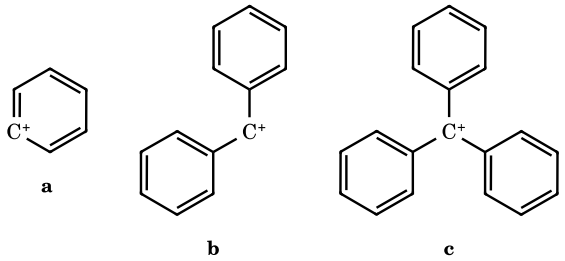
5. $(CH_3)_3C^+$ > $(CD_3)_3C^+$ > $(CT_3)_3C^+$. Now the labelled groups are β to the charge: hyperconjugation dominates, and C–H (weakest, highest zero-point energy) donates best. The note summarises: +I order CT_3 > CD_3 > CH_3; hyperconjugation (+H) order CH_3 > CD_3 > CT_3 (BDE C–T > C–D > C–H); the +H effect wins.



a **b**

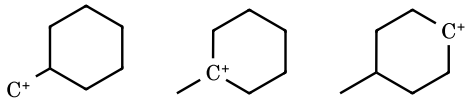
6. As drawn in the note: a > b (phenyl cation > cyclohexyl cation).

⚑ **Check, item 6.** This inequality is the wrong way round on the note's own data: the hydride affinity table (5E.4) gives phenyl 294 and a secondary cation (isopropyl) 249 kcal/mol; a secondary cyclohexyl cation is therefore far *more* stable than phenyl, whose empty orbital is an sp^2 σ orbital in the ring plane with no resonance. I recommend recording: cyclohexyl > phenyl. If your source key says otherwise, treat it as a misprint.



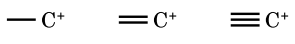
a **b** **c**

7. a < b < c: phenyl cation < diphenylmethyl < trityl (each phenyl delocalises the charge).



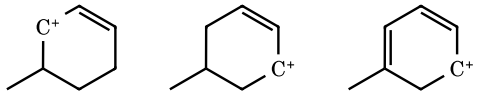
a (1°) **b (3°)** **c (2°)**

8. a < c < b: primary < secondary < tertiary.



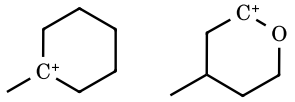
a **b** **c**

9. a > b > c: the more s-character in the carbon bearing the empty orbital (sp^3-like p, sp^2, sp), the less stable the cation.



a◊ **b◊** **c◊**

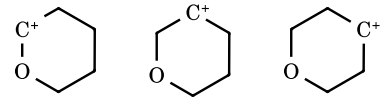
10. c > a > b. c is a cyclohexadienyl (pentadienyl-type, two C=C) cation; a and b are allylic secondary; a has a methyl-bearing carbon next to the cationic carbon (extra hyperconjugation), b only a remote one. ◊ Positions of the methyl and the ring double bonds are my reading of the drawing.



a **b**

11. a < b as printed in the note (cyclic oxocarbenium: b has the + next to O).

⚑ **Check, item 11.** In the second structure the cation sits next to the ring oxygen only if the positive charge is on C2 (drawn at the position adjacent to O). That is the oxocarbenium ion, and it is more stable than the tertiary 1-methylcyclohexyl cation, which agrees with the note. If the cation were on C3 (β to O) the –I effect would make it *less* stable than the tertiary ion; item 12 shows exactly that difference.

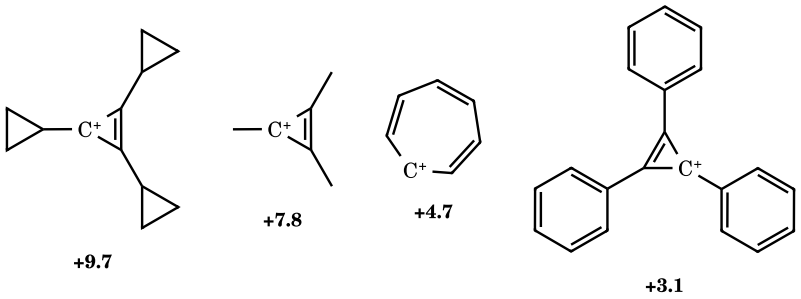


a (C2, +M) **b (C3, –I)** **c (C4, –I)**

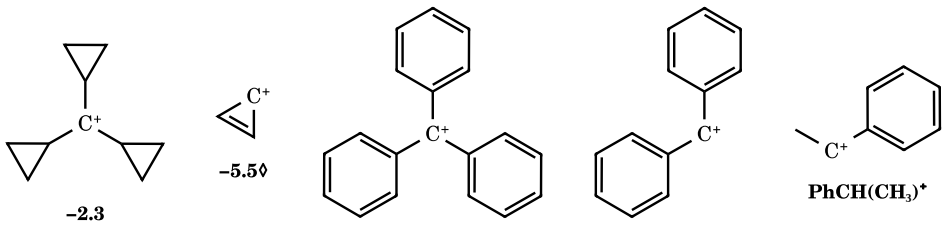
12. a > b > c: oxygen donates its lone pair when the cation is α (C2); at C3 and C4 only the –I effect operates, strongest at C3 (nearer to O).

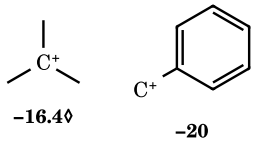
5E.3 'Important order': the pK_{R^+} ladder

pK_{R^+} is defined from the equilibrium $R^+ + H_2O \rightleftharpoons ROH + H^+$ by $pK_{R^+} = \log([R^+]/[ROH]) + H_R$, where H_R is the acidity function that plays the role of pH in strong acid. A cation with a larger pK_{R^+} persists at a more acidic medium, i.e. is more stable. (Your note writes the equilibrium as $H^+ + ROH \rightleftharpoons R^+ + H_2O$; that is the same reaction read in the cation-forming direction, and the formula above is the standard one.)

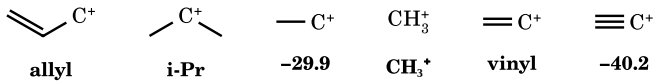


Aromatic cations first: tricyclopropylcyclopropenyl > trimethylcyclopropenyl > tropylium > triphenylcyclopropenyl.





…then tricyclopropylmethyl > (parent cyclopropenyl, value as printed) > Ph_3C^+ > Ph_2CH^+ > $PhCH(CH_3)^+$ > $(CH_3)_3C^+$ > $PhCH_2^+$.



…then allyl > isopropyl > ethyl (−29.9) > methyl > vinyl > ethynyl (−40.2).

> ⚑ **Check (values).** (1) Your ladder gives t-Bu$^+$ = −16.4, but the Carey–Sundberg table on page 6 of the same note gives −15.5 (and McManus's page says 'estimate −15.5' in FSO_3H–SbF_5); I would use −15.5. (2) The tricyclopropylcyclopropenyl cation is +9.7 on the ladder and in the Carey table, but '7.2' on page 5. (3) Tropylium is +4.7 on the ladder and in Carey, '4.2' on page 5. (4) The ladder puts the parent cyclopropenyl cation at −5.5; I cannot confirm this value from the tables you supplied, so it is marked ◊. **Important consequence:** on this ladder t-butyl (−16.4/−15.5) is *above* benzyl (≈ −20), and 2-phenyl-2-propyl (cumyl, −12.3) is above both. So **on your source's data, t-Bu$^+$ is more stable than the benzyl cation**, not less. Earlier parts of this book use the usual 'resonance > hyperconjugation' coaching convention for benzyl vs t-Bu in some answers and flag it; with this note as your source, use t-Bu$^+$ > benzyl. This changes the count in D-27 (below).

5E.4 Gas-phase hydride affinities (kcal mol^{-1}, as in your note)

R–H → R$^+$ + H$^-$; the smaller the number, the more stable the cation. Values are those written on your page (attributed there to March); they are quoted, not re-derived.

Cation	HIA	Cation	HIA	Cation	HIA
CH_3^+	314.6	tropylium	200	$PhCH_2^+$	238
$CH_3CH_2^+$ (bridged)	276.7	CH_2=CH–CH_2^+	256	$PhCH^+CH_3$	225
$(CH_3)_2CH^+$	249.2	CH_2=CH–CH^+–CH_3	239	$PhC^+(CH_3)_2$	220
$(CH_3)_3C^+$	231.9	CH_2=CH–$C^+(CH_3)_2$	230		
CH_2=CH^+ (bridged)	287	CH_3CH=CH–CH^+CH_3	228		
cyclopropenyl	223◊	phenyl	294		

Reading the table: tropylium (200) is by far the most stable, then the α,α-dimethylbenzyl (220) and 1-phenylethyl (225) cations; **t-Bu$^+$ (231.9) is more stable than benzyl (238)**; allyl (256) is less stable than isopropyl (249); the stability of the substituted-allyl series follows substitution (256 → 239 → 230 → 228). Phenyl (294) and vinyl (287) are the worst, as expected for cations whose empty orbital is sp^2/sp. ◊ The value 223 for cyclopropenyl is as written in your note; for an aromatic cation it looks comparatively high next to tropylium (200), so check it against your source before using it numerically.

5E.5 Tabulated pK_{R^+} values from the note

Cation (Carey–Sundberg Table 4.15, as on your page)	pK_{R^+}	Cation	pK_{R^+}
Triphenylmethyl	−6.63	4,4′,4″-tri(dimethylamino)trityl	+9.36
4,4′,4″-trimethyltrityl	−3.56	4,4′,4″-trichlorotrityl	−7.74
4-methoxytrityl	−3.40	4-nitrotrityl	−9.15
4,4′-dimethoxytrityl	−1.24	4,4′,4″-trinitrotrityl	−16.27
4,4′,4″-trimethoxytrityl	+0.82		
Diphenylmethyl	−13.3	2,2′,4,4′,6,6′-hexamethyldiphenylmethyl	−6.6
4,4′-dimethyldiphenylmethyl	−10.4	4,4′-dichlorodiphenylmethyl	−13.96
4,4′-dimethoxydiphenylmethyl	−5.71		
Benzyl	≤ −20	Triphenylcyclopropenyl	+3.1
tert-Butyl	−15.5	2,4,6-trimethylbenzyl	−17.4
2-Phenyl-2-propyl (cumyl)	−12.3	Trimethylcyclopropenyl	+7.8
Tropylium	+4.7	Tricyclopropylcyclopropenyl	+9.7
Tricyclopropylmethyl	−2.3		

Patterns to learn from this table: (i) every p-donor (OMe, NMe_2, Me) raises pK_{R^+} of trityl, every p-acceptor (Cl, NO_2) lowers it; (ii) the three-fold cyclopropyl substitution (−2.3) puts a carbocation above any simple trityl cation; (iii) aromatic cations (tropylium, cyclopropenyl) sit above all acyclic non-heteroatom-stabilised cations; (iv) three methyl groups on benzyl (2,4,6-trimethylbenzyl, −17.4) raise the value relative to benzyl (≤ −20), and six methyls on diphenylmethyl (−6.6 against −13.3) raise it much more: alkyl groups donate to the cationic centre through the ring. The note also lists the ‘cyclopentenyl/cyclohexenyl’ allylic cations from McManus; the entries on your page (% H_2SO_4 for 50 % protonation, then pK) are:

Allylic cation (McManus table, from your page)	% H_2SO_4 (50 %)	pK
1,3,4,4,5,5-Hexamethylcyclopentenyl	33	−1.9
1,3-Dimethylcyclopentenyl	35	−2.1
1-Cyclopropyl-3-methylcyclopentenyl	12	−0.46
1-Methyl-3-phenylcyclopentenyl	50	−3.4
2,4-Dimethyl-3-penten-2-yl	73	−6.3
2,3,4-Trimethyl-3-penten-2-yl	82	−7.7
1,3,5,5-Tetramethylcyclohexenyl	50	−3.4
1,5,5-Trimethylcyclohexenyl	80	−7.3
1,5,5-Trimethyl-3-phenylcyclohexenyl	52	−3.6
1,3-Dicyclopropyl-5,5-dimethylcyclohexenyl	1.2	+0.6
2,6-Dimethyl-3,5-heptadien-2-yl	≈ 4	0
1,1-Diphenylethyl	71	−6.0
Mesitylenonium (1,3,5-trimethyl-2,4-cyclohexadien-1-yl)	94.5	−9.6
1,3,5-Trimethoxy-2,4-cyclohexadienyl	7 M $HClO_4$	−5.1
1,5-Dimethyl-2,4-cyclohexadienyl	$HF–BF_3$	−14
tert-Butyl	$FSO_3H–SbF_5$	estimate −15.5

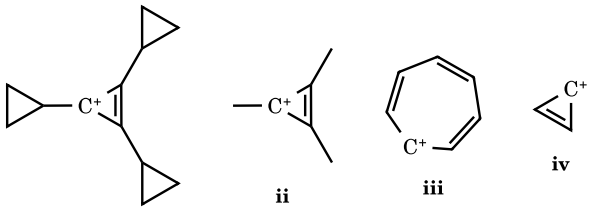
Reading it: cyclopropyl groups on an allylic cation raise its stability far more than methyl groups (1,3-dicyclopropyl, pK +0.6, against a methyl analogue at about −7); a phenyl on the allylic terminus helps (−3.4/−3.6 against −7.3); and a nonplanar or non-resonance-stabilised cation such as t-butyl can only be estimated (−15.5).

5E.6 Examples from the note (March / Carey sets): where they appear in this book

The nine orders printed on your page 3 under ‘Source – Jerry March’ are the same problems as your Exercise 2 and are answered, with reasons, in **Part 5A**. Two points need to be stated here.

- **Substituted cyclopropylmethyl:** 1 > 2 > 3 for the 2,3-dimethyl, 2-methyl and parent cyclopropylmethyl cations, and for the p-cyclopropylbenzyl series: more methyl groups on the ring, more donation (matches Part 5A, Q21 and Q17).
- **⚑ Check, the set ‘benzyl / cyclopropylmethyl / tropylium / branched alkyl’:** your page prints 2 > 3 > 1 > 4, i.e. cyclopropylmethyl above tropylium. The pK_{R^+} data on your own page (tropylium +4.7) put tropylium far above any cyclopropylmethyl or benzyl cation, so I cannot support putting it second; Part 5A gives tropylium > cyclopropylmethyl > benzyl > alkyl. If the examiner's key matches the page, it is the key's choice, not the data's.

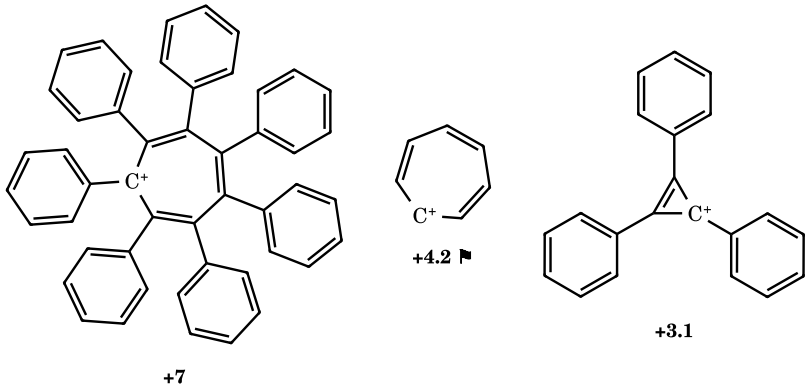
- The sets ‘4 > 2 > 3 > 1’ (cage ortho-ester, dioxanyl, oxanyl, diazinanyl cations), ‘3 > 2 > 1’ (secondary, tertiary and iminium-type sets), ‘1 > 3 > 2’ (decalyl), and ‘3 > 1 > 2’ (alkyl: tertiary > secondary > primary) agree with the reasoning already given in Part 5A.



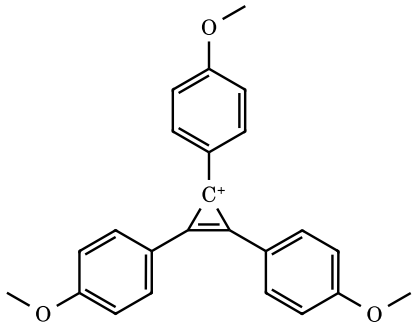
i **ii** **iii** **iv**

Practice item on page 4 (answer ticked in your note): arrange i–iv in decreasing stability: (a) i, ii, iii, iv. Tricyclopropylcyclopropenyl (+9.7) > trimethylcyclopropenyl (+7.8) > tropylium (+4.7) > parent cyclopropenyl.

5E.7 More pK_{R^+} structures from the note (pages 5 and 6)

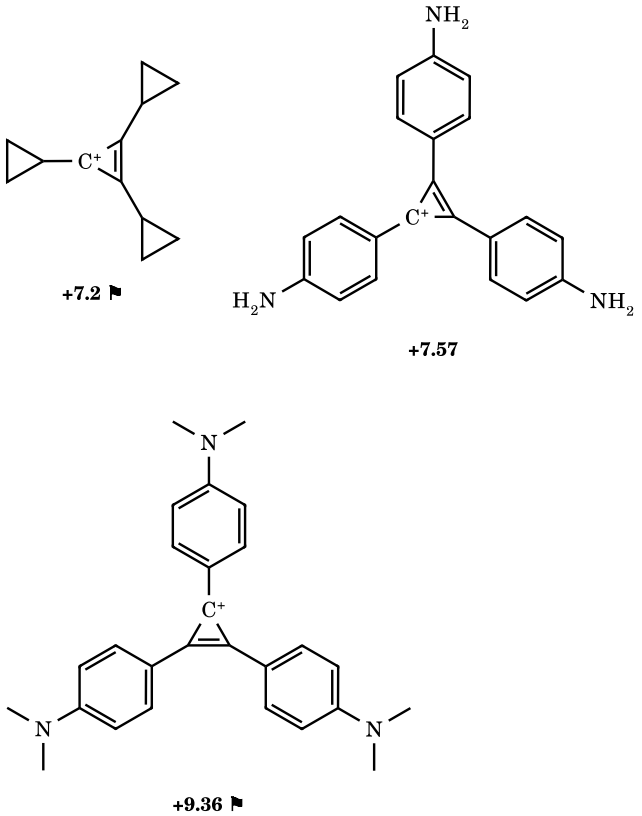


+7 **+4.2** ⚑ **+3.1**



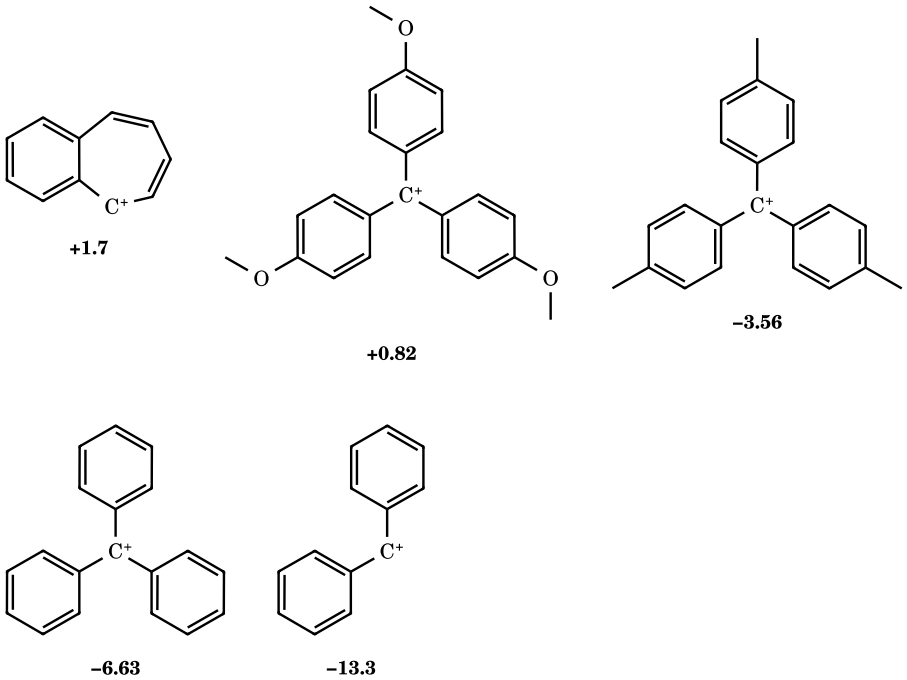
+6.5

Heptaphenyltropylium (+7) is more stable than tropylium (+4.7/4.2): phenyl groups add some conjugation and steric protection. Tris(p-anisyl)cyclopropenyl (+6.5) > triphenylcyclopropenyl (+3.1): p-OMe donates.



Amino-substituted cyclopropenyl cations: the stronger the donor, the larger the pK_{R^+}.

⚑ **Check.** (a) +7.2 for tricyclopropylcyclopropenyl contradicts +9.7 elsewhere (use +9.7, which is the Carey–Sundberg value). (b) +9.36 is, in Carey–Sundberg, the value for the *triarylmethyl* cation with three p-dimethylamino groups (crystal-violet type), not for the cyclopropenyl analogue; confirm which structure your source means before using the number. The trend (donors raise pK) is not in doubt.



Benzotropylium (+1.7, a benzo-fused seven-membered ring cation), tris(p-anisyl)methyl, tris(p-tolyl)methyl, trityl and diphenylmethyl: stability follows the donor strength of the para groups.

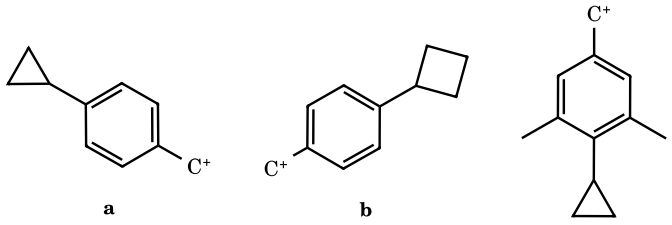
5E.8 Pentavalent carbonium ions and σ-bond resonance

Carbonium ions (five-coordinate carbon): CH_5^+ (methanonium) and $C_2H_7^+$ (ethanonium) contain a three-centre two-electron (3c–2e) C···H···C or C···H···H unit: the bonding pair of a C–H σ bond is shared by three atoms. They are observed in the gas phase and in superacid (CH_5^+ is a true, fluxional species); in ordinary solution chemistry

they appear as transition states or short-lived intermediates (your note writes ‘T.S. or true intermediate’). Do not confuse them with **carbenium** ions (three-coordinate R_3C^+), which are the ions of all the other parts.

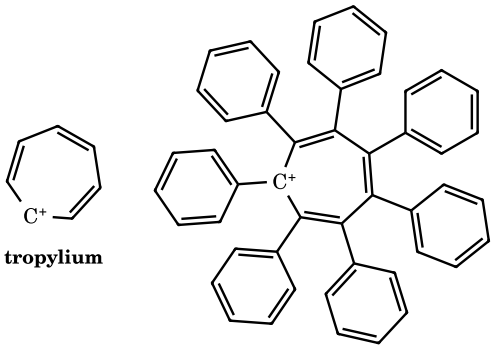
σ-bond resonance (‘dancing resonance’) in the cyclopropylmethyl cation: the C–C σ bonds of cyclopropane are bent and strained, and their electrons are close in energy to a π bond (cyclopropane behaves electronically as a very weak π-donor). In the *bisected* conformation the empty p orbital lies parallel to the plane of the ring bonds, so a ring σ bond can overlap with it. The cation is then a hybrid of cyclopropylcarbinyl ⇌ homoallyl (CH_2=CH–CH_2–CH_2^+) contributors with partial δ^+ on the ring carbons, the three ‘R.H.’ charges drawn in the note. Your note says the effect is ‘even stronger than +H’: that is correct as a statement of stabilisation (cyclopropylmethyl > benzyl on most measures); the description of the C–C bond as ‘weaker than the π bond of an alkene’ is a simplification: it is strained, and its electrons are much easier to donate than an ordinary C–C σ bond, which is the reason it works.

Order from the note: cyclopropylmethyl < 2-methyl-cyclopropylmethyl < 2,3-dimethyl-cyclopropylmethyl (methyl groups on the ring improve the donor); p-cyclopropylbenzyl > p-cyclobutylbenzyl (cyclopropyl donates by σ-resonance, cyclobutyl does not). And p-cyclopropylbenzyl > 3,5-dimethyl-4-cyclopropylbenzyl: the two methyl groups ortho to the cyclopropyl group force the ring out of the bisected arrangement, so the donation is lost (steric inhibition of resonance).



a **b** **c**

a > b (σ-resonance); a > c (steric inhibition of resonance by the flanking methyls).

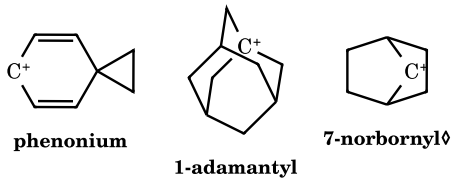


tropylium **heptaphenyl**

Tropylium (4.7) < heptaphenyltropylium (7): see 5E.7. The cation has seven equivalent resonance contributors (aromatic, 6π over 7 carbons).

5E.9 Non-classical cations from the note

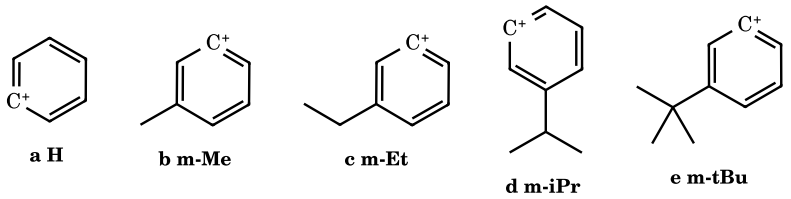
- **Phenonium ion**: β-phenylethyl cation $C_6H_5CH_2CH_2^+$ is bridged by the ring π electrons to give a spiro[2.5]octadienyl (phenonium) cation, with the starred label carbon becoming equivalent to its partner by the shift; the ring is a σ/π donor through its ipso carbon (sketch in your note: ring in dotted outline bonded to both CH_2 carbons).
- **7-Norbornenyl / nortricyclyl system**: the π bond of norbornene participates with a developing cation at C7, giving a delocalised three-centre σ/π ion; the note draws the two limiting structures and the hybrid with dotted partial bonds.
- **2-Norbornyl cation** (page 8): classical ion ⇌ σ-bridged (non-classical) ion, drawn with the C1–C6 bond delocalised to C2 (see Part 4 for the full discussion and the evidence).
- **1-Adamantyl cation**: a bridgehead tertiary cation; it cannot flatten, but is stabilised by hyperconjugation from the three β C–C bonds and is observable in superacid; it solvolyses much slower than t-butyl in comparable media, which is why it is the usual example of a ‘bridgehead’ cation.



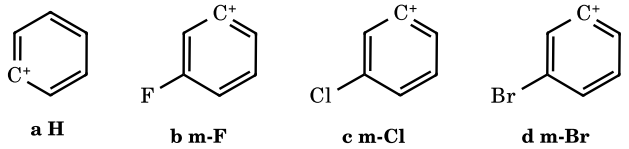
Left to right: the phenonium ion drawn as its spiro cyclohexadienyl (arenium) contributor, the 1-adamantyl cation, and a classical 7-norbornyl cation. ◊ Bridged ions are hybrids that a single line-bond structure cannot draw faithfully (see Part 4).

5E.10 Aryl-cation substituent series (pages 10 and 11)

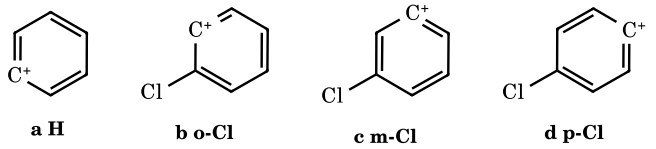
These series are for aryl cations (empty orbital on a ring carbon); the note uses the substituent effects as a guide to the order. Remember what the empty orbital is: an sp^2 σ orbital in the ring plane, orthogonal to the π system, so the π-donor effect (+M) cannot operate by direct conjugation; the orders below therefore mix inductive effects with the textbook 'conjugation-type' rules the note uses. Note the three groups of series: +I alkyl (a < b < c < d < e), halogens (distance and electronegativity), and donors/acceptors (OH, NH_2, OCH_3, NO_2, CCl_3).



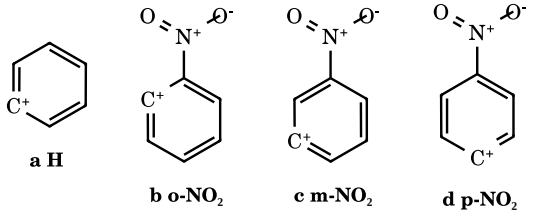
a < b < c < d < e: stability rises with the +I strength of the alkyl group (Me < Et < i-Pr < t-Bu).



a > d > c > b: the –I effect is strongest for F, then Cl, then Br; all are below the unsubstituted ion.

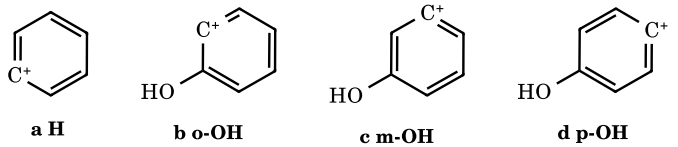


a > d > c > b: –I falls with distance, so the para ion is least destabilised and the ortho the most. (Your page prints the last symbol as 'd'; it should read 'b', a typo.)

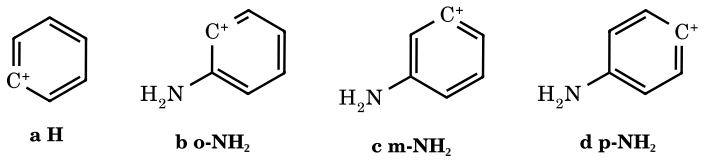


a > c > b > d as printed (m > o > p).

> ⚑ **Check, nitro series.** For chlorine the note uses a pure distance order (p > m > o); for NO_2 it puts p-NO_2 *below* o-NO_2 (treating the –M effect as the extra factor at the para position). If only the in-plane σ cation is considered, –M cannot act and the distance (–I) order p > m > o, as for Cl, would be expected. The note's order is the one to use if your exam source is this note; keep in mind that the two series follow different rules.

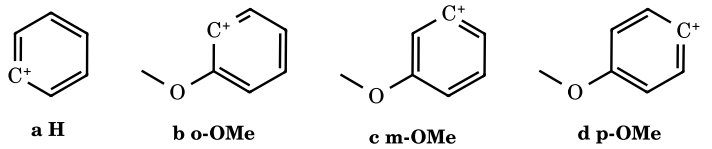


d > b > a > c (OH: ortho –I,+M; meta –I only; para +M). The same series is given for NH_2 (d > b > a > c) and for OCH_3 (d > b > a > c).



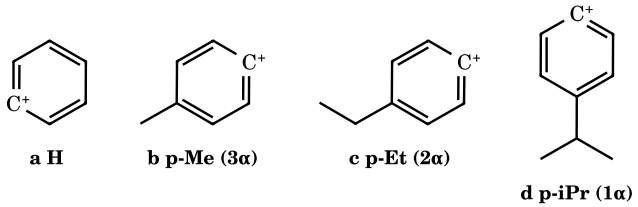
a H **b o-NH_2** **c m-NH_2** **d p-NH_2**

NH_2: d > b > a > c.



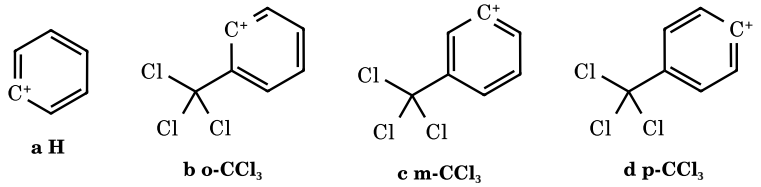
a H **b o-OMe** **c m-OMe** **d p-OMe**

OCH_3: d > b > a > c.



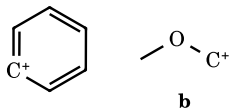
a H **b p-Me (3α)** **c p-Et (2α)** **d p-iPr (1α)**

b > c > d > a: the para alkyl group donates by hyperconjugation, in proportion to its α C–H count (3, 2, 1), all better than hydrogen (Baker–Nathan type order).



a H **b o-CCl_3** **c m-CCl_3** **d p-CCl_3**

CCl_3: a > c > d > b as printed (ortho worst: –I plus loss of hyperconjugation; meta > para as printed).



a **b**

Phenyl cation < CH_2^+–OCH_3 (oxocarbenium; ≈ 294 vs far lower HIA).

5E.11 Carbanion stability (the note ends here)

The last page begins the next topic: carbanion stability ∝ –M, –H (loss of hyperconjugation destabilises less than donating groups destabilise a carbanion; here the note means electron-withdrawing groups), –I, and ∝ 1/(+I), 1/(+H), 1/(+M): the exact mirror of the carbocation list. The carbanion chapter, still to be written, will cover this fully; I have not drawn conclusions from the single page you supplied.

5E.12 Effect on the earlier answer D-27

> **D-27 (heterolytic C–Cl bond energy less than that of t-BuCl).** I answered 4 (acetyl chloride, benzyl, 3-chlorocyclopropene, cyclopropylmethyl) using the coaching convention 'benzyl > tertiary'. With the data of *your note* (pK_{R^+}: t-Bu –16.4/–15.5, benzyl ≈ –20; HIA: t-Bu 231.9, benzyl 238) the benzyl cation is **less** stable than t-Bu^+, so benzyl chloride does **not** count, and the answer becomes **3** (acetyl chloride [acylium, if counted], 3-chlorocyclopropene, cyclopropylmethyl chloride). Which of 3 or 4 your source wants depends on whether it follows the data or the convention; on the data of this note use 3. The same caveat applies to any earlier answer that ranks benzyl above tert-butyl (Part 5A, Q28 uses t-Bu last; its position is already flagged there).

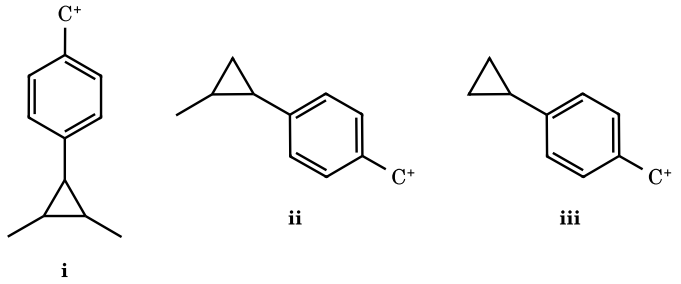
Part 5F Complete Worked Answers: the 'Electronic Displacement Effects' Exercise Book and the Allen Q48–51 Set

This part answers every problem in the 47-page exercise you supplied (Exercises O-I, O-II, S-I, J-Main and J-Advanced) and the Allen Q48–51 stability set that came with the earlier images, so that nothing in those uploads is left out. Questions are paraphrased, not copied, and the reasoning is written fresh. Your PDF prints

an answer key on its last pages; I worked each item out from the chemistry first and then compared. ‡ marks an item where the drawing was too small or ambiguous for me to verify independently, so the answer is the key's; ⚑ marks a place where I disagree with the key or where two of your sources disagree. Structures are redrawn only where they carry the argument.

5F.0 Allen Q48–51 (stability orders, from the first image set)

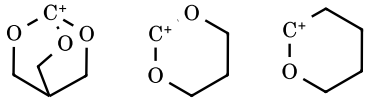
A-48.
Order the p-(cyclopropyl)benzyl cations: (i) 2,3-dimethylcyclopropyl, (ii) 2-methylcyclopropyl, (iii) cyclopropyl.



i ii iii

Answer: (a) i > ii > iii
The cyclopropyl ring donates its bent σ bonds (σ-resonance); each methyl on the ring makes it a better donor.

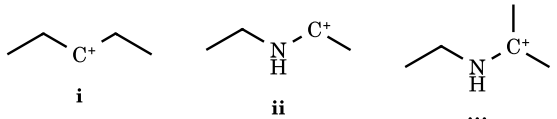
A-49.
Order: (i) the cage cation with three oxygens at the bridgehead, (ii) 1,3-dioxan-2-yl, (iii) tetrahydropyran-2-yl.



i ii iii

Answer: (b) ii > iii > i
Two oxygen lone pairs on a planar carbon (ii) beat one (iii). In (i) the cationic carbon is a bridgehead of a rigid cage, so it cannot become planar and its empty orbital cannot overlap with the oxygen lone pairs: least stable despite three oxygens.

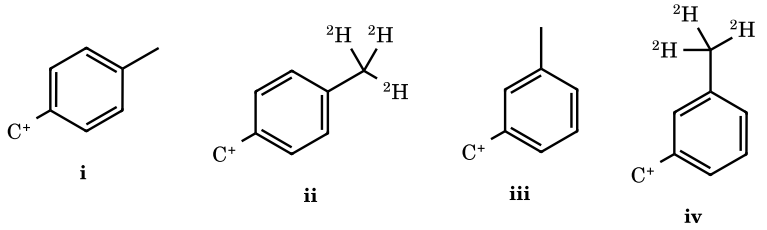
A-50.
Order: (i) a secondary alkyl cation, (ii) a secondary iminium-type cation N–C$^+$H–CH$_3$, (iii) a tertiary iminium-type cation.



i ii iii

Answer: (c) iii > ii > i
A nitrogen lone pair on C$^+$ (+M) is worth far more than hyperconjugation, and the extra methyl on (iii) adds hyperconjugation and +I on top.

A-51.
Order the benzyl cations: (i) p-CH$_3$, (ii) p-CD$_3$, (iii) m-CH$_3$, (iv) m-CD$_3$.



i ii iii iv

Answer: (b) i > ii > iv > iii
Para: hyperconjugation operates, and C–H donates better than C–D (i > ii). Meta: no hyperconjugation is possible, only +I, and D is slightly the better inductive donor (iv > iii). Any para isomer beats any meta one.

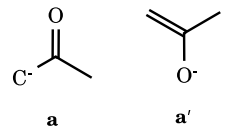
5F.1 Exercise O-I (Q1–Q51)

Q	Answer	Reasoning
1	(D)	False order of –I: an alkynyl group (sp carbon) is more electron-withdrawing than phenyl, so the true order is –C ≡ CH > –Ph > –H. The other three are true (F > Cl > Br > I; –NR$_3^+$ > –NH$_3^+$ > –NO$_2$; F > OH > NH$_2$).

2	(C)	+I order: $-O^-$ > $-CH_2CH_3$ > –D > –H. In (A) CMe_3 is a stronger donor than CH_3; in (B) O^- beats CO_2^-.
3	(B)	Only cyclohexyl (sp^3 carbon) is +I; phenyl, p-tolyl and vinyl are attached through sp^2 carbon and are –I.
4	(A, B, D)	–OH, –COOH and $-OCH_3$ are –I; $-CH_3$ is +I.
5	4 (b, d, f, g)	+I groups are the negatively charged or alkyl ones: O^-, N^-H, COO^-, CH_3. OH, NH_2, COOH, OMe, F, NF_3^+ are –I.
6	(a, c, d, g, j, l, m)	Resonance is intramolecular, involves π (and lone-pair) electrons only, lowers the energy, is not the only stabilising factor, and the hybrid (not any canonical form) describes the molecule; canonical forms are imaginary, the hybrid is real and more stable than any one form. False: b, e, f, h, i, k.
7	(A)	Incorrect statement: resonating (canonical) structures are hypothetical and have no real existence.
8	(A)	A conjugated diene is more stable than an isolated diene, which is more stable than a cumulated one (allene).
9	(A) a > d > c > b ‡	Contribution to the hybrid of CH_2N_2: the form with the negative charge on the terminal (more electronegative) nitrogen and complete octets dominates; forms with a carbanion contribute less.
10	(A, D)	A cyclic 4n+2 delocalisation stabilises (aromatic); a cyclic 4n delocalisation destabilises. A negative charge is better placed on the more electronegative atom, a positive charge on the less electronegative one.
11	(D)	Pair D is keto–enol tautomerism (an H changes position), not resonance.
12	(b, d, e)	The lone pair is in resonance in pyrrole N–H, in the enamine-type N–H, and in the allyl anion. In pyridine, the imine N of (c) and (f) the lone pair is in the ring plane (sp^2), orthogonal to the π system.
13	(b, d, e)	Not involved: the in-plane lone pair on the carbonyl O, the sp lone pair on a nitrile N, and the O lone pair in (e), which is not attached to the π system.
14	(d)	$-NH_3^+$ has neither a lone pair nor a π bond.
15	(c, f)	Groups that donate by resonance have a lone pair on the atom attached to the π system: $-OCOCH_3$ (the O) and $-NHCOCH_3$ (the N).
16	(a, b, c, d, f)	–COOH, –CONHR, –COCl, –CN and an imine C=N withdraw by –M; $-O-CH=CH_2$ donates.
17	(b, c, f)	–NO, $-CH=CH_2$ and –N=NH can accept (–M) or donate (+M) depending on what they are attached to.
18	see figure	Resonance forms drawn below: (a) enolate, (b) vinylogous enolate, (c) benzyl cation (ortho and para forms), (d) cyclohexadienyl cation, (e) phenoxide, (f) vinylogous iminium, (g) oxocarbenium, (h) vinylogous enolate, (i, j) dienyl cations.
19	a-I, b-II, c-II, d-II, e-I	Less stable form: (a) the sextet carbocation; (b) the naphthalene structure with only one benzene sextet; (c, d) the one with the charge on carbon or N instead of O^-; (e) the cycloheptatrienyl-anion (8π, antiaromatic) form.
20	a-I, b-I, c-I, d-I, e-II, f-II	More stable form: neutral rather than charge-separated; aromatic ring kept (phenoxide, pyridine, Cp^-); (e) $CH_2=CH-CH=O$ rather than $^+CH_2-CH=CH-O^-$; (f) all octets with the positive charge on N.
21	(a, e, f, g)	–OH (lone pair), –CH=CH–Ar (π), $-BH_2$ (empty p) and $-PPh_3^+$ (empty d/σ* acceptor, as the key treats it) can take part; the saturated $-CH_2C^-H_2$, $-CH_2C^+H_2$ and benzyl groups cannot.
22	(a) resonance forms; (b) A; (c) C; (d) A and B; (e) B and C; (f) 0; (g) B; (h) B ‡	A, B and C are canonical forms of diazomethane (same atom positions). B ($CH_2=N^+=N^-$) has complete octets and no charge on carbon, so it is the most stable.
23	(a) II, (b) II, (c) II	(a) the charge on N, not C; (b) the enolate with the charge on O; (c) the form with all octets ($C=N^+$).
24	(A) I > II > III > IV	Neutral form first; then the charge-separated form with O^-/O^+ and complete octets; then C^+ forms (sextet) and last the C^-/O^+ form.
25	a-II, b-II, c-II, d-II	Lesser resonance energy: the non-aromatic or less conjugated member of each pair; furan has less aromatic stabilisation than pyrrole (about 16 vs 21 kcal mol^{-1}).
26	(a) and (b)	Equivalent canonical forms and aromaticity both raise the resonance energy.
27	i-I, ii-II, iii-II, iv-I, v-I, vi-II, vii-I, viii-II ‡, ix-II, x-II, xi-II, xii-I, xiii-I, xiv-I, xv-I	Higher RE: pyridine over pyrrole; N is a better donor than O (divinylamine); equivalent forms (amidinate); F over Br (2p–2p overlap); benzyl over allyl radical; naphthalene (61) over phenol (36); two isolated benzenes (2 × 36) over styrene; longer conjugation (ix); linear over cross-conjugated (x); carboxylate over acid; vinyl-O^- over vinyl-OH; benzoate; diphenylmethane over naphthalene; benzene over hexatriene.
28	a-II, b-I, c-I, d-II, e-I	Lesser RE: $HCOO^-$ (2 forms) vs CO_3^{2-} (3); the antiaromatic cyclopropenyl anion vs allyl anion; cyclobutadiene vs butadiene; allyl cation vs aromatic cyclopropenyl cation; benzene vs phenol.

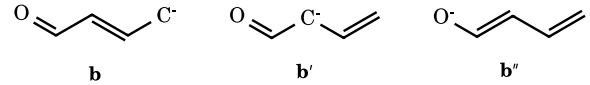
29	(B)	The pentadienyl cation has more resonance than the antiaromatic cyclopentadienyl cation; hexatriene has less than benzene; divinylamine has less than 1-amino-butadiene.
30	(D)	The ring π density is lowest in 1,3-dinitrobenzene (two –M/–I acceptors, nothing donating).
31	(B)	The ring π density is highest in the phenoxide ion (the strongest +M donor).
32	(C) i > iv > ii > iii	Reactivity to electrophiles follows hyperconjugation by the α C–H count: CH_3 (3) > C_2H_5 (2) > i-Pr (1) > t-Bu (0): toluene, ethyl-, isopropyl-, t-butyl-benzene.
33	(C)	Inductive, mesomeric and hyperconjugative effects all operate in the acetyl-methyl-cyclohexene (C=C–C=O gives M, the alkyl groups give H and I).
34	(C)	$CH_3CH{=}CH{-}CO{-}CH_3$ has C=C conjugated with C=O (M), CH_3 on the double bond (H) and the polar C=O (I).
35	(D)	All three statements are correct: σ delocalisation is hyperconjugation, π delocalisation is resonance, permanent σ polarisation is the inductive effect.
36	(A, B, C)	Number of hyperconjugative structures = α C–H bonds: propene 3, 2-methyl-2-butene 9, the t-amyl cation 8. 2-Butyne has 6, not 5.
37	(B)	The ortho-di-t-butyl groups twist the NO_2 out of the ring plane (steric inhibition of resonance), so C–N has less double-bond character and is longer in II.
38	(B) ‡	The two nitro groups flanking Br are twisted out of the ring plane (no resonance, longer C–N); the unhindered nitro group conjugates and has the shorter C–N bond.
39	(C) III > II > I > IV	Single sp^3–N (cyclohexylamine) is longest; aniline N lone pair adds partial double bond; a p-CHO withdraws and adds more; the imine C=N is shortest.
40	(D)	The shortest C1–C2 bond joins the sp^2 carbon of a ring to an sp carbon (C ≡ CH): both shortening factors (s-character, no hyperconjugation).
41	(D)	Longest C=C is in isobutene: two methyls give the most hyperconjugation, which lengthens the double bond.
42	(A)	Shortest C=C is in allene: sp-hybridised central carbon.
43	(B, C, D)	Butadiene, 1,3-cyclohexadiene and naphthalene have alternating bond lengths; benzene has equal bonds.
44	(B) C_2H_6 > C_6H_6 > C_2H_4 > C_2H_2	154, 139, 134, 120 pm.
45	(A) I > II > III > IV	The push–pull ethene I (OMe donor, NO_2 acceptor) has the most single-bond character; then nitroethene, vinyl chloride, ethene.
46	(A, C, D)	C–C > C=C > C ≡ C > C ≡ N; C=C > C=N > C=O; C–C > C=C > C ≡ C > C–H are correct. (B) is not.
47	a-I, b-I, c-II, d-I, e-I, f-I ‡	Bond dissociation energy of the marked bond is lower where it carries more s/less strength: cyclohexene vs ethylene; propyne C–CH_3; urea vs amides etc. The drawings are small, so these follow the key.
48	a-II, b-I, c-I, d-I, e-II, f-II ‡	Stronger bond: C–Cl > C–Br; vinyl–Br > alkyl–Br; acyl–Cl and alkyl–Cl comparison, sp^2–sp^2 > sp^2–sp^3 etc. (as in the key).
49	(D) C^1–H > C^4–H > C^3–H > C^2–H	Vinylic C–H is strongest (sp^2); then the primary sp^3 C–H; then secondary; the allylic C–H is weakest because the radical is resonance stabilised.
50	C > B > A	C–N bond length: the simple enamine (C) has the least double-bond character and is the longest; the dienamine (B) is shorter; the amino-enone (A) is shortest because N donates into the C=C–C=O system.
51	(A) ‡	As in the key: the t-butyl ortho group twists the COOH out of conjugation, so the C–O bond is shorter (less delocalisation into the ring) than in benzoic acid.

Q18 resonance forms



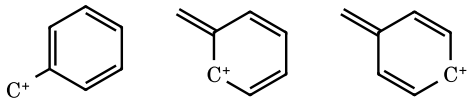
a **a′**

(a) acetone enolate: carbanion form and enolate form



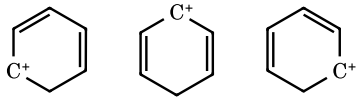
b **b′** **b″**

(b) the charge delocalised over C-4, C-2 and O



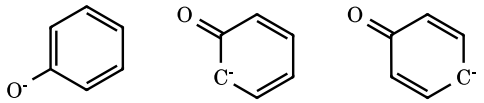
c **c′ (o)** **c″ (p)**

(c) benzyl cation (the ortho form shown once; two ortho and one para)



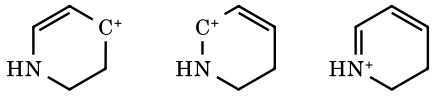
d **d′** **d″**

(d) cyclohexadienyl cation: charge at C-1, C-3, C-5



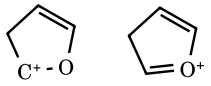
e **e′ (o)** **e″ (p)**

(e) phenoxide: charge at O and at the ortho and para carbons



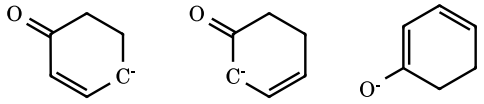
f **f′** **f″**

(f) the vinylogous iminium ion (my reading of the ring: N lone pair donating through C=C)



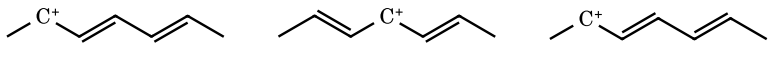
g **g′**

(g) oxocarbenium form (my reading of the ring)



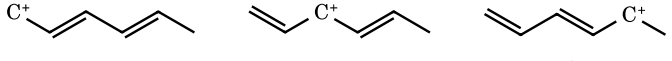
h **h′** **h″**

(h) cyclohexenone-type anion: charge at C-4, C-2, O



i **i′** **i″**

(i) the heptadienyl cation: charge on C-2, C-4, C-6



j **j′** **j″**

(j) the hexadienyl cation: charge on C-1, C-3, C-5

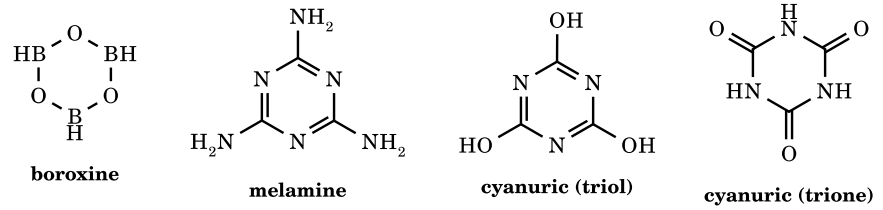
5F.2 Exercise O-II (Q1–Q40)

Q	Answer	Reasoning
1	(i) a, b; (ii) a, c; (iii) b, c, d; (iv) all	(i) the benzo-fused cyclopentadienyl anion (10π) and thiophene; (ii) tropylium and the benzo-fused (isoindenyl-type) anion; (iii) the cyclobutadiene dianion (6π), cyclopropenyl cation and protonated pyridine; (iv) azulene-type, 6π borole-ring and silabenzene systems, borazine and indole. Cyclohexadienyl, cyclopentadienyl cation, cyclobutadiene and the 8π anion are excluded (sp^3 carbon or 4nπ). ‡
2	aromatic: a, b, d, e, g; non-aromatic: c, f	Isoxazole, thiazole, pyrylium, γ-pyrone and cytosine are aromatic; 4H-pyran and 1,2-dihydropyridine have an sp^3 ring atom.

3	(C) II > I > III	Ease of rotation about the exocyclic C=C. Each compound is a fulvene-type system whose dipolar form places a charge on each ring. If both rings become aromatic (II: tropylium$^+$/Cp$^-$, I: cyclopropenyl$^+$/Cp$^-$) the bond has much single-bond character and rotates easily; tropylium is the more stable cation, so II > I. In III the cyclopropenyl ring would carry the anion (antiaromatic), so the bond keeps its double-bond character.
4	(C) III	Lowest rotation barrier: III is the Cp$^-$/cyclopropenyl$^+$ dipole, both rings aromatic. In I the dipole would put a negative charge on an antiaromatic cyclopropenyl ring, and in II it would give an antiaromatic cyclopentadienyl cation.
5	(B)	Cyclooctatetraene (8π, tub-shaped) is not aromatic.
6	(A)	The pyrrolium ion (N–H$_2^+$, sp^3 N, no lone pair in the ring π system) is non-aromatic.
7	see below	(a) $B_3H_3O_3$ is boroxine, a planar B_3O_3 ring with 6π (weakly aromatic); (b) melamine, aromatic (triazine ring, 6π); (c) the trimer of HN=C=O is cyanuric (isocyanuric) acid, drawn as the aromatic triol, though the trione tautomer predominates in the solid.
8	(D) neopentyl	α C–H counts: ethyl 3, n-propyl 2, isobutyl 1, neopentyl 0 (see the correction in Part 5B).
9	(i) c > b > a; (ii) c > b > a; (iii) b > c > a; (iv) d > c > b > a; (v) c > b > a; (vi) b > c > a ⚑; (vii) c > b > a; (viii) c > a > d > b; (ix) b > c > a; (x) c > a > b; (xi) c > a > b; (xii) a > b > c > d; (xiii) b > a > c > d; (xiv) d > e > b > a > c; (xv) a > c > b; (xvi) b > c > a	Worked in full in Part 5B (items 9(i)–(xvi)). For (xiv) the note on your Allen page tags the five cations as a = 2 α-H, b = 6 α-H, c = 1 α-H, d = resonance + 4 α-H, e = resonance + 2 α-H: resonance first, then α-H count. For (vi) the key and the Allen solution disagree (⚑ see Part 5B).
10	(C)	I, II and III are correct (oxocarbenium, secondary > primary, allyl > n-propyl); IV is wrong, since the vinyl cation is less stable than ethyl.
11	(a) I; (b) I; (c) II; (d) II	(a) benzyl > allyl; (b) ethyl > vinyl; (c) the allylic exocyclic cation II > the cation I that is not conjugated with the ring C=C; (d) N-substituted: the α-amino carbenium (iminium) is far more stable than a simple tertiary cation.
12	(C) I > IV > III > II ‡	Norbornyl cations: the least stable is II, the 7-norbornyl cation (bridge carbon, geometrically unable to be planar and not stabilised by σ-bridging). The secondary ones are ordered by hyperconjugation and by exo-σ-participation, with the methyl-bearing neighbour (I, IV) better than the parent III.
13	(A)	Trityl is the most stable (pK_{R^+} −6.6).
14	(A) d < b < c < a	Ethyl < benzyl < p-methylbenzyl < p-methoxybenzyl.
15	(D) 2 > 1 > 3 > 4	NH_2 > OH (+M), then CH_3 (3 α-H) > t-Bu (0).
16	(B) II < III < I	Steric inhibition of resonance leaves III with only the –I effect of NO_2.
17	see below	Increasing stability of anions and radicals. (i) a < b; (ii) d < a < c < b; (iii) b < a; (iv) c < b < a; (v) c < a < b; (vi) a < c < b; (vii) c < b < a; (viii) c < b < a; (ix) b < a; (x) a < c < b < d. Reasons are given in the sub-table that follows.
18	(A)	HC ≡ C$^-$: the lone pair is in an sp orbital (50 % s), the most stable carbanion.
19	(D)	p-Nitrobenzyl anion: the charge delocalises onto the nitro group (–M).
20	(B)	The anion conjugated with the C=O across the ring (a dienolate) is the most stable.
21	(D)	Radicals: CH_2=CH–C$(CH_3)_2$ > allyl > ethyl > vinyl is correct. In (A) $CH_3OCH_2^+$ is more stable than t-Bu$^+$; in (B) isobutene is slightly more stable than cis-2-butene; in (C) the order of the benzyl anions is reversed.
22	(a) IV < I < II < III < V; (b) III < IV < I < II ‡	(a) the bridgehead cage cation is least stable; β-oxygen cations (–I only) next; the allylic and the dioxanyl cation (two O) most. (b) Carbanions next to C=O: aldehyde > ketone > ester > amide (a more electron-poor carbonyl stabilises the enolate better).
23	(B) IV > III > II > I	Radical stability grows with benzylic + allylic character and substitution at the radical centre.
24	(A)	CH_2=CH–CH=CH–CH_3 is conjugated (resonance); CH_3CH=C=CH–CH_3 is cumulated, with no resonance.
25	(a) I; (b) I; (c) II	More substituted alkene: 1-methylcyclohexene; isopropenylcyclopentane (1,1-disubstituted) over allylcyclopentane (mono); the double bond in the six-membered ring over the strained cyclobutene.

26	(a) 4658; (b) 4638; (c) 4632; (d) 4650; (e) 5293	(e) is the C_8 alkene (largest heat); among the C_7 alkenes, the most stable (trisubstituted, c) has the smallest heat and the least stable (1-heptene) the largest.
27	(A) I < III < IV < II	Alkene stability increases with the number of hyperconjugating α C–H bonds and with substitution: 2-butene < isobutene < 2-methyl-2-butene < tetramethylethylene.
28	(B) I and IV	The C–H bonds on the allylic carbons that can overlap with the π bond take part; I and IV are such bonds.
29	(D) I > IV > III > II	For isomers a larger heat of combustion means a less stable isomer: 1-butene > cis-2-butene > trans-2-butene > isobutene.
30	(i) d > c > b > a; (ii) e > c > d > b > a; (iii) b > a; (iv) a > b; (v) b > a; (vi) a > b	Heat of hydrogenation is larger for the less stable (less substituted, unconjugated, cis) alkene.
31	(i) a < b; (ii) e < d < c < b < a; (iii) a < c < b; (iv) a > b > c	1,3-diene < 1,4-diene; tetrasubstituted < … < ethylene; cyclohexene (28) < benzene (50) < cyclohexadiene (55); resonance energy per ring drops from benzene to naphthalene to anthracene.
32	(i) c > b > a; (ii) a > b > c > d; (iii) a > b; (iv) c > b > a	Isomers: the less stable has the larger heat of combustion; different formulae: more carbons, larger heat.
33	a-I, b-I, c-II, d-I	Higher heat of hydrogenation: 1,4-cyclohexadiene (not conjugated), benzene (two further double bonds), 1-butene, the exocyclic alkene.
34	stability d > c > b > a; heat of hydrogenation a > b > c > d	The more aromatic ring retained, the more stable (tetralin-type d best).
35	(C) 57	1,3-Butadiene: 2 × 30 – about 3 kcal mol^{-1} of conjugation energy.
36	(D)	Steric inhibition of resonance only in A (the NMe_2 flanked by two ortho methyls is twisted out of the ring plane); NH_2 is small enough to stay planar.
37	(B)	See Q37 of O-I: C–N is shorter in the unhindered nitrobenzene.
38	(C) ‡	See Q38 of O-I: twisted flanked nitro groups have the longer C–N bonds.
39	(i) c > b > a; (ii) a > b > c; (iii) a > b	Dipole moments: the para isomer of an electron-donor/acceptor pair has parallel bond dipoles in (i) and opposed ones in (ii); the ortho isomer of the pair (ii) has the largest vector sum; N,N-dimethylaniline has a larger moment than the hindered 2,4,6-trimethyl analogue (resonance lost).
40	—	A bridgehead cation of a small rigid bicycle such as norbornane cannot become planar, so the empty p orbital cannot overlap with neighbouring σ bonds; it is also strained (anti-Bredt-type). Larger cages (adamantyl) tolerate it.

O-II Q7: structures



Boroxine, melamine and cyanuric acid (aromatic triol form and the more stable trione form).

O-II Q17: reasons

Item	Order (increasing stability)	Reason
i	p-F-benzyl anion < p-NO_2-benzyl anion	–M of NO_2 delocalises the charge; F is +M/–I and destabilises the anion by lone-pair repulsion
ii	benzyl < p-Cl < p-CN < p-NO_2	electron-withdrawing strength
iii	ethyl anion < CH_2CHO enolate	resonance into C=O
iv	pentan-3-yl anion < ketone enolate < acetylacetonate	one, then two carbonyls
v	benzyl < p-Cl < p-NO_2	as above
vi	cyclopropenyl anion < cycloheptatrienyl anion < cyclopentadienyl anion	antiaromatic < non-planar 8π < aromatic
vii	phenyl < 2-phenylethyl < benzyl radical	in-plane σ radical < primary alkyl radical < delocalised benzyl radical
viii	alkyl < allylic cyclohexenyl < cyclohexadienyl radical	number of conjugated double bonds
ix	b < a	more extended delocalisation (dienyl radical) for a

x	vinyl < cyclohexenylmethyl < allyl < tertiary allyl radical	s-character and delocalisation, then substitution (key order)

5F.3 Exercise S-I (Q1–Q25)

Q	Answer	Reasoning
1	(C)	Cyclopentadienyl anion is aromatic (6π, planar, cyclic).
2	(A, D)	Ethanol has the higher boiling point and forms intermolecular H-bonds; dimethyl ether cannot H-bond to itself.
3	(D)	The nitrile carbon is sp-hybridised: highest s-character and the most electronegative carbon.
4	(C)	The CH_3 group hyperconjugates better than CD_3 (the C–H bond is weaker/longer-ranging than C–D).
5	(A, B, D)	In acetate and formate both C–O bonds are equal and the two resonance structures are equivalent.
6	A → P, S; B → Q, R; C → Q, S; D → P, S	NO_2: –R, –I; O^-: +R, +I; OCH_3: +R, –I; CN: –R, –I.
7	A → P, Q, S; B → R; C → P, S; D → Q, S	–N=O: +M (lone pair), –M (π acceptor), –I; CH_3: +H; $–NHCOCH_3$: +M, –I; $–COOCH_3$: –M, –I.
8	A → R; B → S; C → P, T; D → Q	Ethyl: +I only; NH_3^+: –I only; OH and NH_2: –I and +M; NO_2: –I and –M.
9	A → Q, S; B → P, S; C → P, S; D → R, S	A bicyclic 8π system is non-aromatic/cyclic, naphthalene and tropylium are aromatic and cyclic, and the cyclopentadienyl cation is antiaromatic. ‡
10	(D)	The statement is false when bond a is the C=O (it is longer in the ester, whose O lone pair donates into only one C=O, than in the anhydride, where it is shared); the reason is true. ‡
11	(A)	t-Bu^+ > i-Pr^+ > Et^+ and the number of hyperconjugative structures (9, 6, 3) explains it.
12	(D)	Statement I is false (hyperconjugation in 2-butene slightly lowers the barrier compared with ethylene); statement II is true.
13	(A) I > II > III	Neutral vinyl chloride form first; then the form with the negative charge on carbon and positive on Cl, then the one with positive on C.
14	(A)	Benzyl has more resonance structures (five) than allyl (two).
15	(B)	$^-CH_2–CHO$ and $CH_2=CH–O^-$ are resonance forms; the others are tautomers or different species.
16	(D)	t-Butyl cation is the most stable.
17	(C)	The most substituted, tertiary cation.
18	(C)	The vinyl cation (I) is less stable than ethyl (II).
19	(B)	Structure II is unlikely because nitrogen has only a sextet.
20	(C)	Structure II is unacceptable because nitrogen would have ten valence electrons.
21	(C)	Delocalisation lowers both the potential energy and the electron–electron repulsion.
22	(D) I, IV	Most stable tropylium (I); least stable isopropyl (IV) among those listed.
23	(B)	The cation formed from Ph_3CBr, trityl, is the most stable.
24	(4)	Naphthalene, squaric acid, azulene and tropone are counted aromatic; the pyrrolium and cycloheptatriene are not. ‡
25	(4)	Unstable: cyclobutadiene, pentalene, cyclopentadienone and borole.

5F.4 Exercise J-Main (Q1–Q31)

Q	Answer	Reasoning
1	(1)	$(CH_3)_3C–$ > $(CH_3)_2CH–$ > $CH_3CH_2–$ (+I order).
2	(3)	Formate has two equivalent resonance forms.
3	(3)	Acetonitrile has an sp carbon and an sp^3 one, no sp^2 carbon.
4	(2)	An unpaired electron makes a free radical reactive.
5	(4)	$(CH_3)_2CH•$ < $(CH_3)_3C•$ < $(C_6H_5)_2CH•$ < $(C_6H_5)_3C•$.
6	(1)	CCl_3^- > benzyl > i-Pr^- > t-Bu^-.
7	(1)	Cyclopentadiene has an sp^3 carbon: non-aromatic.
8	(3)	o-Nitrophenol is intramolecularly H-bonded, so it is steam volatile and less soluble.
9	(4)	Antiaromatic: the cyclopentadienyl cation (V) and cyclobutadiene (VI).
10	(3)	CCl_4 is tetrahedral and non-polar.
11	(4)	III > I > II: benzyl > allyl > n-propyl.
12	(2)	p-Dihydroxy and p-dimercapto benzenes have a net dipole because the O–H/S–H bonds are out of the ring plane; the p-dichloro and p-dicyano compounds are symmetric (μ = 0).

13	(4)	4,4-Dimethylcyclohexa-2,5-dienone has the least resonance stabilisation; benzene, furan and pyridine are aromatic.
14	(1)	Alkoxide stability: the nitro-substituted enolate-type C > the nitro-substituted B > A.
15	(1) a < b < c	The trans,trans-2,4-hexadiene is the most stable (smallest heat of combustion), the cis,cis the least.
16	(3)	Vinyl chloride has the shortest C–Cl bond (sp^2 carbon and p–π overlap).
17	(2)	Benzene and the cyclopentadienyl anion are aromatic.
18	(3)	Benzyl and allyl cations are resonance-stabilised; the other two are not conjugated with the cationic carbon.
19	(4)	$CH_3CH_2CH{=}CHCH_2NH_2$ has no conjugation.
20	(3)	Statement I is true; II is false: hyperconjugation in ethyl cation involves a C(sp^3)–H σ bond, not sp^2.
21	(1)	Both statements are correct (weak π bonds, and a C=C is stronger than a C–C).
22	(3)	p-Benzoquinone is the conjugated diketone.
23	(1)	The cyclopropenyl cation is aromatic and the most stable.
24	(2)	The cyclopentadienyl anion and the tropylium cation are aromatic; the other two are not.
25	(4)	$CH_3O{-}CH{=}CH{-}CH{=}CH{-}CH_2^+$: the longest conjugation with an oxygen donor.
26	(3)	[10]Annulene, although 4n+2, is non-planar (the inner H atoms clash), hence non-aromatic.
27	(1 and 2)	Phenalene-type (1) and cyclooctatetraene (2) are not benzenoid.
28	NTA (1); Allen (4)	[6]Annulene is aromatic, [8]annulene is not, but cis-[10]annulene is not planar and so not aromatic: the assertion is wrong and the reason (planarity is required) is right: option 4 (the NTA's published option was 1).
29	(1)	Charge at the position ortho to NH_2 is stabilised by the N lone pair (iminium form); the NTA key follows the nitrogen's +M effect.
30	NTA (2); Allen: bonus	No printed option matches: the correct increasing order is C < A < B < D.
31	(3)	Both correct and R explains A: the delocalised π electrons lie in bonding molecular orbitals, so benzene is more stable than the localised hypothetical cyclohexatriene.

5F.5 Exercise J-Advanced (Q1–Q18)

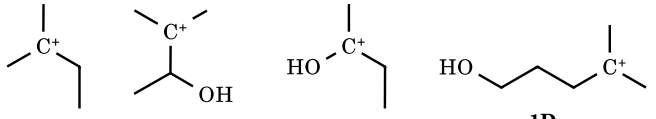
Q	Answer	Reasoning
1	(D) ⚑	The key gives 1,3-butadiene. Tabulated heats of hydrogenation per mole of H_2 are about 28.5 for butadiene (57 for two) and 27.6 for trans-2-butene, which is lower, so by that data (B) would win; the intended argument is conjugation lowering ΔH relative to 1-butene (30.3). Use (D) for the key.
2	(D)	p-Methoxybenzyl: +M of OCH_3.
3	(B) IV < I < II < III	p-Dichlorobenzene (0) < toluene (0.4) < m-dichloro (1.5) < o-dichloro (2.5 D).
4	(C)	The form with a pentavalent nitrogen (O=N=O) is the most unlikely resonance structure.
5	(A)	An aromatic molecule does not have 4nπ electrons.
6	(D)	p-Hydroxybenzoic acid has the higher boiling point (intermolecular H-bonding); the o-isomer is intramolecularly H-bonded. Statement I is false, II is true.
7	(A)	CH_3Cl has the highest dipole moment (1.9 D); CCl_4 is zero.
8	enol resonance	$CH_2{=}C(OH){-}CH_2CH{=}CH_2 \leftrightarrow {}^-CH_2{-}C(=O^+H){-}CH_2CH{=}CH_2$: only the enol C=C–O is conjugated; the other C=C is isolated.
9	(D)	The structure with unlike charges placed far from stabilising atoms and with a sextet is the least stable.
10	(A) ‡	The form with the positive charge and negative charge farthest from the nitro group stabilisation is least stable.
11	(D) I > III > II > IV ‡	The two α-alkoxy cations (oxocarbenium) come first, and of the alkyl cations the branched one (II) beats the more remote primary-type one (IV).
12	(B) I > III > II > IV	Diazomethane forms: all octets and no charge on C first; then the form with N ≡ N^+; the carbanion/cation forms with a sextet last.
13	6	α C–H bonds of the cyclohexyl–$C^+(CH_3)CH_2CH_3$ cation: CH_3 (3) + CH_2 (2) + the ring CH (1) = 6.
14	(B, C)	Cyclobutadiene and cyclopentadienone are unstable (antiaromatic); the others are stable.
15	(A)	t-Bu^+ is stabilised by σ→p(empty) hyperconjugation; 2-butene by σ→π* delocalisation.
16	6	Melamine: three lone pairs on the ring N atoms and three on the NH_2 nitrogens: 6.
17	9	Resonance structures of the naphthoxide ion: 9.
18	5	Aromatic: cyclopropenyl cation, tropylium, cyclopentadienyl anion, phenanthrene and the dihydronaphthalene (its benzene ring): 5.

⚑ **Check, summary.** Disagreements between your sources that I have flagged: O-II Q8 (corrected in Part 5B: the key says neopentyl); O-II Q9(vi) (key vs the Allen solution page); J-Advanced Q1 (key vs tabulated heats of hydrogenation); J-Main Q28 and Q30 (the NTA answer differs from Allen's). Items marked ‡ rely on the key because the drawing was too small for me to verify independently; if you want any of them proved from scratch, send the structure at a larger size.

Part 5G Complete Worked Answers: the Nucleus ‘Carbocation and Free Radical’ Exercise

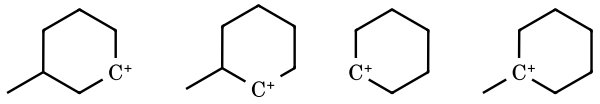
Every question of the exercise you supplied (Exercise I, Q1–53; II-A, Q1–18; II-B, Q1–13; JEE-Main Q1–11; JEE-Advanced Q1–4) is answered here. The PDF prints an answer key; I reasoned each item first and compared. ‡ = I relied on the key because I could not verify the drawing or the arithmetic independently; ⚑ = I disagree with the key or two sources disagree. Questions on *nucleophilicity comparison* (II-A Q8, Main Q3, Advanced Q2) are left out because your topic list excludes it. The stability and rate **orders** found in this exercise are also collected in the master index at the end of Part 5H.

5G.0 Cations drawn for reference



1A **1B** **1C** **1D**

Q1: (C) has the OH on the cationic carbon (oxocarbenium): the most stable.



3A **3B** **3C** **3D**

Q3: only (B) rearranges (a 1,2-H shift from the methyl-bearing neighbour gives the tertiary ion); (D) is already tertiary.

5G.1 Exercise I – mains oriented (Q1–Q53)

Q	Answer	Reasoning
1	(C)	Only C has the OH on C^+ itself (+M, oxocarbenium); in B and D the OH is further away and only withdraws.
2	(C)	Least likely: the 7-norbornyl-type bridge/bridgehead cation, which cannot become planar; trityl, cyclohexenyl (allylic) and ethyl are all possible.
3	(B)	A secondary cation next to a carbon bearing a methyl group shifts H to give the tertiary 1-methylcyclohexyl cation.
4	(D) 4	Dicyclobutylmethyl cation: ring expansion (cyclobutyl → cyclopentyl) and further ring expansions/hydride shifts to the fused tertiary bicycle: four 1,2-shifts in the key. ‡
5	(B) 8	All eight can lower their energy by a 1,2-shift: primary n-propyl (H shift), the two secondary branched cations (H or CH_3 shift), the quaternary-neopentyl type, the methyl-substituted cyclopentyl and spiro cations (ring changes), the β-alkoxy cation (H shift to oxocarbenium) and the secondary benzylic-type cation. ‡
6	(B) $\Delta H_4 > \Delta H_1 > \Delta H_2 > \Delta H_3$	Heterolysis is hardest where the cation is least stable: phenyl (IV) > cyclohexyl (I) > cyclohexenyl-allylic (II) > benzyl (III).
7	(C)	3-Bromocyclopropene ionises (silver bromide, pale yellow) to the aromatic cyclopropenyl cation, which has three equivalent resonance forms. The false statement is that it has the lower dipole moment than bromocyclopropane.
8	(C)	$SbCl_5$ ionises C–Cl to the planar benzylic carbocation, so the optically active chloride racemises.
9	(B) 0	1-Butanol dehydrates by E2 on the protonated alcohol (no primary cation), and the 2-butene results from acid-catalysed isomerisation of 1-butene, not from a 1,2-shift in a carbocation. ‡
10	(A)	Secondary cation next to a quaternary carbon → methyl shifts → tertiary → loses H to give the tetrasubstituted 1,2-dimethylcyclohexene (Zaitsev).
11	(D) ‡	In cold dilute acid the rearranged tertiary cation is trapped by water (low temperature favours substitution over elimination).
12	(A) I < II < III < IV	Dehydration rate follows the stability of the cation formed: phenol does not ionise; cyclohexyl (secondary) < allylic cyclohexenyl < pentadienyl-type.

13	(D)	Dienol–benzene rearrangement: lose water to the tertiary allylic cation, a methyl migrates, and loss of H^+ restores the aromatic ring (3,4-dimethylphenol).
14	(C) ‡	The fastest to dehydrate is the tertiary, allylic alcohol, since the allyl–tertiary cation is the best.
15	(D) ‡	Acid-catalysed dienone/dienol–phenol-type shift and aromatisation, giving the hydroquinone-type product.
16	(D)	Cyclobutyl-substituted cyclopentyl carbinyl cation: the strained cyclobutane expands (ring strain relief) and the alkene forms in the fused tetrasubstituted position.
17	(A)	Meyer–Schuster rearrangement: protonation, loss of water, allenyl/propargyl cation, then water at the other terminus and tautomerisation to the α,β-unsaturated ketone (chalcone).
18	(D) A > B > C > D	Rate with HBr (SN1 through the benzyl cation): p-OMe > H > p-Cl > p-NO_2.
19	(A)	The tertiary alcohol, 2-methylbutan-2-ol, gives the most stable cation.
20	(B)	The double bond most reactive to electrophiles gives the most stabilised cation; b is in the position whose cation is stabilised by the neighbouring ring system. ‡
21	(C)	Secondary cation → methyl shift → tertiary → Zaitsev: the most substituted ring alkene.
22	(B) 2	Two diols are mixed; each rearranges on its own, so there are two products (no crossover products between molecules).
23	(C) II > I > III	Rate-determining step: ionisation of the diol to a cation. The key ranks the mixed methyl/phenyl diol fastest, then the diphenyl diol, and pinacol slowest. ‡
24	(D)	Pinacol rearrangement with ring expansion: the cyclobutane ring bond migrates, giving the 2,2-dimethylcyclopentanone.
25	(B) b > a > c	2-Methoxypropene (enol ether, +M) > isobutene > allyl methyl ether (–I).
26	(D)	X cannot be 1-methylcyclopent-2-ene-type isomer; the others give 1-bromo-1-methylcyclohexane by Markovnikov addition or ring expansion. ‡
27	(D)	HCl does not show the peroxide effect; Markovnikov addition and a hydride shift give the tertiary chloride.
28	(B)	Intramolecular protonation of the enol ether, then the OH adds to the oxocarbenium: the spiroketal.
29	(B) P > R > Q	Rate of electrophilic addition: methyl vinyl ether (+M) > propene > vinyl chloride (–I).
30	(C)	HBr/CCl_4 without rearrangement gives a new stereocentre next to the existing one: a mixture of diastereomers.
31	(A)	Acid hydration and oxymercuration–demercuration are both Markovnikov (same tertiary alcohol); hydroboration gives the anti-Markovnikov alcohol.
32	(A)	Anti addition of Br_2 to a trans-alkene gives a racemic pair of enantiomers: it is optically inactive, so ‘optically active’ is the incorrect statement.
33	(D)	Pinacol coupling then pinacol rearrangement gives a spiro ketone $C_{10}H_{16}O$ with three degrees of unsaturation, not four.
34	(D)	The bromonium ion is opened transannularly by the nitrogen lone pair, giving the bicyclic ammonium-type product.
35	(D)	trans-2-Butene + Br_2 (anti) gives meso-2,3-dibromobutane: inactive; statements (C) and (D) about optical isomers are the incorrect ones (the key lists D).
36	(D)	HBr adds to the double bond of allyl bromide to give 1,2-dibromopropane; the other substrates give gem-dihalides.
37	(B) IV > III > I > II	Triflate > mesylate > acetate > methoxide (strength of the conjugate acid).
38	(C) IV > III > II > I	Wurtz rate follows C–X bond weakness and polarisability: I > Br > Cl > F.
39	(C)	NBS brominates the allylic position of the tetrasubstituted alkene; Na/ether couples the allylic radical sites (Wurtz).
40	(C) III > II > I	Allylic radical substitution: the more substituted the allylic C–H, the weaker the bond and the more stable the radical.
41	(B)	Bromine is highly selective: it abstracts the tertiary H.
42	(C)	The substituted cyclohexane has different kinds of hydrogen and gives more than one monochloride.
43	(B)	The alkyl radical is trigonal planar (sp^2), and that is why chlorination of 2-chloropentane gives 2,3-dichloropentane as a stereoisomer mixture.
44	(A)	Neopentane has only one kind of hydrogen.
45	(A)	2-Bromo-2-methylbutane: the tertiary radical.
46	(C)	Intramolecular Wurtz coupling gives bicyclobutane.
47	(A) II < I < III	C–D is stronger than C–H (primary kinetic isotope effect), so CD_3CD_3 is slowest; neopentane (12 primary H) reacts fastest per molecule.
48	(B, D) ‡	NBS gives the allylic bromide; HBr adds to give the dibromide. The key lists B and D.

49	(C)	Kolbe electrolysis of a succinate gives an alkene; 2,3-dimethylsuccinate gives 2-butene.
50	(B)	The carboxylate is consumed and hydroxide forms at the cathode: pH increases.
51	(B)	Chlorine is unselective: 9 primary H and 1 tertiary H (reactivity about 1 : 4) give mostly isobutyl chloride.
52	(A) 8	Methylcyclopentane gives 4 monochlorides and isopentane gives 4 (without stereoisomers): X + Y = 8.
53	(B) 9 ‡	NBS on 1-methylcyclohexene: allylic substitution at three allylic positions with rearranged radicals; the key counts nine.

5G.2 Exercise II-A (multiple correct, Q1–Q18)

Q	Answer	Reasoning
1	(C)	Isopentane (2-methylbutane) has four kinds of hydrogen: four monochlorides.
2	(A, B)	HI adds to the double bond and the alkyl bromide is displaced via the bromonium-type neighbour; only the products that retain the label pattern in A and B form.
3	(A, B, D)	At 500 °C, chlorination of 1-butene is allylic radical substitution: four products, two of them chiral; the chiral one cannot be made by HCl addition (which would put Cl on the other carbon).
4	(A, C, D)	NBS is an allylic brominating agent; 1-methylcyclopentene has seven allylic H; nine allylic bromides in total; the number of fractions is not six.
5	(B, D)	Wurtz works well for coupling two identical primary-type halides: B and D are symmetric products.
6	(A, B)	HCl adds to 2,4-hexadiene to give 4-chloro-2-hexene (1,2-adduct) and 2-chloro-3-hexene (the other 1,2-/1,4-adducts).
7	(A, B, C, D)	All four naming statements are correct.
8	(B, C, D)	Nucleophilicity statements: excluded from your syllabus list (see the note at the top).
9	(A, D)	Diastereomeric pairs result in (A) bromination of the cyclohexene and (D) radical addition to the optically pure alkene.
10	(A, B)	HBr adds to the quinone-methide-type diene to give the benzylic-type bromides of A and B.
11	(A, C)	Acid-catalysed monoetherisation (1 equivalent) gives two optically active products (diastereomers), a resolvable mixture.
12	(A, B, D)	The tertiary cation from the allyl-substituted phenol can lose H (A), add water (B) or be trapped by the OH to form the dihydrobenzofuran (D).
13	(B)	The spirocyclic cation undergoes ring expansion by a 1,2-alkyl shift and loses H to give the aromatic tetralin-type product.
14	(C)	Pinacol-type coupling of the diketone (Mg) followed by acid: pinacol rearrangement with ring contraction to the spiro ketone.
15	(A, B, C, D)	All four are true (cyclopropane opens with Br_2; bromination more selective than chlorination; hindered phenoxyl radical does not dimerise; the benzylic radical forms faster with Ar = Ph than with p-nitrophenyl).
16	(B) 4	Hydrocarbon major products: Wurtz coupling, Kolbe electrolysis, Zn dehalogenation, and hydroboration–protonolysis give hydrocarbons or alkenes as counted by the key; $Hg(OAc)_2$ and Br_2 do not. ‡
17	(A, B, C)	Transannular Wurtz coupling gives a bicyclic hydrocarbon with a double bond that shows geometrical isomerism; no carbocation is involved.
18	(A, C)	$BrCCl_3$ adds by a radical chain: $CCl_3^{\bullet}$ adds to the terminal carbon of styrene, giving the benzylic radical, which abstracts Br: a mixture of enantiomers.

5G.3 Exercise II-B (Q1–Q13)

Q	Answer	Reasoning
1	(B)	HBr is the one that shows the peroxide effect because both chain-propagation steps are exothermic; ‘stronger acid’ is true but is not the reason.
2	A–S,T; B–P,S,T; C–U; D–Q; E–T,U	Counting the further chlorination products, with chirality.
3	A–R; B–Q,S; C–P,Q; D–Q,S	Carbocation: dehydration of alcohols; carbanion: Wurtz and soda-lime decarboxylation; free radical: Kolbe and Wurtz (radical mechanism in the key); octet complete in one of the intermediates (Wurtz, soda-lime).
4	A–P; B–P; C–Q; D–R	Matching to the energy profiles with one, two, three rearrangement steps (the number of carbocation minima).
5	1-chloro-3-methylbutane	Isopentane is formed by hydrolysis of R–Li and 2,7-dimethyloctane by Wurtz coupling: R = $(CH_3)_2CHCH_2CH_2$–.
6	neopentyl chloride	Reduction gives neopentane (C_5) and Wurtz coupling gives 2,2,5,5-tetramethylhexane.
7	HBr only	For HBr both propagation steps are exothermic (–25.1 and –50.2 kJ mol^{-1}); for HCl the second step is endothermic (+12.6); for HI the first step is endothermic (+46). So only HBr sustains the chain.

8	8 products	Isohexane (2-methylpentane): 1-chloro-2-methylpentane (2 enantiomers), 2-chloro-2-methylpentane (1), 3-chloro-2-methylpentane (2), 4-chloro-2-methylpentane (2) and 1-chloro-4-methylpentane (1).
9	see below	(A) 1-bromo-1-phenylpropane (benzylic cation); (B) 2-iodo-2-methylbutane; (C) trans-1-bromo-2-methylcyclohexane (anti-Markovnikov, radical); (D) the tertiary chloride (Markovnikov).
10	A = trans-2-butene; B = 2-bromobutane	0.7 g per 2.0 g of Br_2 (0.0125 mol) gives M = 56 (C_4H_8); the dibromide is non-resolvable (meso, from the trans alkene); both HBr paths give the same 2-bromobutane because the alkene is symmetric.
11	(a) tertiary chloride; (b) trans-dibromides; (c) dichloride	(a) Markovnikov addition gives the tertiary chloride; (b) anti addition of Br_2 to the ring gives two diastereomeric trans-dibromides; (c) Cl_2/CS_2 gives the vicinal dichloride as racemic diastereomers. ‡
12	reasoning	In CH ≡ C–CH_2–CH=CH_2 protonation of the alkene gives a secondary cation. In CH ≡ C–CH=CH_2 protonation of the alkene end would give a cation next to an alkyne that is destabilised, so the terminal alkyne carbon is protonated to a vinyl cation conjugated with the vinyl group (an allyl-like cation) and bromide adds at the internal carbon: 2-bromo-1,3-butadiene.
13	A = 2,2-; B = 1,3-; C = 1,2-; D = 1,1-dichloropropane	Further chlorination: 2,2- gives one trichloride; 1,3- gives two; 1,1- and 1,2- give three each. The optically active one is 1,2-dichloropropane (chiral), whose 1,1,2-trichloride is chiral and the other two are not.

5G.4 JEE-Main and JEE-Advanced (objective)

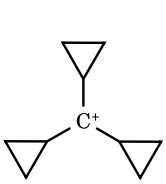
Q	Answer	Reasoning
M1	(3)	Cl^+ (from Cl_2) adds first, giving the intermediate CH_3–CH^+–CH_2Cl, which water attacks.
M2	(4)	HBr with peroxide on 3-methyl-2-pentene gives the anti-Markovnikov bromide with two stereocentres: four stereoisomers.
M3	—	Nucleophilicity order: excluded from your syllabus list.
M4	(2)	Cl_2/H_2O gives the chlorohydrin $CH_3CH(OH)CH_2Cl$.
M5	(1)	Vinyl halides resist substitution; but the reason given (a cation stabilised by π electrons) is wrong, since the vinyl cation is unstable; A is right and R is wrong.
M6	(1)	CF_3–CH=CH_2 gives the anti-Markovnikov product: the CF_3 group makes the secondary cation very unstable.
M7	(4)	DCl adds Markovnikov to the alkyne to give $CH_3C(Cl)$=CHD; DI then adds to give $CH_3C(Cl)(I)CHD_2$.
M8	(3) ‡	The nitrile-stabilised tertiary radical is formed and adds to the terminal alkene carbon.
M9	(1)	Acid-catalysed cyclisation of the diene (geraniol-type) gives the terpineol skeleton.
M10	(3)	Protonation of the cyclopropenone gives the aromatic hydroxycyclopropenyl cation.
M11	(3) ‡	Hydroboration–oxidation gives the alcohol; dilute acid dehydration gives the most substituted (Zaitsev) alkene.
A1	(A, C)	Benzyl chloride from toluene: SO_2Cl_2 (radical) and Cl_2/hv; $SOCl_2$ and NaOCl do not do benzylic chlorination.
A2	—	Nucleophilicity: excluded from your syllabus list.
A3	(D)	One equivalent of HBr on isoprene gives the 1,4-adduct (a trisubstituted alkene) as the major product by the key.
A4	(A, B)	trans-2-butene (i) and cis-2-butene (ii) with Br_2: anti addition in both; (M, O) and (N, P) are diastereomeric pairs.

⚑ **Check.** The printed key assigns the Main Q3 and Advanced Q2 nucleophilicity items which I did not work. For questions marked ‡ I relied on the key because the drawings did not allow independent verification.

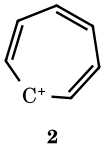
Part 5H Orders and Rules from Your Lecture Notes (Lessons 1–5, Doubt-Clearing Sessions)

The 12 lecture PDFs are screen captures of handwritten board work. I went through all of their pages; the legible, carbocation-relevant content is collected here as orders and rules. Board problems whose answers were never written on the page (for example the 'number of shifts' drawings 7–25 of Carbocation-V and the dehydration problems 1–48) are **not** answered here, because I could not read the structures reliably: send those pages at higher resolution and I will solve them. Conformational analysis (Lesson 1) and nucleophilicity-in-solvents (Doubt session) are outside your topic list, so they are not reproduced.

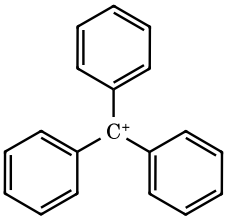
5H.1 Master stability order (Doubt-clearing session)



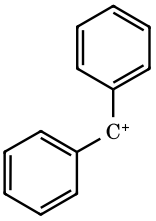
1



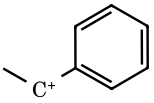
2



3



4

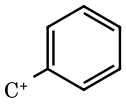


5



6

Board order, left to right: tricyclopropylmethyl > tropylium > Ph_3C^+ > Ph_2CH^+ > $PhCH^+R$ > $(CH_3)_3C^+$ …



7



8



9



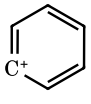
10



11



12



13



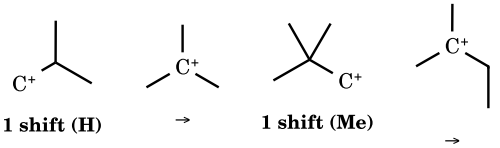
14

… > $PhCH_2^+$ > allyl ≈ isopropyl > ethyl > methyl > vinyl > phenyl > ethynyl. The cyclopentadienyl cation is written as 'can't exist' (antiaromatic).

> ⚑ **Check.** This board order puts tricyclopropylmethyl *above* tropylium; the pK_{R^+} values in your cation notes (tropylium +4.7, tricyclopropylmethyl −2.3) and the hydride affinity (tropylium 200) put tropylium far above it, so this one position on the board disagrees with the data (Part 5A Q93 and Part 5E use tropylium first). It also ranks t-Bu^+ above benzyl, consistent with Part 5E (hydride affinities 231.9 vs 238) but opposite to the coaching convention used in some other answers. The exact positions of allyl versus isopropyl on the board line are not clearly legible; the HIA table gives isopropyl 249 < allyl 256.

5H.2 Rearrangement rules from Carbocation-V

- A rearrangement happens to reach a **more stable** cation (3° > 2° > 1°) and only if the three-membered cyclic transition state (3c–2e bonding, a *1,2-shift*) is accessible. It is spontaneous and fast.
- **Migratory aptitude:** H > Ar > R (bond shifted, steric factor and electron donation all matter); for alkyl groups t-Bu > i-Pr > Et > Me; for isotopes H > D > T; among aryl groups p-OCH_3 (+M) > p-CH_3 (+H) > H > p-Cl (–I).
- Counting shifts (board examples): CH_3^+, $CH_3CH_2^+$ and $(CH_3)_2CH^+$ → **0** shifts (nothing more stable is reachable); $(CH_3)_2CH–CH_2^+$ → **1** (H) giving t-Bu^+; $(CH_3)_3C–CH_2^+$ → **1** (Me) giving t-amyl; $(CH_3)_2CH–CH(CH_3)–CH_2^+$-type chains → **2** (H, H) to the tertiary ion.



1 shift (H) → **1 shift (Me)** →

Two of the board counts redrawn: isobutyl → t-butyl (one hydride shift); neopentyl → t-amyl (one methyl shift).

5H.3 Leaving-group and ionisation rules from Carbocation-IV

Rule from the board	Detail
Leaving tendency	∝ stability of the leaving group ∝ stability of the carbocation formed; weaker base = better leaving group
Halides	I^- > Br^- > Cl^- > F^- (size/polarisability)
Oxyanions	CH_3O^- (not a leaving group) < acetate < mesylate < triflate; tosylate < brosylate < nosylate
Super leaving groups	N_2 (from R–N_2^+, the amine/HNO_2 route) and triflate

Conversions the board shows	R–OH is poor, but R–OH + H^+ → R–OH_2^+ (leaves as H_2O); R–OH + TsCl/BsCl/AcCl/MsCl/TfCl gives R–OTs etc.; R–NH_2 + H^+ → R–NH_3^+ and R–NH_2 + 2 TsCl → R–NTs_2 (leaves as $^-NTs_2$); ethers need H^+ first; H–OH does not ionise to H^+ + OH^- (strong base); CH_3–I does not ionise in water (primary methyl cation); aryl and vinyl halides do not ionise (but ArN_2^+ loses N_2 on heating to give phenol via the unstable aryl cation); R–Cl + $AlCl_3$/$AlBr_3$/$SbCl_5$/$AgNO_3$ gives R^+ (Lewis acids)

5H.4 Reactions of carbocations (Lessons 5 and FR-II)

Type	Rule	Product pattern
Combination with a nucleophile	attack from either face of the planar sp^2 cation	chiral cation: racemic mixture (2 enantiomers) or two diastereomers if another stereocentre exists
Elimination (if a base is present)	loss of a β-H, Saytzeff (more substituted, more stable alkene, via the more stable transition state) is the major product; Hofmann (less substituted) is the minor	stability: Saytzeff > Hofmann; for 2-methylbutane-type cations the more substituted alkene is formed by losing the β-H of the more substituted carbon
Rearrangement	see 5H.2 and Part 7.0	—

5H.5 Dehydration of alcohols (Free-Radical III/IV lessons)

- Conc. H_2SO_4 or conc. H_3PO_4, H^+/Δ gives the alkene as the major product; ethanol: 180 °C → ethylene (elimination), 140 °C → diethyl ether (substitution), 110 °C → ethyl hydrogen sulfate.
- **Rate of dehydration ∝ stability of the carbocation formed** (slow step): 3° > 2° > 1°. The slow step is the loss of water from the protonated alcohol (rate-determining), so the potential-energy diagram has three transition states and one intermediate (the oxonium ion) before the carbocation.
- If both C=C and –OH are present, protonation occurs at the site that gives the more stable carbocation; primary alcohols dehydrate by E2 without a free carbocation, so rearrangement is not really observed for 1° alcohols (the board marks this).
- Yield order of the three products from the sec-pentyl cation: the trans-2-alkene (Saytzeff) > the cis-2-alkene > the 1-alkene (Hofmann).

5H.6 Pinacol rearrangement rules (Free-Radical IV)

- vic-Diols (pinacol) with conc. H_2SO_4 → pinacolone: protonate one OH, lose H_2O (RDS), 1,2-shift, deprotonate the O-protonated ketone.
- Which OH leaves: the one that gives the more stable cation (benzylic/tertiary). Which group migrates: the better migrating group (aryl > H > alkyl on your board; with aryl groups, the more electron-rich). Semipinacol: amino-alcohol + HNO_2 gives the same cation by loss of N_2.
- Worked mechanisms 7.10–7.19 in Part 7.0 cover pinacolone, benzopinacolone, ring expansion/contraction and the dienone–phenol shift.

5H.7 Electrophiles and nucleophiles (Lesson 2) – the part of your topic list

Class	Definition on the board	Examples
Electrophile	electron-loving; an electron-deficient species, electron acceptor; must have an (exactly or momentarily) vacant orbital	—
Positive electrophile	carries a + charge	H^+, D^+, T^+, Cl^+, Br^+, I^+, NO^+, NO_2^+, carbocations. **Alkali-metal cations (Na^+, Li^+, K^+) are not electrophiles.**
Neutral incomplete-octet electrophile (Lewis acid)	electron-deficient atom, no charge	BH_3, B_2H_6, BF_3, BCl_3, $AlCl_3$, $SbCl_5$, carbenes, nitrenes
Complete-octet electrophile	octet complete, but a polarised bond makes one atom δ+ and open to attack	carbonyl compounds (C=O), RCOX, RCOOR′, CO_2, CH_3–I, Br–Br, Cl–Cl, I–Cl, I–N_3, HO–Cl, NO–Cl
Nucleophile	nucleus-loving; electron donor; any species with a lone pair or a loose π bond pair	HO^-, RO^-, CN^-, NH_3, H_2O, alkenes, arenes, carbanions

The board also classifies single-step and multi-step reactions: a single-step reaction goes through one transition state (a pentavalent carbon in the TS) with no intermediate; a multi-step reaction must have at least one intermediate (carbocation, carbanion, free radical) and uses normal tetravalent TS's. Endothermic reactions have $\Delta H > 0$ and the product is less stable; exothermic ones $\Delta H < 0$.

5H.8 Carbocation structure (Doubt-clearing session)

Property	Carbenium ion R_3C^+	Carbonium ion (five-coordinate)

Hybridisation / shape	sp^2, trigonal planar, bond angle 120°, empty p orbital perpendicular to the plane	sp^3d-type description is wrong; it is a 3c–2e bonded species (CH_5^+)
Electron count	sextet: 3 bond pairs, 0 unpaired electrons, 0 lone pairs; diamagnetic ($\mu = \sqrt{n(n+2)} = 0$), spin multiplicity 2S+1 = 1 (singlet)	complete octet counting the 3c–2e pair; highly unstable and reactive
Reactivity	electrophile (Lewis acid); planar, so a nucleophile attacks from both faces (racemisation)	—
Formation	R–LG heterolysis; $H^+ + CH_2$ (carbene) gives CH_3^+; $H^+ + CH_4$ gives CH_5^+; H^+ adds to lone pairs ($H_2O \rightarrow H_3O^+$, $NH_3 \rightarrow NH_4^+$) but not to NaCl-type ionic species	protonation of an alkane

Part 7.0 Carbocation Rearrangements: Thermodynamic and Kinetic Control, Mechanisms, and Named Reactions

A carbocation rearranges by a 1,2-shift of H (hydride), alkyl or aryl with its bonding electron pair to the neighbouring cationic carbon, through a three-centre two-electron (3c–2e) transition state or bridged intermediate. The cation moves to the carbon that lost the migrating group. Your lecture notes divide the subject into two kinds, and the division is useful for every problem:

	Thermodynamic control (TCP): reversible	**Kinetic control (KCP): irreversible**
Where the cation goes	to the *more stable* cation, but each step is reversible, so the mixture reaches the thermodynamically best cation(s)	to a cation that is captured at once by an irreversible step (loss of H^+, formation of C=O by deprotonation, ring-strain relief)
Examples	neopentyl ⇌ t-amyl; 2,2-dimethylcyclohexyl ⇌ 1,2-dimethylcyclohexyl; cycloalkylmethyl ⇌ methylcycloalkyl	pinacol (ends in a ketone); semipinacol/Tiffeneau–Demjanov (HNO_2); dienone–phenol (ends aromatic)
What decides the product	relative stability of the cations (substitution, resonance, α-heteroatom lone pair, ring strain)	what the cation can *do* next: an OH or NH_2 on the new cationic carbon forms $C{=}O^+H$ or $C{=}N^+H_2$ and the sequence cannot go back

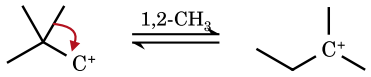
A 1,2-rearrangement runs to completion only if (i) the new cation is substantially better stabilised electronically (more substituted, resonance, adjacent lone pair); or (ii) it is more stable for another reason such as relief of ring strain; or (iii) it is captured in a subsequent irreversible reaction. Primary cations are generally not true intermediates: the shift occurs together with the departure of the leaving group. A shift between two equally stable cations (tertiary ⇌ tertiary) is allowed and reversible, but it is not counted as a 'rearrangement' in counting problems unless it changes the skeleton or the product.

7.0.1 Orders you must know

Order	**Rule**
Migratory aptitude in a reversible 1,2-shift (your note)	Ph < 1° < 2° < 3° < H: the better the migrating group stabilises a positive charge (electron-donating ability), the better it migrates. Hydride is the best, then the more substituted alkyl, and phenyl is the worst here because its migration goes by a less favourable ipso (phenonium) pathway when an easier alkyl/H shift is available.
Migratory aptitude in the pinacol rearrangement (your note)	Ph > H > Me, i.e. aryl first. Among aryl groups, ∝ electron donation: p-OCH_3-C_6H_4 (+M) > p-CH_3-C_6H_4 (+H) > C_6H_5 > p-Cl-C_6H_4 (–I). Always combine it with the stability of the cation formed by the first ionisation.
Which OH ionises first in a pinacol	The OH on the carbon that gives the *more stable cation* leaves (diaryl > aryl,alkyl > dialkyl; tertiary > secondary), and the group that migrates then goes to *that* carbon.
Ring size in expansions	A small ring next to a cationic carbon expands (cyclopropylcarbinyl → cyclobutyl; cyclobutylcarbinyl → cyclopentyl; cyclopentylcarbinyl → cyclohexyl): the driving force is relief of angle/torsional strain (σ-bond shift).
Rate order of pinacol rearrangement (Nucleus Q23)	key: ArMe-diol (II) > diaryl diol (I) > pinacol (III): the faster the first ionisation, the faster the overall reaction.

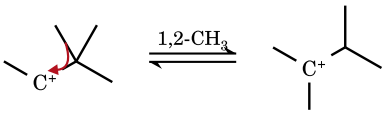
7.0.2 Step-validated 1,2-shifts and named rearrangements

7.1 Neopentyl cation ⇌ tert-amyl cation (methyl shift, TCP)



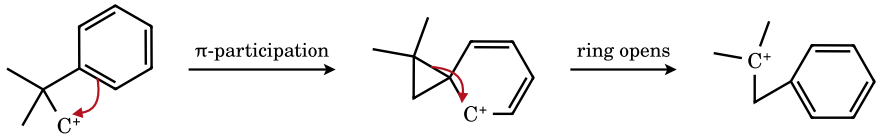
The C–CH_3 σ pair migrates to the primary cation; the cation moves to the quaternary carbon, which becomes tertiary: primary → tertiary is downhill by about 70 kJ mol^{-1}, and the reverse is not observed.

7.2 3,3-Dimethylbut-2-yl cation → 2,3-dimethylbut-2-yl cation (secondary → tertiary)



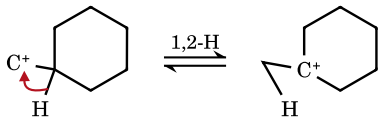
The cation next to a quaternary carbon is the classic setting for a methyl shift: the secondary cation becomes tertiary. This is the reason pinacolyl alcohol with HBr gives 2-bromo-2,3-dimethylbutane.

7.3 Neophyl cation: phenyl shift through the phenonium ion



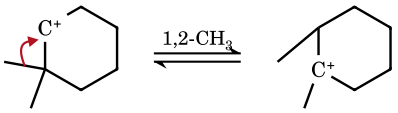
The ring π electrons attack the cationic carbon, giving the bridged phenonium ion (a spiro cyclohexadienyl cation with a three-membered ring), then the C–C bond to the quaternary carbon breaks and the tertiary cation forms. The bridged ion explains retention of configuration in aryl migrations.

7.4 Cyclohexylmethyl cation → 1-methylcyclohexyl cation (hydride shift)



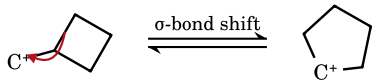
A primary cation next to a tertiary C–H takes the hydride and becomes the tertiary ion; the competing ring expansion to the cycloheptyl cation (secondary) is slower.

7.5 2,2-Dimethylcyclohexyl cation → 1,2-dimethylcyclohexyl cation



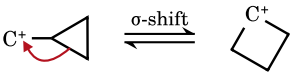
The same reasoning in a ring. The tertiary ion then loses a β-hydrogen to give the most substituted alkene, 1,2-dimethylcyclohexene (Nucleus Q10).

7.6 Cyclobutylmethyl cation → cyclopentyl cation (ring expansion)



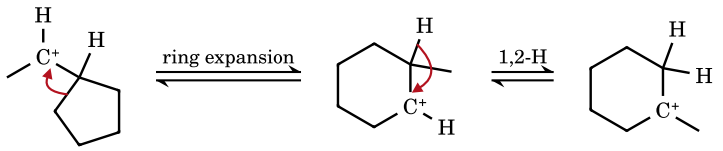
A ring C–C bond migrates to the primary carbon: the four-membered ring becomes five-membered (strain relief ≈ 100 kJ mol^{-1}) and the cation changes from primary to secondary. Ring expansion is driven by strain and by the better cation.

7.8 Cyclopropylcarbinyl cation ⇌ cyclobutyl cation (≈ 48 : 47, with ≈ 5 % homoallyl)



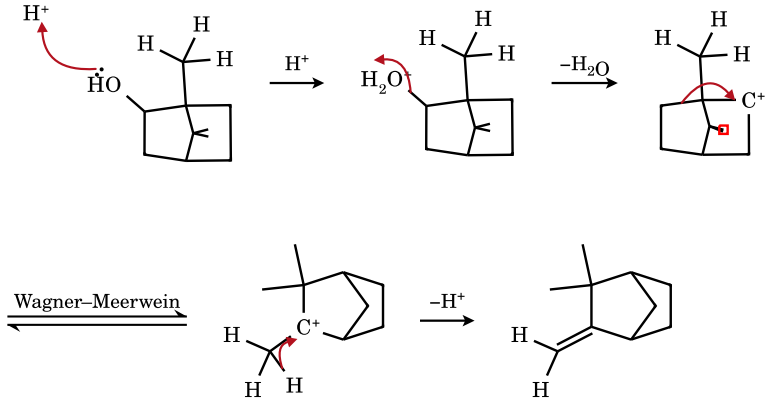
The two ions interconvert rapidly by a σ-bond shift, and solvolysis of either gives the same mixture of cyclopropylcarbinyl (≈ 48 %), cyclobutyl (≈ 47 %) and homoallyl (≈ 5 %) products. The ions are best described as a delocalised bicyclobutonium/non-classical cation (Part 4).

7.7 1-Cyclopentylethyl cation → 1-methylcyclohexyl cation (ring expansion then hydride shift)



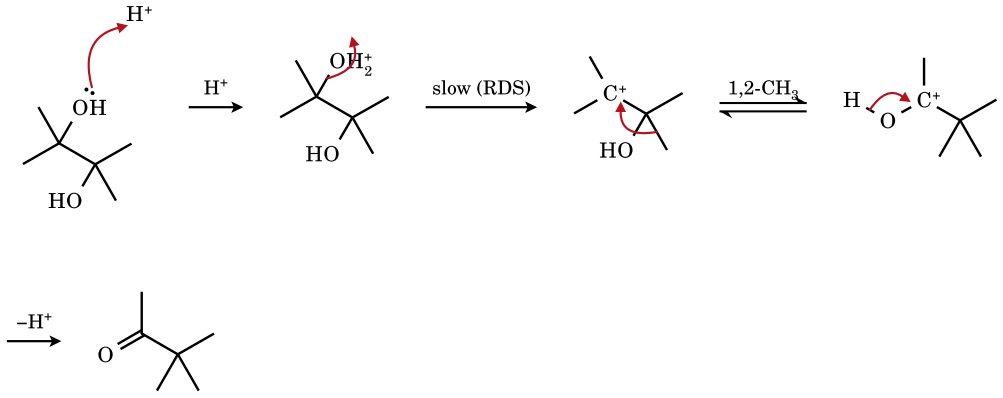
Step 1 moves a ring bond to the cationic carbon and enlarges the ring to cyclohexane (the cation is now the secondary ion next to the methyl-bearing carbon). Step 2 is a hydride shift that puts the charge on the methyl-bearing carbon: tertiary, the global minimum (Kohinoor p. 9; Nucleus Q16 follows the same logic).

7.9 Isoborneol → camphene (Wagner–Meerwein, H^+)



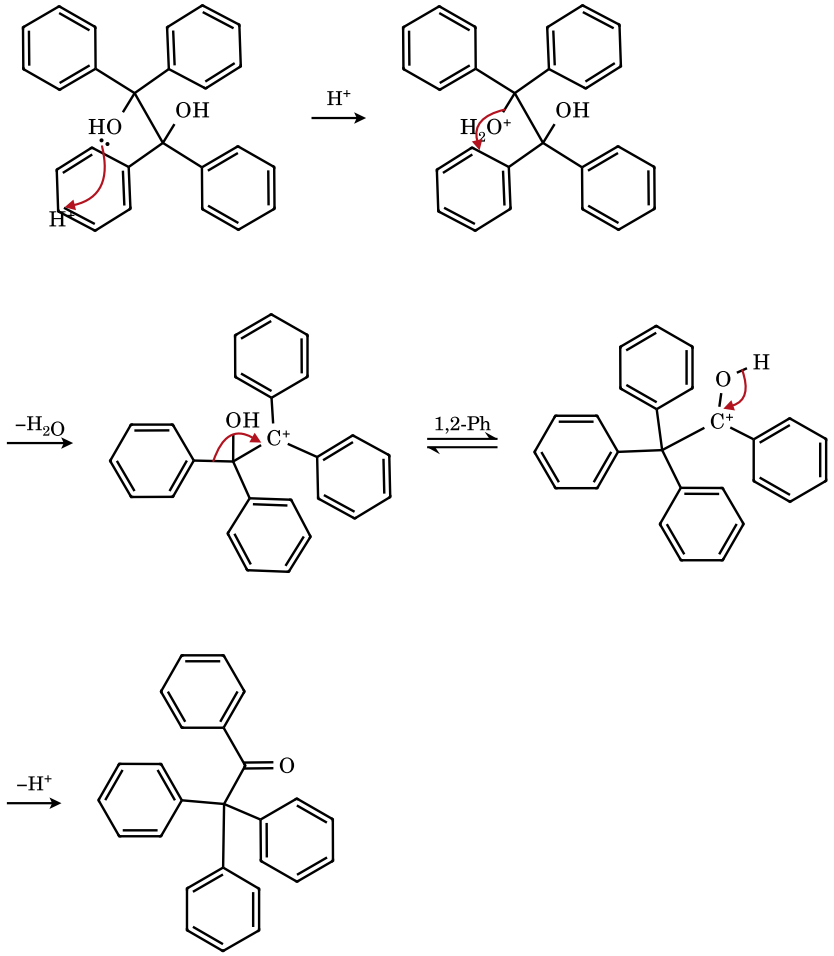
The secondary 2-norbornyl-type cation is formed (as an ion with the OH leaving as water), the C1–C6 σ bond shifts to C2 (Wagner–Meerwein shift) and the cation lands on the bridgehead carbon that bears a methyl group. Loss of a proton from that methyl gives the exocyclic alkene camphene, **not** the bornylene (the endocyclic alkene), as in your page: the proton is removed from the methyl group that can overlap with the empty orbital.

7.10 Pinacol → pinacolone



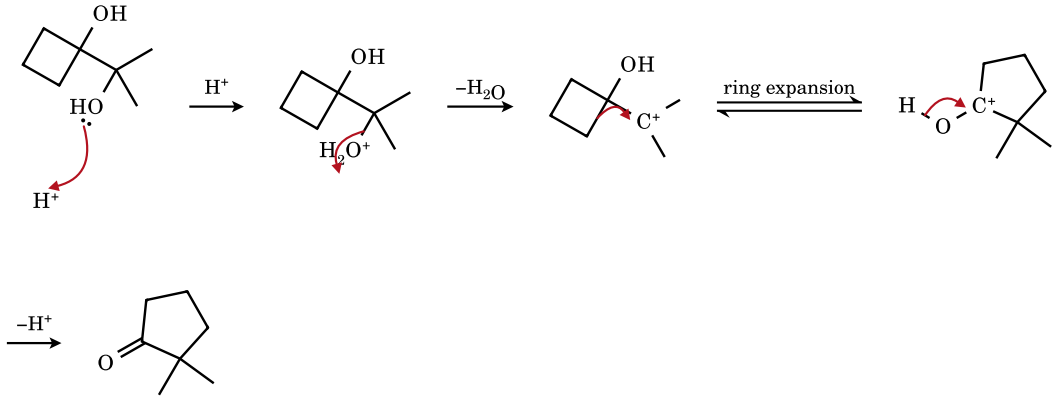
Protonation, loss of water (rate-determining) and a methyl shift give an O-protonated ketone; loss of H^+ (irreversible) gives pinacolone. The lone pair of the remaining OH stabilises the new cation by +M: this is the driving force and it is why the reaction is under kinetic control.

7.11 Benzopinacol → triphenylacetophenone (aryl migration)



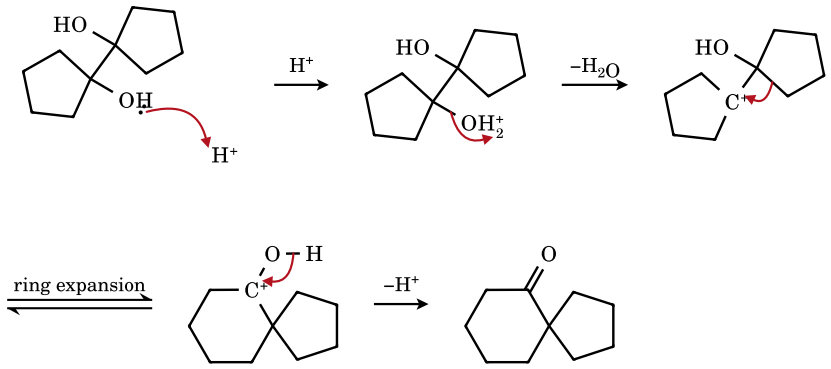
Either OH can ionise (the diaryl cations are equivalent). A phenyl group migrates to the cationic carbon through a phenonium-type transition state; the new cation is stabilised by the OH lone pair and gives triphenylacetophenone (β-benzopinacolone).

7.12 1-(2-Hydroxyprop-2-yl)cyclobutanol → 2,2-dimethylcyclopentanone (pinacol with ring expansion)



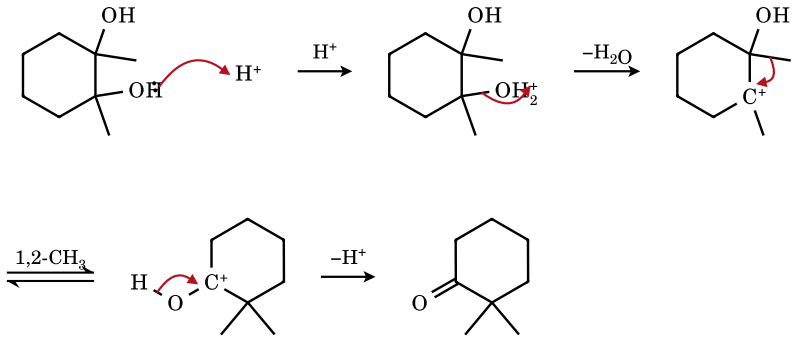
The tertiary open-chain OH ionises (more stable cation than the strained cyclobutyl one). The migrating group is a ring C–C bond: it moves to the cation, enlarging the four-membered ring to five. The adjacent carbon becomes C=O. Nucleus Q24 (answer D).

7.13 1,1′-Bi(cyclopentyl)-1,1′-diol → spiro[4.5]decan-6-one



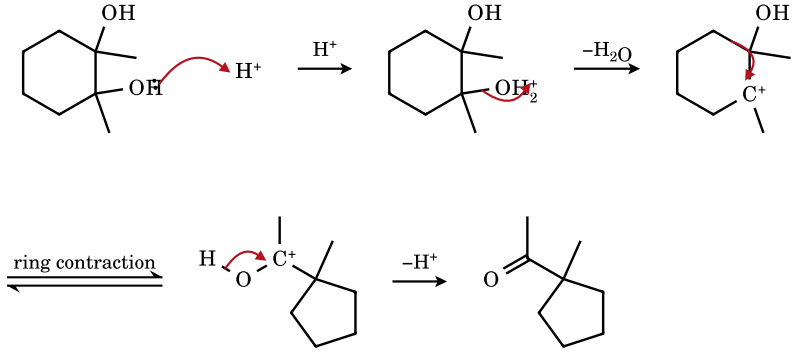
Pinacol coupling of cyclopentanone (Mg–Hg) followed by acid gives the spiro ketone: one ring bond of the other cyclopentane migrates to the cationic carbon, so that ring expands to six and the cationic carbon becomes the spiro centre (Kohinoor items 7 and 12; Nucleus II-A Q14).

7.14 cis-1,2-Dimethylcyclohexane-1,2-diol → 2,2-dimethylcyclohexanone (methyl shift)



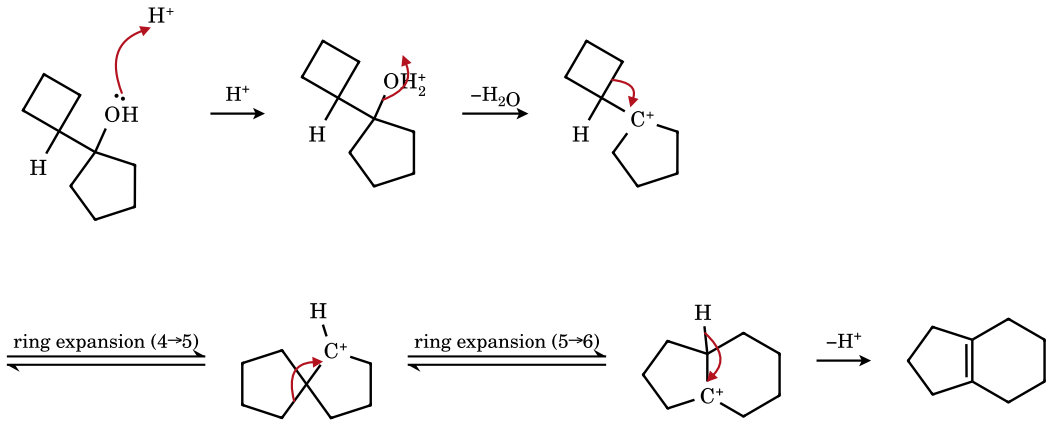
In the cis diol one OH is axial and the other equatorial. The axial OH is the one that leaves, and the group anti-periplanar to it on the neighbouring carbon is the *axial methyl* (the OH on that carbon is equatorial), so the methyl migrates. The product is 2,2-dimethylcyclohexanone.

7.15 trans-1,2-Dimethylcyclohexane-1,2-diol (diequatorial) → 1-acetyl-1-methylcyclopentane (ring contraction)



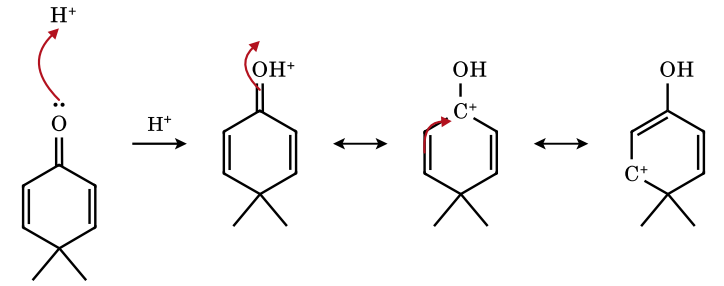
When the leaving OH is equatorial, the antiperiplanar bond is a ring C–C bond, not the methyl: that ring bond migrates, the six-membered ring contracts to five, and an exocyclic acetyl group results. The same compound, different stereochemistry: different product (stereoelectronic control).

7.16 1-Cyclobutylcyclopentanol → bicyclo[4.3.0]non-1(6)-ene (two ring expansions, then E1)



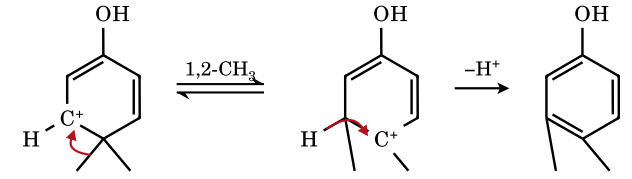
The tertiary cation next to a strained cyclobutane first expands the four-membered ring (giving a spiro[4.4] secondary ion), which then expands the other ring to six (fused tertiary ion at the ring junction); loss of the junction proton gives the tetrasubstituted alkene (Nucleus Q16, option D).

7.17 4,4-Dimethylcyclohexa-2,5-dienone → 3,4-dimethylphenol (dienone–phenol rearrangement)



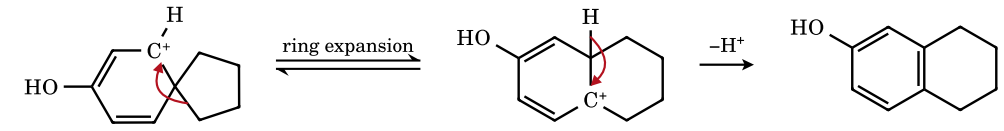
Step 1 protonates the carbonyl oxygen; steps 2–3 move the positive charge to C-4, next to the quaternary carbon. The next step is the methyl shift and the loss of a proton (shown separately in 7.18).

7.18 Dienone–phenol, the shift and the aromatisation



The cation on C-4 is the arenium-like ion with a quaternary neighbour; a methyl group shifts to it, the new cation at C-5 is a fully delocalised hydroxy-arenium ion, and loss of H^+ from C-4 gives the aromatic phenol. The migrating group goes to the adjacent ring carbon (3,4-dimethylphenol).

7.19 Spiro[4.5]deca-6,9-dien-8-one → 5,6,7,8-tetrahydro-2-naphthol (spiro ring expansion)



In the protonated dienone the positive charge sits next to the spiro carbon. A bond of the spiro cyclopentane migrates, expanding that ring to six and converting the ion to a hydroxy-arenium; loss of H^+ restores the aromatic ring. Nucleus II-A Q13 and Kohinoor p. 14.

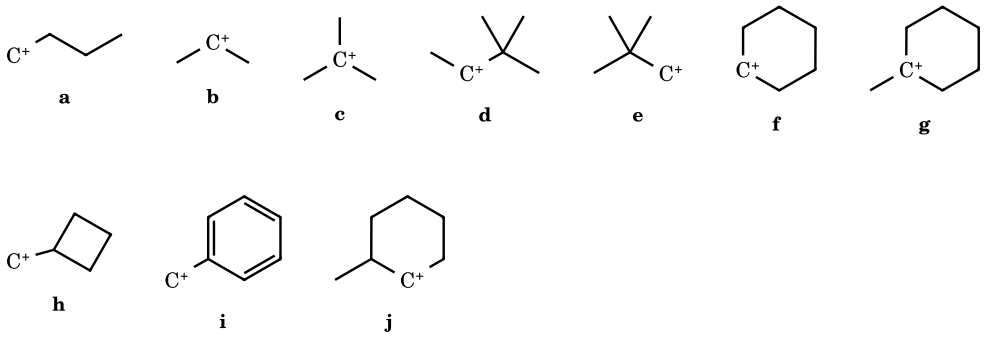
Part 7 Carbocation Rearrangements — Counting Problems (opening section)

Counting problems test whether you can *recognise* a rearrangement before drawing any mechanism. Rules used (JEE-Advanced level): a carbocation rearranges by a 1,2-H, 1,2-alkyl, 1,2-aryl or ring-bond shift **only if the new cation is more stable** (more substituted, conjugated, or less strained) or the new ion is trapped irreversibly; shifts to an equally stable cation (secondary → secondary, tertiary → tertiary) are degenerate or unproductive and are **not** counted as 'rearranging'; tertiary, benzylic and allylic cations do not rearrange to a less stable ion.

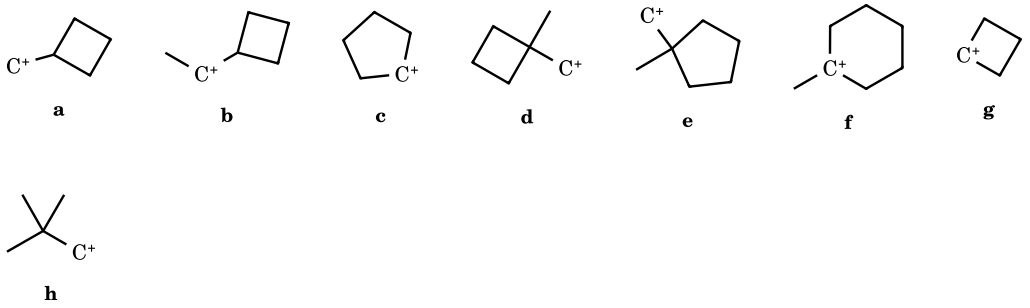
Primary cations are usually not true intermediates: the shift is concerted with the departure of the leaving group, but for counting purposes such a shift is ‘a rearrangement’. The answers below were cross-checked with a small shift-enumeration program (every 1,2-shift of every cation was generated and the product classified) and by hand.

7.1 How many rearrange? How many ring-expand? How many shifts are possible?

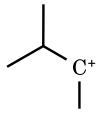
R1. How many of the following carbocations (a–j) will **undergo rearrangement** by a 1,2-shift to give a more stable carbocation? (Integer answer)



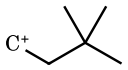
R2. How many of the carbocations (a–h) can undergo **ring expansion** by a single 1,2-shift to a more stable cation (more substituted and/or less strained)? (Integer answer)



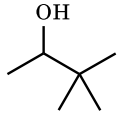
R3. The 3-methylbutan-2-yl cation, $CH_3–CH^+–CH(CH_3)_2$, can undergo different 1,2-shifts. (i) How many **different** carbocations (other than the starting ion itself) can be formed by one 1,2-shift of H or CH_3? (ii) How many of these are **more stable** than the starting ion? Give (i), (ii).



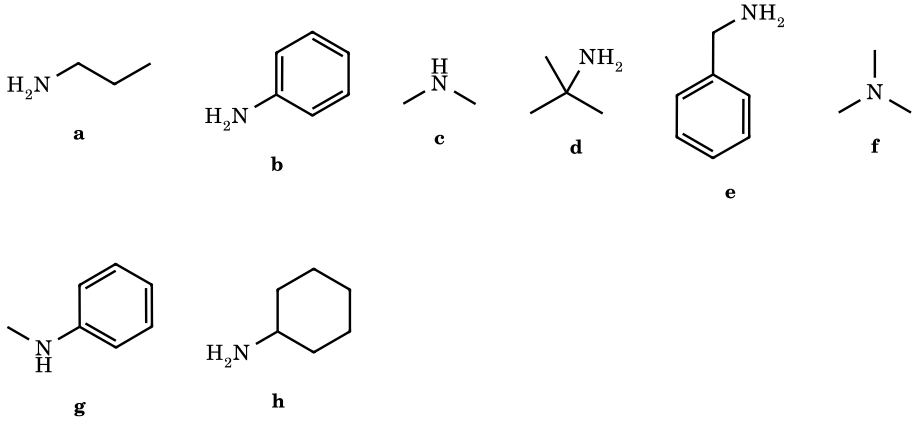
R4. How many 1,2-shifts, at minimum, convert the primary 3,3-dimethylbutyl cation, $(CH_3)_3C–CH_2–CH_2^+$, into the **most stable** carbocation that is reachable? (Integer answer; each hydride, alkyl or ring-bond shift counts one.)



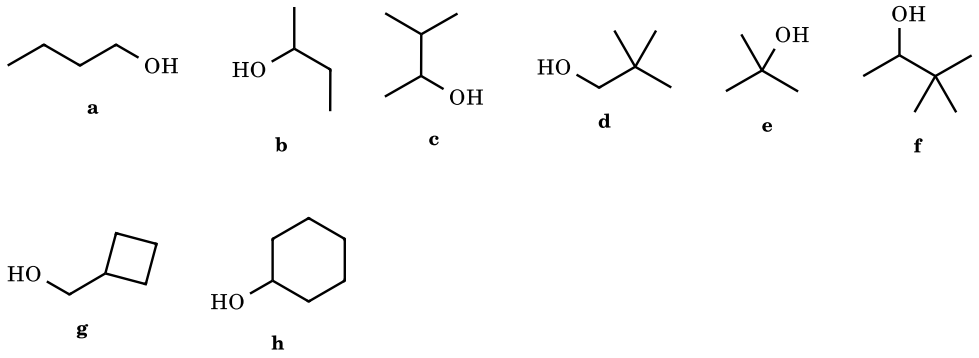
R5. 3,3-Dimethylbutan-2-ol is heated with conc. H_3PO_4 (E1). How many **different alkenes** (structural isomers, counting only constitutional isomers, no stereoisomers arise) can be formed from *all* the carbocations that are intermediates, including the rearranged one? (Integer answer)



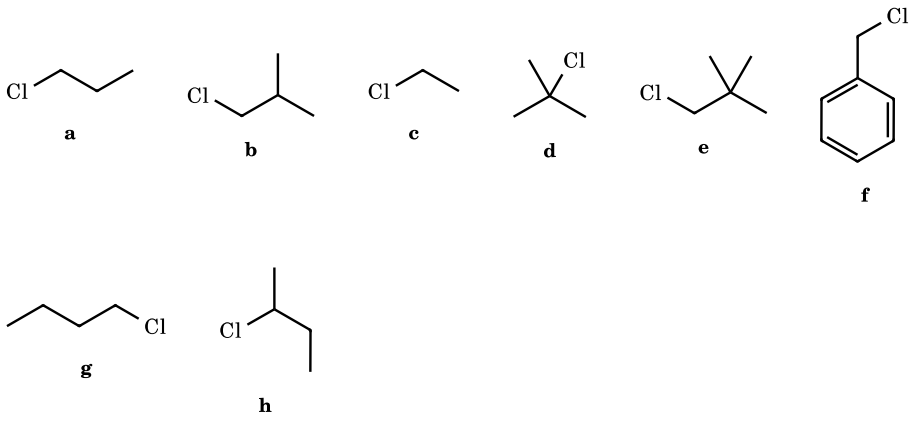
R6. Each compound (a–h) is treated with $NaNO_2$/HCl at 0–5 °C, then the solution is allowed to stand. **How many** of the compounds give a **carbocation** as an intermediate (free, or as a tight ion pair, formed by loss of N_2)? (Integer answer)



R7. Each alcohol (a–h) is heated with conc. HBr. **How many** give the alkyl bromide (or major product) with a **rearranged carbon skeleton or rearranged position of Br** through a carbocation or a concerted shift? (Integer answer)



R8. Benzene is alkylated with each alkyl chloride (a–h) and $AlCl_3$. **How many** give a product in which the carbon skeleton or the point of attachment of the alkyl group **differs from the alkyl chloride used** (a rearranged product as the major product)? (Integer answer)



Part 8 Carbanions: Structure, Generation, Stability and Reactions

A **carbanion** is a species in which a carbon atom carries a formal negative charge and an unshared pair of electrons: R_3C^-. It has **eight electrons** (three bond pairs + one lone pair), a filled octet, and is therefore a *nucleophile and a base* – the mirror image of the carbocation, whose carbon has a sextet and an empty orbital. Carbanions are almost never free in solution: they exist as ion pairs or aggregates with Li^+, Na^+, K^+, MgX^+, and the carbon–metal bond in RLi and RMgX is strongly polarised $C^{\delta-}$–$M^{\delta+}$, so that the organometallic reagent *reacts as if it were* R^-.

8.1 Structure and shape

Carbanion	Carbon hybridisation	Shape	Where the lone pair sits	Consequence
CH_3^-, R_3C^- (saturated)	sp^3	pyramidal (like NH_3), ≈ 107°	sp^3 orbital (25 % s)	inversion barrier very low, so a chiral carbanion racemises rapidly; the carbanion is a strong base
Vinyl, phenyl, cyclopropyl	sp^2	lone pair in an sp^2 orbital in the plane (vinyl is bent/configurationally stable)	33 % s; *not* conjugated with the π system of the ring	more stable than sp^3 carbanions by s-character; **no resonance** in phenyl anion (the lone pair is orthogonal to the π cloud)
Acetylide R–C ≡ C^-	sp	linear	50 % s	the most stable simple carbanion; pK_a (C–H) ≈ 25
Allyl, benzyl, enolate, nitronate, cyclopentadienide	sp^2, planar	the carbanion carbon is planar; lone pair in a p orbital conjugated with π bonds	p orbital (0 % s)	delocalisation into the π system (resonance) is the main stabilising effect

Stereochemistry. An sp^3 carbanion inverts through a planar transition state in a few kcal mol^{-1}, so an optically active centre at a carbanion carbon is lost (racemisation by enolisation of an α-chiral ketone in base is the standard example, 8.5). Inversion is hindered, and the carbanion is configurationally stable, when it is held in a ring (cyclopropyl; bridgehead) or when the lone pair is bound tightly to a metal (RLi at low temperature). A bridgehead carbanion cannot become planar, so it is destabilised by exactly the geometric factor that destabilises a bridgehead carbocation.

8.2 Generation of carbanions

Method	Equation	Comment
Deprotonation of C–H by a base	$R_3C–H + B^- \rightleftharpoons R_3C^- + BH$	the base must be stronger than R_3C^-: compare pK_a(C–H) and pK_a(BH). Enolates (HO^-, RO^-, LDA), acetylides ($NaNH_2$), cyclopentadienide, malonate
Metal insertion into C–X	R–X + 2 Li → R–Li + LiX; R–X + Mg → R–MgX	RMgX and RLi are the most important sources of 'R⁻' (Grignard 1900)
Halogen–metal exchange	R–Br + n-BuLi → R–Li + n-BuBr	gives vinyl-, aryl-lithiums
Decarboxylation	$R–COO^- \rightarrow R^- + CO_2$	soda lime (NaOH/CaO, Δ); easy when R^- is stabilised (β-ketoacids, malonic acids, trichloroacetate, nitroacetate)
Conjugate (Michael) addition	$Nu^- + C=C–C=O \rightarrow Nu–C–C^-=C–O^-$ (enolate)	creates an enolate at the α-carbon
Nucleophilic addition to C=O, C ≡ N	$R^- + R'_2C=O \rightarrow R'_2C(R)–O^-$	the new anion is an alkoxide; carbon-centred carbanions form from CN^- + aldehyde (cyanohydrin anion)
α-Elimination	$HCCl_3 + HO^- \rightarrow {}^-CCl_3 + H_2O \rightarrow :CCl_2 + Cl^-$	carbanion then loses X^- to give a carbene (Part 10)
Electron transfer to a π system	arene + Na → $arene^{\bullet -}$ → carbanion on protonation (Birch)	see the radical-ion section
Fragmentation	$R–C(=O)–CR'_3 \rightarrow$ ... (retro-Claisen, haloform cleavage of CCl_3–C=O)	CCl_3^- is a good leaving group from the tetrahedral intermediate of the haloform reaction

8.3 Acidity of C–H bonds and the stability of the conjugate carbanion

The stability of R^- and the acidity of R–H are the same fact: the more stable the anion, the stronger the acid R–H. Approximate pK_a values (recalled; aqueous or DMSO scale as customary for each class, so only the *order* should be used in an exam) are:

C–H acid	≈ pK_a	Why
CH_4, CH_3CH_3	≈ 50	sp^3 carbanion, no stabilisation
CH_2=CH–H	≈ 44	sp^2: 33 % s
C_6H_6 (aryl C–H)	≈ 43	sp^2, lone pair in the plane, no resonance
CH_2=CH–CH_3 (allylic)	≈ 43	allyl anion (two resonance forms)
$C_6H_5CH_3$	≈ 41	benzyl anion
Ph_2CH_2, Ph_3CH	≈ 32, 31	two and three phenyls delocalise the charge
NH_3	≈ 38 (N–H)	for comparison

HC ≡ CH, RC ≡ CH	≈ 25	sp carbon (50 % s)
$CHCl_3$	≈ 24	–I of 3 Cl, polarisability of Cl
CH_3CO_2Et (α-H)	≈ 25	ester enolate
CH_3COCH_3 (α-H)	≈ 19–20	enolate (charge on O)
fluorene (C-9 H)	≈ 22	cyclopentadienide-type aromatic anion fused to two benzenes
cyclopentadiene	≈ 16	aromatic $C_5H_5^-$ (6π)
$CH_2(COOEt)_2$ (malonate)	≈ 13	two carbonyl groups
CH_3NO_2	≈ 10	nitronate: charge on two oxygens
$CH_3COCH_2COCH_3$	≈ 9	two C=O, planar W-shaped anion

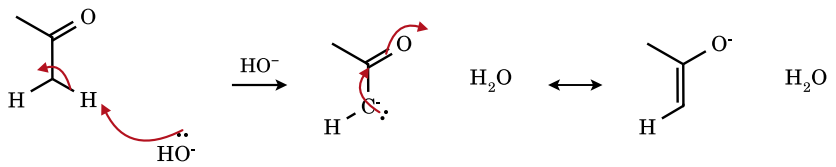
8.4 Factors that stabilise a carbanion

Factor	Effect on carbanion stability	Standard orders
Hybridisation (s-character)	more s-character holds the lone pair closer to the nucleus	$HC\equiv C^- > CH_2{=}CH^- \approx C_6H_5^- > CH_3CH_2^-$ ($sp > sp^2 > sp^3$)
Inductive effect	–I groups stabilise, +I groups destabilise; alkyl groups are +I, so alkyl-substituted carbanions are less stable	$CH_3^- > 1° > 2° > 3°$ (in solution; in the gas phase the order is reversed by polarisability)
Resonance (–M groups)	delocalisation of the lone pair into C=O, C ≡ N, NO_2, aryl	$^-CH_2NO_2 > {}^-CH_2CHO > {}^-CH_2COR > {}^-CH_2COOR > {}^-CH_2CN > PhCH_2^-$ (typical)
Aromaticity	aromatic anions are far more stable; antiaromatic ones far less	$C_5H_5^-$ (6π, aromatic) ≫ allyl; cyclopropenyl anion and cycloheptatrienyl anion (4nπ, antiaromatic) are very unstable
Multiple resonance forms	more delocalisation = more stable	$Ph_3C^- > Ph_2CH^- > PhCH_2^-$
α-Heteroatom	S, P, Si stabilise (polarisability, negative hyperconjugation); O and N destabilise (lone-pair/lone-pair repulsion)	$PhS{-}CH_2^- > CH_3O{-}CH_2^-$; $^-CCl_3 > {}^-CF_3$ (JEE key)
Steric inhibition of resonance	a group twisted out of the plane cannot delocalise the charge	planar 2-nitropropane anion vs a twisted nitro-aryl anion
Solvent and counter-ion	polar aprotic solvents (DMSO, DMF, HMPA) leave the anion free and reactive; protic solvents H-bond to it	relative acidities of C–H acids are measured in DMSO to avoid H-bonding

JEE pitfalls. (i) The phenyl and vinyl carbanions get *no* resonance stabilisation (lone pair in the plane); their stabilisation is s-character only. (ii) Carbanion stability and carbocation stability follow *opposite* substituent rules: alkyl groups stabilise a cation and destabilise an anion; π-acceptors (NO_2, C=O) stabilise an anion and destabilise a cation; π-donors (OMe, NR_2) stabilise a cation and destabilise an anion. (iii) Basicity is the inverse of stability: the least stable carbanion is the strongest base. (iv) The comparison $^-CCl_3$ vs $^-CF_3$ is a trap: F is more electronegative, but its lone pairs repel the carbanion lone pair and F is a poor polarisable group, so the JEE key puts CCl_3^- first.

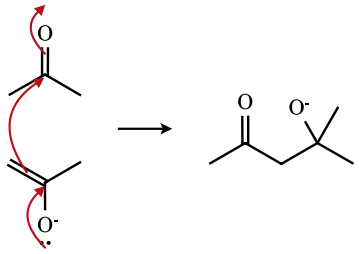
8.5 Mechanisms in which a carbanion is the key intermediate (arrow-validated)

8.1 Acetone + HO^-: the enolate ion (carbanion with resonance)



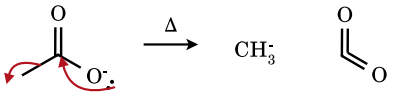
Hydroxide removes the α-proton; the C–H electron pair becomes a lone pair on carbon (carbanion). The lone pair is conjugated with C=O and flows onto oxygen: the enolate is a hybrid of the two forms, with most of the negative charge on oxygen (that is why the α-H of acetone, $pK_a \approx 20$, is 10^{30} times more acidic than that of ethane). The enolate is *ambident*: it reacts at C (alkylation, aldol) or at O (acylation, silylation).

8.2 Aldol addition: enolate + acetone



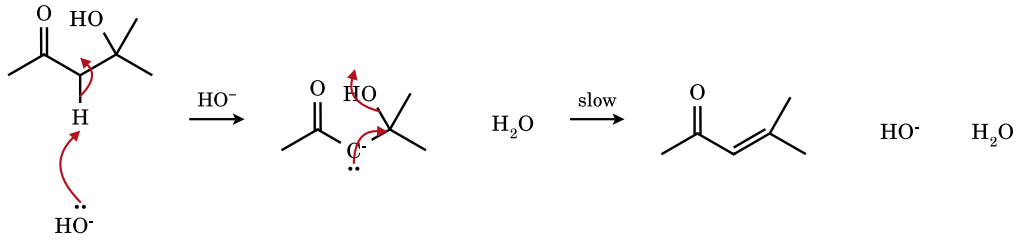
The enolate carbon attacks the carbonyl carbon of a second molecule; the C=O π pair becomes an alkoxide lone pair. Protonation gives 4-hydroxy-4-methylpentan-2-one (diacetone alcohol); this carbanion route builds C–C bonds.

8.5 Soda-lime decarboxylation: sodium acetate → methane



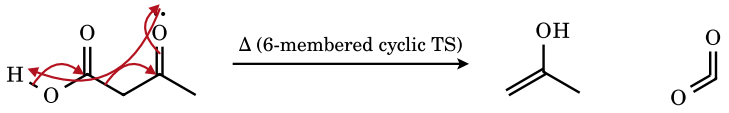
The carboxylate expels CO_2; the C–C σ pair becomes the lone pair of the methyl carbanion, which is protonated by water (or the lattice OH) at once to give CH_4. The reaction is limited to cases where the carbanion is at least somewhat stabilised (or the temperature is very high, as with soda lime).

8.3 E1cB dehydration of an aldol: a carbanion that expels HO^-



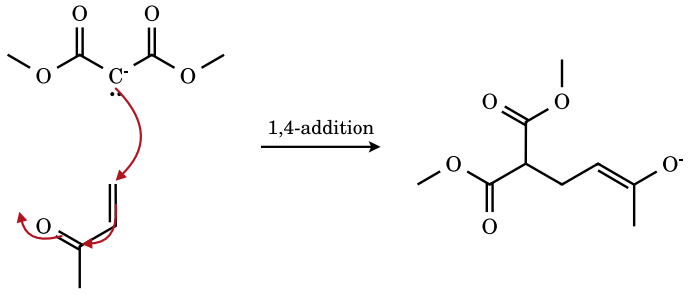
E1cB = *E*limination, *1* molecular, conjugate *B*ase. The α-proton is removed first (carbanion), then the β-leaving group departs in the slow step. This is why aldol condensations proceed by heating even though HO^- is a poor leaving group: the carbanion pushes it out. The product mesityl oxide is conjugated (resonance energy).

8.4 β-Keto acid decarboxylation: a carbanion/enol formed with CO_2 loss



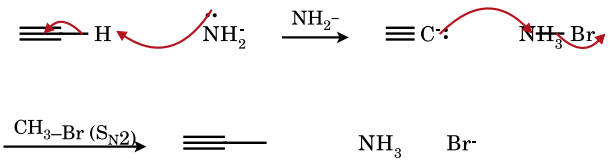
Six electrons move around a six-membered cyclic transition state; the carbanion character appears at C-4 and is immediately delocalised onto the ketone oxygen as the enol, which tautomerises to acetone. This is why malonic acid, acetoacetic acid and β,γ-unsaturated acids lose CO_2 on gentle heating whereas simple acids need soda lime.

8.6 Michael addition of the malonate carbanion to methyl vinyl ketone



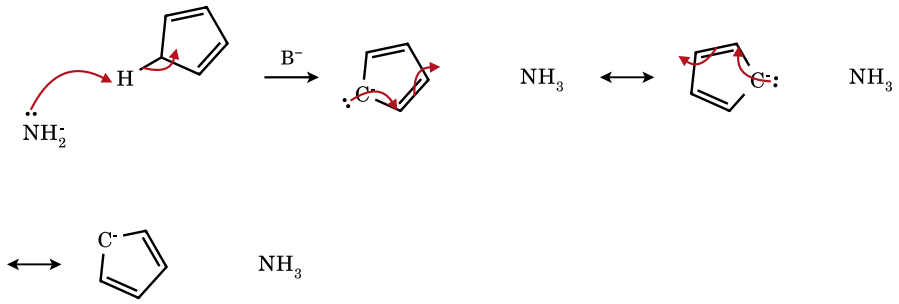
A soft carbanion attacks the β-carbon of the enone (conjugate addition, 1,4) rather than the carbonyl carbon (1,2); the new carbanion is an enolate and is protonated to the 1,5-dicarbonyl. The stronger and harder the nucleophile (RLi), the more 1,2-addition.

8.7 Acetylide formation and alkylation



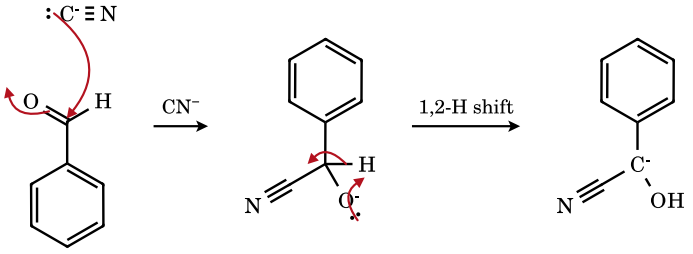
Sodamide (pK_a of NH_3 ≈ 38) is strong enough to deprotonate a terminal alkyne (pK_a ≈ 25) because the sp carbanion is stable; hydroxide is not strong enough (pK_a of H_2O ≈ 16). The acetylide then does an S_N2 on a primary halide. Secondary and tertiary halides give elimination instead because the acetylide is a strong base.

8.8 Cyclopentadiene → cyclopentadienide (aromatic carbanion) and its resonance



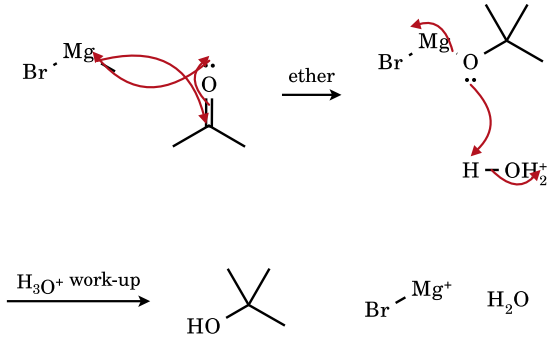
The CH_2 group of cyclopentadiene ($pK_a \approx 16$, a hydrocarbon as acidic as water) loses H^+ to give a planar $C_5H_5^-$ with six π electrons (4n+2): all five carbons are equivalent and each carries one-fifth of the charge. This is the carbanion analogue of tropylium (6π) and the reason ferrocene exists.

8.9 Cyanohydrin carbanion: the acyl-anion equivalent of the benzoin condensation



Cyanide adds to the aldehyde to give an alkoxide; a proton transfer then puts the negative charge on the benzylic carbon bearing CN, where it is stabilised by the nitrile (–M) and the phenyl ring. This carbanion is the 'masked acyl anion' that attacks a second PhCHO (umpolung of the carbonyl carbon), and the loss of cyanide gives benzoin.

8.10 Grignard reagent + acetone → tertiary alcohol (after work-up)



The C–Mg σ pair is the nucleophile (a carbanion in disguise). It adds to the C=O carbon, and the alkoxide oxygen binds magnesium. Aqueous acid protonates the alkoxide and removes Mg^{2+}. Grignard reagents react with CO_2 (→ RCOOH), aldehydes (→ 2° alcohols), HCHO (→ 1° alcohols), epoxides, nitriles (→ ketones after hydrolysis) and esters (→ 3° alcohols, two equivalents). They are destroyed by any acid with a pK_a below that of the C–H bond of R–H (HOH, ROH, RCOOH, RNH_2, R–C ≡ C–H): this is the basis of the Zerewitinoff determination of active hydrogen by CH_4 evolution from CH_3MgI.

8.6 Kinetic and thermodynamic enolates; why optical activity is lost

- **Racemisation by enolisation.** A ketone with an α-stereocentre that carries an H (for example (R)-2-methyl-1-phenylbutan-1-one) racemises in dilute base or acid, because the enol or enolate is planar at that carbon: re-protonation from either face gives both enantiomers. Deuterium from D_2O is exchanged into the α-positions at the same rate ($CH_3COCH_3 \rightarrow CD_3COCD_3$): the proof of a carbanion/enol intermediate.
- **Regioselectivity in unsymmetrical ketones** (2-methylcyclohexanone): LDA, THF, –78 °C removes the more accessible, less substituted α-H irreversibly (*kinetic* enolate, more s-character in the transition state, faster); NaH or NaOEt/EtOH at room temperature equilibrates to the more substituted, more stable enolate (*thermodynamic*). Alkylation with CH_3I then gives 2,6-dimethyl- (kinetic) or 2,2-dimethylcyclohexanone (thermodynamic).
- **Which α-H is most acidic** in $CH_3COCH_2COCH_3$? The central CH_2 ($pK_a \approx 9$): its anion is stabilised by *both* carbonyls, whereas deprotonation at CH_3 is stabilised by one.
- **Intramolecular H-bond and chelation** stabilise the enol form of β-diketones (acetylacetone is ≈ 80 % enol in hexane) and the enolate (planar, W-shaped, O···M^+···O).

8.7 Reactions of carbanions (summary)

Reaction type	Example	Product
Protonation (acid–base)	RMgX + H_2O	R–H + Mg(OH)X
Nucleophilic addition to C=O	RMgX/RLi + RCHO, RCOR, CO_2, HCHO	alcohols, carboxylic acids
Nucleophilic substitution (S_N2)	R′C ≡ C^- + R–X; enolate + R–X	alkyne, α-alkylated carbonyl
Conjugate addition	malonate + enone; R_2CuLi + enone	1,5-dicarbonyl; β-alkyl ketone
Addition–elimination (acyl)	Claisen: enolate + RCOOR′ → β-ketoester	C–C bond, loss of RO^-
Elimination (E1cB)	aldol condensation; β-hydroxy carbonyl dehydration	α,β-unsaturated carbonyl
Wurtz coupling	2 RX + 2 Na → R–R + 2 NaX (via R–Na and S_N2 on a second RX)	alkane (even carbon count)
Decarboxylation	β-keto acid, malonic acid, soda lime	ketone, acetic acid, alkane
α-Elimination → carbene	$CHCl_3$ + KOH; Reimer–Tiemann	dichlorocarbene (Part 10)
Dimerisation by electron transfer	2 $Ph_3C^-Na^+$ (trityl sodium)	stable coloured solutions

8.9 The carbon–metal bond, ion pairs and aggregates: what ‘R^-’ really is in the flask

A free carbanion is almost never the reacting species. Organometallic ‘carbanions’ are polar covalent compounds, and their reactivity depends on how ionic the C–M bond is and on how the metal is solvated and aggregated. The Pauling estimate of ionic character, $1 - \exp(-\tfrac{1}{4}\,\Delta\chi^2)$, with χ(C) = 2.55, shows the trend (values are indicative; real bonds are never fully ionic):

Bond C–M	χ(M)	$\Delta\chi$	≈ % ionic	Reagent behaviour
C–K	0.82	1.73	53	essentially ionic, very reactive (benzylpotassium, KH-generated enolates); insoluble in hydrocarbons
C–Na	0.93	1.62	48	ionic (trityl-sodium, sodium acetylide, Wurtz intermediates); salts
C–Li	0.98	1.57	46	polar covalent, aggregated, soluble in ether/hydrocarbon; RLi are strong bases and nucleophiles
C–Mg	1.31	1.24	32	Grignard reagents: polar covalent; react with C=O, epoxides, CO_2; mild bases
C–Al	1.61	0.94	20	R_3Al: electron-deficient, Lewis acidic, bridged dimers
C–Zn	1.65	0.90	18	R_2Zn, RZnX (Reformatsky): tolerate esters; weak nucleophiles
C–Cu	1.90	0.65	10	R_2CuLi (Gilman): soft nucleophile, 1,4-addition, S_N2 on RX
C–Hg	2.00	0.55	7	R_2Hg: covalent, air-stable, unreactive toward C=O
C–B / C–Si / C–Sn	2.04 / 1.90 / 1.96	0.5–0.65	6–10	covalent; organoboranes and silanes act as carbanion equivalents only after activation (ate complexes, fluoride)

- **Aggregation.** Alkyllithiums exist as clusters held by multicentre C–Li–C bonds: MeLi is a tetramer (Li_4 tetrahedron with a CH_3 over each face) in ether and THF; n-BuLi is a *hexamer* in hydrocarbons, a tetramer in diethyl ether, and tetramer ⇌ dimer in THF; t-BuLi is a tetramer in hydrocarbons and a dimer/monomer in THF; PhLi is a tetramer in ether and a dimer/monomer in THF (values approximate). Strong donors (THF, TMEDA, HMPA, DMPU) break the aggregates into smaller, more reactive species: n-BuLi in hexane deprotonates slowly but n-BuLi/TMEDA deprotonates benzene.
- **Grignard reagents** are governed by the **Schlenk equilibrium**, 2 RMgX ⇌ R_2Mg + MgX_2, with Mg tetrahedrally coordinated by two ether molecules ($RMgX{\cdot}2Et_2O$). In ether the equilibrium lies on the RMgX side for X = Br, I; dioxane precipitates MgX_2·dioxane and drives it to R_2Mg. Formation from RX + Mg proceeds on the metal surface by single-electron transfer (a radical-pair step), which is why Wurtz coupling R–R and radical rearrangements (cyclopropylcarbinyl → homoallyl) are by-products.
- **Ion pairs.** In ethers an enolate or a delocalised carbanion exists as *contact* ion pairs (cation near the carbanion), *solvent-separated* ion pairs, or free ions, in order of increasing dissociation. Better cation solvation ($Li^+ < Na^+ < K^+$ with crown ethers, HMPA, DMSO) gives a ‘naked’, more reactive, less selective anion; this alters rates by factors of 10^3–10^6 and flips C- versus O-alkylation (8.11).
- **Base strength is not the same as nucleophilicity at carbon.** t-BuLi is the stronger base, but n-BuLi adds to C=O and to alkenes more readily because t-BuLi is hindered. LDA is a strong, hindered, *non-nucleophilic* base (pK_a of diisopropylamine ≈ 36).

8.10 Quantitative acidity of C–H: scales, equilibria, and the subtle cases

Two scales are in use. The **aqueous** scale (pK_a in water, defined below ≈ 14 by the solvent) and the **DMSO** scale of F. G. Bordwell (up to ≈ 35, free of H-bonding to the anion). The orders agree, but numerical values differ by 2–10 units, so a question that gives numbers from one scale must not be mixed with the other. The values below are recalled from the literature and should be used for *ranking*, not as exact data.

Acid HA	pK_a (DMSO)	pK_a (water)	Remark
Meldrum's acid	7.3	4.8	anomalously strong C–H acid: the rigid ring holds the two C=O in the Z (s-cis) arrangement, so both are conjugated and the anion is planar; compare dimedone
dimedone	11.2	5.2	cyclic 1,3-diketone (enolises)
$CH_2(CN)_2$	11.1	11.2	two nitriles
acetylacetone	13.3	8.9	enol-H-bonded, mostly enol (water: more acidic since the enol is the form measured)
ethyl acetoacetate	14.2	10.7	β-ketoester
CH_3NO_2	17.2	10.2	nitronate; the 'nitroalkane anomaly' (see below)
$CH_2(CO_2Et)_2$	16.4	13	malonic ester
cyclopentadiene	18.0	16	aromatic anion
indene	20.1	—	benzo-fused; charge partly on the benzene ring
$PhCH_2CN$	21.9	—	benzylic + nitrile
fluorene	22.6	—	biphenyl-bridged cyclopentadienide
acetophenone (CH_3)	24.7	≈ 18	ketone enolate
acetone	26.5	19.3	ketone enolate
PhC ≡ CH	28.8	—	sp C–H
$PhSO_2CH_3$	29.0	—	α-sulfonyl carbanion
EtOAc (α-H)	29.5	25.6	ester enolate
CH_3CN	31.3	≈ 25	nitrile anion (linear, ketenimine-like)
Ph_3CH	30.6	—	trityl anion (propeller, three rings)
Ph_2CH_2	32.2	—	
$CH_3CON(CH_3)_2$	34	≈ 30	amide enolate (weakest carbonyl C–H acid)
DMSO	35.1	—	dimsyl anion
$PhCH_3$	≈ 43	—	benzyl anion
for comparison: H_2O, MeOH, NH_3, Et_2NH	31.4, 29.0, 41, ≈ 44	15.7, 15.5, ≈ 36, ≈ 38	pK_a of the conjugate acid of the base used

- **Reading an equilibrium.** For HA + B^- ⇌ A^- + HB, log K = pK_a(HB) − pK_a(HA). Example (aqueous): acetone (19.3) + EtO^- (EtOH 15.9) → K = $10^{-3.4}$ ≈ 4×10^{-4}, so only 0.04 % of acetone is an enolate in 1 M ethoxide, yet this tiny concentration is enough for the aldol to run; with LDA (pK_a 36), K = $10^{+9.5}$ and deprotonation is quantitative. For the same reason ethyl acetoacetate (10.7) is converted completely to its anion by NaOEt, and this is what drives the Claisen condensation (8.12).
- **Acid strength of α-C–H in carbonyl compounds** falls in the order aldehyde (≈ 17) > ketone (≈ 19–20) > ester (≈ 25) > nitrile (≈ 25) ≈ amide (≈ 30) because the carbonyl's ability to accept the charge (electronegativity of the acyl group, competition from the OR or NR_2 lone pair) falls in the same order. In 1,3-dicarbonyls the second acceptor adds ≈ 10 pK units. A 1,3,5-tricarbonyl or a carbon flanked by three EWG (e.g. triacetylmethane, pK_a ≈ 6; tricyanomethane, ≈ −5) is a strong acid.
- **Kinetic versus thermodynamic acidity.** The thermodynamic acidity measures the stability of the anion; the kinetic acidity measures the rate of removal of H^+. The two can differ when the anion needs reorganisation. The **nitroalkane anomaly**: nitromethane is a thermodynamically stronger acid than acetone by ≈ 9 pK units, but its H is removed ≈ 10^5-fold *more slowly* than that of acetone relative to the Brønsted line, because in the transition state the carbon has rehybridised and charge delocalisation onto the nitro oxygens lags behind proton transfer. Rates of deprotonation of nitroalkanes are therefore slow, and they give the *aci*-form (nitronic acid) first.
- **Hybridisation versus electronegativity.** The s-character argument orders HC ≡ C–H (25), H_2C=CH–H (44), CH_3CH_2–H (50) and even cyclopropane C–H (≈ 46: exocyclic C–H bonds of cyclopropane use ≈ 33 % s orbitals). The same argument explains why HCN (pK_a 9) is acidic and why alkynyl anions persist.
- **Substituent effects.** Electron-withdrawing groups on an aryl ring acidify benzylic C–H of $ArCH_3$ and $ArCH_2CN$; the effect follows σ^- (resonance-enhanced) values: p-NO_2 shifts pK_a of toluene-type C–H by ≈ 10

pK units (4-nitrotoluene ≈ 20 in DMSO versus 43). A twisted nitro group (steric inhibition of resonance) loses most of this: 2,6-dimethyl-4-nitro… series; the ortho effect operates in the same way as for carbocations (Part 4.8), but with opposite sign.

- **Solvent effects.** Protic solvents H-bond to localised, small anions (alkoxides, enolate O) and flatten acidity differences; DMSO and other dipolar aprotic solvents do not, and expose the intrinsic order. Alcohols are 10^4–10^8 times *more* acidic relative to water in DMSO than ketones (see H_2O 31.4 versus acetone 26.5 in DMSO, the reverse of the aqueous order 15.7 < 19.3).

8.11 Enolates: formation, regiochemistry, ambident reactivity

An **enolate** is the conjugate base of a carbonyl compound deprotonated at the α-carbon. It is a resonance hybrid of the carbanion (C^-–C=O) and the oxy-anion (C=C–O^-); the O-form carries most of the charge. Enolates are *ambident nucleophiles* with two reactive sites, O and C.

Factor	Favours C-attack (softer site)	Favours O-attack (harder site)
Electrophile (HSAB)	soft: alkyl iodides/bromides (R–I > R–Br), allylic and benzylic halides, α-halo carbonyls, Michael acceptors, aldehydes (aldol)	hard: R_3SiCl, acyl chlorides and anhydrides (acylation), ROTs, $ROSO_2OR$, R_3O^+ (Meerwein salts)
Leaving group on R–X	I^-, Br^- (soft)	OTs, OSO_2Me, Cl (hard)
Solvent	protic solvents (H-bond the O), non-polar solvents, THF	dipolar aprotic: HMPA, DMSO, DMF (free the O^-)
Counter-ion	Li^+ (tight, covalent O–Li, shields O)	K^+, Cs^+, R_4N^+, crown-ether-complexed K^+ (naked O)
Thermodynamics	C-alkylation gives a C–C bond: the stable (thermodynamic) product	O-alkylation is faster for hard electrophiles (kinetic) and reversible with allyl systems

Regioselectivity: kinetic and thermodynamic enolates

	Kinetic enolate	Thermodynamic enolate
Conditions	LDA (1.0 equiv), THF, −78 °C, ketone added slowly; quench with electrophile (or Me_3SiCl) at −78 °C; strongly basic, irreversible, no excess ketone, no proton source	weaker base in slight deficiency (NaH, KH, NaOEt/EtOH, Et_3N/Me_3SiCl), warm, longer times, excess ketone: reversible proton exchange equilibrates the enolates
Which H is removed	the more *accessible* and more acidic (less substituted) α-H: in 2-methylcyclohexanone, C-6 (CH_2) rather than C-2 (CH)	the one giving the more substituted, more stable (conjugated or more highly substituted) C=C
Product from 2-methylcyclohexanone + MeI	2,6-dimethylcyclohexanone	2,2-dimethylcyclohexanone
Why	bulky LDA removes the less hindered H by a cyclic six-membered Ireland TS faster	the more substituted enolate is lower in energy (alkene hyperconjugation), no kinetic barrier when equilibrating

- **Enolate geometry.** With LDA in THF, ketones with a bulky α′-group give the *E*-enolate (O and the alkyl group trans), esters the *E*-enolate (OLi and Me trans by Ireland's model), and adding HMPA gives the *Z*. The geometry transfers into the stereochemistry of the aldol product via the Zimmerman–Traxler chair (below).
- **Zimmerman–Traxler model.** A metal enolate adds to an aldehyde through a six-membered chair in which M coordinates both oxygens and the aldehyde R sits equatorial: **Z-enolates → syn aldols; E-enolates → anti aldols**. Boron enolates (shorter B–O) show higher selectivity. This is the basis of diastereoselective aldol reactions.
- **Silyl enol ethers** (Me_3SiCl/Et_3N for thermodynamic, LDA/Me_3SiCl for kinetic) are isolable enolate equivalents; they react with electrophiles under Lewis acid catalysis (Mukaiyama aldol, $TiCl_4$) and are regenerated to the specific enolate with MeLi or fluoride ($Bu_4N^+F^-$).
- **Enamines (Stork).** A secondary amine (pyrrolidine) condenses with a ketone to give an enamine, an *uncharged* enolate equivalent (N lone pair pushes into C=C). It C-alkylates with reactive halides or acceptors and the iminium is hydrolysed back to the ketone; the substituted ketone is formed without strong base and without polyalkylation. The enamine's nucleophilic carbon is the β-carbon.
- **Dianions.** Acetoacetate ester with 2 equiv of base (NaH then n-BuLi) forms the dianion; the more basic γ-carbanion reacts with electrophile first (γ-alkylation).
- **α-Halogenation (Hell–Volhard–Zelinsky, α-bromination of ketones).** In base, successive halogenation speeds up (the halogen increases acidity of the remaining H): the haloform reaction of CH_3COR with X_2/NaOH; in acid (enol), halogenation stops at mono because the halogen destabilises the protonated carbonyl needed for the next enolisation.

8.12 Carbanion-based carbon–carbon bond-forming reactions (named reactions)

Name	Carbanion (nucleophile)	Electrophile	Product / key feature
Aldol	enolate of aldehyde/ketone (NaOH, EtOH)	aldehyde or ketone	β-hydroxy carbonyl; Δ → E1cB → α,β-unsaturated carbonyl (condensation). Equilibrium; ketone aldols are unfavourable ($K \approx 10^{-2}$) unless dehydration drives it
Crossed aldol / Claisen–Schmidt	enolate of a ketone (or aldehyde with α-H)	aromatic aldehyde (no α-H)	ArCH=CH–COR (chalcones); only one partner can enolise, so a single product
Knoevenagel	active-methylene compound (malonate, $CH_2(CN)_2$), weak base (piperidine, pyridine)	aldehyde/ketone	alkylidene malonate (α,β-unsaturated); Doebner modification: decarboxylates to cinnamic acid
Perkin	carboxylate-derived enolate of acid anhydride $(CH_3CO)_2O$ + CH_3COONa	aromatic aldehyde	cinnamic acid (E); needs 180 °C
Claisen condensation	ester enolate (NaOEt)	another ester	β-ketoester; *driven by deprotonation of the product* (pK_a 11 versus EtOH 16): requires ≥ 1 equiv base and an α-H with two H on the acylating partner
Dieckmann	diester enolate (intramolecular Claisen)	ester in same molecule	5- or 6-membered cyclic β-ketoester (1,6-/1,7-diesters)
Michael (conjugate) addition	stabilised carbanion (malonate, nitronate, enolate, Gilman reagent)	α,β-unsaturated carbonyl/nitrile/nitro	1,5-dicarbonyl; 1,4-addition (soft nucleophile)
Robinson annulation	Michael donor (a ketone enolate)	methyl vinyl ketone	Michael then intramolecular aldol and dehydration: forms a six-membered ring (steroid/terpenoid synthesis; Wieland–Miescher ketone)
Mannich	enol/enolate of ketone	iminium ion $CH_2{=}N^+R_2$ (from HCHO + R_2NH)	β-aminoketone (Mannich base); eliminates to a vinyl ketone
Reformatsky	zinc enolate $BrZn–CH_2COOEt$ (from $BrCH_2COOEt$ + Zn)	aldehyde/ketone	β-hydroxy ester; Zn enolate does not attack ester (unlike Grignard)
Darzens	carbanion of α-halo ester (NaOEt)	aldehyde/ketone	α,β-epoxy ester (glycidic ester): aldol then intramolecular S_N2
Benzoin	cyanohydrin anion (from CN^- + ArCHO)	second ArCHO	α-hydroxy ketone (umpolung)
Stobbe	succinate-ester enolate	ketone	alkylidene succinic half-ester via γ-lactone
Malonic ester synthesis	diethyl malonate + NaOEt	RX (S_N2)	after hydrolysis and Δ: RCH_2COOH (and $R^1R^2CHCOOH$); two carbons added
Acetoacetic ester synthesis	ethyl acetoacetate + NaOEt	RX	after ketonic hydrolysis and decarboxylation: CH_3COCH_2R (methyl ketones)
Haloform	CX_3^- leaving group	—	$RCOCX_3 + HO^- \rightarrow RCOO^- + CHX_3$: methyl ketone → acid with one carbon less + haloform
Favorskii	enolate of α′-position of an α-haloketone	intramolecular C–X (S_N2)	cyclopropanone intermediate, then ring-opening by RO^- to an ester; ring contraction for cyclic α-halo ketones
Wittig	phosphorus ylide $Ph_3P^+–C^-R_2$	aldehyde/ketone	alkene + $Ph_3P{=}O$; non-stabilised ylides → Z, stabilised → E
Corey–Chaykovsky	sulfonium/sulfoxonium ylide ($CH_2{=}SMe_2$, $CH_2{=}S(O)Me_2$)	C=O or C=C–C=O	epoxide (sulfonium) or cyclopropane (sulfoxonium, with enones)
Grignard / RLi additions	RMgX, RLi	C=O, epoxide, CO_2, nitrile, ester	see 8.13
Wurtz, Wurtz–Fittig, Corey–House	R–Na, Ar–Na, R_2CuLi	R′X	C–C coupling; Corey–House couples a Gilman reagent with R′X (also vinyl/aryl halides) cleanly
Kolbe–Schmitt	phenoxide (ring carbanion character at ortho)	CO_2	salicylic acid (ortho, Na); p-hydroxybenzoic acid with K at higher T
Reimer–Tiemann	phenoxide	$:CCl_2$ (from $CHCl_3$)	salicylaldehyde (Part 10)

8.13 Organometallic reagents as carbanion sources: what they do and do not do

Substrate	Product with RMgX / RLi	Notes
HCHO	1° alcohol (R–CH_2OH)	adds one carbon
R′CHO	2° alcohol	—
R′$_2$C=O	3° alcohol	hindered ketones give enolisation (R^- acts as base) and reduction (β-hydride transfer from RMgX with β-H) instead
CO_2 (dry ice)	R–COOH after H_3O^+	ketone by-product via second addition
epoxide	alcohol with two more carbons (R–CH_2CH_2OH from ethylene oxide)	attack at the less hindered carbon (S_N2)
RC ≡ N	ketone after hydrolysis of the imine anion	one equivalent; stops because the imine anion is not electrophilic
RCOOR′	3° alcohol with two identical R groups (two equiv)	ketone intermediate is more reactive than the ester
RCOCl	ketone (with R_2Cd, R_2CuLi, or 1 equiv at −78 °C)	Grignard over-adds to give 3° alcohol
α,β-unsaturated carbonyl	RMgX/RLi: 1,2-addition; R_2CuLi: 1,4-addition	hard versus soft nucleophile
H–A (HOH, ROH, RCOOH, RNH_2, RC ≡ CH)	R–H + metal salt	the Grignard is destroyed; basis of the Zerewitinoff active-hydrogen determination (CH_4 from CH_3MgI)
RX (itself)	R–R (Wurtz-type) as a side reaction	minimised by slow addition and dilute solutions
O_2	ROMgX → ROH	protect from air

8.14 Stereochemistry and reaction pathways of carbanions (Cram's experiments)

Electrophilic substitution at a saturated carbanion centre, R–M + E^+ → R–E + M^+, can proceed with retention, inversion or racemisation. D. J. Cram's base-catalysed H/D-exchange studies on optically active 2-methyl-3-phenylpropionitrile, 2-methyl-2-phenyl..., and related compounds established the pathways:

Pathway	Meaning	Stereochemical result	Typical conditions
S_E1	carbanion forms and is free of the leaving group before reaction	racemisation (planar/inverting carbanion)	polar solvents (DMSO, MeOH/NaOMe), delocalised carbanions
S_E2 (front)	electrophile attacks from the same side as the leaving group (cyclic TS)	retention	organometallics in non-polar solvents; RLi, RMgX (retention is the rule)
S_E2 (back)	attack from the side opposite to M	inversion	rare; strong dipolar systems
S_Ei	internal return (intimate ion pair, H delivered by the conjugate acid within the ion pair)	retention, 'isoinversion' or 'isoracemisation' depending on solvent (Cram: t-BuOH/t-BuOK: retention; DMSO: racemisation)	ion-pair collapse with proton from the solvent

- **Configurational stability** increases for carbanions that cannot become planar or are bound to metal: cyclopropyl and vinyl lithiums, bridgehead anions (in bicyclo[2.2.1]), α-oxy and α-amino lithiums at −78 °C (dipole-stabilised, chelated by the heteroatom). Alkyllithiums such as sec-butyllithium racemise slowly above −40 °C.
- **Hydrogen–deuterium exchange** is the standard proof of a carbanion: a ketone in D_2O/OD^- exchanges its α-H at a rate equal to its racemisation (when chiral) and its halogenation rate in base (zero-order in halogen), showing the same enolate intermediate (Ingold, Hughes).
- **Carbanions and rearrangements.** Aryl groups can migrate to a carbanion centre through a spiro (Meisenheimer-like) intermediate: the Grovenstein–Zimmerman rearrangement, the carbanion analogue of the neophyl rearrangement (Part 9) and the 1,2-aryl shift of cations (Part 7). Alkyl and hydride shifts to carbanion centres do not occur. Other carbanion rearrangements: [1,2]-Wittig (R–O–CH_2R′ + BuLi → alkoxide of R–CH(R′)–OH by a radical-pair path), Stevens ([1,2] shift in ammonium ylides), Brook (C → O silyl migration), Favorskii (8.12) and the benzilic-acid rearrangement (with hydroxide as the nucleophile).

8.15 Ylides: carbanions stabilised by an adjacent onium centre

An **ylide** has a carbanion adjacent to a positively charged heteroatom: Ph_3P^+–C^-HR (phosphonium), Me_2S^+–C^-H_2 (sulfonium), $Me_2S(O)^+$–C^-H_2 (sulfoxonium), R_3N^+–C^-H_2 (ammonium). The carbanion is stabilised by electrostatic attraction and by negative hyperconjugation (n_C → $\sigma^*_{P–C}$, not by 3d overlap), and ylides are far more

stable than simple carbanions: the pK_a of $Ph_3P^+CH_3$ is ≈ 22 (DMSO), so n-BuLi or NaH/DMSO or $NaNH_2$ deprotonates it.

Ylide	Substituent on carbanion	Reactivity	Product with RCHO
non-stabilised (R = H, alkyl)	alkyl	very reactive, air-sensitive; made with BuLi/$NaNH_2$ (salt-free: more Z)	**Z**-alkene, kinetic control, irreversible oxaphosphetane formation
semi-stabilised (R = aryl, vinyl)	Ph, $CH{=}CH_2$	moderate	E/Z mixtures
stabilised (R = CO_2Et, COR, CN)	EWG delocalises the charge; ylide is isolable, weak nucleophile; reacts with aldehydes only (ketones sluggish)	reversible oxaphosphetane formation, thermodynamic control	**E**-alkene
Schlosser modification	—	PhLi at −78 °C interconverts the betaine/oxaphosphetane	**E** even from non-stabilised ylides
Horner–Wadsworth–Emmons (phosphonate carbanion)	$(EtO)_2P(O)–C^-H–CO_2Et$	more nucleophilic than the Wittig ylide; water-soluble phosphate by-product	E-α,β-unsaturated esters

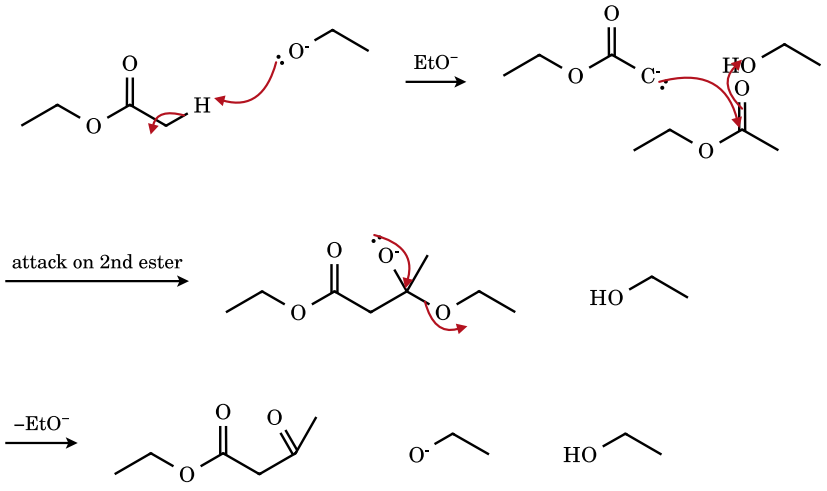
8.16 Choosing a base: which proton is removed first?

A base removes the most acidic proton in the molecule first (thermodynamic acidity, with kinetic modifications above). Compare the pK_a of the conjugate acid of the base with those of each acidic H:

Base	pK_a of conjugate acid (≈ , water or DMSO as marked)	Deprotonates (examples)
$NaHCO_3$, Na_2CO_3	6.4 (H_2CO_3), 10.3 (HCO_3^-)	carboxylic acids, phenols (Na_2CO_3); not alcohols
Et_3N, pyridine, piperidine	10.7, 5.2, 11.1	1,3-dicarbonyls (Knoevenagel), generates enamines
NaOH, KOH	15.7	phenols, 1,3-dicarbonyls, nitroalkanes, aldehydes/ketones (aldol equilibrium)
NaOEt, KOtBu	16–19 (t-BuOH ≈ 18)	ester/ketone α-H (equilibrium), E2 eliminations
NaH, KH	≈ 35 (H_2)	irreversible deprotonation of ketones, alcohols, esters
$NaNH_2$	≈ 38 (NH_3)	terminal alkynes, ketones, esters; benzyne from ArX
LDA, LiHMDS	36, 26 (HMDS)	kinetic enolates (LDA); LiHMDS for sensitive substrates
n-BuLi, s-BuLi, t-BuLi	≈ 50 (butane)	directed ortho-metalation of arenes, vinylic H, terminal alkynes, THF is attacked above −20 °C
Grignard RMgX	≈ 45–50	acidic H (water, alcohol, acid, alkyne) and C=O addition

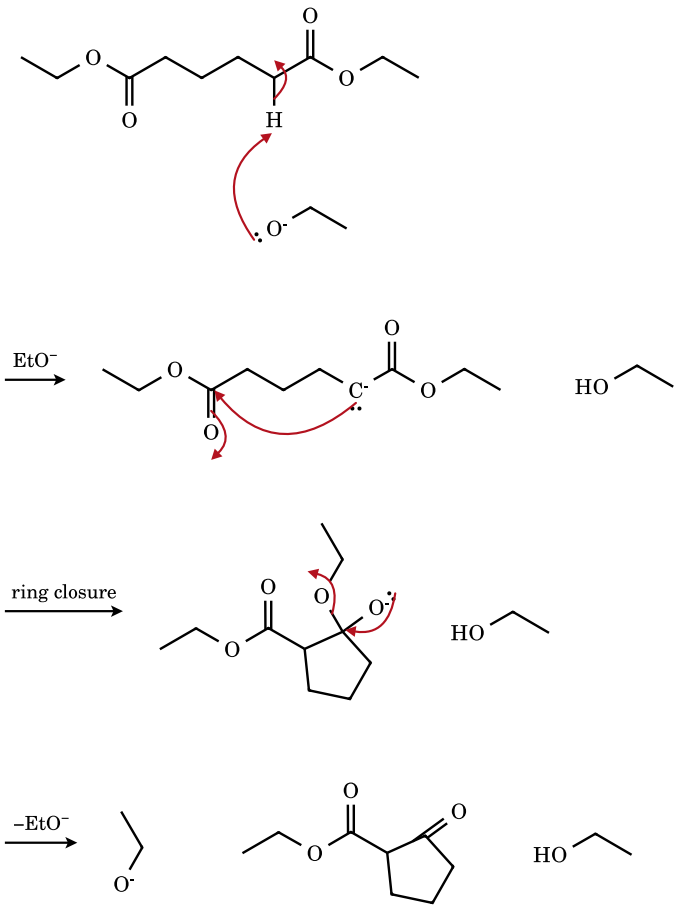
Typical errors in exam problems. (1) Treating the less substituted enolate as ‘always the major one’: it is kinetic only (LDA, −78 °C); NaOEt/EtOH gives the more substituted. (2) Forgetting that the Claisen condensation needs a stoichiometric base because the β-ketoester product is deprotonated and that this is why it fails for esters with only one α-H. (3) Adding a Grignard to a molecule containing OH, COOH, NH, C ≡ C–H or a carbonyl/ester elsewhere without protection. (4) Using the carbocation rule that alkyl groups stabilise: for carbanions they *destabilise*. (5) Ranking CF_3^- above CCl_3^- on electronegativity alone. (6) Assuming conjugate addition for every nucleophile: hard nucleophiles (RLi, RMgX) add 1,2; soft (cuprates, enolates, CN^-, thiolates, amines) add 1,4. (7) Ignoring that an aryl C–H or a vinylic C–H is removed only by bases stronger than NH_3.

8.11 Claisen condensation: ester enolate + ester → β-ketoester



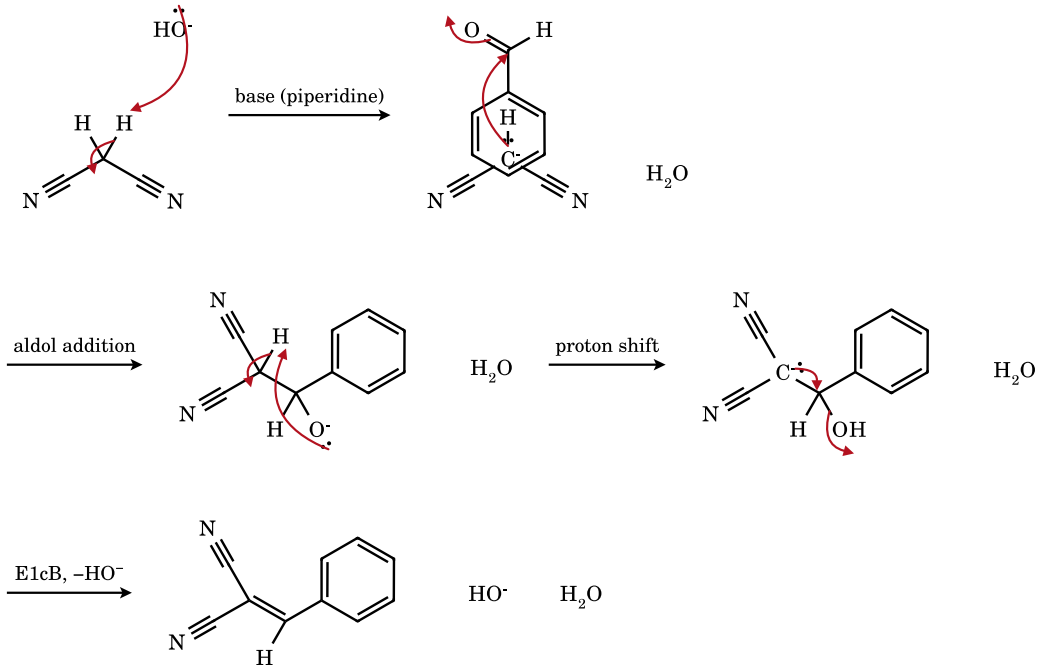
The ester enolate adds to the carbonyl carbon of a second ester molecule to give a tetrahedral alkoxide; reforming C=O expels ethoxide (addition–elimination, as for any acyl substitution). The β-ketoester ($pK_a \approx 11$) is far more acidic than ethanol (16) and is deprotonated to its stabilised enolate by the ethoxide, which pulls the unfavourable equilibrium to the right; acidification at the end gives the neutral β-ketoester. **Therefore the stoichiometric base is a reagent, and an ester with only one α-hydrogen cannot be driven to product in this way** (use a stronger base, e.g. Ph_3CNa).

8.12 Dieckmann cyclisation: diethyl adipate → 2-ethoxycarbonylcyclopentanone



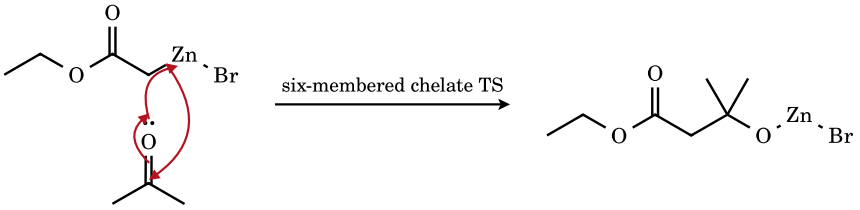
The intramolecular Claisen condensation forms five- and six-membered rings (1,6- and 1,7-diesters give cyclopentanones and cyclohexanones; 1,4- and 1,5-diesters would give strained rings and are not used; the reverse ring-opening is possible). As in the Claisen condensation, deprotonation of the cyclic β-ketoester drives the reaction. Hydrolysis and decarboxylation (8.4) then give cyclopentanone.

8.13 Knoevenagel condensation: malononitrile + benzaldehyde → benzylidenemalononitrile



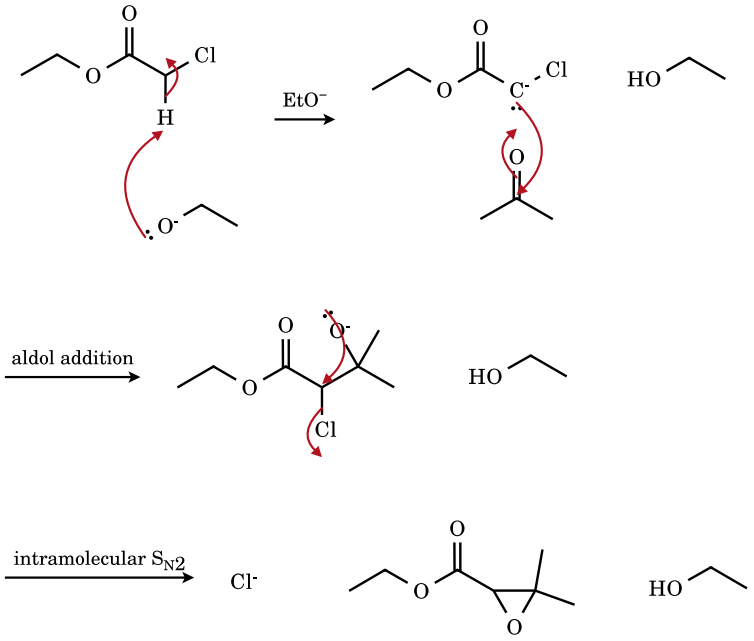
A weak base suffices because the CH_2 is flanked by two nitriles (pK_a 11). Addition to the aldehyde, proton transfer to give a new carbanion (stabilised by two CN and now β to a leaving OH), and loss of hydroxide (E1cB) give the conjugated alkene irreversibly. With malonic acid (Doebner), pyridine as base: decarboxylation follows and cinnamic acids are formed.

8.14 Reformatsky reaction: organozinc enolate + acetone



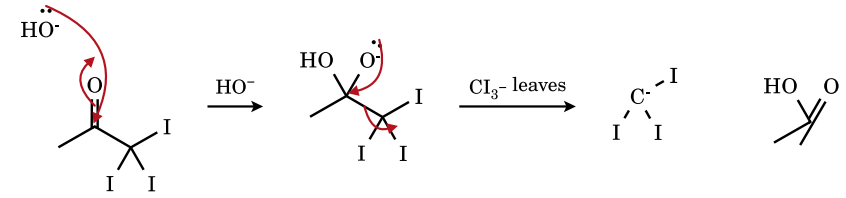
Zinc inserts into the C–Br bond of ethyl bromoacetate to give a carbon-bound (or O-bound) organozinc whose carbanion character is tempered by the C–Zn bond (≈ 18 % ionic), so it adds to aldehydes and ketones but not to esters — the reason a Grignard reagent could not be used. Acid work-up releases the β-hydroxy ester (dehydration gives the α,β-unsaturated ester).

8.15 Darzens glycidic ester condensation



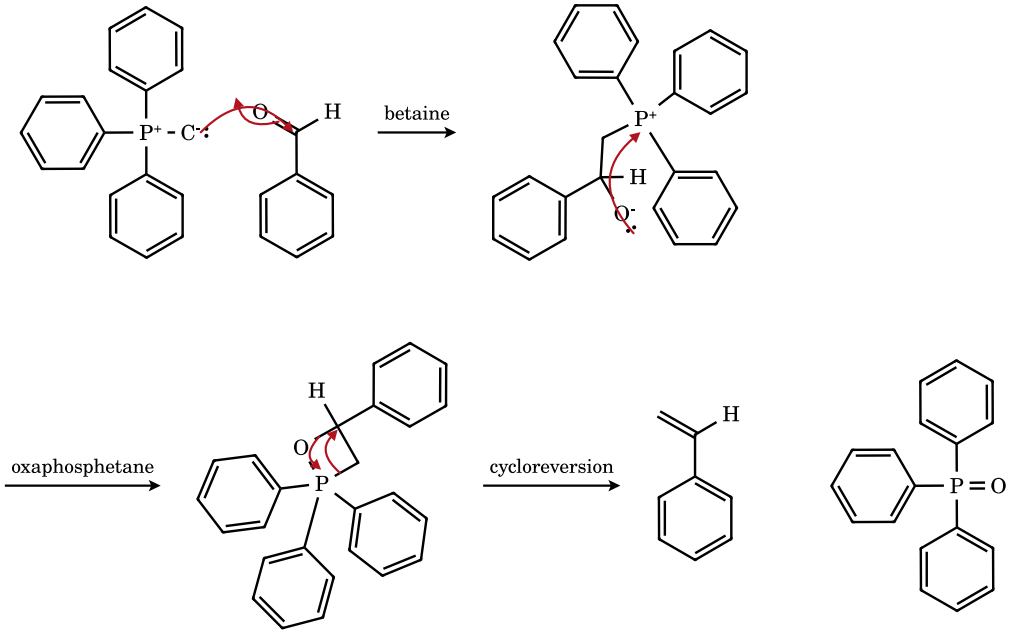
The carbanion of the α-chloro ester adds to the ketone; the resulting alkoxide, positioned on the carbon next to C–Cl, performs an intramolecular S_N2 (3-exo-tet) displacing chloride to give an epoxide. The glycidic ester is hydrolysed and decarboxylated to an aldehyde or ketone with one more carbon than the starting carbonyl.

8.16 Haloform reaction, final step: triiodoacetone + HO⁻



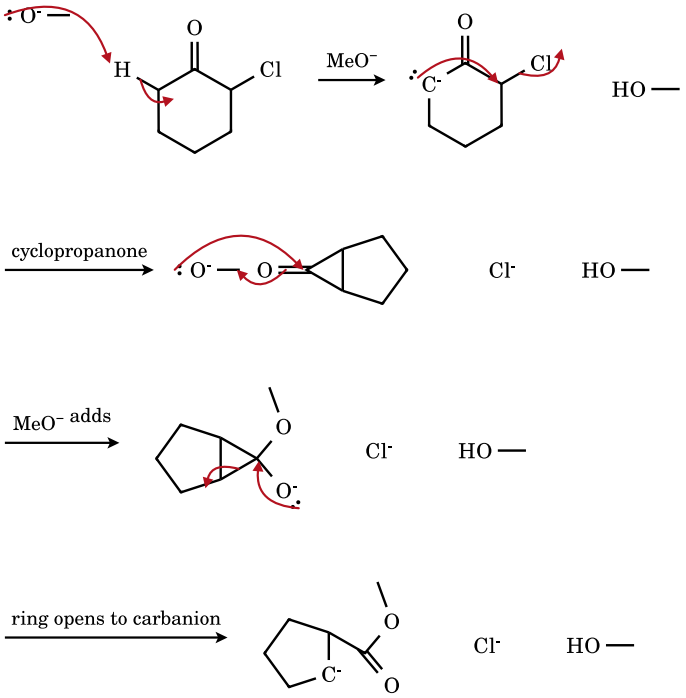
Under basic conditions each successive halogenation is faster than the last, so a methyl ketone is converted to R–CO–CX_3. Hydroxide adds to the carbonyl, and the tetrahedral intermediate expels the best available carbanion, $^-CX_3$ (stabilised by the –I effect and polarisability of three halogens; pK_a of $CHCl_3$ ≈ 24 in the leaving group role), which protonates to the haloform (CHI_3, yellow ppt: the iodoform test for CH_3CO–, $CH_3CH(OH)$–). The carboxylic acid has one carbon fewer than the ketone.

8.17 Wittig reaction: methylenetriphenylphosphorane + benzaldehyde (via oxaphosphetane)



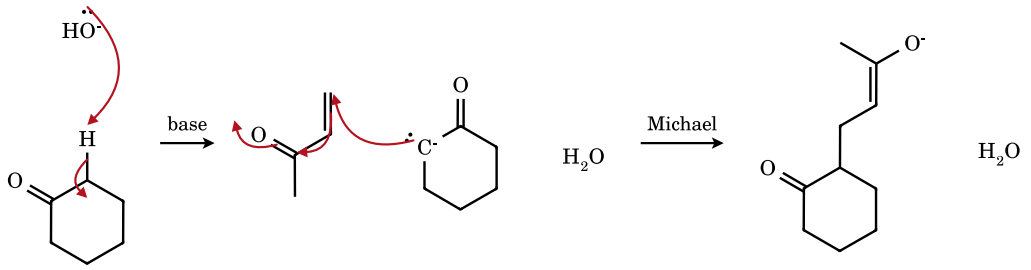
The ylide carbon attacks the aldehyde carbonyl; the alkoxide closes onto phosphorus to give a four-membered oxaphosphetane (modern evidence favours direct [2+2] formation with no discrete betaine for salt-free ylides), which collapses in a retro-[2+2] to the alkene and triphenylphosphine oxide. The strength of the P=O bond (≈ 130 kcal mol^{-1}) is the driving force. Non-stabilised ylides: kinetic *cis* oxaphosphetane → Z-alkene; stabilised: reversible, trans → E.

8.18 Favorskii rearrangement: 2-chlorocyclohexanone + MeO^- → methyl cyclopentanecarboxylate



The base removes the α′-hydrogen (on the carbon *without* chlorine) to give an enolate whose carbon displaces chloride intramolecularly (3-exo-tet) to give a strained bicyclic cyclopropanone. Methoxide adds to the very electrophilic cyclopropanone C=O and the three-membered ring opens at the bond that gives the more stable carbanion (here the secondary carbanion; for unsymmetrical systems the bond that gives the more stabilised, usually more substituted-hindered-less carbanion cleaves). Protonation gives the ester with the ring contracted by one carbon. Evidence: the same cyclopropanone is formed from the isomeric α-chloroketone (labelling scrambles the two carbons).

Robinson annulation: Michael addition → intramolecular aldol → dehydration



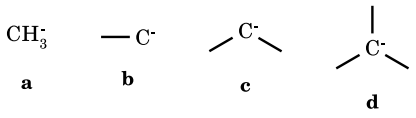
Step 1 of the Robinson annulation: the cyclohexanone enolate adds in a 1,4-fashion to methyl vinyl ketone, giving the enolate of 2-(3-oxobutyl)cyclohexanone (a 1,5-diketone).

Step 2: the methyl ketone enolate (CH_2–C=O, formed by deprotonation of the CH_3) attacks the ring C=O intramolecularly (6-exo-trig): a bicyclic β-hydroxyketone. **Step 3**: E1cB loss of HO^- gives the conjugated enone $\Delta^{1,9}$-2-octalone (4,4a,5,6,7,8-hexahydro-2(3H)-naphthalenone). Net: a six-membered ring and one new C=C are built from a ketone and MVK (use the Wieland–Miescher ketone from 2-methylcyclohexane-1,3-dione in steroid synthesis).

Malonic ester synthesis (arrow-validated alkylation in 8.7 for the alkynyl case): NaOEt converts $CH_2(CO_2Et)_2$ into its carbanion (pK_a 13), which is alkylated by R–X (S_N2: primary or methyl halides; secondary halides give some elimination; aryl and tertiary halides fail); a second alkylation with R′X is possible; hydrolysis (NaOH, then H_3O^+) gives the substituted malonic acid, which decarboxylates on heating (cyclic TS, 8.4) to RCH_2COOH. With 1,2-dibromoethane and two equivalents of malonate the product is cyclopropane-1,1-dicarboxylate (Perkin ring closure). **Acetoacetic ester synthesis** is the same with $CH_3COCH_2CO_2Et$; the final decarboxylation of the β-ketoacid gives methyl ketones CH_3COCH_2R. Dialkylation gives $CH_3COCHRR'$.

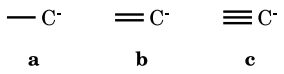
8.20 Problems on carbanions

CA-1.
Arrange in decreasing stability (solution): (a) CH_3^-, (b) $CH_3CH_2^-$, (c) $(CH_3)_2CH^-$, (d) $(CH_3)_3C^-$.



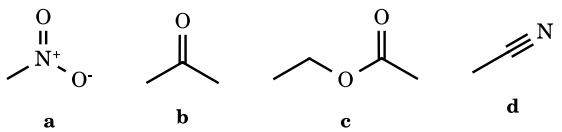
Answer: a > b > c > d
Alkyl groups are +I and push electron density onto the carbon that already holds a lone pair, destabilising it. (In the gas phase polarisability reverses the order; the JEE key uses the inductive argument.)

CA-2.
Arrange in decreasing stability: (a) $CH_3CH_2^-$, (b) $CH_2{=}CH^-$, (c) $HC{\equiv}C^-$.



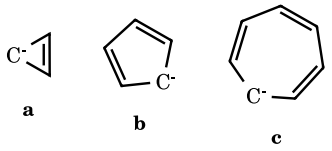
Answer: c > b > a
Stability follows the s-character of the orbital holding the lone pair: sp (50 %) > sp^2 (33 %) > sp^3 (25 %). Equivalently, pK_a of the parent C–H ≈ 25, 44, 50.

CA-3.
Arrange in decreasing acidity of the marked H: (a) CH_3NO_2, (b) CH_3COCH_3, (c) $CH_3COOC_2H_5$, (d) CH_3CN.



Answer: a > b > d > c (JEE-type key; DMSO pK_a: 17.2, 26.5, 31.3, ≈ 30)
The nitronate delocalises the charge onto two oxygens and the N^+ pulls strongly; the ketone enolate has one O; the ester enolate is less stable because the OR group donates into C=O and competes. ⚑ Between nitrile and ester the experimental DMSO values are close (31 vs ≈ 30); keys vary, so check the source order before answering a question that contains both.

CA-4.
Which of the following carbanions is aromatic? (a) cyclopropenyl anion, (b) cyclopentadienyl anion, (c) cycloheptatrienyl anion, (d) cyclooctatetraene dianion.

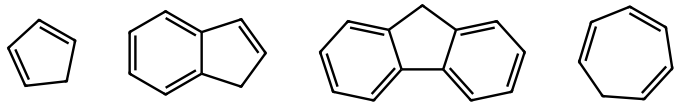


a **b** **c**

Answer: (b) and (d)
Cyclopentadienide has 6π electrons (4n+2, n = 1) in a planar ring; the cyclooctatetraene dianion (10π, planar) is also aromatic. The cyclopropenyl anion (4π) and cycloheptatrienyl anion (8π) are antiaromatic.

CA-5.
Why is the pK_a of cyclopentadiene (≈ 16) about 25 units below that of cyclopentane (≈ 50)?
Answer: aromatic stabilisation of the anion
The conjugate base of cyclopentane is a localised sp^3 carbanion. Removing H^+ from the CH_2 of cyclopentadiene gives a planar $C_5H_5^-$ ring with six π electrons delocalised over five carbons (aromatic): this accounts for most of the difference.

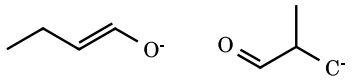
CA-6.
Arrange in decreasing acidity: (a) cyclopentadiene, (b) indene, (c) fluorene, (d) cycloheptatriene.


a **b** **c** **d**

Answer: a > b > c > d (≈ 16, 20, 22.6, 36)
Cyclopentadienide is a simple aromatic anion; in indenide and fluorenide the negative charge is also partly shared with the benzo ring(s), which costs some benzene aromaticity, so they are weaker acids. Cycloheptatriene would give an antiaromatic 8π anion and is the weakest.

CA-7.
Which pair of carbanions is correctly ordered by stability? (A) CCl_3^- > CF_3^- (B) $PhCH_2^-$ > Ph_2CH^- (C) $CH_3OCH_2^-$ > $PhSCH_2^-$ (D) $(CH_3)_3C^-$ > CH_3^-
Answer: (A)
(B) is reversed (more phenyls delocalise more); (C) is reversed (S stabilises, O destabilises by lone-pair repulsion); (D) is reversed (+I destabilises). (A) follows the JEE key (F lone-pair repulsion; Cl polarisability).

CA-8.
A nucleophile attacks the β-carbon of CH_2=CH–CHO (Michael addition). Identify the carbanion intermediate and say why β- rather than α-attack occurs.



β-attack

α-attack

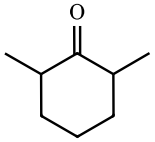
Answer: attack at the β-carbon gives the enolate
β-Attack puts the negative charge on the α-carbon, adjacent to C=O: the enolate (charge shared with oxygen). α-Attack would put the charge on the β-carbon, which is isolated from the carbonyl and unstabilised. The more stable carbanion is formed, as always.

CA-9.
$CHCl_3$ is treated with KOH. What is formed first, and what does it lose?
Answer: $^-CCl_3$ then $:CCl_2$
HO^- removes the single H (pK_a ≈ 24 is acidic enough thanks to –I of Cl and polarisability), giving the trichloromethyl carbanion, which expels Cl^- (α-elimination) to give dichlorocarbene. This is the first step of the Reimer–Tiemann and carbylamine reactions.

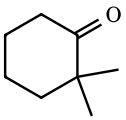
CA-10.
Sodium acetate is heated with soda lime. Name the product and the intermediate.
Answer: methane via CH_3^-
$CH_3COO^- \rightarrow CH_3^- + CO_2$; the unstabilised methyl carbanion is protonated at once. (Contrast the Kolbe electrolysis of the same salt, which goes through $CH_3\bullet$ radicals and gives ethane.)

CA-11.
Optically active 2-phenylbutanal racemises on standing in dilute NaOH. Explain.
Answer: planar enolate (carbanion)
The α-H, which is on the stereocentre, is acidic (adjacent C=O and Ph). Its removal gives a planar enolate in which the stereocentre is destroyed; protonation from either face gives both enantiomers.

CA-12.
2-Methylcyclohexanone is treated (i) with LDA at −78 °C, then CH_3I, and (ii) with NaOEt/EtOH at 25 °C, then CH_3I. Give the major product of each.



(i)



(ii)

Answer: (i) 2,6-dimethylcyclohexanone; (ii) 2,2-dimethylcyclohexanone
LDA is a bulky, strong base: it removes the more accessible H irreversibly (kinetic enolate, less substituted). NaOEt is a weaker base: the enolates equilibrate and the more substituted (more stable, thermodynamic) enolate is alkylated.

CA-13.
How many moles of methane are evolved when 1 mol of $HOCH_2CH_2–C \equiv CH$ reacts with excess CH_3MgI?
Answer: 2
Both the O–H and the acetylenic C–H are acidic enough ($pK_a \approx 16$ and 25, far below that of $CH_4 \approx 50$) to protonate CH_3^-. Counting active hydrogens: 2. (The Zerewitinoff method.)

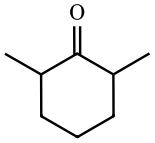
CA-14.
In which of the following will the carbanion that is formed be stabilised by resonance? (a) phenyl anion, (b) benzyl anion, (c) vinyl anion, (d) acetylide, (e) cyclohexadienyl-type enolate.
Answer: (b) and (e)
Only (b) and (e) have the lone pair in a p orbital parallel to a π system. The lone pair of the phenyl and vinyl anions is in an sp^2 orbital in the plane (orthogonal to π), and the acetylide lone pair is in an sp orbital along the axis.

CA-15.
Match the carbanion with its source: (A) $CH_3–$, (B) $^-CCl_3$, (C) enolate, (D) $RC \equiv C^-$; (P) aldol reaction, (Q) Reimer–Tiemann, (R) soda-lime decarboxylation, (S) $NaNH_2$ + alkyne.
Answer: A–R; B–Q; C–P; D–S
Soda lime gives CH_3^- from acetate; $CHCl_3$/KOH gives CCl_3^- and then dichlorocarbene; base + carbonyl gives the enolate for the aldol; $NaNH_2$ deprotonates a terminal alkyne.

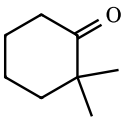
CA-16.
How many of the following can be converted completely into their anion by sodium ethoxide in ethanol? (pK_a of EtOH ≈ 16): acetone (19.3), ethyl acetoacetate (10.7), diethyl malonate (13), nitromethane (10.2), ethyl acetate (25.6), cyclopentadiene (16), phenylacetylene (≈ 29).
Answer: 3 (ethyl acetoacetate, diethyl malonate, nitromethane)
Quantitative deprotonation needs pK_a(HA) at least about 2 units below 16: only acetoacetate, malonate and nitromethane qualify. Cyclopentadiene (16) is half-deprotonated; acetone, ethyl acetate and the alkyne are deprotonated to a negligible extent (but acetone still reacts through the 10^{-4} fraction of enolate).

CA-17.
Using the aqueous pK_a values, calculate the equilibrium constant for $CH_3COCH_3 + EtO^- \rightleftharpoons CH_3COCH_2^- + EtOH$ (pK_a: acetone 19.3, ethanol 15.9).
Answer: $K = 10^{-3.4} \approx 4 \times 10^{-4}$
$\log K = pK_a(EtOH) - pK_a(acetone) = 15.9 - 19.3 = -3.4$. In 1 M EtO^- the enolate fraction is about 0.04 %: too small for stoichiometric enolate chemistry (use LDA) but sufficient for a catalytic aldol.

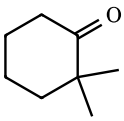
CA-18.
Predict the product of 2-methylcyclohexanone with (a) LDA/THF, −78 °C, then CH_3I, (b) NaOEt/EtOH, 25 °C, then CH_3I, (c) Me_3SiCl/Et_3N, DMF, 130 °C (then MeLi, MeI).



(a)



(b)



(c)

Answer: (a) 2,6-dimethyl; (b) 2,2-dimethyl; (c) 2,2-dimethyl
(a) kinetic enolate at C-6; (b) the equilibrating base gives the more substituted enolate; (c) the thermodynamic silyl enol ether (more substituted) is regenerated to the same enolate by MeLi.

CA-19.

Which is the stronger acid, Meldrum's acid (pK_a 4.8) or dimedone (5.2)? Both have two C=O flanking a CH_2. Explain the difference from acyclic malonic ester (13).

Answer: Meldrum's acid, marginally; both far stronger than diethyl malonate

In diethyl malonate the ester C=O is deactivated by donation from the OR lone pair, and the two esters can adopt the E (s-trans) conformation, which gives poorer overlap and dipole repulsion in the anion. In Meldrum's acid the cyclic 1,3-dioxane-4,6-dione holds both carbonyls in the Z (s-cis) arrangement, forcing a planar anion with full conjugation and no competition between the ring oxygens, hence ≈ 8 pK units more acidic.

CA-20.

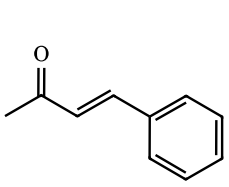
Ethyl acetate is treated with 1 equiv NaOEt, then acid. Name the product. Why does ethyl 2,2-dimethylpropanoate (pivalate) with 1 equiv NaOEt not undergo a Claisen condensation, and why does ethyl isobutyrate fail under the same conditions?

Answer: ethyl acetoacetate; pivalate has no α-H; isobutyrate gives a product without an acidic H

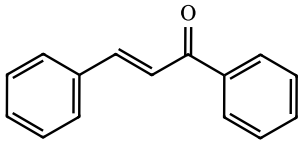
Ethyl pivalate (t-$BuCO_2Et$) has no α-hydrogen, so no enolate. Ethyl isobutyrate gives the condensation product ($(CH_3)_2CH–CO–C(CH_3)_2CO_2Et$), which has no H between the two carbonyls and therefore cannot be deprotonated to drive the equilibrium: the (unfavourable) equilibrium lies on the side of starting materials. A stronger base (Ph_3CNa, LDA) is needed.

CA-21.

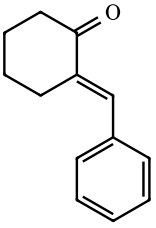
Give the products of the crossed aldol of benzaldehyde with (a) acetone (excess), (b) acetophenone, (c) cyclohexanone (1:1), under NaOH/EtOH, heat.



(a)



(b)



(c)

Answer: (a) benzalacetone; (b) chalcone; (c) 2-benzylidenecyclohexanone

Benzaldehyde has no α-H and so cannot enolise; the ketone enolises and adds to PhCHO; E1cB dehydration gives a conjugated enone (Claisen–Schmidt). With excess PhCHO, acetone gives dibenzalacetone.

CA-22.

How many different products (including the self-condensation products) can form when CH_3CHO and CH_3CH_2CHO are mixed with dilute NaOH (aldol only, no dehydration, ignore stereoisomers)?

Answer: 4

Each aldehyde can act as the enolate or as the electrophile: CH_3CHO + CH_3CHO; CH_3CH_2CHO + CH_3CH_2CHO; enolate of CH_3CHO + CH_3CH_2CHO; enolate of CH_3CH_2CHO + CH_3CHO. Four distinct β-hydroxy aldehydes (a statistical mixture, which is why crossed aldols with two enolisable partners are avoided).

CA-23.

Complete: (a) CH_3COCH_3 + I_2/NaOH (excess), (b) $C_6H_5COCH_3$ + NaOCl, (c) CH_3CH_2OH + I_2/NaOH. Which give a yellow precipitate?

Answer: (a) $CHI_3 + CH_3COO^-$; (b) $CHCl_3 + C_6H_5COO^-$; (c) $CHI_3 + HCOO^-$; yellow CHI_3 from (a) and (c)

Methyl ketones and ethanol (oxidised in situ to acetaldehyde) give the iodoform test; (b) with NaOCl gives chloroform (not yellow). The acid always has one carbon fewer than the methyl ketone.

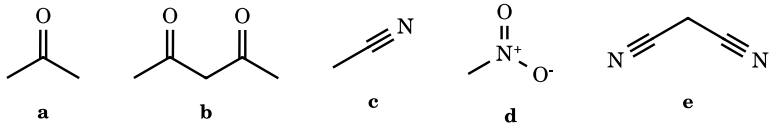
CA-24.

A Grignard reagent CH_3MgI (excess) is added to each of (a) $HOCH_2CH_2COCH_3$, (b) $HC\equiv C–CH_2–COOEt$. Count moles of CH_4 evolved per mole of substrate before addition occurs.

Answer: (a) 1; (b) 1

Active hydrogens are protonated by CH_3^-: (a) the O–H (1); (b) the acetylenic C–H (pK_a 25), (the ester then reacts through addition, not acid–base): 1. The Zerewitinoff count gives the number of active H.

CA-25.
Arrange in order of increasing acidity of the marked hydrogen: (a) CH_3COCH_3, (b) $CH_3COCH_2COCH_3$ (central), (c) CH_3CN, (d) CH_3NO_2, (e) $CH_2(CN)_2$.



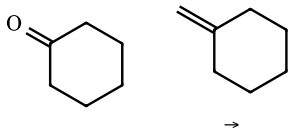
a **b** **c** **d** **e**

Answer: c < a < d ≈ e < b (aqueous pKa ≈ 25, 19.3, 10.2, 11.2, 8.9)
A single acceptor: nitrile (≈ 25) < ketone (19) < nitro (10). Two acceptors: malononitrile (11) and acetylacetone (9, aqueous). The DMSO ordering of the two close pairs is d (17.2) < e (11.1) < b (13.3) with the order of b and e depending on scale: check the scale given in the question.

CA-26.
Why is the CH_2 between the two C=O groups of acetylacetone (pK_a 9) more acidic than the CH_3 groups (≈ 20), and what does base then give on alkylation with 1 equiv CH_3I?
Answer: central position; 3-methylpentane-2,4-dione
The central carbanion is delocalised over two C=O (and both oxygens carry charge), whereas a terminal carbanion is stabilised by only one. The central anion is the one formed and alkylated. With 2 equiv of a very strong base (the dianion) a terminal CH_3 would react first.

CA-27.
State the product and the stereochemical outcome of the Wittig reaction of (a) benzaldehyde with $Ph_3P{=}CHCH_3$ (salt-free), (b) benzaldehyde with $Ph_3P{=}CHCO_2Et$.
Answer: (a) mainly (Z)-1-phenylpropene; (b) (E)-ethyl cinnamate
(a) non-stabilised ylide: irreversible, kinetically controlled formation of the cis-oxaphosphetane leading to Z-alkene. (b) a stabilised ylide forms the oxaphosphetane reversibly, allowing the more stable trans (E) alkene to form.

CA-28.
Cyclohexanone is converted into methylenecyclohexane in one step. Which reagent? Why will $CH_3MgI/H^+/\Delta$ not give a clean result?



→

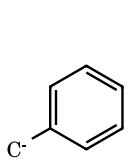
Answer: $Ph_3P{=}CH_2$ (Wittig)
The Grignard route gives 1-methylcyclohexanol which dehydrates (E1) to the endocyclic, more substituted 1-methylcyclohexene, not the exocyclic alkene. The Wittig reaction puts the double bond exactly where the C=O was.

CA-29.
An optically active ketone $CH_3CH(C_2H_5)COC_6H_5$ loses optical activity in $NaOH/H_2O$. A second ketone, with a quaternary α-carbon, does not. Explain, and say what rate law governs the racemisation.
Answer: planar enolate; rate = k[ketone][HO^-], independent of any halogen
The stereocentre carries an acidic α-H; the enolate is planar. The quaternary α-carbon has no H and cannot enolise at that carbon. Halogenation in base and racemisation have the same rate (zero order in halogen), proving the common enolate.

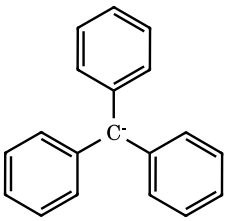
CA-30.
Compare the products of the reaction of cyclohexanone with (a) CH_3MgI then H_3O^+; (b) pyrrolidine then CH_3I then H_3O^+; (c) LDA then CH_3I; (d) NaOEt, CH_3I (excess).
Answer: (a) 1-methylcyclohexanol (C=O attack); (b) 2-methylcyclohexanone (Stork enamine alkylation at α-C); (c) 2-methylcyclohexanone; (d) mixture up to 2,2,6,6-tetramethylcyclohexanone
The Grignard reagent attacks carbon of C=O; the enamine and enolate act as nucleophiles at the α-carbon. NaOEt is a weak equilibrating base so enolates alkylate repeatedly (polyalkylation); LDA/enamine give clean monoalkylation.

CA-31.
Identify the electrophile and the carbanion in each of: (a) Perkin reaction, (b) Knoevenagel, (c) benzoin condensation, (d) Reformatsky, (e) Stork enamine alkylation, (f) Michael addition of diethyl malonate to acrylonitrile.
Answer: see reasoning
(a) enolate of acetic anhydride (from CH_3COONa) + ArCHO; (b) $CH_2(CN)_2$ carbanion + ArCHO; (c) cyanohydrin carbanion + ArCHO; (d) $BrZn{-}CH_2CO_2Et$ + ketone; (e) enamine β-carbon + RX or enone; (f) malonate carbanion + $CH_2{=}CH{-}C{\equiv}N$ (1,4-addition giving $NC{-}CH_2CH_2{-}CH(CO_2Et)_2$).

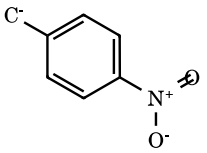
CA-32.
Which carbanion is most stable? Rank (a) $PhCH_2^-$, (b) $(Ph)_3C^-$, (c) p-O_2N–$C_6H_4CH_2^-$, (d) cyclopentadienide, (e) CH_3^-.



a



b



c



d

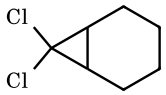


e

Answer: d > c > b > a > e (acid pK_a ≈ 16, 20, 31, 41, 50)
Cp^- is aromatic; the p-nitrobenzyl anion delocalises onto NO_2 (–M); trityl has three phenyls, benzyl one; methyl has none. (⚑ the position of (c) relative to (b) is scale-dependent: the p-nitrobenzyl anion is more stable by ≈ 10 pK units than benzyl and comparable with trityl.)

CA-33.
Which of the following reactions will proceed to completion? (A) $CH_3COOEt + EtO^-$; (B) $CH_3COCH_2COOEt + EtO^-$; (C) $PhC\equiv CH + HO^-$; (D) $CH_3NO_2 + HO^-$.
Answer: (B) and (D)
Use pK_a: acetoacetate 10.7 and nitromethane 10.2 are much more acidic than EtOH (16) or water (15.7), so deprotonation is complete. Ethyl acetate (25.6) and phenylacetylene (≈ 29, DMSO) are not deprotonated appreciably.

CA-34.
Chloroform reacts with solid KOH in a non-polar solvent with a phase-transfer catalyst ($BnEt_3N^+Cl^-$) in the presence of cyclohexene. Name the product and the role of the catalyst.



Answer: 7,7-dichlorobicyclo[4.1.0]heptane; the catalyst carries HO^- (or CCl_3^-) into the organic phase
Makosza interfacial mechanism: HO^- deprotonates $CHCl_3$ at the interface; $Q^+CCl_3^-$ moves into the organic phase, loses Cl^- to give $:CCl_2$, which adds to the alkene.

CA-35.
The conjugate acid of which base has a pK_a high enough to deprotonate benzene? Which of NaOH, NaOEt, $NaNH_2$, n-BuLi/TMEDA, LDA can do so?
Answer: n-BuLi/TMEDA only (pK_a of benzene ≈ 43)
NH_3 (≈ 38) and diisopropylamine (36) are too weak as conjugate acids; butane (≈ 50) is strong enough. TMEDA breaks up the BuLi hexamer, making it kinetically capable.

CA-36.
Racemic 2-bromobutane forms a Grignard reagent which is hydrolysed with D_2O. Is the 2-deutero-butane optically active if one starts from enantiopure (S)-2-bromobutane?
Answer: no, racemic
Grignard formation proceeds through radical intermediates at the metal surface (single-electron transfer), which are planar and lose configuration; the secondary alkyl Grignard also inverts configurationally on the time scale of hours at room temperature. In contrast, vinyl and cyclopropyl Grignards and many organolithiums at −78 °C retain configuration.

CA-37.
Which of the following do **not** take place? (a) $CH_3CH_2CH_2CH_2Li + CH_3CH_2OH \rightarrow$ butane; (b) RMgBr + R′COOH → R–H; (c) RMgBr + aldehyde with a COOH group → alcohol (without protection); (d) PhMgBr + CO_2 → PhCOOH.
Answer: (c) does not give the clean alcohol
The Grignard reagent first deprotonates the COOH (acid–base is faster than addition) and consumes one equivalent; the carboxylate then does not react further, so addition to the aldehyde needs excess reagent or protection of the acid. (a), (b), (d) occur.

CA-38.
Convert (i) $CH_3COCH_3 \rightarrow (CH_3)_2C{=}CH{-}COCH_3$ (mesityl oxide), (ii) $CH_3CHO \rightarrow CH_3CH{=}CH{-}CHO$, (iii) HCHO + CH_3CHO → ? (NaOH; crossed aldol), (iv) PhCHO + $(CH_3CO)_2O/CH_3COONa/\Delta$ → ?
Answer: (i) aldol then E1cB; (ii) crotonaldehyde; (iii) acrolein or (with excess HCHO) pentaerythritol precursors; (iv) cinnamic acid (Perkin)
In (iii) formaldehyde has no α-H and is an excellent electrophile; with acetaldehyde in excess HCHO and base the three α-H are all hydroxymethylated and a final crossed Cannizzaro reduction by HCHO gives pentaerythritol $C(CH_2OH)_4$. (iv) The enolate of the anhydride attacks PhCHO and elimination/hydrolysis gives E-cinnamic acid.

CA-39.

A student claims that carbanions rearrange like carbocations by 1,2-hydride and 1,2-alkyl shifts. Evaluate.

Answer: incorrect

A 1,2-shift needs a three-centre, two-electron TS with an *empty* orbital to receive the pair; the carbanion carbon already has a filled lone pair, so such a shift would require a four-electron three-centre TS (forbidden/high barrier). Aryl and vinyl groups migrate to carbanions through spiro/cyclopropane-like intermediates (Grovenstein–Zimmerman), by using their π systems.

CA-40.

α-Elimination versus β-elimination: why does $CHCl_3 + HO^-$ give $:CCl_2$ while $CH_3CH_2Cl + HO^-$ gives $CH_2{=}CH_2$?

Answer: no β-H in $CHCl_3$; the anion CCl_3^- is stable and eliminates from the same carbon

Chloroform has only one carbon, so a β-elimination is impossible. Its carbanion is stabilised by three Cl (–I, polarisability), long-lived enough to lose Cl^- (α). Ethyl chloride has a β-H and an E2 pathway.

CA-41.

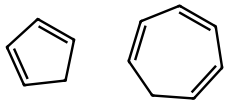
Using the Zimmerman–Traxler model, say which aldol diastereomer is formed by the Z-lithium enolate of 3-pentanone with benzaldehyde.

Answer: the syn aldol

In the chair six-membered TS the aldehyde Ph occupies the equatorial position; with a Z-enolate the two substituents that become the stereocentres end up syn in the zig-zag drawing. E-enolates give anti.

CA-42.

Which has the more acidic C–H, 1,3-cyclopentadiene ($\approx$ 16) or 1,3-cycloheptatriene ($\approx$ 36)? Explain, and also predict where CH_3Li would remove a proton from 1,3,5-cycloheptatriene.



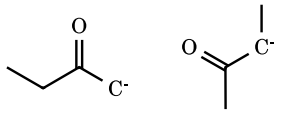
a **b**

Answer: cyclopentadiene; at the CH_2 (C-7)

Cp^- is aromatic; the cycloheptatrienyl anion would have 8π (antiaromatic) in a planar ring and is destabilised (the pK_a of 36 corresponds to a non-planar, non-aromatic anion). Both lose H from the sp^3 carbon.

CA-43.

A carbanion is generated from $CH_3COCH_2CH_3$ (2-butanone). Name the two possible enolates and the conditions that select each.



A **B**

Answer: A: less substituted (kinetic, LDA –78 °C); B: more substituted (thermodynamic, NaOEt/EtOH)

A arises from removal of a CH_3 hydrogen; B from the CH_2. LDA gives mostly A (kinetic); thermodynamic conditions (equilibrating base) give mostly B, with E/Z isomers.

CA-44.

Why is an α-C–H of an ester ($pK_a \approx 25$) less acidic than the α-C–H of the corresponding ketone ($\approx$ 20), when the ester carbonyl carbon is the more electrophilic?

Answer: OR lone-pair donation into C=O

The ester C=O is already stabilised by resonance with the alkoxy oxygen (ROC=O ⇔ $RO^+{=}C{-}O^-$); therefore its carbonyl accepts the carbanion's charge less well than a ketone's. The same effect makes amide α-H still less acidic ($\approx$ 30).

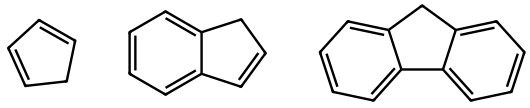
CA-45.

(JEE-type) How many of the following compounds will liberate methane on treatment with CH_3MgI? CH_3OH, CH_3COOH, CH_3CONH_2, CH_3COCH_3, $HC \equiv CH$, $C_6H_5NH_2$, CH_3COOCH_3, H_2O.

Answer: 6 (CH_3OH, CH_3COOH, CH_3CONH_2, $HC \equiv CH$, $C_6H_5NH_2$, H_2O)

Each has an active H more acidic than CH_4. Acetone does not liberate CH_4 because the Grignard adds to its C=O (a small amount of enolisation may occur), and the ester is attacked at C=O. Acetamide has two N–H (2 equivalents).

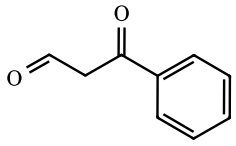
CA-46.
(IChO-style) The pK_a values of cyclopentadiene, indene and fluorene (DMSO) are 18, 20 and 22.6. Explain why benzo-fusion lowers the acidity although the conjugate base keeps the cyclopentadienide ring.



Answer: each fused benzene ring competes for the π electrons
In Cp^- all six π electrons are shared by five equivalent carbons. In indenide and fluorenide, part of the charge must be delocalised into a benzene ring, which costs some of that ring's aromatic stabilisation, and the cyclopentadienide ring is perturbed. Each benzo-fusion therefore destabilises the anion by ≈ 2 pK units relative to Cp^-.

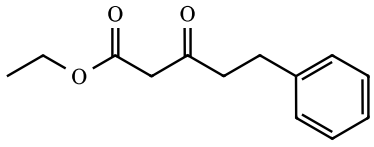
CA-47.
(JEE Advanced, integer) Propanal undergoes self-aldol addition (dilute NaOH, 5 °C). How many stereoisomers does the aldol (3-hydroxy-2-methylpentanal) have, and how many geometrical isomers does the dehydration product have?
Answer: 4 and 2
The aldol has two stereocentres (C-2 and C-3, no symmetry): $2^2 = 4$ stereoisomers. Dehydration gives 2-methylpent-2-enal, with a trisubstituted C=C bearing different groups on each carbon: E and Z, so 2.

CA-48.
(GATE/CSIR style) Acetophenone is treated with ethyl formate and NaOEt, then acid. Give the product and say why formate is a good acylating partner.



Answer: benzoylacetaldehyde (3-oxo-3-phenylpropanal), largely as its enol
Crossed Claisen condensation: the acetophenone enolate attacks HCOOEt, whose carbonyl is unhindered and cannot itself enolise (no α-H), so a single product forms. The product is a 1,3-dicarbonyl (pK_a ≈ 7–9), which is deprotonated by ethoxide and drives the reaction.

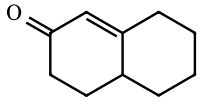
CA-49.
Ethyl acetoacetate is treated with (i) NaH (1 equiv), (ii) n-BuLi (1 equiv), (iii) $PhCH_2Br$, (iv) H_3O^+. Predict the product and explain the site of alkylation.



Answer: ethyl 3-oxo-5-phenylpentanoate (γ-alkylation)
NaH removes the most acidic C-2 proton; n-BuLi then removes a proton from the terminal CH_3 (C-4) to give a dianion. In the dianion the less stabilised, more basic carbanion (C-4) is the more nucleophilic and reacts first with the halide. Acid work-up reprotonates C-2.

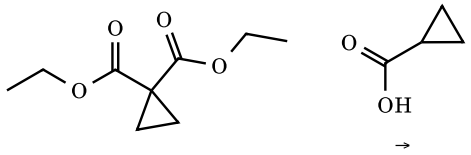
CA-50.
Arrange in order of increasing rate of E1cB elimination of HO^- from β-hydroxy compounds: (a) $CH_3CH(OH)CH_2COCH_3$, (b) $CH_3CH(OH)CH_2COOEt$, (c) $CH_3CH(OH)CH_2CON(CH_3)_2$, (d) $CH_3CH(OH)CH_2NO_2$.
Answer: c < b < a < d
E1cB rate follows the ease of forming the carbanion (acidity of the α-H): amide (≈ 30) < ester (≈ 25) < ketone (≈ 20) < nitro (≈ 10 in water). Hydroxide is a poor leaving group, but the stabilised carbanion makes expulsion feasible.

CA-52.
Cyclohexanone is treated with methyl vinyl ketone and NaOEt/EtOH, then heated. Give the bicyclic product and name the three steps.



Answer: $\Delta^{1(9)}$-octalone (4,4a,5,6,7,8-hexahydronaphthalen-2(3H)-one); Michael, intramolecular aldol, E1cB dehydration
The cyclohexanone enolate adds 1,4 to MVK to give the 1,5-diketone; the methyl ketone enolate attacks the ring C=O (6-exo) to give a β-hydroxy ketone, which loses water by E1cB to give the conjugated enone (Robinson annulation).

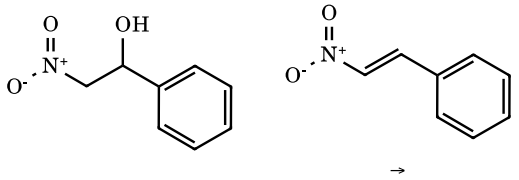
CA-53.
Diethyl malonate (2 equiv) with NaOEt and 1,2-dibromoethane gives a cyclic diester. Name it and the product after hydrolysis and heating.



Answer: diethyl cyclopropane-1,1-dicarboxylate; cyclopropanecarboxylic acid
One malonate carbanion displaces bromide (S_N2); the remaining acidic C–H is deprotonated and the carbanion closes the three-membered ring (3-exo-tet). Hydrolysis and decarboxylation of the gem-diacid give the mono-acid.

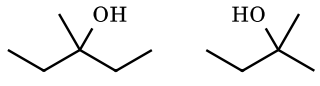
CA-54.
Diethyl 3-methyladipate (diethyl 3-methylhexanedioate) undergoes Dieckmann cyclisation. Which regioisomeric β-keto ester is expected to predominate, and why?
Answer: 2-ethoxycarbonyl-4-methylcyclopentanone (major) over the 3-methyl isomer
Enolate formation at C-5 (remote from the methyl) is less hindered; it attacks the C-1 ester to give the 4-methyl isomer. Enolate formation at C-2 (next to the methyl-bearing C-3) gives the 3-methyl isomer. The product with the acidic ring C–H is deprotonated, which drives either closure.

CA-55.
Nitromethane reacts with benzaldehyde and base (Henry reaction). Give the adduct and the compound formed on dehydration, and explain why only catalytic base is needed.



Answer: 2-nitro-1-phenylethanol; β-nitrostyrene
The nitronate ($pK_a \approx 10$) adds to the aldehyde; the alkoxide is protonated by nitromethane, regenerating the nitronate. Dehydration (E1cB) gives the conjugated nitroalkene.

CA-56.
Which alcohol is formed from (a) ethyl acetate + 2 EtMgBr, (b) ethyl propanoate + 2 CH_3MgBr? Are they the same?

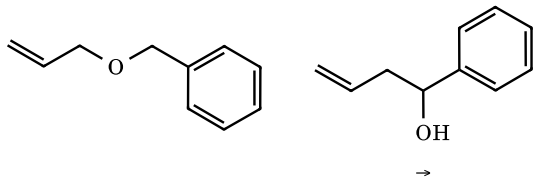


a **b**

Answer: (a) 3-methylpentan-3-ol; (b) 2-methylbutan-2-ol; different
Two equivalents of R^- add to the ester carbonyl; the acyl group supplies one substituent and the OEt is lost after the first addition. (a) acyl = CH_3, R = Et twice; (b) acyl = Et, R = CH_3 twice.

CA-57.
(JEE integer) How many of these give a yellow precipitate with I_2/NaOH: ethanol, propan-2-ol, acetophenone, pentan-3-one, acetaldehyde, propanal, butan-2-one, methyl acetate?
Answer: 5
Ethanol and propan-2-ol (oxidised in situ to CH_3CHO and acetone), acetophenone, acetaldehyde and butan-2-one contain $CH_3CO–$ or $CH_3CH(OH)–$. Pentan-3-one, propanal and methyl acetate do not.

CA-58.
Benzyl allyl ether is treated with n-BuLi at −78 °C and warmed. Give the product and name the rearrangement.



Answer: 1-phenylbut-3-en-1-ol via [2,3]-Wittig rearrangement
BuLi removes the benzylic proton; the carbanion attacks the terminal alkene carbon in a five-membered cyclic transition state while the C–O bond to the allyl group breaks, giving a homoallylic alkoxide.

Part 9 Free Radicals: Structure, Generation, Stability and Reactions

A **free radical** is a species with an unpaired electron: $R_3C^{\bullet}$ has seven valence electrons on carbon (three bond pairs + one unpaired electron). It is **neutral**, paramagnetic (spin ½, doublet; μ = √3 BM), usually short-lived (diffusion-controlled dimerisation, k ≈ 10^9–10^{10} L mol^{-1} s^{-1}), and is formed by **homolysis** – each atom keeps one electron of the bond – which is drawn with a **single-barbed (fish-hook) arrow**, while heterolysis uses a double-barbed arrow. Free radicals are electron-deficient in the sense that the carbon is one electron short of an octet, so they behave as weak electrophiles towards electron-rich partners and as nucleophiles towards electron-poor ones; their reactivity is dominated by the strength of the bond formed, not by charge.

9.0 Homolysis and bond dissociation energy

Homolysis A–B → $A^{\bullet}$ + $B^{\bullet}$ needs energy equal to the bond dissociation energy D(A–B); it is promoted by heat (Δ), light (hν), or an initiator that decomposes at a low temperature. Representative values (kJ mol^{-1} in parentheses are in kcal mol^{-1}; recalled approximate figures):

Bond	D (kcal mol^{-1})	Bond	D (kcal mol^{-1})
H–CH_3	105	H–F	136
H–CH_2CH_3 (primary)	101	H–Cl	103
H–$CH(CH_3)_2$ (secondary)	98.5	H–Br	87.5
H–$C(CH_3)_3$ (tertiary)	96.5	H–I	71
H–$CH_2CH{=}CH_2$ (allylic)	88	F–F	38
H–CH_2Ph (benzylic)	89.8	Cl–Cl	58
H–$CH{=}CH_2$ (vinylic)	111	Br–Br	46
H–C_6H_5 (phenyl)	112	I–I	36
CH_3–CH_3	90	CH_3–Cl / –Br / –I	85 / 70 / 57
RO–OR (peroxide)	≈ 35–38	N ≡ C–$C(CH_3)_2$–N=N– (AIBN, N–C)	≈ 31 (loses N_2)

9.1 Structure and geometry

- **Alkyl radicals** $R_3C^{\bullet}$ are **planar or very shallow pyramids** ($CH_3^{\bullet}$ is planar, sp^2 with the unpaired electron in a p orbital; t-Bu• is slightly pyramidal). The inversion barrier is tiny, so a radical centre is racemised – radical halogenation at a chiral carbon gives the racemate. The geometry lies between the planar carbocation and the pyramidal carbanion.
- **Vinyl radicals** are bent (σ radical, sp^2 orbital in the plane, inverts rapidly); **aryl radicals** are σ radicals with the unpaired electron in the plane of the ring (orthogonal to the π cloud, no resonance); $\mathbf{CF_3^{\bullet}}$ is pyramidal (≈ 110°) because the electronegative fluorines increase the s-character of the carbon orbitals that bind them, leaving more p-character in the half-filled orbital.
- **π radicals** (allyl, benzyl, trityl) are planar with the unpaired electron delocalised over several atoms. The spin density is shown by ESR hyperfine couplings (benzyl: ortho/para positions).
- **Stable radicals** exist when the unpaired electron is delocalised and the centre is hindered: $Ph_3C^{\bullet}$ (Gomberg 1900, in equilibrium with its quinoid dimer), 2,4,6-tri-tert-butylphenoxyl, DPPH, galvinoxyl, TEMPO (a nitroxide), Fremy's salt. NO, NO_2 and O_2 (triplet, a biradical) are also paramagnetic.

9.2 Stability of free radicals

Stability is measured by the bond dissociation energy of the C–H bond that yields the radical (smaller D = more stable radical). Three effects, in order of importance: **(i) resonance delocalisation** (allyl, benzyl, trityl, α-carbonyl, α-N, α-O); **(ii) hyperconjugation** of the adjacent C–H/C–C σ bonds into the singly occupied orbital (3° > 2° > 1° > $CH_3^{\bullet}$); **(iii) hybridisation** (sp^2 σ radicals such as vinyl and phenyl are unstabilised).

Order	Statement
Alkyl radicals	3° > 2° > 1° > $CH_3^{\bullet}$ (more α-C–H/C–C bonds donate electron density into the half-filled orbital; D(C–H) 96.5 < 98.5 < 101 < 105)
Delocalised radicals	$Ph_3C^{\bullet}$ > $Ph_2CH^{\bullet}$ > $PhCH_2^{\bullet}$ ≳ $CH_2{=}CH{-}CH_2^{\bullet}$ > 3° alkyl > 2° > 1° > $CH_3^{\bullet}$ > $CH_2{=}CH^{\bullet}$ ≈ $C_6H_5^{\bullet}$ (σ radicals least stable)
α-Heteroatom (captodative)	a radical flanked by a donor and an acceptor (e.g. •C(OR)(CN)) is specially stabilised; α-O and α-N lone pairs also stabilise (three-electron two-orbital interaction)

Substituted benzyl radicals	stabilised by both donors and acceptors (spin delocalisation, no charge separation) – so radical stability does **not** follow σ^+ the way cation stability does; the rate of side-chain chlorination, $ArCH_3 \rightarrow ArCH_2Cl$, is only weakly affected by Ar (but p-NO_2 slows it slightly because the radical-abstracting chlorine atom is electrophilic: your note's 'Ar = phenyl faster than p-nitrophenyl')

Do not transfer carbocation reasoning blindly. (1) Radicals do **not** rearrange by 1,2-H or 1,2-alkyl shifts (the three-centre TS would need 3 electrons and has a high barrier); the neopentyl radical stays neopentyl. They do undergo 1,2-aryl (neophyl), 1,2-vinyl, 1,2-acyl, 1,2-Cl/Br shifts and β-scission. (2) Radical stability is a *bond-energy* property, not a charge property: both +M and –M substituents stabilise it. (3) The allylic radical is stabilised by resonance *and* allylic substitution gives a mixture of regioisomers (resonance) – see 9.5.

9.3 Generation of free radicals

Method	Example	Notes
Thermolysis (Δ)	ROOR → 2 RO•; R–N=N–R → 2 R• + N_2; PhCOO–OOCPh → 2 PhCOO• → 2 Ph• + 2 CO_2	weak O–O (35–38 kcal mol^{-1}) and C–N bonds; AIBN at 60–80 °C
Photolysis (hν)	Cl_2 → 2 Cl•; Br_2 → 2 Br•; R–I → R• + I•; ketones (Norrish)	light of the correct energy; used in radical halogenation
Redox (single electron transfer)	Fe^{2+} + H_2O_2 → Fe^{3+} + HO^- + HO• (Fenton); Cu^+ + ArN_2^+ → Cu^{2+} + Ar• + N_2 (Sandmeyer); Na + ketone → ketyl radical anion	metal ions and electrodes
Electrolysis (Kolbe)	2 $RCOO^-$ → 2 RCOO• + 2e^- → 2 R• + 2 CO_2 → R–R	anodic oxidation of carboxylate; products from mixtures are statistical
Atom abstraction by a radical	X• + R–H → H–X + R•	the chain-carrying step of halogenation
Addition of a radical to a π bond	X• + C=C → X–C–C•	polymerisation, anti-Markovnikov addition
Fragmentation	RCOO• → R• + CO_2; R–CH_2–CH_2• → R• + CH_2=CH_2 (β-scission)	decarboxylation and ring opening

9.4 Reactions of free radicals

Reaction	Scheme	Example / use
Atom (H) abstraction	R• + H–X → R–H + X•	propagation in halogenation; selectivity 3° > 2° > 1°
Addition to C=C	R• + CH_2=CHY → R–CH_2–C•HY	chain polymerisation; adds to the less hindered carbon to give the more stable radical
Halogen abstraction	R• + X–X → R–X + X•	second propagation step
Combination (termination)	R• + R• → R–R	diffusion controlled; ethane from 2 CH_3•
Disproportionation	2 CH_3CH_2• → CH_3CH_3 + CH_2=CH_2	H transfers from one radical to another
β-Scission / fragmentation	R–CH_2–C•H_2 → R• + CH_2=CH_2; RCOO• → R• + CO_2	ring-opening of cyclopropylcarbinyl radical (k ≈ 10^8 s^{-1}, a 'radical clock')
Rearrangement	1,2-aryl (neophyl), 1,2-vinyl, 1,2-acyl; **no 1,2-H or alkyl shift**	$PhC(CH_3)_2CH_2$• → $PhCH_2C$•$(CH_3)_2$
Oxidation / reduction (electron transfer)	R• + Cu^{2+} → R^+ + Cu^+; R• + e^- → R^-	Sandmeyer, $S_{RN}1$

9.5 Radical halogenation of alkanes: thermochemistry and selectivity

Mechanism (chain). *Initiation* X_2 → 2 X•. *Propagation* X• + R–H → HX + R• (1); R• + X_2 → R–X + X• (2). *Termination* any two radicals combine. A single photon can cause thousands of cycles (quantum yield > 1) – the proof of a chain mechanism. Inhibitors (O_2, hydroquinone, I_2, BHT) trap the chain-carrying radicals by forming less reactive radicals; a short induction period is observed.

Halogen	Step (1): X• + CH_4 ΔH	Step (2): CH_3• + X_2 ΔH	Overall	Reactivity / selectivity
F_2	–31 (105 – 136)	≈ –72	≈ –103	explosive; non-selective (early TS)
Cl_2	+2 (105 – 103)	–27	–25	fast; poorly selective (1° : 2° : 3° ≈ 1 : 3.8 : 5)
Br_2	+17.5 (105 – 87.5)	–24	–6.5	slow; highly selective (≈ 1 : 82 : 1600)
I_2	+34 (105 – 71)	–21	+13	step (1) too endothermic: no iodination (HI even reduces R–I)

Hammond postulate: chlorine's step (1) is nearly thermoneutral, so its transition state is early and resembles the reactants (little radical character developed, so little discrimination between C–H types); bromine's step (1) is endothermic by 17 kcal mol^{-1}, so its TS is late, resembles the alkyl radical, and is sensitive to radical stability. **Probability × reactivity** gives the product ratio: product % ∝ (number of H of that type) × (relative reactivity). Propane with Cl_2: 6 × 1 : 2 × 3.8 = 6 : 7.6 → 44 % 1-chloro, 56 % 2-chloro; with Br_2: 6 : 164 → 3.5 % : 96.5 %. Isobutane with Cl_2: 9 × 1 : 1 × 5 = 9 : 5 → 64 % isobutyl chloride, 36 % t-butyl chloride (Nucleus Q51: the *primary* chloride predominates because there are nine primary H). **Isotope effect:** C–D bonds (zero-point energy lower, D(C–D) ≈ 1.2 kcal mol^{-1} higher than D(C–H)) are abstracted more slowly, so CD_3CD_3 chlorinates more slowly than CH_3CH_3 (primary kinetic isotope effect k_H/k_D ≈ 1.3 for Cl•, ≈ 4.6 for Br•), and a tertiary C–H of neopentane-type alkanes ranks by the number of H.

9.6 Counting problems in radical halogenation

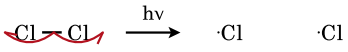
Substrate	Structural monochlorides (type of H)	Including stereoisomers
CH_3CH_3, neopentane, cyclopentane, benzene (side-chain none)	1	1
propane	2 (1°, 2°)	2
n-butane	2	3 (2-chlorobutane is chiral)
isobutane	2	2
2-methylbutane (isopentane)	4	6 (1-chloro-2-methyl- and 2-chloro-3-methylbutane are chiral)
2,3-dimethylbutane	2	3
n-pentane	3	5
methylcyclopentane	4 (CH_3, C1, C2, C3)	10 (2- and 3-chloro each as cis/trans, each chiral)
2-methylpentane (isohexane)	5	8 (see Nucleus II-B Q8)

9.7 Allylic and benzylic halogenation; the HBr peroxide effect; polymerisation

- **NBS (N-bromosuccinimide), CCl_4, peroxide/hv**: allylic and benzylic bromination. NBS supplies a very low, steady concentration of Br_2 (NBS + HBr → succinimide + Br_2), so Br• abstracts the weak allylic C–H (88 kcal mol^{-1}) and Br_2 reacts with the allylic radical faster than it can add across C=C by the ionic pathway. An unsymmetrical allylic radical gives **two regioisomers** (both radical termini carry spin): 1-butene → 3-bromo-1-butene + 1-bromo-2-butene. Count the distinct allylic hydrogens first (Nucleus II-A Q4: 1-methylcyclopentene has 7 allylic H and gives 9 allylic bromides when stereoisomers are included).
- **Anti-Markovnikov addition of HBr (peroxide effect, Kharasch)**: RO• + HBr → ROH + Br•; Br• adds to the *terminal* carbon of the alkene to give the more stable (secondary) radical; the radical abstracts H from HBr to give the 1-bromoalkane and a new Br•. HCl does not show the effect because step (2) (R• + HCl) is endothermic by ≈ +12.6 kJ mol^{-1}; HI does not because step (1) (I• adds to C=C) is endothermic (+46 kJ mol^{-1}) and I• tends to combine (data of Nucleus II-B Q7).
- **Radical polymerisation** (initiation, propagation, termination): R• + CH_2=CHY → RCH_2C•HY → …; used for polyethylene (LDPE, 1000–2000 atm, O_2/peroxide), polystyrene, PVC, polyacrylonitrile, teflon. Head-to-tail addition is dominant because it forms the more stable radical.
- **Wurtz and Kolbe**: Wurtz coupling 2 R–X + 2 Na → R–R + 2 NaX has both carbanion (R–Na) and radical features; Kolbe electrolysis of 2 RCOONa gives R–R, 2 CO_2, H_2 (cathode) and NaOH (the pH increases, Nucleus Q50). A mixture of two different carboxylates gives three alkanes (R–R, R′–R′, R–R′), statistically 1 : 1 : 2.

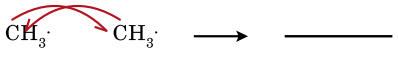
9.8 Mechanisms (fish-hook arrows; each step electron-count validated)

9.1 Initiation: homolysis of chlorine



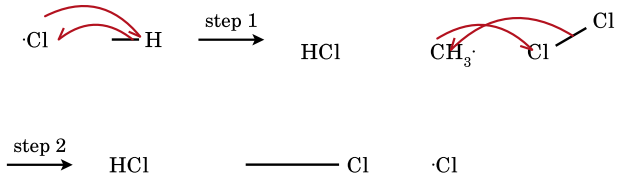
Each chlorine receives one electron of the Cl–Cl bond; one fish-hook for each electron. Light of wavelength ≈ 350 nm supplies the 58 kcal mol^{-1}.

9.3 Termination by combination; disproportionation



Two radicals pair their electrons. Termination steps are rare in the propagating chain because radical concentrations are about 10^{-8} M; they end the chain when they occur.

9.2 Chain propagation: chlorination of methane



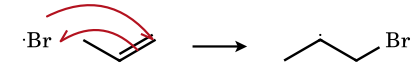
Step 1: the chlorine radical pairs its electron with one electron of the C–H bond to form H–Cl; the other electron of the C–H bond stays on carbon (CH_3•). Step 2: CH_3• pairs with one electron of Cl–Cl to form C–Cl; the other electron remains on the second Cl, which is the chain carrier again. The two steps repeat 10^3–10^6 times per photon.

9.4 Disproportionation of two ethyl radicals



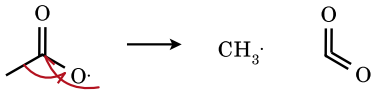
One radical abstracts a β-hydrogen from the other: the abstracted H forms the new C–H bond of ethane and the remaining C–H electron pairs with the radical electron to give the C=C of ethene. Disproportionation/combination ratio for ethyl radicals ≈ 0.13.

9.5 Peroxide effect, step 1: Br• adds to the terminal carbon of propene



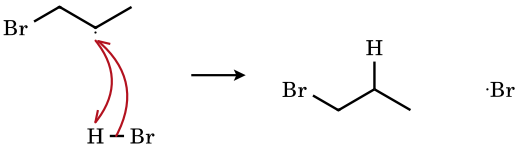
Bromine adds to the CH_2 terminus: the new radical is secondary (CH_3–C•H–CH_2Br), more stable than the primary radical that would form by attack at C-3. The regiochemistry is set by the stability of the radical, and is the reverse of Markovnikov addition of H^+.

9.8 Kolbe electrolysis: decarboxylation of the acyloxy radical



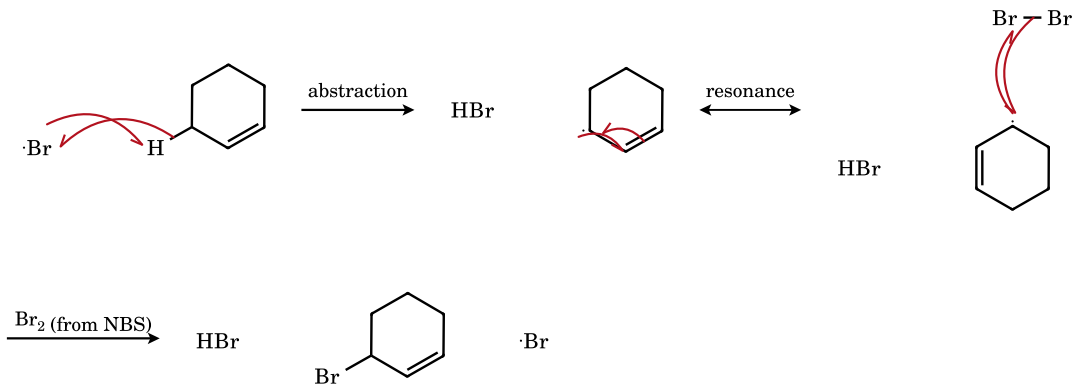
Anodic oxidation removes one electron from acetate ($CH_3COO^- \rightarrow CH_3COO$•). The acyloxy radical loses CO_2 in a β-scission: the O-radical electron and one electron of the C–C σ bond make the second C=O π bond, and the other σ electron stays on CH_3•. Two CH_3• then combine (9.3) to give ethane.

9.6 Peroxide effect, step 2: chain transfer



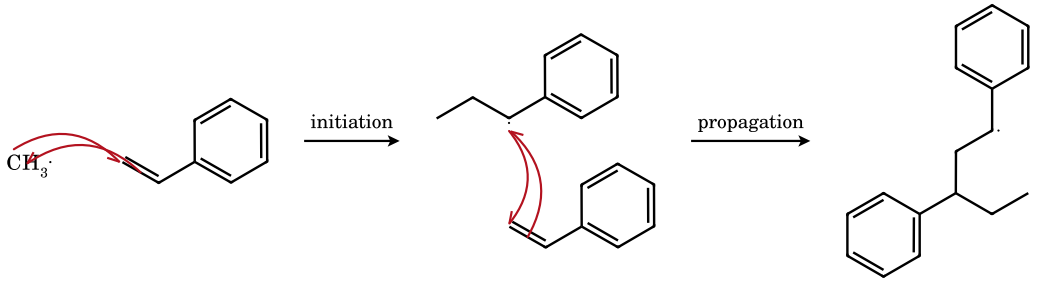
The secondary radical takes the H atom of HBr (the weak H–Br bond, 87.5 kcal mol^{-1}, makes this step exothermic by 50 kJ mol^{-1}), giving 1-bromopropane and regenerating Br•.

9.7 Allylic bromination of cyclohexene with NBS: H abstraction, allylic resonance, Br_2 transfer



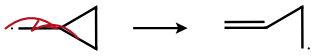
Br• removes the allylic H (88 kcal mol^{-1}); the allyl radical is delocalised over C-2 and C-5 (the two forms are equivalent for cyclohexene, so one product, 3-bromocyclohexene). With an unsymmetrical alkene the two resonance forms give two bromides. The low [Br_2] supplied by NBS prevents the competing ionic addition to the double bond.

9.9 Radical polymerisation of styrene: initiation and first propagation step



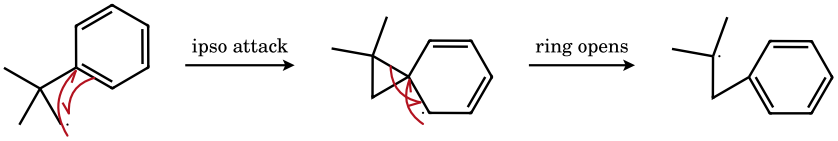
The initiator radical adds to the CH_2 end of the vinyl monomer so that the new radical is benzylic (resonance-stabilised); the same regiochemistry is repeated in each propagation step (head-to-tail polymer). Chain transfer and termination by combination/disproportionation end the chain.

9.10 Radical clock: cyclopropylcarbinyl → but-3-enyl (β-scission, k ≈ 10^8 s^{-1})



The unpaired electron and one electron of the strained ring C–C bond form a C=C π bond; the other electron of the broken bond becomes the new radical. Strain relief (27 kcal mol^{-1}) drives the opening. The reaction is used as a 'clock': if a radical trapped by an H-donor gives the ring-opened product, its lifetime was longer than 10^{-8} s.

9.11 Neophyl rearrangement: 1,2-aryl shift via a spiro cyclohexadienyl radical



The radical centre attacks the ipso carbon of the neighbouring phenyl group, forming a three-membered spiro intermediate (cyclohexadienyl radical delocalised over the ring); the ring bond C–C$(CH_3)_2$ then breaks to relieve strain and re-aromatise, leaving the more stable tertiary radical. This is the only common 1,2-shift for radicals; a hydrogen or methyl does **not** migrate (the TS would have no π participation).

9.9 Radical substitution of aromatics and related reactions

- **Sandmeyer** (ArN_2^+ + CuX → ArX + N_2) and **Gattermann** (Cu powder) proceed by single-electron transfer from Cu(I) to the diazonium ion to give Ar• (σ radical) which abstracts X from Cu(II)X_2.
- **Gomberg–Bachmann**: ArN_2^+ + Ar′H/NaOH → Ar–Ar′ (aryl radical adds to the arene to give a cyclohexadienyl radical, then loses H•).
- **Hunsdiecker**: RCOOAg + Br_2 → R–Br + CO_2 + AgBr (acyloxy radical; one carbon less).
- **$S_{RN}1$**: ArX + e^- → $ArX^{\bullet -}$ → Ar• + X^-; Ar• + Nu^- → $ArNu^{\bullet -}$ → (electron transfer to ArX) ArNu: a radical chain that explains aryl substitution without an electron-withdrawing group.
- **Hofmann–Löffler–Freytag**: an N-centred radical abstracts the δ-H through a six-membered TS (1,5-HAT) and gives a pyrrolidine after cyclisation.

9.11 Electronic structure of carbon radicals: SOMO, spin density, EPR

- **Orbital picture.** In a planar sp^2 alkyl radical the unpaired electron occupies a p orbital (the SOMO, singly occupied molecular orbital). The SOMO is higher than the filled bonding MOs and lower than the empty antibonding ones, so a radical can interact both with a donor LUMO-type partner and with an acceptor HOMO-type partner. A carbon radical with only three σ bonds is isoelectronic with the planar BH_3 framework plus one electron, and with the amine N–H inversion potential flattened (radical inversion barrier < 1 kcal mol^{-1} for CH_3•; ≈ 0.6 kcal mol^{-1} for t-Bu• to pyramidalise).
- **Pyramidalisation increases with electronegative and small substituents.** CH_3• and alkyl radicals: planar or nearly so; •CF_3: pyramidal (≈ 17° out-of-plane, inversion barrier ≈ 3 kcal mol^{-1}); •CH_2F, •CHF_2 progressively pyramidal; cyclopropyl radical: pyramidal (the ring angle strain favours more p-character in the ring bonds and more s-character in the SOMO; barrier ≈ 3 kcal mol^{-1}); •CH_2OH, •CH_2NH_2: nearly planar with the lone pair donating into the SOMO (three-electron two-orbital interaction; the radical is stabilised by ≈ 5–7 kcal mol^{-1}).

- **σ-radicals** (vinyl, aryl, acyl, alkynyl) have the odd electron in an sp^x orbital in the nodal plane of the π system; they are not stabilised by conjugation, so C–H BDEs are large (vinyl 111, phenyl 112, ethynyl 133 kcal mol^{-1}) and the radicals are very reactive. The acyl radical R–C•=O is a σ radical, bent at carbon; it loses CO (decarbonylation) when R• is stable (e.g. R = t-Bu, benzyl).
- **EPR (ESR) spectroscopy** detects unpaired electrons. The hyperfine splitting a_H from n equivalent protons gives n + 1 lines of binomial intensity, and the magnitude is proportional to the spin density on the carbon bearing H (McConnell equation, $a_H = Q \cdot \rho$, $Q \approx -23$ G). Data (approximate): CH_3•: 1:3:3:1 quartet, $a_H \approx 23$ G; ethyl: α-H (2) ≈ 22 G and β-H (3) ≈ 27 G, giving a triplet of quartets (12 lines); benzyl: ortho (2) ≈ 5 G, meta (2) ≈ 1.8 G, para (1) ≈ 6 G, α (2) ≈ 16 G, showing the spin density is delocalised to ortho/para positions; allyl radical: terminal H 13–14 G, central H 4 G, with 3 sets of lines (2 + 2 + 1 H). The large β-H coupling is hyperconjugation (the C–H bond overlapping the SOMO) and depends on the dihedral angle as $\cos^2\theta$ (Heller–McConnell), which allows conformational analysis of radicals.
- **Spin trapping.** Very reactive radicals are detected by adding them to a nitrone or nitroso compound (PBN, DMPO), giving a persistent nitroxide whose EPR spectrum identifies the trapped radical.
- **CIDNP** (chemically induced dynamic nuclear polarisation): NMR signals appear in enhanced absorption or emission during a radical-pair reaction, which is proof of a radical-pair mechanism (e.g. decomposition of benzoyl peroxide, Stevens and Wittig rearrangements).

9.12 Stability of radicals: bond dissociation energies and radical stabilisation energy

Radical stability is measured thermodynamically by the C–H bond dissociation energy of its parent, or by the **radical stabilisation energy RSE** = D(CH_3–H) − D(R–H). The values below are recalled from standard tables (kcal mol^{-1}); they are accurate to a few units and are used for *ranking*.

Radical	D(R–H)	RSE	Origin of the stabilisation
•CH_3	105	0	reference
CH_3CH_2• (1°)	101	4	hyperconjugation with 3 β C–H
$(CH_3)_2CH$• (2°)	98.5	6.5	6 β C–H
$(CH_3)_3C$• (3°)	96.5	8.5	9 β C–H, and release of strain on moving from sp^3 to sp^2
CH_2=CH–CH_2•	88	17	allylic resonance (two equivalent forms), loses ≈ 13 kcal mol^{-1} from the π-delocalisation net of rotation barrier
$C_6H_5CH_2$•	89.8	15	resonance with benzene ring (four forms), partial loss of aromatic π energy
$(C_6H_5)_2CH$•	≈ 84	21	two aryl groups (nonplanar)
$(C_6H_5)_3C$•	≈ 81	24	three aryl groups, sterically twisted propeller; persists in solution, dimerises through the para position of one ring (Gomberg dimer)
•CH_2OH, •CH_2OCH_3	≈ 96	9–10	n(O) → SOMO donation (3-electron 2-orbital)
•CH_2COCH_3, •CH_2CN, •CH_2COOR	≈ 92–94	11–13	SOMO conjugation with π* (C=O, C ≡ N)
C_6H_5• (σ)	112	−7	none; destabilised by high s-character
CH_2=CH• (σ)	111	−6	none
HC ≡ C•	133	−28	50 % s-character
$CH_3C(=O)$• (acyl)	≈ 89	≈ 16	partly carbonyl-conjugated σ/π hybrid

- **The radical is not a charge, so both donors and acceptors stabilise it.** The *captodative* effect: a carbon carrying simultaneously an electron donor (OR, NR_2) and an acceptor (CN, COR) is more stable than the sum of the individual effects (e.g. •C(CN)(OR)H), as both SOMO interactions apply (donor lone pair → SOMO, SOMO → acceptor π*).
- **Why RSE for benzyl is lower than for allyl** despite more resonance forms: delocalisation into the ring costs some aromatic stabilisation energy; the experimental difference between the BDEs is small (≈ 2 kcal mol^{-1}) and a JEE question that asks for the order normally expects *benzyl > allyl* by the number of resonance structures (benzyl: 5 including the radical form; allyl: 2). ⚑ The data order is allyl ≈ benzyl; tell which the exam uses.
- **Radical versus cation versus anion.** Alkyl substituents stabilise radicals (weakly) and cations (strongly) and *destabilise* anions. Resonance with π systems stabilises all three. The order of stability of alkyl radicals (3° > 2° > 1° > CH_3) is the same as for carbocations, but the spread is a quarter as large: ΔD = 8.5 versus ΔHCIA ≈ 40 kcal mol^{-1} for cations.
- **Steric effects.** Bulky groups (t-butyl) raise radical stability by kinetic persistence (no combination): the 2,4,6-tri-t-butylphenoxyl radical, TEMPO (a nitroxide), DPPH, galvinoxyl and the triarylmethyls are

persistent or even indefinitely stable. A stable radical can be bottled; a persistent radical lives long because its decay by self-reaction is slow; a transient radical decays at the diffusion limit.

Stable and persistent radicals

Radical	Structure and why it persists	Use
Triphenylmethyl (trityl)	$Ph_3C^\bullet$; steric hindrance to dimerisation at the central carbon; the dimer (Gomberg 1900) is a quinoid compound formed by attack at the para-carbon of one ring (≈ 2 % radical at 0.1 M, 25 °C, in benzene)	first radical ever observed; yellow, reversible with O_2 (gives peroxide)
Nitroxides: TEMPO	2,2,6,6-tetramethylpiperidin-1-oxyl: N–O• with three-electron N∴O bond and no α-hydrogens (no disproportionation)	oxidation catalyst (alcohol → carbonyl), spin label, radical trap in living polymerisation (NMP)
DPPH	2,2-diphenyl-1-picrylhydrazyl: spin delocalised over the hydrazyl N–N and the picryl ring	radical scavenger assay; EPR standard (g = 2.0036)
Galvinoxyl; Fremy's salt	O-centred radicals delocalised through quinone methide; $(KSO_3)_2NO^\bullet$	inhibitor, oxidant (Teuber reaction)
2,4,6-tri-t-butylphenoxyl	ArO•; the t-butyl groups block ortho/para coupling; green, stable for days	model of antioxidant radical
Molecular oxygen	triplet O_2 (ground state is a diradical •O–O•)	traps carbon radicals at diffusion rate; inhibits radical chains and starts autoxidation
NO, NO_2	odd-electron molecules; NO is a persistent radical trap	atmospheric chemistry, nitrosation

9.13 Reactivity and selectivity: the Hammond postulate for hydrogen abstraction

For X• + R–H → X–H + R•, the transition state is *early* (reactant-like) when the step is strongly exothermic and *late* (product-like) when it is endothermic (Hammond). A late TS has substantial radical character on carbon, so the stability of R• is felt strongly in the activation energy; an early TS barely feels it. Hence the more reactive reagent is the less selective.

Halogen radical	ΔH (kcal mol^{-1}) for CH_4	E_a for CH_4	TS	Relative rate per H: 1° : 2° : 3°	Selectivity
F•	–32	≈ 1	very early	1 : 1.2 : 1.4 (25 °C)	almost none
Cl•	+1	3.8	early	1 : 3.8 : 5.0 (25 °C)	low
Br•	+17	18	late	1 : 82 : 1640 (127 °C)	very high
I•	+34	≈ 34	very late	H abstraction does not occur	—

- **Product distribution** = (number of H of that type) × (relative reactivity) / Σ. Propane + Cl_2/hv: 6 × 1 = 6 (1-chloro) versus 2 × 3.8 = 7.6 (2-chloro) → 44 : 56. Isobutane + Cl_2: 9 × 1 = 9 versus 1 × 5.0 = 5 → 64 : 36 (primary chloride is the major product). Isobutane + Br_2: 9 × 1 versus 1 × 1640 → > 99 % tert-butyl bromide. Butane + Br_2: 6 × 1 versus 4 × 82 → 98 % 2-bromobutane.
- **Bromine is selective because the first step is endothermic:** the TS resembles R•, so the energy difference between 1°, 2° and 3° radicals (≈ 2–3 kcal mol^{-1}) appears almost fully in ΔE_a.
- **Chlorine is less selective for two reasons:** an early TS (low E_a) and a high collision-controlled rate. In solvents that complex Cl• (benzene, CS_2, alkenes) the Cl•–π complex is more selective (3° : 1° rises from 5 to 50 in 8 M benzene).
- **Polar effects.** The TS has charge-transfer character: Cl• (electrophilic) prefers electron-rich C–H and avoids H on carbons bearing EWG: chlorination of CH_3CH_2COCl gives mainly the β-chloride ($CH_3CHClCOCl$ is minor), and chlorination of 1-chloropropane gives 1,2-, 1,3- and 1,1-dichloro products in a ratio 50 : 30 : 10 : 10 (approx.), not statistical. Bromine radical is even more discriminating and reacts with toluene at the benzylic position at least 10^5 times faster than with methane.
- **Isotope effects.** k_H/k_D for H abstraction: Cl• ≈ 1.3–1.5 (small, early TS, little C–H stretching), Br• ≈ 4.6 (large, late TS), but the free energy of the abstracted C–H dominates. A competition experiment between $C_6H_5CH_3$ and $C_6H_5CD_3$ brominated with NBS gives k_H/k_D = 4.9, proving C–H cleavage is rate-determining and Br• (not Br_2 ionic path) is the chain carrier.
- **Why fluorination and iodination are not used:** F_2 reacts violently (the C–H abstraction is –32 kcal mol^{-1}, and homolysis of F_2 needs only 38), leading to C–C cleavage; with I_2 the second step is fine but the first (I• + R–H → HI + R•) is +34 kcal mol^{-1} uphill, and HI reduces the alkyl iodide back to RH (RI + HI ⇌ RH + I_2), so iodoalkanes are made by halide exchange (Finkelstein) rather than radical iodination.

9.14 Radical addition to C=C; regiochemistry, polarity and Baldwin–Beckwith rules

- **Regiochemistry.** A radical X• adds to the less hindered carbon of a C=C so that the new radical is the more stable one (benzylic, allylic, tertiary, α to C=O). Addition to styrene occurs at CH_2; addition to methyl acrylate

at CH_2 (giving the α-ester radical); to 1,3-butadiene at C-1 (giving the allylic radical).

- **Polar effects in radical addition.** Nucleophilic radicals (alkyl, α-hydroxyalkyl, α-aminoalkyl) add faster to electron-poor alkenes (acrylonitrile, acrylates), electrophilic radicals ($•CF_3$, $•CCl_3$, Br•, RS•, malonyl radicals) add faster to electron-rich alkenes (vinyl ethers, enamines). This 'polar effect' explains copolymerisation behaviour: styrene (donor) and maleic anhydride (acceptor) copolymerise in strict alternation.
- **Thiyl, bromine, and CCl_3 addition**: Br• and RS• add reversibly; the equilibrium allows cis→trans isomerisation of alkenes by catalytic Br• or RS• (add–rotate–eliminate), the basis of the trans-fat formation and the Z/E isomerisation of oleic to elaidic acid. $CBrCl_3$ and CCl_4 add across alkenes by chain: $Cl_3C•$ + CH_2=CHR → $Cl_3C–CH_2–C•HR$, which abstracts Br (from $CBrCl_3$) to give $Cl_3C–CH_2–CHBr–R$ (Kharasch addition).
- **Anti-Markovnikov addition of HBr (peroxide effect).** Only HBr adds by the radical chain: HCl is too strong an H-donor bond (103 kcal mol^{-1}; the step R• + H–Cl is endothermic by ≈ 4–7); HI adds to the alkene, but I• then adds reversibly and is too unreactive (I• + alkene is endothermic, ΔH ≈ +4): the chain does not propagate. HBr is the only hydrogen halide for which both propagation steps are exothermic or thermoneutral (Br• + alkene ≈ −5, R• + HBr ≈ −8 to −10 kcal mol^{-1}).
- **Cyclisation of ω-alkenyl radicals (Baldwin–Beckwith).** The 5-hexenyl radical cyclises 5-*exo* ($k ≈ 2 × 10^5$ s^{-1} at 25 °C) in preference to 6-*endo* ($k ≈ 4 × 10^3$ s^{-1}), 50 : 1, though the six-membered radical is the more stable; this is stereoelectronic: the chair-like TS of the 5-exo path places the SOMO in better overlap with π* (Beckwith–Houk model). Methyl substituents at C-5 reverse the selectivity (6-endo) by hindering the exo attack. 4-pentenyl (4-exo) and 3-butenyl cyclisations are unfavourable; 6-heptenyl gives 6-exo, 7-endo slowly. Cyclisation onto C ≡ C and C=O is also known (Bu_3SnH-mediated radical cyclisations used in natural-product synthesis).
- **Intermolecular versus intramolecular competition** is controlled by concentration of the H-donor: slow addition of Bu_3SnH (syringe pump, 0.02 M) favours cyclisation; high [Bu_3SnH] gives reduction without ring closure. This is the basis of using a 'radical clock' cyclisation to time a reaction.

9.15 Radical chain reactions: initiation, propagation, termination, inhibition

Stage	Description	Examples
Initiation	formation of the first radicals, usually by homolysis of a weak bond	Cl–Cl (hv, 58); RO–OR (≈ 35); PhCO–O–O–COPh (≈ 30); AIBN (N–C, 31, loses N_2); Br–Br; N–Br of NBS (via HBr + NBS → Br_2); photolysis of ketones (Norrish I, II); Fenton: Fe^{2+} + H_2O_2 → Fe^{3+} + HO^- + HO•
Propagation	each step consumes one radical and produces another; the number of radical produced is conserved	X• + R–H → HX + R•; R• + X_2 → RX + X•. The **kinetic chain length** (steps per initiation) is 10^3–10^6; quantum yield Φ = molecules of product per photon is then ⩾ 1
Termination	two radicals combine or disproportionate	R• + R• → R–R; 2 $CH_3CH_2•$ → CH_3CH_3 + CH_2=CH_2 (disproportionation); radical + wall; radical + inhibitor
Inhibition / retardation	a species that reacts with chain carriers to give unreactive radicals; induction period	O_2, I_2, quinones, phenols (BHT, α-tocopherol), nitroxides, DPPH, NO

- **Test for a chain.** (i) quantum yield Φ ⩾ 1; (ii) rate is first order in [initiator]$^{1/2}$ (because termination is second order in radicals: $R_i = 2k_t[R•]^2$); (iii) strongly inhibited by traces of O_2, hydroquinone or galvinoxyl; (iv) accelerated by hν or initiator; (v) insensitive to polar solvents (a radical rather than ionic mechanism).
- **Why only a small amount of peroxide is needed:** each initiating radical produces a long chain. Peroxide is used catalytically (0.1–1 mol %).
- **Why radical addition to alkenes is not driven by hν alone for HBr:** thermochemistry of the two propagation steps decides whether a chain is possible (HF, HCl, HI fail).
- **Autoxidation** (aerial oxidation) is a radical chain: R• + O_2 → ROO• ($k ≈ 10^9$ $M^{-1}s^{-1}$); ROO• + R–H → ROOH + R• (the rate-determining step, selective for weak C–H bonds: benzylic, allylic, tertiary, α to O). Examples: cumene → cumene hydroperoxide (industrial phenol/acetone synthesis, the Hock process), peroxide formation in ethers and THF (isopropyl ether is dangerous), drying of linseed oil, rancidity of fats (allylic C–H of polyunsaturated acids), ageing of rubber. Antioxidants (BHT, vitamin E) donate H• to ROO•, converting it to an unreactive ArO•.

9.16 Radical polymerisation

Vinyl monomers CH_2=CHX polymerise by a radical chain: **initiation** (In• + CH_2=CHX → $In–CH_2–C•HX$), **propagation** (head-to-tail addition to the less substituted carbon, so the growing radical is always α to X) and **termination** (combination or disproportionation, or chain transfer to monomer, solvent or polymer).

Termination by combination gives chains twice the kinetic chain length.

Polymer	Monomer	Notes
Polyethylene (LDPE)	$CH_2=CH_2$	high pressure (1000–3000 atm), 200 °C, trace O_2 or peroxide; branching through 1,5-backbiting (intramolecular H transfer via a six-membered TS) gives short-chain branches (butyl)
Polystyrene	$PhCH=CH_2$	AIBN or BPO, 60–80 °C; atactic
PVC	$CH_2=CHCl$	chain transfer to monomer dominates
PAN (Orlon, Acrilan)	$CH_2=CHCN$	
PMMA (Perspex, Lucite)	$CH_2=C(CH_3)COOCH_3$	termination mostly by disproportionation; Trommsdorff (gel) effect
Teflon (PTFE)	$CF_2=CF_2$	persulfate initiator, aqueous
Neoprene	$CH_2=CCl–CH=CH_2$ (chloroprene)	1,4-addition through allylic radical
Buna-S (SBR)	butadiene + styrene (3 : 1)	copolymer, emulsion
Buna-N	butadiene + acrylonitrile	oil-resistant rubber
Polyvinyl acetate	$CH_2=CHOCOCH_3$	hydrolysed to PVA

- **Why polymerisation is thermodynamically allowed:** a C=C π bond (≈ 63 kcal mol^{-1}) is replaced by a σ bond (≈ 88), $\Delta H \approx -20$ to -25 kcal mol^{-1} per addition; ΔS is negative, so there is a ceiling temperature ($T_c = \Delta H/\Delta S$), above which depolymerisation wins (α-methylstyrene: $T_c = 61$ °C).
- **Head-to-tail regiochemistry** is due to the preference for attack at the unsubstituted CH_2 (steric access and radical stability).
- **Inhibitors and retarders** (hydroquinone, BHT) are added to stabilise monomers during storage; O_2 is the most common inhibitor in practice.
- **Controlled radical polymerisation:** ATRP (R–X + Cu(I) ⇌ R• + Cu(II)X), RAFT, NMP (TEMPO-capped chain ends) keep the radical concentration very low with a dynamic equilibrium between active radicals and dormant species, giving narrow molecular-weight distributions.

9.17 Fragmentation and rearrangement of radicals

Process	Equation	Comment
β-Scission (C–C)	$R–CH_2–C•H_2 \rightleftharpoons R• + CH_2=CH_2$	reverse of radical addition; important in thermal cracking (~500 °C) and in the combustion and pyrolysis of hydrocarbons; the thermodynamically favoured direction depends on ΔH (about +21 for ethyl → ethene + H•)
Decarboxylation of acyloxy	$RCOO• \rightarrow R• + CO_2$	very fast for alkanoyloxy ($k \approx 10^9$ s^{-1}), slow for aroyloxy (10^6; benzoyloxy partly abstracts or adds to arenes)
Decarbonylation	$RC•=O \rightarrow R• + CO$	easy when R• is stabilised (tertiary, benzylic); the aldehyde chain reaction R–CHO → R–H + CO
Alkoxy radical β-scission	$R_3C–O• \rightarrow R• + R_2C=O$	releases the most stable alkyl radical; alkoxy radical from tert-butyl hydroperoxide gives $CH_3•$ + acetone; used for ring expansion and the Barton nitrite photolysis
1,5-Hydrogen atom transfer (1,5-HAT)	RO• ... H–$C_δ$ via a six-membered chair-like TS	Barton nitrite reaction (nitrite ester photolysis; δ-C–H → oxime), Hofmann–Löffler–Freytag (N-chloroamine, δ-H, pyrrolidine), 1,5-backbiting of PE chains; 1,2-H shifts of radicals do not occur (cf. carbocations)
1,2-Aryl migration (neophyl)	$PhC(CH_3)_2CH_2• \rightarrow •C(CH_3)_2CH_2Ph$	via a spirocyclohexadienyl radical (3-membered bridged); 1,2-vinyl, 1,2-acyl (Dowd–Beckwith), 1,2-Cl, Br shifts (via bridged radicals) also happen; alkyl and H do not
Ring opening of cyclopropylcarbinyl	$•CH_2–cPr \rightarrow CH_2=CH–CH_2CH_2•$	$k \approx 10^8$ s^{-1} at 25 °C: 'radical clock'; cyclobutylcarbinyl 5×10^3 s^{-1}
Dowd–Beckwith ring expansion	ring-bearing $CH_2•$ attached to a ketone α-carbon; adds to C=O then ring-opens	a ring-expansion of β-ketoesters by one or more CH_2 via alkoxy radical intermediate (3-exo attack on C=O)

9.18 Other named radical reactions and their key features

Reaction	Reagents / conditions	Radical intermediate and outcome
Wohl–Ziegler bromination	NBS, CCl_4, hν or peroxide	Br• abstracts allylic/benzylic H; R• + Br_2; Br_2 supplied in tiny concentration from NBS + HBr; allylic transposition for unsymmetrical alkenes

Hunsdiecker–Borodin	RCOOAg + Br_2, CCl_4	RCOOBr → RCOO• → R• + CO_2; R–Br with one carbon less; fails for tertiary acids in some cases and requires dry salt
Kolbe electrolysis	$RCOO^-$, anode (Pt), MeOH, high current density	RCOO• → R•; 2R• → R–R; mixed Kolbe gives all three couplings
Barton decarboxylation	N-acyloxy-2-thiopyridone ester + Bu_3SnH, hν	R• formed by radical-chain fragmentation; reduced to R–H
Sandmeyer	ArN_2^+ + CuCl/CuBr/CuCN	single-electron transfer Cu(I) → ArN_2^+ gives Ar• + N_2; Ar• + CuX_2 → ArX + CuX
Gomberg–Bachmann	ArN_2^+ + Ar′H, NaOH	Ar• adds to Ar′H (ortho/para to substituents, almost regardless of electronic effect), cyclohexadienyl radical, –H•: biaryls in low yield
Pschorr cyclisation	o-substituted diazonium of a stilbene/biarylamine, Cu	intramolecular radical arylation → phenanthrene/fluorenone
Kharasch addition	CCl_4, $CBrCl_3$, ICH_2COOR + alkene, peroxide	R• adds to alkene; adduct radical abstracts X
Anti-Markovnikov HBr	HBr, ROOR	Br• chain (9.14)
Pinacol coupling (Mg/Hg, SmI_2)	ketone + $2e^-$	ketyl radical anion dimerises → 1,2-diol (see Part 13)
Acyloin condensation	RCOOR′ + Na, toluene reflux, then TMSCl	ester radical anion couples to α-diketone → enediolate → α-hydroxy ketone; high yield for ring closure of medium and large rings
Birch reduction	arene + Na/NH_3(l), ROH	arene radical anion, protonation, second electron, protonation: 1,4-cyclohexadiene (Part 13)
Hofmann–Löffler–Freytag	N-chloroamine, H_2SO_4, hν; then base	aminium radical, 1,5-HAT, Cl abstraction, S_N2 ring closure: pyrrolidine
Barton nitrite photolysis	R–CH_2–O–N=O, hν	RO• 1,5-HAT, trapped by NO; δ-nitroso → oxime; used for functionalising the unactivated C-18 methyl of steroids (aldosterone synthesis)
Fenton / Haber–Weiss	Fe^{2+} + H_2O_2	HO• (very reactive, abstracts any C–H, adds to arenes: hydroxylation); oxidative damage in biology
Autoxidation, Hock process	cumene + O_2	cumyl radical → cumyl-OO• → cumene hydroperoxide; acid-catalysed rearrangement (Part 7 type, aryl migration to O) gives phenol + acetone

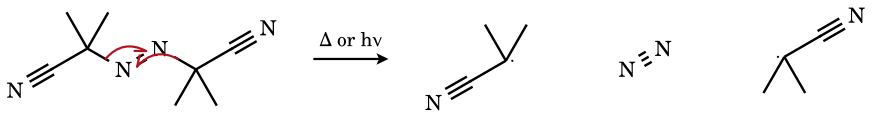
9.19 Stereochemistry of radical reactions

- Radical formation at a stereocentre usually gives a planar (or rapidly inverting) radical, so the product is racemic: bromination of (S)-3-methylhexane at C-3 gives racemic 3-bromo-3-methylhexane.
- Radical addition to a cycloalkene typically gives *trans*-addition products because the second group is delivered anti to the first group already on the ring, e.g. HBr/peroxide addition to 1-methylcyclohexene gives mainly trans-1-bromo-2-methylcyclohexane ([Br•] adds to C-2 equatorial first; H-transfer occurs from the opposite face).
- Radical cyclisation of 2-substituted 5-hexenyl radicals gives mainly cis-1,3 / trans-1,2 product (Beckwith chair rule): 1- or 3-substituents give cis-, 2- or 4-substituents give trans-disubstituted cyclopentanes.
- In a cage, a radical pair can recombine with retention of configuration (Wittig [1,2], Stevens, Barton nitrite) when the pair is held in the solvent cage faster than it diffuses apart.
- A radical α to an existing stereocentre is attacked from the less hindered face (1,2-asymmetric induction), as in the Giese addition to chiral radicals; the selectivity improves at low temperature.

Typical errors in radical questions. (1) Writing H or alkyl 1,2-shifts for radicals. (2) Giving HBr anti-Markovnikov addition for HCl or HI. (3) Counting each H as equally reactive in chlorination of an unsymmetrical alkane; use (number of H) × (relative rate). (4) Treating the radical as charged and applying –I/+I arguments: use bond energies and delocalisation. (5) Forgetting that allyl radicals give both regioisomers (NBS on 1-heptene gives 3-bromo- and 1-bromo-hept-2-ene). (6) Using AIBN/peroxide as a reactant: it is an initiator (catalytic). (7) Assuming a radical chain is operating in the presence of O_2/quinone; those inhibit it. (8) Treating the Sandmeyer reaction as nucleophilic substitution (it is radical-nucleophilic, SRN1-like, with SET from Cu(I)).

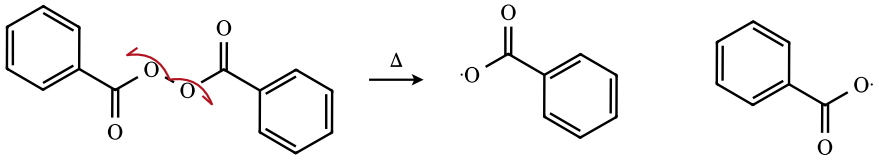
9.20 Additional validated mechanisms

9.12 Initiation: homolysis of AIBN with loss of N_2



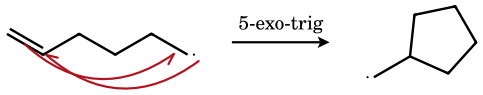
Both C–N bonds break; the two electrons that would remain on N each pair to give the very strong N ≡ N bond (226 kcal mol^{-1}), which is why AIBN decomposes readily at 60–80 °C. One electron of each broken σ bond stays with its carbon, giving two 2-cyano-2-propyl radicals. Each fish-hook ends at the same N–N bond: two arrows = one extra bond order.

9.13 Decarboxylation of the benzoyloxy radical (from benzoyl peroxide)



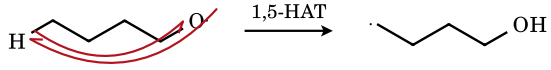
The weak O–O bond (≈ 30 kcal mol^{-1}) breaks homolytically. Benzoyloxy radicals lose CO_2 only slowly (k ≈ 10^6 s^{-1}) so many add to the monomer or arene before decarboxylating, unlike alkanoyloxy radicals, which lose CO_2 within nanoseconds.

9.14 5-Exo cyclisation of the 5-hexenyl radical



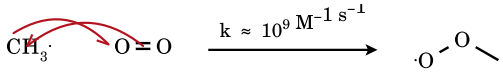
The SOMO attacks the nearer carbon of the alkene (C-5), forming a five-membered ring; the π electron pair splits, one forming the new σ bond and the other left on C-6 (primary, exocyclic cyclopentylmethyl radical). The product radical is less stable than the 6-endo secondary alternative, yet 5-exo is faster by a factor of ≈ 50 (stereoelectronic control, Beckwith). With Bu_3SnH the radical is reduced to methylcyclopentane.

9.15 1,5-Hydrogen atom transfer in an alkoxy radical (Barton / Hofmann–Löffler type)



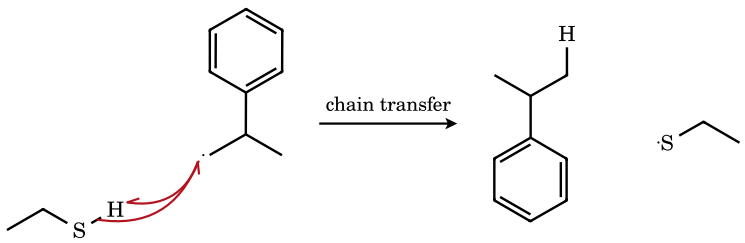
The oxygen radical reaches the δ-hydrogen through a six-membered, chair-like transition state; the O–H bond formed (105 kcal mol^{-1}) is stronger than the C–H bond broken (99), so the step is exothermic by ≈ 6 kcal mol^{-1}, with a very low barrier (k ≈ 10^7 s^{-1}). The new carbon radical is trapped by NO (Barton) or Cl (Hofmann–Löffler–Freytag).

9.16 Autoxidation: carbon radical traps O_2



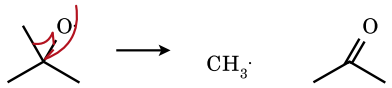
Triplet O_2 is a diradical, so it adds to a carbon radical at nearly the diffusion-controlled rate to give the peroxyl radical R–O–O•. The peroxyl radical then abstracts H from another molecule of R–H (slow, rate-determining) to give R–OOH and propagate the chain. This is why O_2 inhibits other radical chains (it traps R• faster than R• can react with the substrate) and why ethers, aldehydes and alkenes slowly oxidise in air.

9.17 Reduction by chain transfer: alkyl radical + thiol (RS–H) in radical polymerisation



Thiols have weak S–H bonds (≈ 87 kcal mol^{-1}), so they are excellent H-atom donors; a growing polymer radical abstracts the thiol hydrogen to give a dead chain and a thiyl radical, which re-initiates polymerisation. This ‘chain transfer’ reduces the average molecular weight (mercaptans are used as molecular-weight regulators in SBR production, e.g. dodecyl mercaptan).

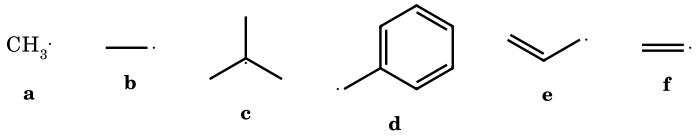
9.18 β-Scission of the tert-butoxy radical



The oxygen radical electron and one electron of an adjacent C–C σ bond form the C=O π bond (the stronger C=O compensates for the lost C–C), while the other electron stays on CH_3•. This competes with H-abstraction by t-BuO•: in non-H-donor solvents at higher temperatures it gives acetone and CH_3• (the source of methane/ethane in t-BuOOt-Bu photolysis).

9.21 Problems on free radicals

FR-1.
Arrange in decreasing stability: (a) CH_3•, (b) CH_3CH_2•, (c) $(CH_3)_3C$•, (d) $PhCH_2$•, (e) CH_2=CH–CH_2•, (f) CH_2=CH•.



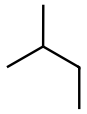
Answer: d > e > c > b > a > f
Benzyl (resonance over the ring) > allyl (two forms) > tertiary (hyperconjugation, 9 C–H) > primary > methyl > vinyl (σ radical, 33 % s). BDE of the parent C–H: 89.8, 88, 96.5, 101, 105, 111 kcal mol^{-1} (⚑ benzylic and allylic BDEs are within 2 kcal mol^{-1}; the key follows 'more resonance forms').

FR-2.
Arrange the C–H bonds in decreasing bond strength: (a) C_6H_5–H, (b) CH_2=CH–H, (c) CH_3–H, (d) $(CH_3)_3C$–H, (e) $PhCH_2$–H, (f) CH_2=$CHCH_2$–H.
Answer: a > b > c > d > e > f
The stronger the C–H, the less stable the radical formed: 112, 111, 105, 96.5, 89.8, 88 kcal mol^{-1}.

FR-3.
Propane is chlorinated and brominated (light). Using relative reactivities (1° : 2° = 1 : 3.8 for Cl; 1 : 82 for Br per H), find the percentage of 2-halopropane.
Answer: Cl: 56 %; Br: 96.5 %
Cl: (2 × 3.8)/(6 × 1 + 2 × 3.8) = 7.6/13.6 = 56 %. Br: (2 × 82)/(6 + 164) = 164/170 = 96.5 %. Bromine is more selective because its H-abstraction TS is late and resembles the radical.

FR-4.
Isobutane is chlorinated with Cl_2 in diffuse sunlight. Which chloride is the major product?
Answer: (B) isobutyl chloride (≈ 64 %)
9 primary H × 1.0 = 9; 1 tertiary H × 5.0 = 5. The numerical advantage of the nine primary H outweighs the reactivity of the single tertiary H (with Br_2 the tertiary bromide would predominate: 9 : 1600).

FR-5.
How many monochlorinated products (without stereoisomers) can be obtained from 2-methylbutane, and how many with stereoisomers?

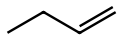


Answer: 4; 6
H types: CH_3–C (6 H, 1-chloro-3-methylbutane... see below); C(CH_3)H (1 H); CH_2 (2 H); CH_3–CH_2 (3 H). Four structural isomers: 1-chloro-2-methylbutane (chiral), 2-chloro-2-methylbutane, 2-chloro-3-methylbutane (chiral), 1-chloro-3-methylbutane. Two are chiral, so 4 + 2 = 6 with stereoisomers.

FR-6.
Which alkane gives only one monochloro derivative? (a) n-pentane, (b) isopentane, (c) neopentane, (d) cyclopentane, (e) 2,2,3,3-tetramethylbutane.
Answer: (c), (d), (e)
All H are equivalent in neopentane (12 H), cyclopentane (10 H) and hexamethylethane (18 H).

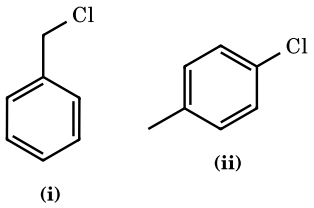
FR-7.
Explain why HBr adds to propene anti-Markovnikov in the presence of peroxide while HCl and HI do not.
Answer: thermochemistry of the two propagation steps
Step 1 (X• + C=C): ΔH = −25.1 (Br), −67 (Cl), +46 (I) kJ mol^{-1}; step 2 (R• + HX): ΔH = −50.2 (Br), +12.6 (Cl), −117.1 (I) kJ mol^{-1}. Only for HBr are both steps exothermic. For HCl step 2 is endothermic (C–H bond formed is not strong enough to pay for breaking the strong H–Cl bond); for HI step 1 is endothermic (the weak C–I bond) and I• mostly dimerises.

FR-8.
NBS/peroxide is reacted with 1-butene. Name the products.



Answer: 3-bromobut-1-ene and 1-bromobut-2-ene
Br• removes an allylic H from C-3 giving CH_2=CH–C•H–CH_3 ⇔ •CH_2–CH=CH–CH_3. Br_2 reacts at either terminus. The secondary bromide has a chiral centre (racemic); the primary bromide is a mixture of E and Z (E major).

FR-9.
Toluene is treated with (i) Cl_2, light, (ii) Cl_2, Fe. Name the main products.

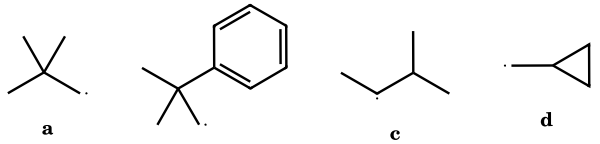


(i) (ii)

Answer: (i) benzyl chloride (then benzal chloride, benzotrichloride); (ii) o- and p-chlorotoluene
(i) Light gives Cl•, which abstracts the benzylic H (89.8 kcal mol^{-1}): a radical chain at the side chain. (ii) Fe gives $FeCl_3$, a Lewis acid that makes Cl^+: electrophilic substitution on the ring.

FR-10.
Electrolysis of (i) potassium propanoate and (ii) an equimolar mixture of sodium acetate and sodium propanoate. Give the hydrocarbons.
Answer: (i) n-butane; (ii) ethane, propane, n-butane (≈ 1 : 2 : 1)
Kolbe: 2 $RCOO^-$ → R–R + 2 CO_2 + 2e^-. With a mixture the radicals CH_3• and C_2H_5• combine at random: CH_3–CH_3, C_2H_5–C_2H_5 (butane), CH_3–C_2H_5 (propane, statistically twice as probable). The solution becomes alkaline (HO^- forms at the cathode).

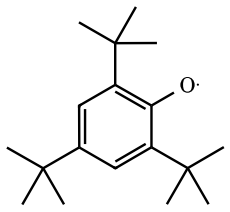
FR-11.
Which of the following radicals will undergo a 1,2-shift? (a) neopentyl, (b) 2,2-dimethyl-2-phenylethyl (neophyl), (c) 3-methyl-2-butyl, (d) cyclopropylcarbinyl (ring opening).



a b c d

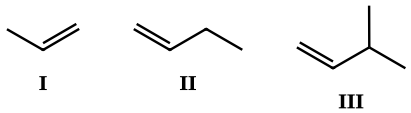
Answer: (b) 1,2-aryl; (d) is a ring-opening (β-scission), not a shift
Radicals do not undergo 1,2-H or 1,2-methyl shifts, so (a) and (c) do not rearrange (unlike the corresponding cations). The neophyl radical undergoes 1,2-phenyl shift to the tertiary radical via the spiro intermediate.

FR-12.
2,4,6-Tri-tert-butylphenoxyl radical is stable and does not dimerise. Why?



Answer: delocalisation + steric protection
The unpaired electron is delocalised over O and the ortho and para carbons; these positions are blocked by tert-butyl groups, so the radical cannot dimerise through C–O or C–C bonds without large strain.

FR-13.
Rank in decreasing rate of free-radical allylic substitution (NBS): (I) $CH_3CH=CH_2$, (II) $CH_3CH_2CH=CH_2$, (III) $(CH_3)_2CHCH=CH_2$.



I II III

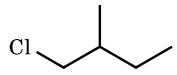
Answer: III > II > I (Nucleus Q40)
III has a tertiary allylic H (weakest allylic C–H, most stable radical); II has a secondary allylic H; I a primary allylic H.

FR-14.
Match: (A) peroxides, (B) NBS, (C) AIBN, (D) HI; with (P) source of Br_2 at low concentration, (Q) source of RO•, (R) source of alkyl radicals + N_2, (S) reducing agent for R–I.
Answer: A–Q; B–P; C–R; D–S
Dialkyl or diacyl peroxides give alkoxy/acyloxy radicals; NBS maintains low [Br_2]; AIBN gives $(CH_3)_2C•CN + N_2$; HI reduces alkyl iodides to alkanes (the reverse of iodination).

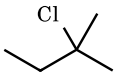
FR-15.
Why does the photochlorination of ethane-d_6 proceed more slowly than that of ethane?
Answer: primary kinetic isotope effect
The C–D bond has a lower zero-point energy and is stronger (≈ 1.2 kcal mol^{-1}), so the H-abstraction step, which is rate-determining, is slower (k_H/k_D ≈ 1.3–2 for Cl•; Nucleus Q47: II < I < III with neopentane the fastest).

FR-16.
An optically active alkane $R^1R^2R^3C–H$ is brominated at the stereocentre by NBS. What is the stereochemical outcome?
Answer: racemic bromide
The radical formed at the stereocentre is planar (or rapidly inverting), so Br_2 attacks from either face with equal probability.

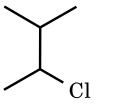
FR-17.
Chlorination of 2-methylbutane (C_5H_{12}) at 25 °C: relative rates per H are 1° : 2° : 3° = 1 : 3.8 : 5. Calculate the percentage of each monochloro constitutional isomer.



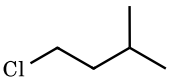
1-chloro-2-methylbutane



2-chloro-2-methylbutane



2-chloro-3-methylbutane



1-chloro-3-methylbutane

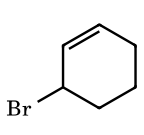
Answer: 27.8 % : 23.1 % : 35.2 % : 13.9 %
Weighted counts: the two CH_3 on C-2 (6 H × 1 = 6) → 1-chloro-2-methylbutane; tertiary C-2 H (1 × 5 = 5) → 2-chloro-2-methylbutane; CH_2 (2 × 3.8 = 7.6) → 2-chloro-3-methylbutane; terminal CH_3 (3 × 1 = 3) → 1-chloro-3-methylbutane. Total 21.6; shares 6/21.6, 5/21.6, 7.6/21.6, 3/21.6.

FR-18.
How many monochlorination products (including stereoisomers) can be formed from 2-methylbutane? How many constitutional isomers are chiral?
Answer: 4 constitutional isomers, 6 including stereoisomers; 2 of the constitutional isomers are chiral
1-Chloro-2-methylbutane (stereocentre C-2: 2 enantiomers) and 2-chloro-3-methylbutane (stereocentre C-2: 2 enantiomers) are chiral; 2-chloro-2-methylbutane and 1-chloro-3-methylbutane are achiral. 2 + 2 + 1 + 1 = 6.

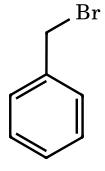
FR-19.
At 127 °C, bromination of propane gives 2-bromopropane almost exclusively (97 %), while chlorination gives a 45 : 55 mixture. Explain with the Hammond postulate and give the energy diagrams in words.
Answer: late versus early transition state
Br• + R–H is endothermic (+17 for CH_4; ≈ +7 for secondary C–H), so the TS resembles the alkyl radical and the 6.5-kcal gap between 1° and 2° radicals appears almost fully in ΔE_a (≈ 2–3 kcal mol^{-1}). Cl• + R–H is exothermic for 1°/2° C–H, an early TS in which little radical character has developed, so the barrier difference is only ≈ 0.5 kcal mol^{-1}.

FR-20.
Why does HBr but not HCl or HI add to propene by the radical chain? Give ΔH of the two propagation steps.
Answer: only HBr has two exothermic propagation steps
Approximate ΔH (kcal mol^{-1}) for X• + CH_2=$CHCH_3$ and R• + HX: Cl: −22 and +3 (second step is endothermic); Br: −5 and −10 (both favourable); I: +4 and −26 (first step is endothermic, reversible, and the I• adds reversibly so no chain develops). Hence the 'peroxide effect' is restricted to HBr.

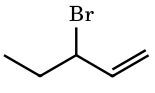
FR-21.
NBS/CCl_4/hν is used with each of: (a) cyclohexene, (b) toluene, (c) 1-pentene, (d) propene, (e) isobutane, (f) methylcyclohexane. Give the major product or products.



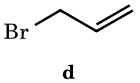
a



b



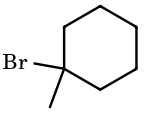
c



d



e

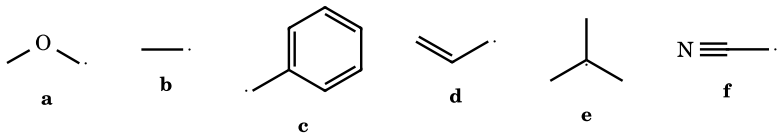


f

Answer: (a) 3-bromocyclohexene; (b) benzyl bromide; (c) 3-bromo-1-pentene and 1-bromo-2-pentene (mixture); (d) allyl bromide; (e) t-butyl bromide; (f) 1-bromo-1-methylcyclohexane
NBS brominates the weakest C–H: allylic, benzylic, or tertiary. For 1-pentene the allyl radical CH_2=CH–C•H–CH_2CH_3 ⇔ •CH_2–CH=CH–CH_2CH_3 reacts at either end; the internal (more substituted, more stable) alkene 1-bromo-2-pentene predominates.

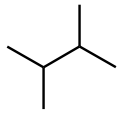
FR-22.
In the Hunsdiecker reaction of silver 3-phenylpropanoate with Br_2, the product is $PhCH_2CH_2Br$. Write the radical steps and say how the product differs from the acid in carbon count.
Answer: $PhCH_2CH_2Br$: one carbon less than the acid
$RCOOAg + Br_2 \rightarrow RCOOBr + AgBr$; $RCOOBr \rightarrow RCOO^\bullet + Br^\bullet$ (initiation); $RCOO^\bullet \rightarrow R^\bullet + CO_2$; $R^\bullet + RCOOBr \rightarrow RBr + RCOO^\bullet$ (propagation). The acid $PhCH_2CH_2COOH$ has 9 carbon, the bromide $PhCH_2CH_2Br$ has 8.

FR-23.
Arrange the following radicals in increasing order of stability: (a) $CH_3OCH_2^\bullet$, (b) $CH_3CH_2^\bullet$, (c) $PhCH_2^\bullet$, (d) $CH_2{=}CHCH_2^\bullet$, (e) $(CH_3)_3C^\bullet$, (f) $^\bullet CH_2CN$.



Answer: b < e < a < f < d ≈ c (⚑ d vs c)
RSE values: 4, 8.5, ≈ 9–10 (α-oxygen lone pair), ≈ 11–12 (cyano conjugation), 17, 15. Position of c versus d depends on the data source; most JEE keys put benzyl above allyl on resonance count.

FR-24.
A hydrocarbon C_6H_{14} gives only two monochlorination products. Identify it and the ratio of the products if relative reactivities are 1 : 3.8 : 5.



Answer: 2,3-dimethylbutane; $(CH_3)_2CHCH(CH_3)_2$: 12 primary H : 2 tertiary H → 12 × 1 : 2 × 5 = 12 : 10
The two halves are identical, with 12 primary and 2 tertiary H. 1-chloro-2,3-dimethylbutane : 2-chloro-2,3-dimethylbutane = 54.5 % : 45.5 %. (2,2-dimethylbutane has three products; hexane has three.)

FR-25.
Optically active (S)-1-phenylethyl chloride is treated with Bu_3SnD / AIBN. What is the stereochemistry of the product $PhCHDCH_3$?
Answer: racemic
$Bu_3Sn^\bullet$ abstracts Cl to give the planar (benzylic) radical; deuterium transfer from Bu_3SnD occurs from either face with equal probability, giving a racemate.

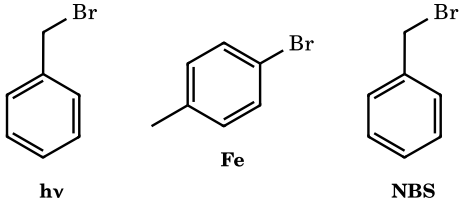
FR-26.
Account for the fact that trans-2-butene is formed from cis-2-butene on adding a trace of Br_2 and light (no net addition).
Answer: Br• adds reversibly, the radical rotates, and Br• is eliminated
$Br^\bullet$ + cis-$CH_3CH{=}CHCH_3 \rightarrow CH_3C^\bullet H{-}CHBrCH_3$. Free rotation about the C–C single bond (≈ 1000 times faster than Br• loss) before Br• leaves gives the more stable trans isomer. Catalytic Br• or RS• thus equilibrates alkenes to the thermodynamic mixture.

FR-27.
Predict the product of the addition of HBr to (a) propene and (b) 3,3,3-trifluoropropene, in the presence of peroxide; (c) in the dark with no peroxide. Explain the (b) result.
Answer: (a) 1-bromopropane; (b) $CF_3CH_2CH_2Br$; (c) 2-bromopropane for propene (ionic)
(a) Br• adds at C-1, giving the secondary radical. (b) $CF_3CH{=}CH_2$: Br• adds at the CH_2 terminus to give $CF_3{-}C^\bullet H{-}CH_2Br$ or at CH to give the primary radical; the CF_3 group destabilises the adjacent radical and cation (both polar and field effect), so the experimental product is $CF_3CH_2CH_2Br$ by either mechanism. (c) Ionic addition follows Markovnikov (2-bromopropane).

FR-28.
The rate of polymerisation of styrene initiated by AIBN is proportional to $[AIBN]^{1/2}$ and [styrene]. Derive this from the steady-state hypothesis.
Answer: $R_p = k_p[M](f\,k_d[I]/k_t)^{1/2}$
Steady-state: rate of initiation = rate of termination: $2fk_d[I] = 2k_t[M^\bullet]^2 \Rightarrow [M^\bullet] = (fk_d[I]/k_t)^{1/2}$. Rate of propagation = $k_p[M^\bullet][M]$. Substituting gives $R_p \propto [M][I]^{1/2}$. The half-order in initiator is the signature of bimolecular radical termination.

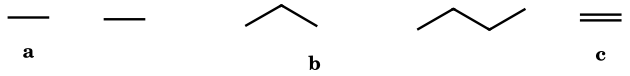
FR-29.
What is the average degree of polymerisation if the kinetic chain length is 500 and termination is (a) entirely by combination, (b) entirely by disproportionation?
Answer: (a) 1000; (b) 500
Combination joins two kinetic chains in one macromolecule (DP = 2ν); disproportionation gives two molecules, each having one kinetic chain (DP = ν). PMMA terminates mainly by disproportionation; polystyrene by combination.

FR-30.
Identify the product of benzoyl peroxide-initiated bromination of the following: toluene + Br_2 (hν), toluene + Br_2 (Fe), toluene + NBS. State what changes.



Answer: benzyl bromide (side chain, radical); o- and p-bromotoluene (ring, ionic); benzyl bromide
Light or radical initiator and non-polar solvent: radical substitution at the weakest C–H (benzylic, 89.8). A Lewis acid ($FeBr_3$) generates Br^+ equivalents for electrophilic aromatic substitution.

FR-31.
Write the products of the Kolbe electrolysis of (a) CH_3COONa, (b) a 1 : 1 mixture of CH_3COONa and CH_3CH_2COONa, (c) sodium succinate $NaOOC–CH_2CH_2–COONa$.



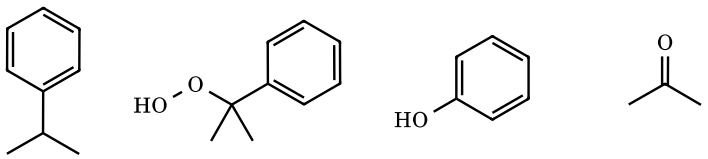
Answer: (a) ethane; (b) ethane + propane + butane (1 : 2 : 1 by statistics); (c) ethene
(a) 2 $CH_3\bullet \rightarrow C_2H_6$. (b) $CH_3\bullet/C_2H_5\bullet$ couple randomly: 25 % $CH_3–CH_3$, 50 % $CH_3CH_2CH_3$ (propane), 25 % butane. (c) the diradical $\bullet CH_2CH_2\bullet$ forms ethene.

FR-32.
How many monobromo products are possible on treating methylcyclohexane with Br_2/hν? Which one will predominate and why?
Answer: 5 constitutional isomers; the tertiary bromide
H types: CH_3 (1°), C-1 tertiary, C-2 (×2), C-3 (×2), C-4 → 5 constitutional isomers (counting C-2/C-6 and C-3/C-5 as pairs: five different environments in total). Br• discriminates 1600 : 82 : 1, so 1-bromo-1-methylcyclohexane forms almost exclusively.

FR-33.
State and explain the product of the following: (a) $CH_3CH_2CH=CH_2$ + HBr (ROOR); (b) $CH_3CH=CHCH_3$ + HBr (ROOR); (c) $CH_2=CH–CH=CH_2$ + HBr (ROOR).
Answer: (a) 1-bromobutane; (b) 2-bromobutane; (c) 1-bromo-2-butene (mainly) and 3-bromo-1-butene
(a) Br• adds to the terminal CH_2. (b) symmetric alkene gives only 2-bromobutane. (c) Br• adds at C-1 to give a resonance-stabilised allylic radical $CH_2Br–CH\bullet–CH=CH_2 \leftrightarrow CH_2Br–CH=CH–C\bullet H_2$; H-transfer at either end gives 3-bromo-1-butene (kinetic, 1,2-adduct) and 1-bromo-2-butene (1,4, thermodynamic).

FR-34.
A neophyl-type radical $PhC(CH_3)_2CH_2\bullet$ rearranges, but $CH_3C(CH_3)_2CH_2\bullet$ (neopentyl) does not. Why? Which other groups can migrate?
Answer: Ph, vinyl, acyl, Cl/Br can; H and alkyl cannot
Migration needs a low-energy bridged (three-membered) radical TS. For phenyl, the π system provides a stabilised spiro-cyclohexadienyl radical; for H or alkyl, the bridging TS has three centres and three electrons with no stabilisation, and the barrier is much higher than that for intermolecular abstraction. Neopentyl radical therefore abstracts H or fragments instead.

FR-35.
Cumene (isopropylbenzene) is treated with O_2 at 100 °C with a trace of initiator, then with acid. Give the radical chain and the final products.



Answer: cumene hydroperoxide, then phenol + acetone (Hock process)
In• + $PhCH(CH_3)_2$ → $PhC\bullet(CH_3)_2$ (tertiary benzylic); + O_2 → $PhC(CH_3)_2OO\bullet$; + cumene → $PhC(CH_3)_2OOH$ + cumyl radical. In acid, the hydroperoxide protonates, loses water as the phenyl migrates from C to O (cationic, Part 7-type shift) to give an oxocarbenium ion, which hydrolyses to phenol and acetone.

FR-36.
Why is the termination step R• + R• → R–R rare in a propagating chain even though it has a rate constant of 10^9 M^{-1} s^{-1}?

Answer: radical concentration is ≈ 10^{-8} M
Rate of termination = $k[R•]^2 = 10^9 \times 10^{-16} = 10^{-7}$ M s^{-1}, but propagation = $k_p[R•][M] \approx (10^2–10^4) \times 10^{-8} \times 1$ M = 10^{-6}–10^{-4} M s^{-1}. Propagation is first order in radical concentration and thus beats the second-order termination until [R•] is high (as in the Trommsdorff gel effect, where termination is slowed by high viscosity).

FR-37.
Which of the following compounds is the best radical inhibitor? (a) benzene, (b) hydroquinone, (c) cyclohexane, (d) benzoyl peroxide, (e) cumene.

Answer: (b) hydroquinone
It donates an H atom (O–H BDE ≈ 80 kcal mol^{-1}) to give a resonance-stabilised semiquinone radical that does not propagate; (d) is an initiator; (a), (c) have strong C–H bonds; (e) is easily autoxidised.

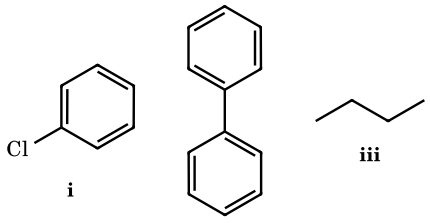
FR-38.
In the HBr addition to 1-hexene catalysed by peroxide, the chain length is about 10^4. How many HBr molecules are consumed per radical formed by peroxide? What changes if 1 % hydroquinone is added?

Answer: 10^4; the reaction is inhibited (induction period)
Each initiating radical starts a chain that adds ≈ 10^4 molecules before termination. Hydroquinone scavenges chain-carrying radicals, so the reaction does not start until it is consumed (induction period), and the product reverts to the ionic Markovnikov addition (2-bromohexane), a slower reaction.

FR-39.
Name the stable free radical among (a) CH_3•, (b) Ph_3C•, (c) TEMPO, (d) t-BuO•, (e) Cl•, and say which of these gives a stable but reactive-solution radical (Gomberg 1900).

Answer: (c) TEMPO is stable (isolable); (b) trityl persists in solution but dimerises on concentration
(c) The four methyl groups block the α-positions, so there is neither disproportionation (no α-H) nor combination; the odd electron is delocalised over N–O. (b) persistent: steric hindrance prevents C–C coupling at the central carbon; the dimer is formed through the para position of one ring.

FR-40.
Complete: (i) Ph–$N_2^+Cl^-$ + Cu powder → ?; (ii) Ph–N_2^+ + C_6H_6 + NaOH → ?; (iii) $CH_3CH_2COO^-$ (anodic oxidation) → ?



i

ii

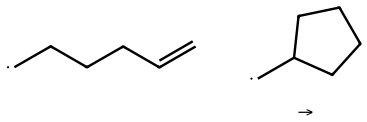
iii

Answer: (i) chlorobenzene (Gattermann); (ii) biphenyl (Gomberg–Bachmann); (iii) butane (Kolbe)
All three involve a radical (Ph• or CH_3CH_2•). In (i) the Cu gives an electron to the diazonium and Cu(II)Cl_2 delivers Cl.

FR-41.
How many different monobromo-alkenes can in principle form on allylic bromination of 1-methylcyclohexene with NBS (constitutional isomers only)? Which positions are allylic?

Answer: 6
Allylic hydrogens are on the CH_3, C-3 and C-6, giving three allylic radicals, each delocalised over two carbons: CH_3-radical → 1-(bromomethyl)cyclohexene and 2-bromo-1-methylenecyclohexane; C-3 radical → 3-bromo-1-methylcyclohexene and 1-bromo-1-methylcyclohex-2-ene; C-6 radical → 6-bromo-1-methylcyclohexene and 2-bromo-1-methylcyclohex-1-ene-type isomer in which the double bond has moved to C-1=C-6. Six distinct constitutional isomers; the most stable (and so usually the major) are those retaining a trisubstituted double bond.

FR-42.
The 5-hexenyl radical can cyclise in two ways. Name them, give the major product and the reason.


→

Answer: 5-exo (cyclopentylmethyl) major; 6-endo (cyclohexyl) minor
5-exo-trig versus 6-endo-trig. The exo mode has better SOMO–π* overlap in a chair-like TS; the ratio is about 98 : 2 at 25 °C. A methyl at C-5 raises the exo barrier sterically and reverses the preference.

FR-43.

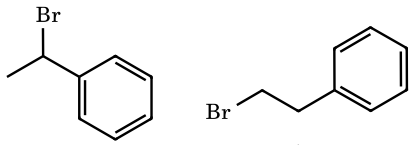
1-Bromobicyclo[2.2.2]octane is used as a substrate for radical (Bu_3SnH) reduction and as a substrate for S_N1 solvolysis. Which proceeds easily, and why?

Answer: radical reduction proceeds; S_N1 does not

A bridgehead carbocation cannot become planar and is highly destabilised, so S_N1 is extremely slow. A radical only needs a modest degree of delocalisation and tolerates pyramidal geometry (radical inversion barriers are tiny), so the bridgehead radical forms readily from the bromide with $Bu_3Sn^\bullet$ ($\approx 10^5\ M^{-1}\ s^{-1}$).

FR-44.

Account for the ratio of products 1-bromo-1-phenylethane : 2-bromo-1-phenylethane = > 99 : 1 in the NBS bromination of ethylbenzene.



major **minor**

Answer: benzylic H is 10^3 times more reactive than the terminal CH_3 H

The secondary benzylic C–H (BDE $\approx$ 85.4) is 12–15 kcal mol^{-1} weaker than the terminal C–H, and $Br^\bullet$ selects the weak bond.

FR-45.

Using bond energies, show that $Cl^\bullet + CH_4 \rightarrow HCl + {}^\bullet CH_3$ is nearly thermoneutral while $Br^\bullet + CH_4 \rightarrow HBr + {}^\bullet CH_3$ is endothermic by 17 kcal mol^{-1}, and use this to explain why Br_2 is more selective than Cl_2.

Answer: $\Delta H(Cl) = 105 - 103 = +2$ (+1); $\Delta H(Br) = 105 - 87.5 = +17.5$

$\Delta H = D(C–H) - D(H–X)$. The late (endothermic) TS of bromination resembles the alkyl radical product, so differences in radical stability are felt in the activation energy (Hammond), making bromination selective; the early TS of chlorination does not.

FR-46.

(JEE Advanced style, integer) How many of the following will give the same single monohalogenation product upon radical bromination with one equivalent of Br_2/hv: ethane, neopentane, cyclohexane, benzene, toluene (side-chain), 2,2,3,3-tetramethylbutane, adamantane?

Answer: 5 (ethane, neopentane, cyclohexane, toluene, 2,2,3,3-tetramethylbutane; adamantane gives two isomers, tertiary and secondary; benzene does not react by radical substitution)

Ethane (1 type of H), neopentane (all 12 H equivalent), cyclohexane (12 equivalent H), toluene (only the benzylic H react), hexamethylethane (18 equivalent H). Adamantane has 12 secondary and 4 tertiary H, with $Br^\bullet$ giving mainly tertiary (1-bromoadamantane).

FR-47.

2-Methylbutane is brominated with Br_2/hv at 127 °C (relative reactivity per H: 1° : 2° : 3° = 1 : 82 : 1640). Calculate the percentage of 2-bromo-2-methylbutane among the monobromides.

Answer: $\approx$ 90.5 %

H count: 9 primary (two CH_3 on C-2 plus the C-4 CH_3), 2 secondary, 1 tertiary. Weights: $9 \times 1 = 9$; $2 \times 82 = 164$; $1 \times 1640 = 1640$. Fraction tertiary = 1640/1813 = 0.905.

FR-48.

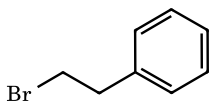
Using D(C–H) = 105 (CH_3–H), D(Cl–Cl) = 58, D(H–Cl) = 103, D(CH_3–Cl) = 85 kcal mol^{-1}, calculate ΔH for each propagation step and for the overall chlorination of methane.

Answer: step 1 +2; step 2 –27; overall –25 kcal mol^{-1}

Step 1: $CH_4 + Cl^\bullet \rightarrow CH_3^\bullet + HCl$: 105 − 103 = +2. Step 2: $CH_3^\bullet + Cl_2 \rightarrow CH_3Cl + Cl^\bullet$: 58 − 85 = −27. Overall = 105 + 58 − 103 − 85 = −25. The slightly endothermic first step is the rate-determining one ($E_a \approx 4$).

FR-49.

Styrene is treated with HBr in the presence of benzoyl peroxide. Give the product, name the intermediate radical and explain why no 1-bromoethylbenzene forms.

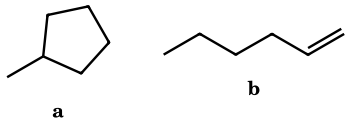


Answer: (2-bromoethyl)benzene via the benzylic radical

$Br^\bullet$ adds to the terminal CH_2 to give $PhC^\bullet H–CH_2Br$, a benzylic radical delocalised over the ring. Attack at the other carbon would give a primary radical with no stabilisation. The benzylic radical abstracts H from HBr (chain transfer). Only HBr has both propagation steps favourable.

FR-50.
(Barton–McCombie) A secondary alcohol is converted to its methyl xanthate ROC(=S)SMe and then treated with Bu_3SnH/AIBN. Give the product and the driving force.
Answer: R–H (deoxygenation); formation of a strong C=O (and Sn–S) bond
$Bu_3Sn^\bullet$ adds to the thiocarbonyl sulfur to give a stabilised radical, which fragments with C–O cleavage to $R^\bullet$ and Bu_3SnS–C(=O)SMe (new C=O bond). $R^\bullet$ then abstracts H from Bu_3SnH, regenerating $Bu_3Sn^\bullet$. It deoxygenates secondary alcohols that would eliminate under ionic conditions.

FR-51.
6-Bromohex-1-ene is treated with Bu_3SnH/AIBN at (a) 0.01 M and (b) 1 M Bu_3SnH. Which product predominates in each case?

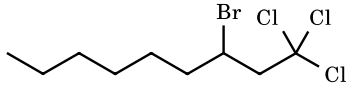


Answer: (a) methylcyclopentane; (b) hex-1-ene
The 5-hexenyl radical either cyclises (first-order, $k \approx 2 \times 10^5\ s^{-1}$) or is reduced by Bu_3SnH (second-order, $k \approx 2 \times 10^6\ M^{-1}\ s^{-1}$). At 0.01 M the reduction rate is $\approx 2 \times 10^4\ s^{-1}$, so cyclisation wins; at 1 M it is $\approx 2 \times 10^6\ s^{-1}$, so direct reduction wins. Rate constants are approximate.

FR-52.
Calculate ΔH for $Br^\bullet + (CH_3)_3C–H$ and $Cl^\bullet + (CH_3)_3C–H$ using D(C–H) = 96.5, D(H–Br) = 87.5, D(H–Cl) = 103 kcal mol^{-1}, and say what this predicts about selectivity.
Answer: +9 ($Br^\bullet$); –6.5 ($Cl^\bullet$)
ΔH = D(C–H) − D(H–X). Endothermic abstraction by $Br^\bullet$ gives a late, radical-like transition state, so it discriminates strongly among 1°/2°/3° C–H; exothermic abstraction by $Cl^\bullet$ has an early transition state and low selectivity.

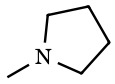
FR-53.
In a radical polymerisation $R_p = 1 \times 10^{-4}\ M\ s^{-1}$ and $R_i = 1 \times 10^{-8}\ M\ s^{-1}$. Find the kinetic chain length and the number-average degree of polymerisation if termination is (a) all by combination, (b) all by disproportionation.
Answer: $\nu = 10^4$; (a) 2×10^4; (b) 10^4
$\nu = R_p/R_i = 10^4$ monomer units per initiating radical. Combination joins two growing chains (DP = 2ν); disproportionation gives one dead chain per radical (DP = ν). Chain transfer is ignored.

FR-54.
1-Octene is heated with bromotrichloromethane and a peroxide. Give the product and the two propagation steps.



Answer: 3-bromo-1,1,1-trichlorononane
$Cl_3C^\bullet$ adds to the terminal CH_2 to give the secondary radical, which abstracts Br from $BrCCl_3$ (weak C–Br bond), regenerating $Cl_3C^\bullet$ (Kharasch addition).

FR-55.
N-Chloro-N-methylbutylamine is irradiated in conc. H_2SO_4 and then made basic. Give the product and the key radical step.



Answer: N-methylpyrrolidine (Hofmann–Löffler–Freytag)
The N-centred aminium radical abstracts the δ-hydrogen through a six-membered transition state (1,5-HAT); the C-radical takes Cl from another N-chloroammonium (chain). On basification the free δ-chloroamine cyclises by intramolecular S_N2.

FR-56.
How does BHT (2,6-di-tert-butyl-4-methylphenol) inhibit autoxidation, and why does the resulting radical not propagate the chain?
Answer: it donates its O–H hydrogen to $ROO^\bullet$; the aryloxyl radical is stabilised and hindered
$ROO^\bullet + ArOH \rightarrow ROOH + ArO^\bullet$. The O–H bond is weak ($\approx$ 80 kcal mol^{-1}), the unpaired electron is delocalised into the ring, and the ortho tert-butyl groups block attack, so $ArO^\bullet$ neither abstracts H from RH nor adds to O_2; it terminates a second $ROO^\bullet$.

FR-57.
(Integer) NBS/CCl_4/peroxide is used with (E)-hex-2-ene. How many constitutional isomeric monobromoalkenes can form (ignore stereoisomers)?
Answer: 4
Allylic hydrogens are on C-1 and C-4. The C-4 radical (⇔ C-2) gives 4-bromohex-2-ene and 2-bromohex-3-ene; the C-1 radical (⇔ C-3) gives 1-bromohex-2-ene and 3-bromohex-1-ene. The C-4 pathway predominates (gives disubstituted alkenes).

FR-58.
Arrange R–X in decreasing rate of reduction by Bu_3SnH/AIBN: R–Cl, R–Br, R–I. Explain with bond energies.
Answer: R–I > R–Br > R–Cl
The rate-limiting halogen abstraction by $Bu_3Sn^{\bullet}$ is easier for weaker C–X bonds (CH_3–I 57, –Br 70, –Cl 85 kcal mol^{-1}) and also forms the strong Sn–X bond.

Part 10 Carbenes: Structure, Generation, Singlet and Triplet States, Reactions

A **carbene** is a neutral species containing a divalent carbon atom with only six valence electrons: R_2C: (two bonds + two non-bonding electrons). It is a Lewis acid (empty orbital) and a Lewis base (lone pair) at the same atom, and is therefore ambiphilic; most simple carbenes behave as **electrophiles** because the empty orbital dominates. Carbenes are reaction intermediates of very short lifetime (CH_2: ≈ 10^{-9} s in solution) except for stabilised cases (N-heterocyclic carbenes, Arduengo 1991).

10.1 Singlet and triplet carbenes

Property	Singlet carbene	Triplet carbene
Electrons in the two non-bonding orbitals	both in one orbital (σ lone pair, spin-paired); the p orbital is empty	one electron in each of two orbitals (σ and p), spins parallel (↑ ↑)
Spin multiplicity 2S + 1	1 (S = 0)	3 (S = 1)
Hybridisation / shape	sp^2, **bent**; H–C–H ≈ 102–103° for CH_2	sp (linear, 180°) or sp^2 bent with a wider angle; H–C–H ≈ 130–136° for CH_2 (ground state, $^3B_1$)
Magnetic behaviour	diamagnetic	paramagnetic (two unpaired electrons; ESR active)
Character	like a carbocation (empty p) and a carbanion (lone pair): ambiphilic, concerted reactions	like a diradical: stepwise radical reactions
Addition to a C=C	**stereospecific, syn** (concerted, one step): cis-alkene → cis-cyclopropane; trans → trans	**not stereospecific** (stepwise through a 1,3-diradical that can rotate): cis and trans mixture
C–H insertion	concerted, with retention of configuration at the carbon that is attacked	H-abstraction then radical recombination (loss of stereochemistry)
Examples	:CCl_2, :CF_2, :CBr_2, :$C(OR)_2$, :$C(NR_2)_2$, Arduengo carbene, the carbenoids; CH_2 generated by photolysis of CH_2N_2 in solution (first singlet)	ground-state :CH_2 (triplet is ≈ 9 kcal mol^{-1} lower), :CHPh, :CPh_2 (diarylcarbenes), :C: (C atom has a triplet ground state)

What decides the spin state? A singlet is favoured when the substituents are **π-donors** (X–: F, Cl, Br, OR, NR_2) because their lone pair donates into the empty p orbital (stabilising the singlet by delocalisation, 2p–2p overlap best for F, N, O) and σ-withdrawing groups lower the energy of the σ orbital (more s-character in the lone pair). A triplet is favoured by substituents that are π-conjugating without lone pairs (aryl groups: Ph_2C: is a triplet) or sterically demanding groups that widen the angle, and by σ-donors. The 1,2-shift reactions (Wolff, hydride) that occur at carbenes are almost always singlet processes.

> **Stability order of dihalocarbenes (singlet):** :CF_2 > :CCl_2 > :CBr_2 > :CI_2 by π-back-donation of the 2p lone pair of F into the empty p orbital of carbon (effective 2p–2p overlap); Cl, Br, I overlap poorly (3p, 4p, 5p with 2p). ⚑ Do not confuse with the *electrophilicity*/ease of formation: :CCl_2 forms most easily from $CHCl_3$ because Cl^- is a better leaving group and the CCl_3^- anion is more stable than CF_3^-.

10.2 Generation of carbenes

Method	Equation	Comment
α-Elimination of HX from a haloform	$CHCl_3 + KOH \rightarrow {}^-CCl_3 + H_2O \rightarrow :CCl_2 + Cl^-$	carbanion first, then loss of halide; the basis of Reimer–Tiemann, carbylamine and dichlorocyclopropane formation
Photolysis/thermolysis of diazomethane (diazo compounds)	$CH_2N_2 \rightarrow :CH_2 + N_2$; $R_2C{=}N_2 \rightarrow R_2C: + N_2$	the irreversible loss of the superb leaving group N_2; singlet first, then triplet by collisions

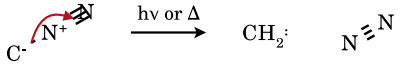
Photolysis of ketene	$CH_2{=}C{=}O \rightarrow :CH_2 + CO$	gas phase source of methylene
Tosylhydrazone salts (Bamford–Stevens/Shapiro)	$R_2C{=}N{-}NH{-}Ts$ + base → $R_2C{=}N{-}N^-{-}Ts \rightarrow R_2C: + N_2 + Ts^-$	diazo compound is formed in situ
Simmons–Smith carbenoid	$CH_2I_2 + Zn(Cu) \rightarrow I{-}CH_2{-}ZnI$	not a free carbene (carbenoid): CH_2 is transferred to alkenes in a concerted way; stereospecific, OH-directed
Diazirines, epoxides (deoxygenation), metal-carbene complexes	diazirine → carbene + N_2	photolabelling
Wolff rearrangement precursor	$R{-}CO{-}CHN_2 \rightarrow R{-}CO{-}CH: + N_2 \rightarrow R{-}CH{=}C{=}O$ (ketene)	Arndt–Eistert homologation

10.3 Reactions of carbenes

Reaction	Example	Outcome
Cyclopropanation of alkenes	$:CCl_2$ + cyclohexene → 7,7-dichlorobicyclo[4.1.0]heptane; CH_2I_2/Zn–Cu + alkene	syn addition; carbene is electrophilic so electron-rich alkenes react faster ($Me_2C{=}CMe_2 > Me_2C{=}CH_2 > CH_2{=}CH_2 > CH_2{=}CH{-}CN$)
C–H insertion	$:CH_2 + CH_3CH_3 \rightarrow CH_3CH_2CH_3$	indiscriminate (carbene is so reactive it inserts into primary, secondary, tertiary C–H nearly statistically)
Reimer–Tiemann	PhOH + $CHCl_3$ + NaOH → salicylaldehyde (major, ortho) + p-hydroxybenzaldehyde (minor)	phenoxide attacks $:CCl_2$ through the ortho carbon (chelation by Na^+), then hydrolysis of $ArCHCl_2$ to ArCHO. With CCl_4 in place of $CHCl_3$ the product is salicylic acid
Carbylamine (Hofmann isocyanide) test	$RNH_2 + CHCl_3 + 3\ KOH \rightarrow R{-}N{\equiv}C + 3\ KCl + 3\ H_2O$	primary amines only (aliphatic or aromatic); isocyanide has an offensive smell (the test). Secondary and tertiary amines do not react
Wolff rearrangement	$ArCOCHN_2 \rightarrow [ArCO{-}CH:] \rightarrow Ar{-}CH{=}C{=}O$	1,2-aryl/alkyl shift to the carbene carbon gives a ketene; trapping with H_2O gives $ArCH_2COOH$ (Arndt–Eistert)
Ring expansion (Ciamician–Dennstedt)	pyrrole + $:CCl_2$ → 3-chloropyridine	dihalocyclopropane intermediate opens with ring expansion
Buchner reaction	benzene + $:CHCO_2Et$ (from N_2CHCO_2Et) → cycloheptatriene ester	addition to benzene then electrocyclic ring opening
Dimerisation	$2\ :CPh_2 \rightarrow Ph_2C{=}CPh_2$	triplet carbenes in the absence of traps
Reaction with Lewis bases (ylide formation)	$:CH_2 + R_3N$, R_2S, Ph_3P → ylides	the lone-pair donor attacks the empty p orbital

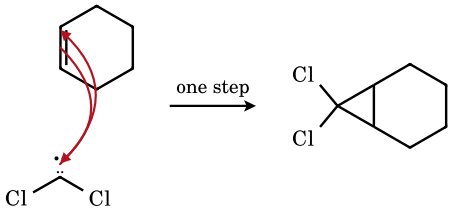
10.4 Mechanisms (arrow-validated)

10.1 Diazomethane → methylene + N_2



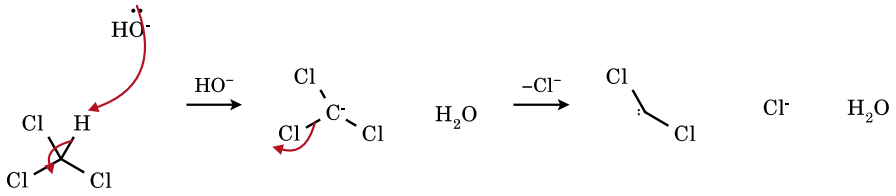
The C–N σ pair leaves with N_2; the carbanion lone pair that was on carbon becomes the lone pair of the carbene. The product is first a singlet methylene (lone pair + empty p), which decays to the triplet ground state.

10.3 Dichlorocarbene + cyclohexene: concerted syn-cyclopropanation (singlet)



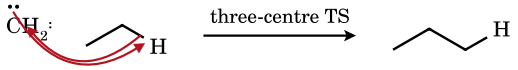
The alkene π electrons attack the empty p orbital of the carbene while the carbene lone pair forms the second C–C bond: both bonds form simultaneously on the same face of the alkene, so the stereochemistry of the alkene is retained (stereospecific syn addition). The product is 7,7-dichlorobicyclo[4.1.0]heptane (a *cis*-fused, achiral meso compound).

10.2 α-Elimination: chloroform + HO^- → $:CCl_2$



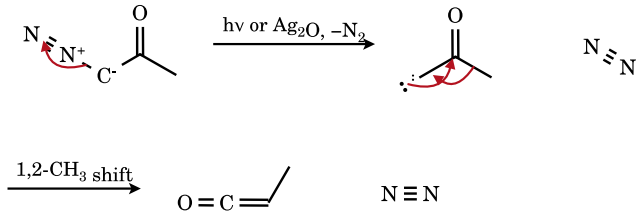
Hydroxide removes the single proton of $CHCl_3$ ($pK_a \approx 24$) to give $^-CCl_3$; the carbanion expels chloride from the same carbon (α-elimination) to give dichlorocarbene. The lone pair on carbon is the carbene lone pair; the empty p orbital is the electrophilic site.

10.4 Carbene C–H insertion: $:CH_2$ + ethane → propane



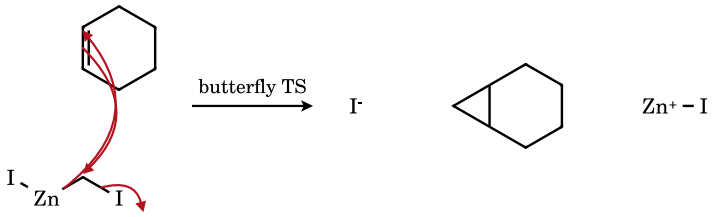
In the singlet insertion the C–H σ electrons attack the empty p orbital of the carbene while the carbene lone pair accepts the hydrogen: a single three-centre transition state with retention at the attacked carbon. Triplet carbenes insert by abstraction–recombination, which loses stereochemistry.

10.5 Wolff rearrangement: α-diazoketone → ketene



Loss of N_2 gives an α-ketocarbene; the substituent on the carbonyl carbon migrates with its bonding pair to the electron-deficient carbene carbon while the carbene lone pair forms the C=C of the ketene (the migrating group moves to the carbon that was the carbene). Water traps the ketene as the carboxylic acid with one more carbon (Arndt–Eistert chain extension): R–COCl → R–CO–CHN_2 → R–CH_2COOH.

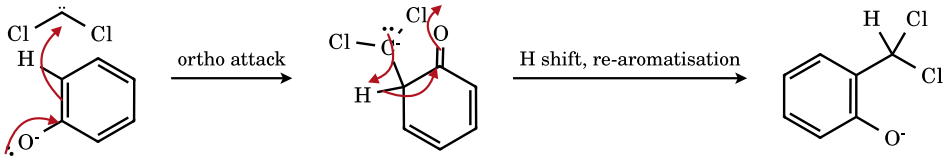
10.6 Simmons–Smith: iodomethylzinc iodide + cyclohexene → norcarane



The carbenoid is not a free carbene: the CH_2 is delivered, with simultaneous loss of iodide, from the carbon bound to zinc to both alkene carbons in a three-centre 'butterfly' transition state. The addition is stereospecific syn and is accelerated by an allylic OH group that coordinates zinc (the directing effect), which free $:CH_2$ would not show.

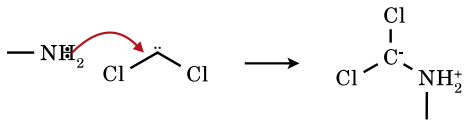
10.5 The Reimer–Tiemann and carbylamine reactions in detail

10.7 Reimer–Tiemann: phenoxide attacks dichlorocarbene at the ortho carbon



The phenoxide ring is the nucleophile (the negative charge is shared with the ortho and para carbons). It attacks $:CCl_2$; the resulting dienone carbanion regains aromaticity by a proton shift to give the o-(dichloromethyl)phenoxide. Hydroxide then hydrolyses the benzal chloride ($ArCHCl_2$) twice to $ArCH(OH)_2$ → ArCHO. The ortho product predominates because Na^+ chelates phenoxide O and the carbene; the para isomer is the minor product.

10.8 Carbylamine reaction, first step: primary amine + dichlorocarbene



The nitrogen lone pair attacks the empty p orbital of the carbene to give a zwitterion (ammonium/carbanion). Two successive deprotonation–dechlorination steps (base removes N–H, Cl^- leaves) then convert $R–NH_2^+–CCl_2^-$ into $R–N^+ \equiv C^-$, the isocyanide: the isocyanide carbon is itself a carbene-like divalent carbon ($R–N^+ \equiv C^- \leftrightarrow R–N{=}C{:}$).

10.7 Electronic structure in depth: why CH_2 is a triplet and CF_2 a singlet

The valence orbitals of a bent carbene are a σ orbital (an sp^2-like hybrid on carbon, holding the non-bonding pair in the singlet) and a nearly pure p orbital perpendicular to the molecular plane. The two non-bonding electrons can occupy them in two ways: **singlet** ($\sigma^2\ p^0$, spins paired, $^1A_1$) or **triplet** ($\sigma^1\ p^1$, spins parallel, $^3B_1$). The singlet–triplet energy gap ΔE_{ST} = E(singlet) − E(triplet) decides the chemistry.

Carbene	Ground state	ΔE_{ST} (kcal mol^{-1}, recalled)	H–C–H / X–C–X angle	Remark
$:CH_2$	triplet ($^3B_1$)	triplet lower by ≈ 9	≈ 134° (T), ≈ 102° (S)	the singlet is formed first from CH_2N_2 photolysis, then decays to the triplet in a dilute gas phase
:CHCl, :CHBr	singlet	singlet lower by a few	—	halogen lone pair donates into the empty p
$:CF_2$	singlet	singlet lower by ≈ 56	≈ 105°	the strongly π-donating, σ-withdrawing F raise the p orbital and lower the σ orbital
$:CCl_2$	singlet	singlet lower by ≈ 13	≈ 109°	electrophilic carbene, most used in the laboratory
:CH–OH, $:C(OCH_3)_2$	singlet	singlet lower by ≥ 10–25	—	O lone pair → empty p orbital
$(R_2N)_2C$: (NHC)	singlet	singlet lower by ≈ 80	N–C–N 101–105°	stable, isolable (Arduengo 1991); nucleophilic
$:CPh_2$	triplet	triplet lower by ≈ 3	≈ 150°	phenyls delocalise the unpaired electron (benzylic radical-like) and widen the angle; diphenylcarbene is persistent in a rigid matrix (EPR)
fluorenylidene	triplet (≈ degenerate)	≈ 1	—	both states react; photolysis of 9-diazofluorene gives mixtures
:C=C=C (vinylidene), $:C{=}CH_2$	singlet	—	linear-like	rearranges to acetylene by 1,2-H shift
HC ≡ C–C–H (propargylene)	triplet	—	bent	diradical character

- **Rule.** A carbene is singlet when the σ–p gap is large (> 2 eV, ≈ 46 kcal mol^{-1}), triplet when it is small (< 1.5 eV). Substituents that **raise the p orbital** (π-donors: N, O, S, halogens, through donation of a lone pair into the empty p orbital) and those that **lower the σ orbital** (σ-acceptors/electronegative: F, Cl, OR, NR_2 through –I) both stabilise the singlet. Conjugating, bulky or σ-donating groups (H, alkyl, aryl, silyl) narrow the gap and favour the triplet.
- **Bond angle follows multiplicity.** A smaller angle (≈ 100–110°, more p-character in the σ lone pair, singlet) versus a larger angle (≈ 130–180°, triplet: the linear limit of a carbon with two sp bonds and two p electrons in degenerate p orbitals is a $^3\Sigma^-$ triplet).
- **Physical consequences.** Singlet carbenes react in concerted, stereospecific steps (zwitterionic character: lone pair nucleophilic, p orbital electrophilic); triplet carbenes react as diradicals (H-atom abstraction, stepwise additions, loss of stereochemistry). Triplet carbenes are paramagnetic and give EPR signals (zero-field splitting D ≈ 0.4 cm^{-1} for CH_2, 0.4 for Ph_2C) which is how diphenylcarbene was identified (Wasserman, Hutton, 1962; first triplet organic carbene seen by ESR).
- **Heavier analogues.** Silylenes (SiH_2: singlet, gap 21 kcal mol^{-1}), germylenes, stannylenes and plumbylenes are singlet ground states with ever-growing stability (inert-pair effect); $SiCl_2$, $SnCl_2$, $PbCl_2$ are isolable.

10.8 Generation of carbenes: a complete catalogue

Method	Precursor and conditions	Notes and examples
Diazoalkane decomposition	$R_2C{=}N_2$, hν, Δ, or metal (Cu, $Rh_2(OAc)_4$, $Pd(OAc)_2$)	CH_2N_2 (from N-methyl-N-nitrosourea or Diazald + base) → $:CH_2 + N_2$; with metals, metal carbenoids (more selective, bear no free CH_2); explosive and toxic, handle dilute in ether
α-Diazoketones	$RCOCHN_2$, Ag_2O/H_2O (Arndt–Eistert) or hν	acylcarbene → Wolff rearrangement to ketene → homologated acid/ester/amide (RCOOH → RCH_2COOH via $RCOCl + CH_2N_2$)
Diazirines	3H-diazirines (cyclic, 3-membered, C(N=N)); hν (350 nm)	small, stable photolabile source of carbenes (aryl-CF_3-diazirine) used in photoaffinity labelling; also isomerises to the linear diazo compound
Tosylhydrazone salts	$R_2C{=}N{-}NHTs$ + NaOMe/Δ (Bamford–Stevens); $R_2C{=}N{-}NHTs$ + 2 RLi (Shapiro)	aprotic solvent: carbene (then alkene by 1,2-H shift); protic solvent: carbocation by protonation of the diazo compound; Shapiro: vinyl-lithium via α-elimination of sulfinate, no carbene
Ketene photolysis	$CH_2{=}C{=}O$, hν (λ < 330 nm)	→ $:CH_2$ + CO (historic method for $^3CH_2$ in the gas phase)
α-Elimination of haloforms	CHX_3 + KOH/tBuOK, or PTC ($BnEt_3N^+Cl^-$, 50 % NaOH)	CCl_3^- → $:CCl_2 + Cl^-$; $CHBr_3$ → $:CBr_2$; CHI_3 would give $:CI_2$ but is not used
Decarboxylation of trihaloacetate	$Cl_3CCOONa$, 80–90 °C (DME) or $ClCF_2COONa$, 180 °C	$Cl_3C{-}COO^-$ → $CCl_3^- + CO_2$ → $:CCl_2$; neutral conditions, no HO^-
Organomercurials (Seyferth)	$PhHgCCl_2Br$, $PhHgCCl_3$, $PhHgCF_3$/NaI, Δ	neutral, base-free source of $:CCl_2$, :CBrCl, $:CF_2$; mercury is toxic, rarely used now
Reduction of gem-dihalides	CH_2I_2 + Zn(Cu) (Simmons–Smith); $CH_2I_2 + Et_2Zn$ (Furukawa); CH_2Cl_2/BuLi	carbenoid ICH_2ZnI (not free $:CH_2$); stereospecific syn, directed by allylic OH; $ClCH_2Li$ at −100 °C
Fragmentation of strained rings	cyclopropenes, oxiranes (deoxygenation), thiiranes, epoxy-ketones	cyclopropene → vinylcarbene (reversible); photolysis of bicyclic azirines/cyclopropenones: PhC ≡ CPh via ring opening, giving :C=O
Photolysis of ketones/alkenes	alkenes (Hg-sensitised) → vinylidenes; allene; 1,2-shifts	$H_2C{=}C$: from acetylene by 1,2-H shift (reverse of the alkyne ⇌ vinylidene rearrangement)
Dehydrohalogenation of vinyl halides	$R_2C{=}CHX$, KOtBu	1,1-elimination gives vinylidene $:C{=}CR_2$, 1,2-H shift → alkyne (Fritsch–Buttenberg–Wiechell)
Metal-bound carbenes	$R_2C{=}N_2$ + LnM (Rh, Cu, Pd, Ru); Fischer-type $(CO)_5W{=}C(OMe)Ph$	no free carbene; metal–carbene complex transfers CR_2; important in catalytic cyclopropanation, C–H functionalisation, olefin metathesis (Grubbs, Schrock)
Deprotonation of azolium salts	imidazolium/thiazolium + base (KOtBu, NaH)	NHC; thiazolium carbene is the active catalyst of thiamine (vitamin B_1) and benzoin/Stetter reactions
Reaction of alkyl dihalide with base	CH_2Cl_2, $NaNH_2$ / CH_2Cl_2, KOtBu (weak)	:CHCl; the haloform route is easier than for CH_2Cl_2 because CCl_3^- is more stable

10.9 Reactions of carbenes: a systematic survey

(a) Addition to C=C: cyclopropanation, with Skell's stereochemical test

- **Singlet carbenes add in a single concerted, stereospecific syn step** (non-linear approach in which the alkene π electrons enter the empty p orbital while the carbene lone pair attacks the other carbon): cis-2-butene gives cis-1,2-dimethylcyclopropane; trans-2-butene gives only the trans isomer (Skell 1956; Doering).
- **Triplet carbenes add in two steps**: the first bond gives a 1,3-diradical whose spin must flip before ring closure; internal rotation about C–C competes with closing, so both cis- and trans-cyclopropanes arise from either alkene isomer (with a stereospecificity that falls as the pressure of the gas falls and the lifetime of the diradical rises). Photosensitised (benzophenone) photolysis of diazomethane gives $^3CH_2$: cis-2-butene → 1:1 cis/trans.
- **Dichlorocarbene is an electrophile**: its addition is faster to electron-rich alkenes. Approximate order of reactivity: $Me_2C{=}CMe_2$ > $Me_2C{=}CHMe$ > $Me_2C{=}CH_2$ > cis-RCH=CHR > $RCH{=}CH_2$ > $CH_2{=}CH_2$ > electron-poor alkenes (acrylonitrile is almost inert); dichlorocarbene adds to a styrene double bond faster than to a simple alkene. Unsubstituted methylene ($^1CH_2$) is far less selective (add to every π bond and insert into every C–H).
- **Diastereoselectivity.** Simmons–Smith cyclopropanation is directed syn to a hydroxyl or ether group on an allylic carbon (the zinc coordinates oxygen); free-carbene additions are controlled by sterics (attack from the less hindered face); bicyclic alkenes (norbornene) give exo products.

- **Ring expansion by dihalocarbene.** Indole + $CHCl_3$/NaOEt → 3-chloroquinoline; pyrrole → 3-chloropyridine (Ciamician–Dennstedt rearrangement): the dihalocyclopropane intermediate opens with expulsion of Cl^-. Cyclic alkenes yield the cyclopropane that can be ring-opened to a ring-expanded cycloalkene (e.g. 7,7-dihalonorcarane + Ag^+ → 2-halocycloheptenyl cation: used to make cycloheptene derivatives).

(b) Insertion into σ bonds

- **Singlet CH_2 inserts into C–H bonds indiscriminately** (1° : 2° : 3° ≈ 1 : 1.2 : 1.5 per H in solution, recalled): pentane + CH_2N_2/hν gives hexane, 2-methylpentane and 3-methylpentane in roughly statistical ratio. The insertion has a three-centre TS; the configuration at the carbon is retained when chiral (insertion at a chiral centre proceeds with retention, in concerted insertion).
- **Triplet CH_2 reacts by abstraction–recombination**: $^3CH_2$ + R–H → •CH_3 + R• then recombination; thus the products show scrambling and radical by-products (ethane).
- **X–H insertion** (O–H, N–H, S–H, Si–H, Sn–H): ROH + :CH_2 → $ROCH_3$ via an oxonium ylide (R–O^+H–CH_2^-) and a 1,2-proton shift; with metal carbenoids this is the basis of $Rh_2(OAc)_4$-catalysed O–H and N–H insertion (β-lactam synthesis in the penem route, Merck thienamycin synthesis).
- **Intramolecular C–H insertion** by a rhodium carbenoid builds five-membered rings with high selectivity (1,5-insertion preferred; tertiary > secondary > primary C–H; Taber, Doyle). Free carbenes give mixtures; catalysts control site selectivity.
- **C–C insertion** of carbenes is rare. **C–halogen insertion**: halocarbenes insert into C–X to give gem-dihalides through ylide formation.

(c) Ylide formation (Doyle–Kirmse, Stevens, carbonyl ylides)

A carbene reacts with a lone pair on O, S, N or halogen to give an onium ylide, which can (i) undergo a proton shift (insertion product), (ii) a [2,3]-sigmatropic rearrangement if an allyl group is on the heteroatom (Doyle–Kirmse: allyl sulfide + diazoacetate/Rh → homoallyl sulfide), (iii) a [1,2]-Stevens shift, (iv) cycloaddition with a dipolarophile if the ylide is a carbonyl ylide (from carbonyl + carbene: [3+2] with alkenes gives tetrahydrofurans), or (v) a Corey–Chaykovsky epoxide when a sulfur ylide is prepared by deprotonation.

(d) Rearrangements of carbenes

Process	Equation	Comment
1,2-Hydride shift	R–CH_2–C̈–R′ → RCH=CHR′	very fast, k ≈ 10^7–10^{10} s^{-1}; the reason alkylcarbenes (CH_3CH:) rarely give intermolecular reactions, and why Bamford–Stevens in aprotic solvent gives alkenes
1,2-Alkyl/aryl shift	R–C̈–$C(R')_3$ → alkene	slower than hydride; neopentylidene gives 2-methyl-2-butene via methyl shift
Wolff rearrangement	RCO–C̈–R′ → R(R′)C=C=O	acylcarbene → ketene (the migrating R retains configuration); basis of Arndt–Eistert homologation and ring contraction of cyclic α-diazoketones (e.g. 2-diazocyclohexanone → cyclopentanecarboxylic acid)
Cyclopropylcarbene → cyclobutene	ring expansion	also cyclobutylidene → methylenecyclopropane; used to interconvert strained rings
Vinylcarbene ⇌ cyclopropene	—	photolysis of cyclopropenes gives vinylcarbenes; vinylcarbene → cyclopropene thermally; used to make furans and indenes
Aryl carbene ring expansion	PhC̈H → cycloheptatetraene (strained cyclic allene)	phenylcarbene/benzocyclopropene interconversion; trapped by amines (Wentrup)
Vinylidene → acetylene	R_2C=C: → RC ≡ CR (1,2-shift)	Fritsch–Buttenberg–Wiechell
Bridgehead/bicyclic carbene fragmentation	carbene loses N_2 then adds across a nearby π bond	used to build tricyclic skeletons (Buchner's ester)

(e) Reactions with arenes: Buchner ring expansion

Carbenes add to benzene to give the norcaradiene (bicyclo[4.1.0]hepta-2,4-diene), which opens in a thermally allowed, disrotatory 6π electrocyclic reaction to cycloheptatriene (Buchner 1885: ethyl diazoacetate + benzene, Cu or Rh). Electron-withdrawing substituents at C-7 stabilise the norcaradiene valence tautomer (equilibrium). In intramolecular variants (Buchner–Curtius–Schlotterbeck), α-diazoketones with aryl tethers give azulenones.

10.10 Carbenoids: the 'free carbene' is the exception

- A **carbenoid** is a species with carbene-like reactivity but in which the divalent carbon is bound to a metal and leaving group: $LiCH_2Cl$, ICH_2ZnI, [Rh]=$CHCO_2Et$, $MeCCl_2Li$ (Doering–LaFlamme). It transfers CR_2 in a

single concerted step (the metal delivers the carbon with its leaving group still partially bound), with selectivity patterns different from the free carbene.

- **Simmons–Smith** (Zn–Cu couple, CH_2I_2, ether): I–CH_2–Zn–I (zinc carbenoid) adds to the alkene through a three-centre 'butterfly' TS to give the cyclopropane with retention of alkene geometry. Reaction with electron-rich alkenes is faster; allylic alcohols and ethers direct the addition syn. Furukawa modification: Et_2Zn replaces Zn(Cu). Charette: enantioselective with chiral dioxaborolane ligands.
- **Doering–LaFlamme allene synthesis**: 1,1-dihalocyclopropane + RLi (or Mg) → carbenoid → cyclopropylidene → ring opening to the allene (via disrotatory electrocyclic opening). Combined with dihalocarbene addition, it converts an alkene $R_2C{=}CR_2$ into an allene $R_2C{=}C{=}CR_2$ with one more carbon.
- **$Rh_2(OAc)_4$ and $Cu(acac)_2$-catalysed decompositions** of diazo compounds proceed via a metal carbenoid ($M{=}CR_2$); chiral Rh_2(S-DOSP)$_4$ and Cu–bis(oxazoline) give enantioselective cyclopropanations (Davies, Doyle, Pfaltz; Nobel-related work in asymmetric catalysis). The first commercial application was the synthesis of the pyrethroid insecticide chrysanthemic ester (Aratani, Sumitomo).
- **Fischer carbenes** $(CO)_5Cr{=}C(OMe)R$: a low-oxidation-state metal with π-acceptor ligands makes the carbene carbon electrophilic (attack by nucleophiles at C), whereas **Schrock carbenes** (high-oxidation-state early metals, e.g. Ta(=CHtBu)) are nucleophilic at C. **Grubbs/Schrock olefin-metathesis catalysts** are metal alkylidenes: [2+2] with an alkene gives a metallacyclobutane which cycloreverts to a new alkene and a new carbene (Chauvin mechanism; Nobel Prize 2005).

10.11 Stable carbenes: from curiosity to catalysts

- **Wanzlick** (1960s) proposed N-heterocyclic carbenes; **Arduengo** isolated the first crystalline NHC (1,3-di-adamantylimidazol-2-ylidene, 1991) by deprotonation of the imidazolium chloride with NaH/DMSO. **Bertrand** reported the first stable acyclic carbene (phosphino-silyl, 1988) and cyclic (alkyl)(amino)carbenes (CAAC).
- **Why NHCs are stable.** (i) Two adjacent nitrogen atoms donate their lone pairs into the empty carbene p orbital (π-donation, aromatic six-π-electron imidazole ring), (ii) the electronegative nitrogens inductively stabilise the σ lone pair, which raises the singlet–triplet gap to ≈ 80 kcal mol^{-1}, (iii) bulky N-substituents (adamantyl, 2,6-diisopropylphenyl, mesityl) prevent dimerisation to the tetra-aminoethylene (Wanzlick equilibrium).
- **Reactivity of NHCs**: strongly σ-donating and nucleophilic; excellent ligands for metals (Grubbs II, Pd–PEPPSI), organocatalysts for the benzoin condensation (aryl aldehyde → benzoin), Stetter (aldehyde + Michael acceptor), umpolung of aldehydes; the same pattern as thiamine pyrophosphate (thiazolium ylide) in pyruvate decarboxylase and transketolase.
- **Link to Part 8:** an NHC adds to PhCHO to give the Breslow intermediate (an enaminol), the acyl-anion equivalent. This is the NHC equivalent of the cyanohydrin anion (mechanism 8.9) used for the benzoin condensation: both achieve 'umpolung' (polarity inversion) of the carbonyl carbon.

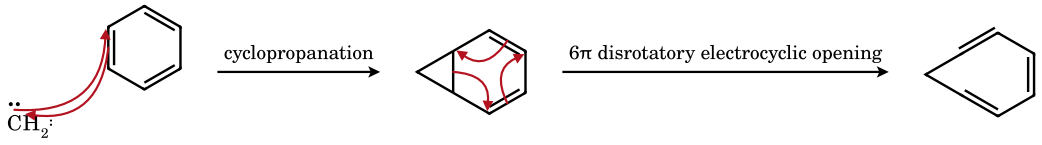
10.12 Evidence for carbenes and experimental techniques

- **Trapping** with alkenes (cyclopropane), with O–H/N–H/Si–H, or with a cheletropic acceptor; the stereochemistry reveals the spin state (Skell test).
- **Matrix-isolation IR/UV** (argon, 10 K) identifies the carbene directly; **ESR** detects triplets ($^3CH_2$ by Wasserman, Ph_2C: by Hutton).
- **Laser flash photolysis** gives lifetimes: Ph_2C: ≈ 1 μs in solution, reacting with methanol (O–H insertion) at ≈ 10^8 M^{-1} s^{-1} (singlet) and with O_2 at 10^9 (triplet); the product ratio with alcohol versus with O_2/alkene identifies the state in equilibrium.
- **Isotope and cross-over experiments**: Hine showed that $:CCl_2$ from $CHCl_3/HO^-$ is formed in a reversible deprotonation (H/D exchange of $CHCl_3$ in D_2O/OD^- is faster than hydrolysis) followed by the rate-limiting loss of Cl^- (so CCl_3^- is an intermediate; the rate law is first order in HO^- and in $CHCl_3$, retarded by added Cl^-: common-ion effect, the signature of S_N1-like rate-determining α-elimination).
- **Hydrolysis of $CHCl_3$ by the carbene path** gives formate and CO: $:CCl_2 + 2\ HO^- \rightarrow CO + 2\ Cl^- + H_2O$ (and, with more hydroxide, $HCOO^-$).

Typical errors in carbene questions. (1) Calling a carbene 'electron-deficient with a positive charge': it is neutral with a sextet. (2) Assuming all carbenes give stereospecific syn addition; only singlets do. (3) Giving a carbocation product under Bamford–Stevens conditions: it depends on the solvent. (4) Forgetting that the Reimer–Tiemann reaction yields mainly the *ortho* aldehyde (via the phenoxide), and that para is minor. (5) Writing :CCl_2 inserting into C–H bonds: dichlorocarbene prefers addition to C=C. (6) Using :CH_2 from CH_2Cl_2/base in exam questions where Simmons–Smith is meant: CH_2I_2/Zn–Cu gives a carbenoid, not free methylene. (7) Mistaking Wolff for a carbocation 1,2-shift: the carbene α to carbonyl rearranges concertedly with loss of N_2; a tertiary R group migrates with retention.

10.13 Additional validated mechanisms

10.9 Buchner ring expansion: benzene + :CH_2 → norcaradiene → cycloheptatriene



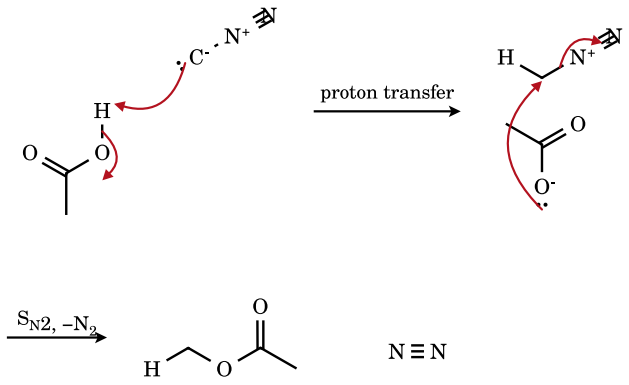
Methylene adds to one π bond of benzene (cyclopropanation) to give norcaradiene, a cyclohexadiene with a cyclopropane fused at C-1/C-6. The cyclopropane σ bond shared with the six-membered ring and the two double bonds form a 6π electron system (σ + π + π); thermally, a disrotatory electrocyclic ring-opening gives the seven-membered cycloheptatriene (the two valence tautomers interconvert; electron-withdrawing groups at C-7 favour norcaradiene).

10.10 Triplet methylene adds stepwise: diradical intermediate, then ring closure after spin flip



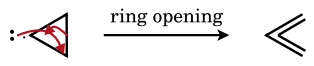
In the triplet the two non-bonding electrons have parallel spin, so the first bond leaves a 1,3-diradical that cannot close until a spin flip (intersystem crossing) converts it into a singlet. During this time (10^{-10}–10^{-9} s) rotation about the former C=C bond can occur, so cis- and trans-2-butene both give a mixture of cis- and trans-1,2-dimethylcyclopropane: the lack of stereospecificity is the experimental signature of the triplet (Skell).

10.11 Diazomethane esterification: no carbene, a protonation / S_N2 sequence



The nucleophilic carbon of CH_2N_2 is protonated by the acid ($pK_a \lesssim 10$ needed), giving the methyldiazonium ion – an extremely good alkylating agent (leaving group N_2) – which is attacked by the carboxylate in an S_N2 step. Diazomethane therefore methylates carboxylic acids and phenols quantitatively under mild conditions, though no free CH_2 appears; it is the same compound that gives :CH_2 on photolysis.

10.12 Doering–LaFlamme: cyclopropylidene → allene (disrotatory electrocyclic ring opening)



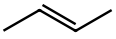
The cyclopropylidene (formed from a 1,1-dihalocyclopropane and BuLi or Mg through the α-halo-lithium carbenoid) opens by breaking the distal C–C bond of the ring. The carbene lone pair and the σ electrons of the C2–C3 bond are converted into the two C=C π bonds of the allene: the process is a two-electron ring-opening that releases the cyclopropane strain (27 kcal mol^{-1}) and converts a carbene into a stable cumulene. Together with the addition of :CBr_2 to an alkene, it converts C=C into C=C=C with one additional carbon.

10.14 Problems on carbenes

CE-1.
How many valence electrons does the carbon atom of a carbene have, and how many are non-bonding?
Answer: 6; 2 non-bonding
Two electrons in each of the two σ bonds, plus two non-bonding electrons (a lone pair in the singlet; two unpaired electrons in the triplet) = 6 around carbon, an incomplete octet.

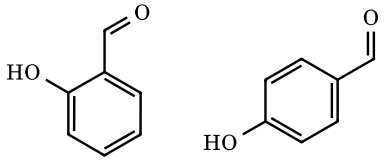
CE-2.
Which statements about singlet methylene are correct? (A) sp^2, bent, H–C–H ≈ 103°; (B) paramagnetic; (C) shows stereospecific syn addition to alkenes; (D) has an empty p orbital.
Answer: (A), (C), (D)
Singlet CH_2 has a filled sp^2 lone pair and an empty p orbital; it is diamagnetic (all electrons paired). Triplet methylene is paramagnetic with H–C–H ≈ 136°.

CE-3.
The addition of singlet $:CBr_2$ to cis-but-2-ene gives how many products, and what are they? What happens with a triplet carbene?



Answer: singlet: one (cis-1,1-dibromo-2,3-dimethylcyclopropane, meso); triplet: two (cis and trans)
Concerted syn addition retains the cis relationship of the two methyl groups, giving the achiral meso cyclopropane. A triplet carbene adds stepwise through a 1,3-diradical that can rotate before the second bond closes, giving cis and trans (the trans is chiral, as a racemate).

CE-4.
Phenol is heated with $CHCl_3$ and aqueous NaOH, and then acidified. Name the major and minor organic products and the electrophile.



major **minor**

Answer: salicylaldehyde (major) and p-hydroxybenzaldehyde (minor); electrophile $:CCl_2$
Hydroxide converts $CHCl_3$ into $:CCl_2$ (electrophilic carbene) and phenol into phenoxide (activated ring). Ortho attack is favoured by Na^+ chelation. Replace $CHCl_3$ by CCl_4 and the product is salicylic acid.

CE-5.
Which of the following give a positive carbylamine test? (a) CH_3NH_2, (b) $(CH_3)_2NH$, (c) $C_6H_5NH_2$, (d) $(CH_3)_3N$, (e) CH_3CONH_2, (f) $C_6H_5NHCH_3$.
Answer: (a) and (c)
The sequence $R–NH_2 + :CCl_2 \rightarrow R–N^+ \equiv C^-$ needs two N–H bonds that can both be lost. Secondary (b, f) and tertiary (d) amines lack them and amides (e) are not nucleophilic enough.

CE-6.
Arrange in decreasing stability: $:CF_2$, $:CCl_2$, $:CBr_2$, $:CI_2$.
Answer: $:CF_2 > :CCl_2 > :CBr_2 > :CI_2$
π back-donation from the halogen lone pair into the empty carbon p orbital is best for the 2p orbital of fluorine (similar size and energy); it decreases as the halogen orbital becomes larger and more diffuse. (Part 5D, problem D-38, statement vi.)

CE-7.
Name the carbene (or carbenoid) and the product when cyclohexene reacts with (a) CH_2I_2 + Zn–Cu, (b) $CHBr_3$ + KOtBu, (c) CH_2N_2 + hν.
Answer: (a) ICH_2ZnI (carbenoid) → norcarane; (b) $:CBr_2$ → 7,7-dibromonorcarane; (c) $:CH_2$ → norcarane (plus C–H insertion products)
Each adds one carbon across the double bond to give a bicyclo[4.1.0]heptane. Free $:CH_2$ from diazomethane also inserts into C–H bonds and gives a mixture (low selectivity); the Simmons–Smith carbenoid is the clean method.

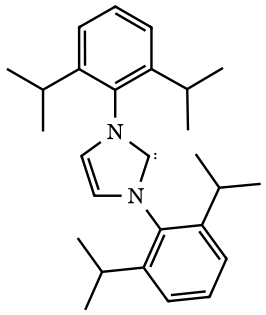
CE-8.
What is the product of the Arndt–Eistert sequence starting from CH_3COOH? (i) $SOCl_2$; (ii) CH_2N_2; (iii) Ag_2O, H_2O.
Answer: propanoic acid, CH_3CH_2COOH
$CH_3COCl + CH_2N_2 \rightarrow CH_3COCHN_2$; Ag_2O gives the carbene, Wolff shift gives $CH_3CH=C=O$ (methylketene), water gives CH_3CH_2COOH: homologation by one CH_2.

CE-9.
Which of the following is not a method of generating a carbene? (A) $CHCl_3$ + KOH, (B) CH_2N_2 + hν, (C) $CH_2{=}C{=}O$ + hν, (D) CH_3Cl + $AlCl_3$.
Answer: (D)
CH_3Cl + $AlCl_3$ gives CH_3^+ (a carbocation, Friedel–Crafts), not a carbene.

CE-10.
Hydrolysis of $CHCl_3$ with aqueous alkali would be expected to give HCOOH via $:CCl_2$. Write the intermediate sequence.
Answer: $CHCl_3 \rightarrow CCl_3^- \rightarrow :CCl_2 \rightarrow HCCl(OH)\ldots \rightarrow HCOO^-$
Hydroxide removes the proton (carbanion), Cl^- is lost (carbene), the carbene is attacked by HO^- and the resulting dichlorohydrin/formyl species loses two Cl^- to give formate (or CO + HCl). This is the same carbene that does the Reimer–Tiemann reaction.

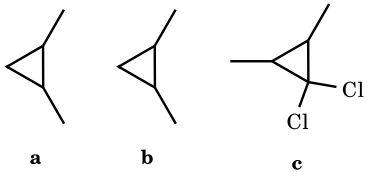
CE-11.
Why is the triplet the ground state of methylene while the singlet is the ground state of $:CF_2$?
Answer: π-donation and electronegativity
In CH_2 the two non-bonding orbitals (σ and p) are close in energy (≈ 9 kcal mol^{-1}), so Hund's rule puts one electron in each. In $:CF_2$ the fluorine lone pairs donate into the empty p orbital and raise its energy, while the σ-withdrawing F lowers the energy of the σ orbital: the gap widens (≈ 56 kcal mol^{-1}) and the singlet becomes the ground state.

CE-12.
A stable, crystalline carbene is made from an imidazolium salt and a base. Which features make it stable?



Answer: two π-donating N atoms and bulky N-substituents
The two nitrogen lone pairs donate into the empty p orbital (singlet stabilised, amidine-like resonance) and the σ-withdrawing N atoms lower the σ lone pair energy; the large aryl groups prevent dimerisation to the tetraaminoethylene. Arduengo (1991).

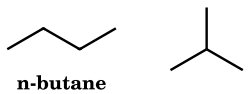
CE-13.
Predict the products and the stereochemistry: (a) cis-2-butene + CH_2N_2/hν (direct photolysis, dilute in liquid), (b) cis-2-butene + CH_2N_2/hν/benzophenone, (c) trans-2-butene + $CHCl_3$/KOtBu.



a **b** **c**

Answer: (a) cis-1,2-dimethylcyclopropane only (plus C–H insertion products); (b) mixture of cis and trans; (c) trans-1,1-dichloro-2,3-dimethylcyclopropane (stereospecific)
(a) singlet CH_2 (direct photolysis): one-step, stereospecific syn addition. (b) triplet sensitisation: stepwise diradical, scrambling of stereochemistry. (c) $:CCl_2$ is a singlet; the trans-alkene stereochemistry is preserved.

CE-14.
How many monobromo/mono-methylated insertion products (C_{n+1} alkanes) are formed when singlet methylene reacts with propane? Estimate the ratio assuming statistical insertion (1° : 2° = 1 : 1.2 per H).


n-butane

isobutane

Answer: 2 products: n-butane : isobutane = 6 : 2.4 = 71 : 29 (statistical would be 75 : 25)
Insertion into a 1° C–H (6 H) gives n-butane; into the 2° C–H (2 H) gives isobutane (2-methylpropane). Weights 6 × 1 = 6 and 2 × 1.2 = 2.4.

CE-15.

Account for the fact that $CHCl_3$ + OD^- in D_2O exchanges H for D faster than it hydrolyses, and that the hydrolysis is retarded by added Cl^-.

Answer: reversible carbanion formation followed by rate-limiting loss of Cl^-

$CHCl_3$ + HO^- ⇌ CCl_3^- + H_2O (fast, reversible: D exchange); CCl_3^- → :CCl_2 + Cl^- (slow); :CCl_2 + H_2O/HO^- → CO + $HCOO^-$. Added chloride pushes the second step backwards (common-ion effect), exactly as in an S_N1 reaction (Hine, 1950s).

CE-16.

Arrange in decreasing order of stability of the singlet over the triplet: :CH_2, :CHF, :CF_2, :$C(OCH_3)_2$, :CPh_2.

Answer: :CF_2 ≈ :$C(OCH_3)_2$ > :CHF > :CH_2 (triplet ground) > :CPh_2 (triplet ground, widest angle)

π-donation from lone pairs on F and O into the empty p orbital plus inductive stabilisation of the σ orbital favour the singlet; two such substituents act cumulatively. H and alkyl do not; conjugating phenyls favour the triplet.

CE-17.

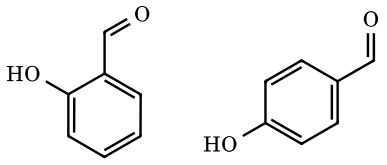
An unknown carbene gives 100 % cis-cyclopropane from cis-2-butene in solution but a 1:1 cis/trans mixture in the gas phase at low pressure. Interpret.

Answer: singlet carbene that relaxes (intersystem crossing) to the triplet before reacting

In solution, collisions with solvent quickly deactivate the energy but the singlet reacts at once: stereospecific. At low pressure the initially formed singlet has a long time before reacting and collisions induce ISC to the triplet ground state, giving the stepwise, non-stereospecific path (Skell/Bader). The cross-over from stereospecific to non-stereospecific with dilution shows that the singlet is the initial state.

CE-18.

Phenol is heated with chloroform and NaOH. (a) Name the main product, (b) say why para isomer is minor, (c) identify the electrophile and write the intermediate before hydrolysis.



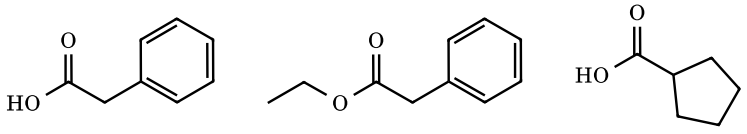
main **minor**

Answer: (a) salicylaldehyde (2-hydroxybenzaldehyde); (b) ion-pair / coordination of Na^+ with the phenoxide and carbene directs ortho; (c) :CCl_2; the intermediate is the dichloromethyl cyclohexadienone which loses Cl^- to give the o-quinone methide-type chloride that is hydrolysed

Phenoxide is a strong activator; dichlorocarbene (electrophile) attacks the ortho (and para) carbon; Na^+ coordination of the oxygen and the incoming carbene favour ortho (ortho : para ≈ 3 : 1). Hydrolysis of the $ArCHCl_2$ (aryl-$CHCl_2$ with phenoxide → o-quinone methide → ArCHO) gives the aldehyde. With CCl_4 the product is the carboxylic acid (salicylic acid) and with $CHBr_3$ the same aldehyde.

CE-19.

Give the products: (a) PhCOCl + CH_2N_2 (excess), then $Ag_2O/H_2O/\Delta$; (b) the same with C_2H_5OH in place of water; (c) 2-diazocyclohexanone, hv, H_2O.



a **b** **c**

Answer: (a) phenylacetic acid; (b) ethyl phenylacetate; (c) cyclopentanecarboxylic acid

Arndt–Eistert: acid chloride + CH_2N_2 → diazoketone (second equivalent removes HCl as CH_3Cl); Ag^+ catalyses Wolff rearrangement to a ketene PhCH=C=O; water or ethanol adds. In (c) the Wolff rearrangement contracts the ring by one carbon.

CE-20.

Explain why diazomethane methylates RCOOH and phenols (ArOH) but not alcohols (ROH) in the absence of a catalyst (BF_3 or HBF_4).

Answer: pK_a threshold for protonating CH_2N_2

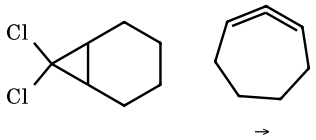
The first step is proton transfer to carbon. The conjugate acid $CH_3N_2^+$ is a very strong acid; CH_2N_2 is protonated significantly only by acids with pK_a ≲ 10 (carboxylic acids, phenols, enols, 1,3-dicarbonyls); alcohols (pK_a ≈ 16) are too weak. A Lewis-acid catalyst makes the alcohol react.

CE-21.
Which of the following gives a free carbene? (a) CH_2I_2 + Zn(Cu); (b) CH_2N_2 + hν; (c) $CHCl_3$ + tBuOK; (d) $Ph_2C{=}N_2$ + Δ; (e) CH_2I_2 + Et_2Zn; (f) CH_2N_2 + CH_3COOH.
Answer: (b), (c), (d)
(a) and (e) give the zinc carbenoid ICH_2ZnI; (f) is protonation of CH_2N_2 and S_N2 (ester formation). The others lose N_2 or Cl^- to produce $:CH_2$, $:CCl_2$ and Ph_2C: respectively.

CE-22.
How many different constitutional isomers of C_7H_{16} can be formed by insertion of singlet $:CH_2$ into the C–H bonds of 2-methylpentane?
Answer: 5
2-Methylpentane $(CH_3)_2CH{-}CH_2{-}CH_2{-}CH_3$ has five sets of non-equivalent hydrogens (C-1/C-1′, C-2, C-3, C-4, C-5). Insertion into each gives a different heptane: C-1–H → 3-methylhexane; C-2–H → 2,2-dimethylpentane; C-3–H → 2,3-dimethylpentane; C-4–H → 2,4-dimethylpentane; C-5–H → 2-methylhexane. All five are different constitutional isomers.

CE-23.
A student states that all carbenes are intermediates that add to alkenes with retention of alkene stereochemistry. Give two counterexamples.
Answer: triplet carbenes; carbenoids of different type
(1) $^3CH_2$ and Ph_2C: (triplet) add stepwise and scramble stereochemistry. (2) Metal carbenes (Fischer type) undergo [2+2]/metathesis rather than cyclopropanation. Also singlet carbenes with electron-poor alkenes can give nucleophilic ylide-type additions.

CE-24.
What is the product when bicyclo[4.1.0]heptane-7,7-dichloride (7,7-dichloronorcarane) is treated with methyllithium?



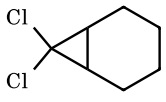
→

Answer: 1,2-cycloheptadiene (strained cyclic allene, via the carbenoid and the cyclopropylidene), trapped as a dimer or cycloadduct
Li/Cl exchange gives the carbenoid, loss of LiCl gives the bicyclic cyclopropylidene, which opens (Doering–LaFlamme) to the strained cyclic allene (1,2-cycloheptadiene) – isolable only as dimer or Diels–Alder adduct; with a more substituted ring the allene is isolable (eight-membered or larger).

CE-25.
Write a balanced equation for the complete alkaline hydrolysis of dichlorocarbene by hydroxide, and state the carbon-containing product.
Answer: $:CCl_2 + 2\ HO^- \rightarrow CO + 2\ Cl^- + H_2O$ (and $HCOO^-$ with more hydroxide)
Hydroxide attacks the carbene carbon; loss of chloride and a proton shift give HO–C̈–Cl-type species that lose HCl to carbon monoxide. Under strongly basic conditions CO is further hydrated to formate ($CO + HO^- \rightarrow HCOO^-$). The formation of CO and formate in the hydrolysis of chloroform was part of the evidence for $:CCl_2$ (Hine).

CE-26.
Dichlorocarbene is added to buta-1,3-diene. How many constitutional isomers form from one equivalent of carbene? How many stereoisomers are possible when two equivalents add to give the bis-cyclopropane?
Answer: 1 constitutional isomer (1,1-dichloro-2-vinylcyclopropane, as a racemate); the bis-adduct has 3 stereoisomers (R,R; S,S; meso)
Carbenes add 1,2 to a conjugated diene (no 1,4-addition). Both double bonds of butadiene are equivalent, so one monoadduct forms. In the bis-adduct 2,2,2′,2′-tetrachloro-bicyclopropyl the two ring carbons attached to each other are stereocentres: (R,R), (S,S) and the meso (R,S) form.

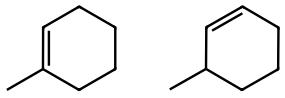
CE-27.
What are the intermediate and the final product of heating the sodium salt of trichloroacetic acid in the presence of cyclohexene?



Answer: $CCl_3^- \rightarrow :CCl_2$; 7,7-dichloronorcarane
Decarboxylation of Cl_3CCOO^- gives the trichloromethyl carbanion without needing HO^-, so a base-sensitive alkene or ester can be cyclopropanated under neutral conditions (Wagner, 1954).

CE-28.
Compare the stability and the reaction type of a carbene, a carbocation, a carbanion and a radical in a table (valence electrons on carbon, charge, geometry, typical role).
Answer: see table
CH_3^+: 6 e, + charge, planar sp^2, electrophile; $CH_3^\bullet$: 7 e, neutral, planar-shallow pyramid, radical; CH_3^-: 8 e, – charge, pyramidal sp^3, nucleophile/base; :CH_2: 6 e, neutral, bent (singlet 102°, triplet 134°), ambiphilic (singlet) or diradical (triplet). Full comparison in Part 14.

CE-29.
How does the Bamford–Stevens reaction of a ketone tosylhydrazone differ from the Shapiro reaction, and how do the products differ for 2-methylcyclohexanone tosylhydrazone?



BS **Shapiro**

Answer: Bamford–Stevens (NaOMe, Δ): more substituted alkene (carbene, 1,2-H shift, thermodynamic: 1-methylcyclohexene); Shapiro (2 RLi): less substituted alkene (3-methylcyclohexene) via the less hindered vinyl-lithium
In the BS reaction a diazo compound/carbene (or carbocation in protic solvent) leads to the thermodynamic Zaitsev-type alkene; in the Shapiro reaction the dianion removes the more accessible, α′ proton syn to the N–Ts group and extrudes sulfinate and N_2 to give the vinyllithium (no carbene), which is protonated to the kinetic, less substituted alkene.

CE-30.
(Assertion–Reason) Assertion: dichlorocarbene reacts with propene faster than with acrylonitrile. Reason: dichlorocarbene is an electrophile. Choose.
Answer: both true; the reason explains the assertion
:CCl_2 has a vacant p orbital and attacks the electron-rich π bond; the CN group withdraws electron density from CH_2=CH–CN, so the reaction is slow.

CE-31.
A chiral ketone R(R′)CH–CO–CHN_2 with R and R′ different is subjected to the Wolff rearrangement. What is the configuration of the migrating carbon in the product acid R(R′)CH–CH_2COOH?
Answer: retention
The migrating group moves from the carbonyl carbon to the carbene carbon with its pair of electrons while retaining configuration, as in all 1,2-shifts to an electron-deficient centre with a concerted three-centre TS (Part 7).

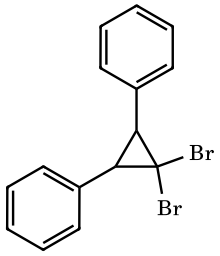
CE-32.
Predict the product of the intramolecular reaction of 1-diazo-5-phenylpentan-2-one with $Rh_2(OAc)_4$ (Buchner) and name the ring formed.
Answer: a bicyclic azulenone (5–7 fused) via carbene attack on the phenyl ring
The rhodium carbenoid adds to the arene ring in an intramolecular manner (Buchner reaction) forming the norcaradiene which opens to the cycloheptatriene fused to a cyclopentanone: bicyclo[5.3.0]decatrienone (the azulenone skeleton).

CE-33.
Why does CH_2Cl_2 with KOtBu give less carbene than $CHCl_3$?
Answer: $CHCl_2^-$ is a less stable carbanion than CCl_3^-
The acidity of the substrate depends on the number of chlorines (pK_a ≈ 24 for $CHCl_3$, ≈ 32 for CH_2Cl_2), and the loss of Cl^- from $CHCl_2^-$ gives the more reactive, less stable :CHCl.

CE-34.
What is the major difference in the products obtained from 1-hexene by (a) CH_2N_2/Cu_2Cl_2 and (b) CH_2N_2/hv?
Answer: (a) butylcyclopropane (selective, carbenoid); (b) butylcyclopropane plus C–H insertion products (isomeric heptenes and alkanes)
The copper carbenoid is selective for the C=C bond; free $^1CH_2$ from photolysis reacts with every C–H bond and gives a mixture.

CE-35.
Methylene generated from the photolysis of ketene (CH_2=C=O) in the gas phase reacts with a large excess of N_2 (inert gas) at high pressure. Which spin state reacts and why?
Answer: triplet
Collisions with the inert gas relax $^1CH_2$ (ISC) to the triplet ground state (≈ 9 kcal below) before reaction. The cis-2-butene products become a mixture of cis and trans dimethylcyclopropanes as the pressure rises (Frey).

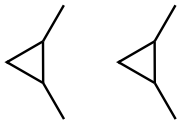
CE-36.
Write the product of treating trans-stilbene with $CHBr_3$/50 % NaOH/PTC and then hot with Ag^+. Name the types of reaction.



Answer: 1,1-dibromo-trans-2,3-diphenylcyclopropane; Ag^+ ionises it to a cyclopropyl cation that ring-opens (disrotatory) to a substituted allyl cation (→ 2-bromo-1,3-diphenylallyl derivatives)
Cyclopropanation by $:CBr_2$ followed by silver-assisted electrocyclic ring-opening (Woodward–Hoffmann disrotatory) of the cyclopropyl cation: a stereospecific route to allylic bromides.

CE-37.
Rank the following by relative rate of addition of $:CCl_2$ (fastest first): ethylene, 2-methyl-2-butene, 1-hexene, tetramethylethylene, cis-2-butene.
Answer: tetramethylethylene > 2-methyl-2-butene > cis-2-butene > 1-hexene > ethylene
The electrophilic carbene reacts faster with more highly alkyl-substituted, electron-rich C=C bonds (hyperconjugative donation): the relative rate roughly tracks the number of alkyl groups (≈ $1 : 10 : 10^2$… in the Doering series).

CE-38.
Predict the products (count stereoisomers) of CH_2I_2/Zn–Cu with (a) cis-2-butene, (b) trans-2-butene.



a **b**

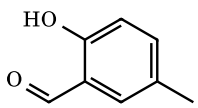
Answer: (a) one product, meso cis-1,2-dimethylcyclopropane; (b) trans-1,2-dimethylcyclopropane as a racemate (two enantiomers)
The carbenoid adds in one concerted syn step, so alkene geometry is retained. The cis product has a mirror plane (meso, achiral); the trans product has two stereocentres and no mirror plane, so it is chiral and forms as a 1 : 1 mixture of enantiomers.

CE-39.
Acetyl chloride is treated with excess CH_2N_2, and the product is heated with Ag_2O in water. Identify each product and the carbene-related step.
Answer: diazoacetone, then propanoic acid
CH_3COCl + 2 CH_2N_2 → CH_3COCHN_2 + CH_3Cl + N_2. Ag^+ promotes loss of N_2 to an acylcarbene, which undergoes the Wolff rearrangement (CH_3 migrates) to the ketene $CH_3CH{=}C{=}O$; water adds to give CH_3CH_2COOH (one carbon added to the acid).

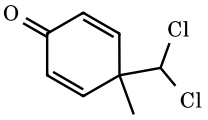
CE-40.
Dichlorocarbene reacts with norbornene. Give the major product and state its stereochemistry.
Answer: exo-3,3-dichlorotricyclo[3.2.1.$0^{2,4}$]octane
Attack occurs from the less hindered exo face of the strained double bond, giving the exo-fused dichlorocyclopropane. The addition is syn, as for any singlet carbene.

CE-41.
A thiazolium salt catalyses the benzoin condensation of benzaldehyde. Identify the carbene and the key intermediate, and say what is inverted.
Answer: thiazol-2-ylidene (NHC); the Breslow intermediate; the polarity of the carbonyl carbon (umpolung)
Base deprotonates C-2 of the thiazolium to give the NHC. It adds to PhCHO; proton transfer gives the Breslow enaminol, in which the former carbonyl carbon is nucleophilic (acyl-anion equivalent). It attacks a second PhCHO and the catalyst is expelled, giving benzoin.

CE-42.
p-Cresol undergoes the Reimer–Tiemann reaction. Give the normal product and the ‘abnormal’ product.



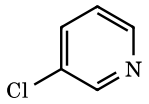
normal



abnormal

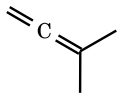
Answer: 2-hydroxy-5-methylbenzaldehyde; 4-(dichloromethyl)-4-methylcyclohexa-2,5-dienone
Ortho attack of :CCl_2 on the phenoxide followed by hydrolysis gives the aldehyde. Para attack at the methyl-bearing carbon cannot rearomatise, so the dichloromethyl dienone is isolated.

CE-43.
Pyrrole is treated with $CHCl_3$/NaOEt. Give the product and the key intermediate.



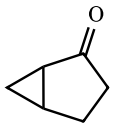
Answer: 3-chloropyridine; a 6,6-dichloro-2-azabicyclo[3.1.0]hexene (dichlorocyclopropane)
(Ciamician–Dennstedt) :CCl_2 adds across the C2=C3 bond; the cyclopropane opens with loss of Cl^- and ring expansion to the six-membered aromatic pyridine. Indole gives 3-chloroquinoline.

CE-44.
1,1-Dibromo-2,2-dimethylcyclopropane is treated with MeLi. Give the product.



Answer: 3-methylbuta-1,2-diene
Li/Br exchange and loss of LiBr give the cyclopropylidene, which opens (Doering–LaFlamme) by cleaving the C2–C3 bond to the allene.

CE-45.
1-Diazohex-5-en-2-one (CH_2=$CHCH_2CH_2COCHN_2$) is treated with $Cu(acac)_2$. Give the product.



Answer: bicyclo[3.1.0]hexan-2-one
The copper carbenoid adds intramolecularly to the C=C, forming a cyclopentanone ring fused to a cyclopropane.

CE-46.
Why are aryl–CF_3 diazirines preferred over aryl azides for photoaffinity labelling?
Answer: they give a carbene that inserts into C–H/X–H bonds indiscriminately
Photolysis ($\approx$ 350 nm) of the diazirine gives a carbene that inserts non-selectively into nearby bonds. Aryl azides give nitrenes that ring-expand to azepines and react mainly with nucleophiles, and the CF_3 group suppresses isomerisation of the diazirine to the diazo compound.

Part 11 Nitrenes: Structure, Generation, Reactions and the Rearrangements that Mimic Them

A **nitrene** (azene, imene) is the nitrogen analogue of a carbene: R–N:, a **monovalent nitrogen with six valence electrons** (one bond pair + two non-bonding pairs-worth of electrons = a lone pair and two further electrons). It is electron-deficient (sextet on N), neutral, and highly reactive. Because nitrogen is more electronegative than carbon, nitrenes are somewhat less electrophilic than carbenes of the same substituent, but they undergo the same family of reactions (insertion, addition, rearrangement).

11.1 Singlet and triplet nitrenes

	Triplet nitrene (ground state for most R–N, NH)	**Singlet nitrene**
Electron arrangement	lone pair in the s-type orbital; **one electron in each of two degenerate p orbitals**, spins parallel (↑ ↑): like O_2 and CH_2 (triplet)	lone pair in an sp-type orbital and a second pair in one p orbital; the other p orbital is **empty**

Magnetism	paramagnetic (S = 1; ESR)	diamagnetic
Reactions	radical-like: H-abstraction then recombination; dimerisation to R–N=N–R; non-stereospecific addition to alkenes	concerted: stereospecific aziridine formation (syn); C–H insertion with retention; rearrangements (acyl → isocyanate)
Generated first from	photolysis/thermolysis of azides gives singlet nitrene first; intersystem crossing to the triplet competes (fast for alkyl and aryl nitrenes)	most acyl and sulfonyl nitrenes in solution

The singlet–triplet gap of NH is ≈ 36 kcal mol^{-1} (triplet lower), much larger than that of CH_2 (9), so triplet is the ground state of almost all nitrenes. Aryl nitrenes can ring-expand to dehydroazepines (see 11.4). Also compare: **nitrenium ions** R_2N^+ (six electrons, positive charge; analogues of carbenium ions) and **nitrenoids** (N–X species that transfer N like a nitrene without free R–N:).

11.2 Generation of nitrenes

Method	Equation	Remarks
Azide decomposition	R–N=N^+=N^- → R–N: + N_2 (hν or Δ)	the standard source (alkyl, aryl, acyl, sulfonyl azides); N_2 is the leaving group
α-Elimination from N-halo or N-sulfonyloxy compounds	R–NH–X (X = Br, OSO_2Ar) + base → R–N: + HX	Hofmann bromamide and Lossen rearrangements are of this type (but see 11.3)
Deoxygenation of nitro/nitroso compounds	Ar–NO_2 + 2 $P(OEt)_3$ → Ar–N: + 2 $O=P(OEt)_3$ (Cadogan)	nitrene cyclises onto a neighbouring ring (2-nitrobiphenyl → carbazole)
Iminoiodinanes	PhI=NTs + Cu(I) or $Rh_2(OAc)_4$ → metal-nitrene → aziridine	transition-metal-catalysed nitrene transfer
Photolysis of isocyanates or diazirines	R–N=C=O → R–N: + CO (hν)	less common

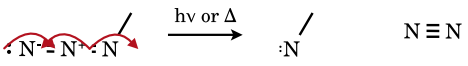
11.3 Rearrangements that proceed through nitrene-like (nitrenoid) steps

Four named reactions end in the same **isocyanate** R–N=C=O by a **1,2-shift of R from C to N** with simultaneous loss of a leaving group from N; each is classically drawn through an acyl nitrene R–CO–N:, but the migration and the loss of the leaving group are **concerted** (no free nitrene is formed), which explains the retention of configuration at the migrating carbon. Hydrolysis of the isocyanate gives the amine R–NH_2 with **one carbon less**:

Name	Substrate	Reagents	Leaving group from N	Product
Hofmann bromamide	$RCONH_2$	Br_2 + 4 NaOH	Br^-	R–NH_2 (1° amine, one C less) + Na_2CO_3
Curtius	$RCON_3$ (from RCOCl + NaN_3)	Δ, then H_2O	N_2	R–NH_2 (or R–NH–COOR′ in ROH)
Lossen	RCONH–OCOR′ (hydroxamic acid ester)	base, Δ	R′COO^-	R–NH_2
Schmidt	RCOOH + HN_3 (conc. H_2SO_4)	Δ	N_2	R–NH_2

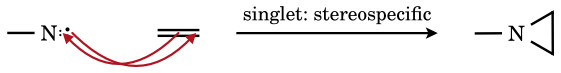
Migratory aptitude in these shifts is aryl ≈ tertiary alkyl > secondary > primary > methyl; the migrating group retains its configuration; the reaction works for R = alkyl, aryl and gives an amine in which N is attached to the same carbon that was attached to C=O. (**Contrast** the *Beckmann* rearrangement of an oxime (R_2C=N–OH → amide), in which the group *anti* to the leaving group migrates: C → N with loss of water, a carbocation-like (nitrilium) pathway.) The Stieglitz rearrangement of trityl hydroxylamine and N-chloroamines is the all-carbon-to-N shift with a positively charged (nitrenium-like) nitrogen.

11.1 Alkyl azide → nitrene + N_2



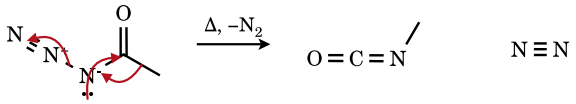
The azide is a resonance hybrid R–N=N^+=N^- ⇔ R–N^-–N^+ ≡ N. Loss of N_2 leaves the nitrogen attached to R with only one bond and six electrons: the nitrene. The N ≡ N is the thermodynamic driving force (≈ 226 kcal mol^{-1} bond energy).

11.2 Singlet methylnitrene + ethene → N-methylaziridine (concerted)



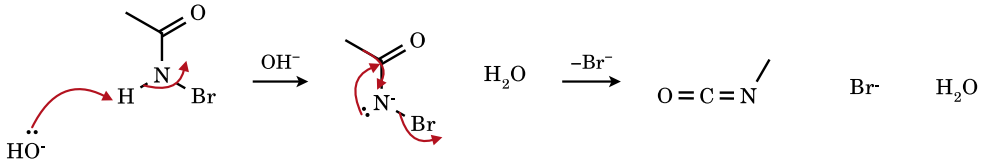
As for carbene cyclopropanation, the alkene π pair attacks the empty p orbital of the nitrene while the nitrene lone pair forms the second C–N bond: a one-step syn addition giving the aziridine with retention of alkene geometry. A triplet nitrene would give a mixture of cis and trans aziridines through a 1,3-diradical.

11.3 Curtius rearrangement: acyl azide → isocyanate (concerted, no free nitrene)



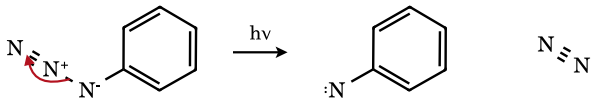
In the acyl azide the methyl group migrates from the carbonyl carbon to the adjacent nitrogen *as* N_2 leaves; the nitrogen lone pair makes the new C=N π bond. All three arrows act in one step (a concerted 1,2-shift with retention of the migrating group's configuration). Methyl isocyanate is trapped by water: R–NHCOOH → R–NH_2 + CO_2.

11.4 Hofmann bromamide: N-bromoamide anion → isocyanate



Br_2/NaOH brominates the amide nitrogen; a second HO^- removes the remaining N–H (the proton is acidic: $pK_a \approx 9$) to give the bromoamide anion. The methyl group then migrates to nitrogen as bromide leaves: the isocyanate. Hydroxide adds to R–N=C=O to give the carbamate, which loses CO_2: overall CH_3CONH_2 → CH_3NH_2 (one carbon less).

11.5 Aryl azide: ring expansion (dehydroazepine) after nitrene formation



Photolysis gives the singlet phenylnitrene, which can close to a benzazirine and then ring-expand to a 1-aza-1,2,4,6-cycloheptatetraene (didehydroazepine); in the presence of diethylamine this is trapped as a 2-diethylamino-3H-azepine. In the absence of traps the triplet nitrene dimerises to azobenzene. (Structure shown as the nitrene only.)

11.4 Reactions of nitrenes: summary

Reaction	Example	Product
Addition to C=C (aziridination)	PhI=NTs, $Cu(OTf)_2$, styrene; EtO_2C–N: + alkene	N-substituted aziridine; singlet: stereospecific; triplet: not
C–H insertion	alkyl/aryl/sulfonyl nitrene + R–H	amine (N–H bond forms at the C–H carbon); intramolecular insertion gives pyrrolidines, indolines
Rearrangement	R–CO–N: → R–N=C=O (Hofmann, Curtius, Lossen)	isocyanate → amine, urethane, urea
Dimerisation	2 ArN: → Ar–N=N–Ar	azo compound (triplet nitrenes)
Ring expansion of aryl nitrenes	PhN: → azirine → azepine	3H-azepines
Hydrogen abstraction	triplet nitrene + solvent R–H → R–NH• + R•	R–NH_2 after H abstraction
Cadogan cyclisation	2-nitrobiphenyl + $P(OEt)_3$	carbazole

11.6 Electronic structure of nitrenes, nitrenoids and nitrenium ions

- **Valence electron count.** N has five valence electrons. In R–N: one is in the R–N σ bond and four are non-bonding: a total of six electrons around N (sextet), two of them in the σ-bonding pair. The four non-bonding electrons occupy an s-like (sp) orbital and two degenerate p orbitals (p_x, p_y) perpendicular to the R–N axis. Three electronic states arise from placing two electrons in the two p orbitals: the ground-state triplet $^3\Sigma^-$ (↑ in p_x, ↑ in p_y), and two closed-shell-like singlets ($^1\Delta$, $^1\Sigma^+$). For NH: triplet X $^3\Sigma^-$, singlet a $^1\Delta$ at ≈ 36 kcal mol^{-1}, b $^1\Sigma^+$ at ≈ 60.
- **Why the gap is larger than in carbenes.** The two p orbitals of R–N: are exactly degenerate (the R–N axis is a C_∞ axis for a monovalent N), so Hund's rule places the electrons in separate orbitals with parallel spin; in a bent carbene the σ and p orbitals differ in energy and the singlet is competitive. Nitrene triplets are

ground states for alkyl, aryl and NH; singlets are ground states only when strongly stabilised (e.g. N-amino-nitrenes $R_2N–N$: 'aminonitrenes', and some acyl/imino nitrenes by lone-pair donation).

- **Substituent effects.** π-Donating R groups (NR_2, OR, X) donate to an empty p orbital of the singlet and so reduce the gap: aminonitrenes ($R_2N–N$:) are singlets; the nitrene of the acyl type RC(=O)–N: is a triplet in the gas phase but the singlet is lower than expected because C=O π overlaps one p orbital. Aryl nitrenes: triplet ground state by 14–18 kcal mol^{-1} for phenylnitrene (the unpaired electron is delocalised into the ring, giving a strong ESR signal of the triplet in a matrix). Alkyl nitrenes: triplet ground state, but a singlet alkylnitrene undergoes 1,2-H shift to an imine so fast ($k > 10^{10}$ s^{-1}) that it is rarely seen.
- **Nitrenoids.** Many 'nitrene' reactions are really transfers from metal-bound or N–X nitrogen with no free R–N:. Examples: iminoiodinanes PhI=NTs, chloramine-T and bromamine-T with Cu, Rh, Fe or Mn catalysts (metal–imido/nitrenoid intermediates, Du Bois C–H amination, Sharpless aminohydroxylation with Os); N-chloroamines + base; azides + metal. The metal-bound nitrenoid is more selective than a free nitrene and can be rendered enantioselective.
- **Nitrenium ions.** R_2N^+ (six electrons, positive charge, isoelectronic with a carbene) are singlets (or triplets for diarylnitrenium) formed by heterolysis of N–X (N-chloroamines + Ag^+), protonation of N-hydroxy-/N-acyloxyarylamines (the metabolic activation of aromatic amine carcinogens: ArN(OH)Ac → ArN^+Ac, which alkylates DNA at guanine C-8) and in Bamberger rearrangement of N-phenylhydroxylamine in aqueous acid (PhNHOH + H^+ → $PhNH^+$ → p-aminophenol by attack of water at the para position). Nitrenium ions are delocalised (iminium–carbocation resonance: $Ar–N^+R$ ⇔ cyclohexadienyl-type cation with positive charge at ortho and para carbons), which is why nucleophiles attack the ring.

11.7 Generation of nitrenes: a complete catalogue

Method	Precursor and conditions	Notes
Alkyl/aryl azides	$R–N_3$, hν (254–300 nm) or Δ (> 150 °C; aryl azides 100–140 °C)	the cleanest source; hazard: low-MW azides are shock-sensitive and explosive (rule of six: ≥ 6 carbons per azide); N_2 is the leaving group; initially singlet, relaxes to triplet in solution
Acyl azides	$RCON_3$, Δ (60–100 °C) → isocyanate directly; hν → acylnitrene (partly)	concerted Curtius; the same azide photolysed gives nitrene insertion products besides isocyanate
Sulfonyl and phosphoryl azides	$ArSO_2N_3$, hν or Δ; $(RO)_2P(O)N_3$	sulfonylnitrenes insert into C–H, add to alkenes (aziridines) and to arenes; DPPA (diphenyl phosphoryl azide) for the Shioiri–Yamada one-pot Curtius
Vinyl azides	$R_2C=CR–N_3$, Δ or hν	give 2H-azirines (Neber-type ring closure) via vinylnitrene or concerted loss of N_2; azirines are versatile building blocks
Azidoformates	$ROCON_3$, hν or Δ (or Rh_2)	carbalkoxynitrenes: aziridines and C–H insertion (carbamates of amines)
N-Aminophthalimide / N-amino heterocycles	$Pb(OAc)_4$ or $PhI(OAc)_2$ oxidation	Atkinson: N-phthalimidonitrene (singlet) adds stereospecifically to alkenes: a high-yield aziridination
α-Elimination of N-halo / N-sulfonyloxy	R–NH–X + base; $ArSO_2O–NHR$	Hofmann (Br), Lossen (carboxylate), Tiemann (sulfonates); nitrenoid
Deoxygenation of nitroarenes / nitroso	$ArNO_2$ + $P(OEt)_3$, Δ (Cadogan–Sundberg); ArNO + PPh_3	arylnitrene → carbazoles, indoles (Sundberg: 2-nitrostyrene), benzimidazoles, 2H-indazoles
Iminoiodinanes + metals	PhI=NTs/Cu(I)Cl or $Rh_2(OAc)_4$, Ru-porphyrin	metallonitrene transfer: aziridination of alkenes (Evans), C–H amination (Du Bois/Che: sulfamate esters + $Rh_2(esp)_2$/$PhI(OAc)_2$ → cyclic sulfamidates)
Photolysis of isocyanates, azirines, tetrazoles	RNCO → R–N: + CO; azirine ⇌ vinylnitrene	less common
Thermolysis of triazoles/tetrazoles	5-substituted tetrazoles → imidoyl nitrene	heterocyclic nitrene chemistry; ring-chain isomerism

11.8 Reactions of nitrenes in detail

(a) Insertion into C–H bonds (and into N–H, O–H, Si–H)

- Singlet nitrenes insert in a single step with retention of configuration at the carbon (like singlet carbenes); triplet nitrenes abstract H and the resulting radical pair recombines (loss of configuration, mixtures).
- **Selectivity:** sulfonyl- and carbalkoxy-nitrenes (electrophilic) favour tertiary > secondary > primary C–H (≈ 1 : 7 : 30–100 for metal nitrenes), and activated C–H (benzylic, allylic, α to O) are the most reactive.
- **Intramolecular C–H amination** builds pyrrolidines (1,5-insertion through a six-membered TS) and indolines from azides: the basis of C–H functionalisation in total synthesis (Du Bois: carbamates give oxazolidinones; sulfamates give oxathiazinanes; each the key step of tetrodotoxin and manzacidin syntheses).

(b) Addition to C=C: aziridination, and why the alkene geometry tells the spin state

Singlet nitrenes add syn and stereospecifically: cis-alkene → cis-aziridine. Triplet nitrenes add stepwise to a 1,3-diradical that closes after intersystem crossing, so geometry is partly lost. With arenes, nitrenes give azepines (ring expansion), by aziridination of one arene bond (azanorcaradiene ⇌ azepine, the nitrogen analogue of the Buchner reaction).

(c) Dimerisation, abstraction, and fragmentation

- Two triplet nitrenes dimerise to the azo compound: 2 ArN: → ArN=NAr (azobenzene from phenylnitrene; the major product of triplet aryl nitrenes in the absence of traps).
- Triplet nitrenes abstract hydrogen from the solvent to give the aminyl radical R–NH•, then R–NH_2 (aniline from phenyl azide in cyclohexane).
- Alkyl nitrenes undergo a 1,2-H or 1,2-alkyl shift to an imine (R–CH_2–N: → R–CH=NH), the nitrogen version of the carbene 1,2-H shift; acylnitrenes → isocyanates; vinyl nitrenes → azirines; azidoalkenes → nitriles by fragmentation; α-azido ketone → loss of N_2 gives α-imino ketone.

11.9 The nitrogen-migration family: scope, mechanism, stereochemistry, applications

The four reactions below (and the related Beckmann, Stieglitz and Tiemann rearrangements) all convert a C–R bond into an N–R bond by a 1,2-shift to an electron-deficient nitrogen. In the **Hofmann–Curtius–Lossen–Schmidt** family the migrating terminus is the carbonyl carbon and the product is an isocyanate; the amine formed after hydrolysis has **one carbon fewer** (CO_2 lost). In the **Beckmann** and **Schmidt (ketone)** variants the product is an amide/lactam with the carbonyl carbon retained.

Reaction	Substrate	Reagents	Group lost from N	Product	Key features
Hofmann	$RCONH_2$	Br_2/NaOH (NaOBr); or NaOCl; $Pb(OAc)_4$; $PhI(OCOCF_3)_2$	Br^-	RNH_2 (+ CO_3^{2-})	4 NaOH + Br_2 per amide; isocyanate trapped by ROH gives carbamates (modified Hofmann: NBS/DBU/MeOH)
Curtius	$RCON_3$	Δ; or RCOOH + DPPA/Et_3N (one pot)	N_2	RNCO → RNH_2, $RNHCO_2R'$, RNHCONHR′	mildest, no strong base; compatible with sensitive groups; Boc-protected amines with t-BuOH
Lossen	RCONHOCOR′ (O-acyl hydroxamate) or $ArSO_2ONHCOR$	base or Δ	$R'COO^-$	RNH_2	hydroxamic acids from ester + NH_2OH; useful when halogen oxidant is undesirable
Schmidt (acid)	RCOOH	HN_3, conc. H_2SO_4, Δ	N_2	RNH_2	acylium + HN_3 → protonated acyl azide
Schmidt (ketone)	$R_2C{=}O$	HN_3, H_2SO_4 (or $TiCl_4$)	N_2	amide RCONHR or lactam	cyclohexanone → caprolactam (nylon-6); anti group to N_2 migrates
Beckmann	$R_2C{=}N{-}OH$ (oxime)	H_2SO_4, PCl_5, TsCl/py, $SOCl_2$, PPA	H_2O	amide RCONHR′	group *anti* to the leaving OH migrates (stereospecific); nitrilium ion; fragmentation when R is stabilised (abnormal Beckmann)
Stieglitz	R_3C–NH–OH or N-chloroamine	Ag^+, PCl_5	H_2O or Cl^-	imine $R_2C{=}NR$	carbon → nitrogen shift to nitrenium; trityl hydroxylamine
Tiemann	amidoxime $RC(=NOH)NH_2$ + $ArSO_2Cl$	base	$ArSO_3^-$	N-substituted cyanamide	—

- **Evidence against a free acylnitrene in Hofmann/Curtius reactions in solution.** (i) Thermolysis of acyl azides gives isocyanate quantitatively with no insertion or addition products from the nitrene; (ii) photolysis of the same azide gives nitrene products (C–H insertion, aziridines) besides isocyanate; (iii) the rate of Curtius rearrangement correlates with migratory aptitude of R and is accelerated by electron-releasing R, which implies migration is concerted with N_2 loss; (iv) migration occurs with retention of configuration at the migrating carbon (Wallis and Nagel, 1931: optically active 2-methyl-2-phenylbutanamide → optically active amine); (v) crossover experiments show no free acylnitrene.
- **Migratory aptitude.** tertiary alkyl > cyclohexyl ≈ secondary alkyl ≈ benzyl ≈ aryl > primary alkyl > methyl (electron-rich groups migrate best). Mixed alkyl–aryl behaviour depends on conditions: in Hofmann, p-methoxy aryl > aryl; ⚑ for JEE state: ‘the group attached to C=O migrates’ and ignore competition; there is only one migrating group in an amide $RCONH_2$.

- **The carbon-count rule.** $RCONH_2$ → RNH_2; for the dicarboxylic diamide $H_2NCO–(CH_2)_n–CONH_2$ the product is $H_2N–(CH_2)_n–NH_2$ (two carbons less in total); for phthalimide → anthranilic acid (o-aminobenzoic acid) (industrial route to anthranilate esters and indigo, 4 NaOH + Br_2/NaOCl); for succinimide → β-alanine; for phthalamide → o-phenylenediamine… (with 2 Hofmann steps).
- **The Hofmann elimination is a different reaction** (E2 of a quaternary ammonium hydroxide, giving the least substituted alkene). Do not confuse 'Hofmann bromamide degradation' and 'Hofmann exhaustive methylation/elimination'.
- **Industrial and synthetic uses.** Curtius: biologically active carbamates (Boc/Cbz amines from acids), the Tamiflu synthesis; Hofmann: production of anthranilic acid; Schmidt: caprolactam alternative route (HN_3 is dangerous; Beckmann with H_2SO_4 is used on the 10^6-tonne scale); Lossen: amino-acid-derived isocyanates; isocyanates are monomers of polyurethanes (TDI, MDI: made industrially from phosgene, whose accident was the Bhopal MIC release).

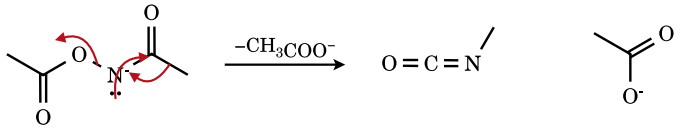
11.10 Choosing the right rearrangement: a decision table

Question asks for…	Think of	Check
amine with one carbon less than the amide	Hofmann (Br_2/KOH)	$RCONH_2$ must be a primary amide
amine from acid chloride via azide	Curtius	NaN_3 then Δ then H_2O
carbamate/urethane R–NH–COOR′	Curtius or Hofmann in the presence of ROH	isocyanate + R′OH
amide from ketone with HN_3	Schmidt	larger/aryl group migrates more easily
amide from oxime	Beckmann	anti group migrates; nylon-6 from cyclohexanone oxime
aziridine from alkene + sulfonyl azide / iminoiodinane + Cu	nitrene transfer	stereochemistry retained for singlet
carbazole from 2-nitrobiphenyl + $P(OEt)_3$	Cadogan reductive cyclisation	aryl nitrene inserts into the neighbouring C–H
N-chloramine + hν, H_2SO_4 → pyrrolidine	Hofmann–Löffler–Freytag	radical 1,5-HAT, not nitrene (Part 9)
p-aminophenol from N-phenylhydroxylamine	Bamberger rearrangement	nitrenium ion

Typical errors in nitrene/rearrangement questions. (1) Writing the Hofmann reaction with a tertiary or N-substituted amide – an N–H on the nitrogen is required to form the bromoamide anion. (2) Forgetting that the product amine is formed after loss of CO_2 (as carbonate in alkaline medium), and that the amine has the same R as the amide. (3) Applying Hofmann to a nitrile or an ester. (4) Believing the migrating group in Beckmann is the larger group: it is the one *anti* to the leaving group. (5) Treating the Curtius as proceeding through a free nitrene. (6) Confusing nitrene (neutral, six electrons, monovalent) with nitrile oxide/azide/amide anion. (7) Writing Hofmann bromamide reaction for alcohol/oxime. (8) Forgetting that aryl azide photolysis in the presence of amine gives azepines, not anilines.

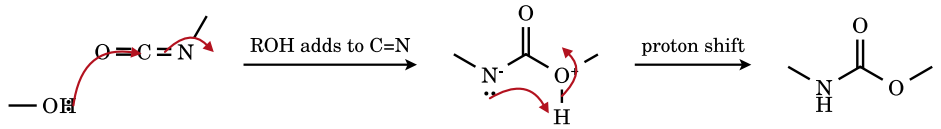
11.11 Additional validated mechanisms

11.6 Lossen rearrangement: O-acylhydroxamate anion → isocyanate



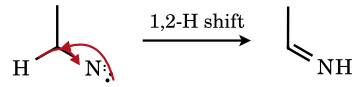
After deprotonation of the N–H (pK_a ≈ 8–9), the methyl group migrates from carbonyl carbon to nitrogen as carboxylate leaves, exactly as bromide leaves in the Hofmann reaction (11.4) and N_2 in the Curtius reaction (11.3). The leaving-group ability (N_2 > carboxylate ≈ Br^- in the anion) controls the temperature required: Curtius 60–100 °C, Lossen 25–100 °C with base.

11.7 Isocyanate trapped by methanol: carbamate (urethane) formation



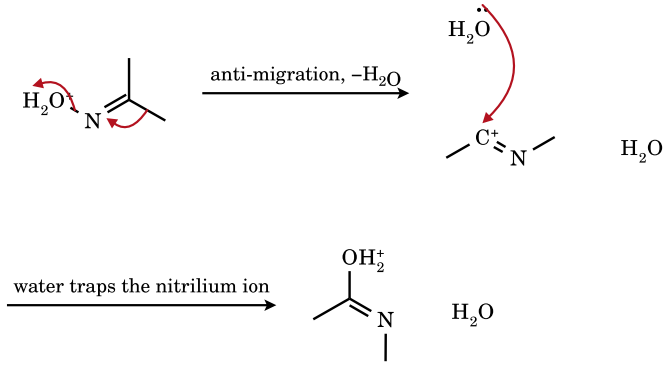
The alcohol oxygen attacks the electrophilic isocyanate carbon (the C=N π bond is the more reactive, giving an amidate anion), and a proton transfer yields the carbamate $CH_3NHCO_2CH_3$. With water the carbamic acid decarboxylates to the amine; with an amine the product is a urea. The same chemistry is the basis of polyurethane and polyurea manufacture.

11.8 Alkylnitrene → imine (1,2-hydride shift)



The C–H bond pair migrates to the electron-deficient nitrogen and a lone pair of nitrogen forms the C=N π bond. The shift is very fast (> 10^{10} s^{-1}) for singlet alkylnitrenes, which is why alkyl azides photolysed at low temperatures give imines rather than insertion products.

11.9 Beckmann rearrangement: anti-methyl migrates in protonated acetone oxime



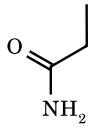
Protonation converts OH into the leaving group H_2O. As water leaves, the group *anti* to it (here a methyl) migrates from carbon to nitrogen, forming the nitrilium ion $CH_3–C^+=N–CH_3$ ⇔ $CH_3–C\equiv N^+–CH_3$. Water adds to carbon, and after a proton transfer the amide N-methylacetamide results. Contrast the Hofmann/Curtius: there the migrating group is on the carbonyl carbon and the leaving group is on the same nitrogen; here the migrating group is the one in the anti position on the C=N.

11.12 Problems on nitrenes

NI-1.
How many electrons does the nitrogen of a nitrene R–N: have in its valence shell? Is it a Lewis acid or base?
Answer: 6; both (lone pair + empty/half-filled p orbitals); mainly electrophilic
N contributes one electron to the N–R bond; the remaining four are two non-bonding electrons in each of two orbitals in the singlet arrangement (a pair in one orbital, a vacant p) = six electrons around nitrogen: a sextet. The lone pair makes it a base, the incomplete octet makes it a Lewis acid.

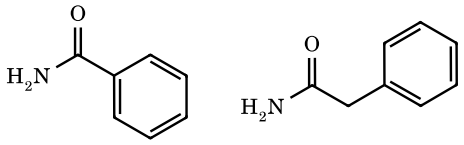
NI-2.
Identify the reactive intermediate in each: (a) $RCONH_2$ + Br_2 + KOH; (b) $RCON_3$, Δ; (c) $R–CH_2–N_3$, hν; (d) $CHCl_3$ + KOH.
Answer: (a) acyl nitrene (nitrenoid; concerted isocyanate formation), (b) same, (c) alkylnitrene, (d) dichlorocarbene
(a) and (b) are drawn through R–CO–N:, but experimentally the R-shift and loss of Br^- or N_2 occur together. In (c) photolysis expels N_2 from the azide to give $R–CH_2–N$:. (d) is the carbene reaction of Part 10.

NI-3.
Propanamide is treated with Br_2/KOH. Name the product and say how the carbon count changes.



Answer: ethanamine (ethylamine), one carbon less
Hofmann degradation: the ethyl group migrates from C to N to give ethyl isocyanate, which is hydrolysed and decarboxylated to $CH_3CH_2NH_2$ + CO_3^{2-}.

NI-4.
Benzamide → ? and phenylacetamide → ?; (Br_2/NaOH).



Answer: aniline; benzylamine
The group attached to the carbonyl carbon migrates to nitrogen: phenyl in benzamide (→ $PhNH_2$) and benzyl in phenylacetamide (→ $PhCH_2NH_2$). The migrating carbon keeps its configuration (R-$CH(CH_3)CONH_2$ optically active → retained).

NI-5.
A chiral amide (S)-$CH_3CH(C_2H_5)CONH_2$ undergoes Hofmann degradation. What is the stereochemical result?
Answer: retention of configuration (the amine is (S)-sec-butylamine in the CIP sense)
The migrating carbon moves with its bonding pair in a 3-centre transition state and never becomes planar; no inversion and no racemisation occur.

NI-6.
Which is not an intermediate or product in the Curtius rearrangement of RCOCl? (A) $RCON_3$, (B) RN=C=O, (C) RNH_2, (D) R–CH_2NH_2.
Answer: (D)
RCOCl + NaN_3 → $RCON_3$ → RN=C=O + N_2 → (H_2O) RNH_2 + CO_2. The amine has one carbon less than the acid; R–CH_2NH_2 would have the same carbon count.

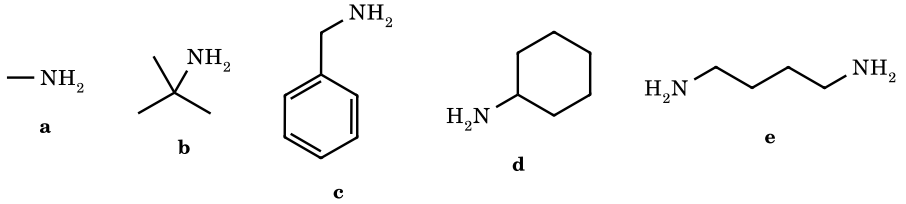
NI-7.
Compare the Curtius, Hofmann, Lossen and Schmidt reactions by giving the leaving group lost from nitrogen.
Answer: N_2; Br^-; carboxylate; N_2
Curtius (from acyl azide): N_2; Hofmann (N-bromoamide anion): Br^-; Lossen (O-acylhydroxamate anion): $R'COO^-$; Schmidt (carboxylic acid + HN_3, protonated acyl azide): N_2. All give the isocyanate by a 1,2-shift.

NI-8.
Phenyl azide is irradiated in the presence of excess diethylamine. Which product is expected, and what does it show about the intermediate?
Answer: 3H-azepine (2-diethylamino-3H-azepine); ring-expanded didehydroazepine
The singlet phenylnitrene cyclises to a benzazirine, which opens to the strained cyclic ketenimine; the amine traps it. If the amine is absent the triplet nitrene dimerises to azobenzene.

NI-9.
How does the stereochemistry of aziridine formation distinguish singlet from triplet nitrenes?
Answer: singlet: stereospecific; triplet: loss of stereochemistry
Singlet nitrene adds in one concerted step (syn); triplet nitrene adds one bond at a time forming a 1,3-diradical that can rotate before ring closure, so cis- and trans-alkenes give the same cis/trans aziridine mixture.

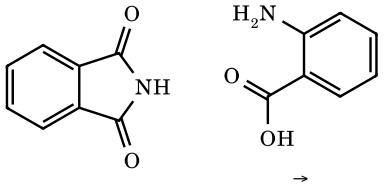
NI-10.
Why is the triplet the ground state for most nitrenes?
Answer: two degenerate p orbitals each hold one electron (Hund's rule)
In R–N: the two non-bonding p orbitals on nitrogen are degenerate (cylindrical symmetry about the R–N axis), so the lowest-energy arrangement places one electron in each with parallel spins, exactly as in O_2 or the NH radical.

NI-11.
Give the product of Hofmann bromamide degradation of (a) CH_3CONH_2, (b) $(CH_3)_3CCONH_2$, (c) $C_6H_5CH_2CONH_2$, (d) cyclohexanecarboxamide, (e) $H_2NCO(CH_2)_4CONH_2$.



Answer: (a) CH_3NH_2; (b) t-$BuNH_2$; (c) benzylamine; (d) cyclohexylamine; (e) 1,4-diaminobutane (putrescine)
R–$CONH_2$ → R–NH_2; for the diamide both $CONH_2$ groups are degraded, leaving H_2N–$(CH_2)_4$–NH_2 (adipamide has 6 C: two are lost as carbonate).

NI-12.
Phthalimide is treated with Br_2 and excess aqueous NaOH. Identify the product after acidification, and state its use.



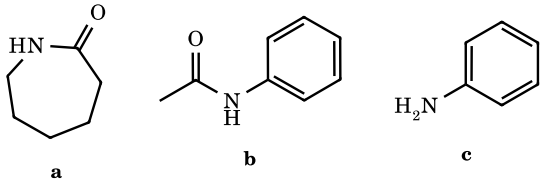
Answer: anthranilic acid (2-aminobenzoic acid)
NaOH opens the imide to the phthalamate; the remaining $CONH_2$ undergoes Hofmann degradation to give NH_2 on the ring while the carboxylate stays. Anthranilic acid is the starting material for indigo, methyl anthranilate (perfumery) and benzyne generation (via diazotisation to benzenediazonium-2-carboxylate).

NI-13.
Convert (a) CH_3CH_2COOH to $CH_3CH_2NH_2$ by two methods, (b) CH_3CH_2COOH to $CH_3CH_2CH_2NH_2$.
Answer: (a) (i) $SOCl_2$; NaN_3; Δ; H_2O (Curtius), or (ii) $SOCl_2$; NH_3; Br_2/KOH (Hofmann); (b) reduce acid or its amide to the alcohol/amine: $CH_3CH_2COOH \rightarrow CH_3CH_2CONH_2 \rightarrow LiAlH_4 \rightarrow CH_3CH_2CH_2NH_2$
Hofmann/Curtius remove one carbon; to preserve the carbon count use $LiAlH_4$ reduction of the amide (or nitrile). To add one carbon: acid → alcohol → bromide → CN^- → $LiAlH_4$.

NI-14.
An optically active amide, (R)-2-phenylbutanamide, is subjected to Hofmann degradation. State the product, its configuration and why optical purity is retained.
Answer: (R)-1-phenylpropylamine (retention); no racemisation
The migrating carbon never becomes a free cation or radical; it is bound to both C and N in the three-centre transition state throughout, so its configuration is retained (the change of CIP descriptor depends only on priority changes: here NH_2 > Ph > Et, so the descriptor remains R).

NI-15.
Which of the following can undergo Hofmann bromamide degradation? (a) CH_3CONH_2, (b) $CH_3CONHCH_3$, (c) $CH_3CON(CH_3)_2$, (d) $CH_3COOC_2H_5$, (e) CH_3CN, (f) $HCONH_2$.
Answer: (a) only (and arguably (f) gives NH_3/HNCO but not a meaningful amine)
Only primary amides $RCONH_2$ have two N–H bonds: one is replaced by Br, the other deprotonated to the bromoamide anion that rearranges. N-Substituted amides lack the second N–H, and (d), (e) have no N–H at all.

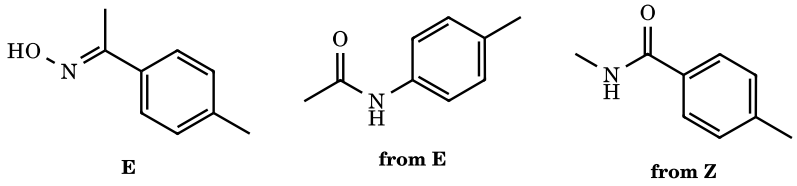
NI-16.
Name the product of Schmidt reaction of (a) cyclohexanone + HN_3/H_2SO_4, (b) acetophenone + HN_3/H_2SO_4, (c) benzoic acid + HN_3/H_2SO_4.



Answer: (a) ε-caprolactam; (b) acetanilide (phenyl migrates); (c) aniline
(a) ring expansion: the ring carbon migrates to nitrogen of the iminodiazonium ion. (b) the group that migrates is the one anti to N_2^+ in the preferred geometry; with aryl alkyl ketones the aryl group migrates to give an N-aryl amide. (c) acid → amine with one carbon less.

NI-17.
Compare the Beckmann rearrangement of cyclohexanone oxime with the Schmidt reaction of cyclohexanone. What do they have in common, and which is used industrially for nylon-6?
Answer: same product (ε-caprolactam); Beckmann
Both produce a seven-membered lactam by ring-carbon migration to nitrogen with loss of a leaving group (H_2O or N_2). Industrially the Beckmann (oleum) is used because HN_3 is toxic and explosive. ε-Caprolactam is polymerised (ring-opening, 250 °C) to nylon-6.

NI-18.
In the Beckmann rearrangement of the E-oxime of 4-methylacetophenone the group anti to OH is the aryl group. Write the product and predict the product from the Z-oxime.



E from E from Z

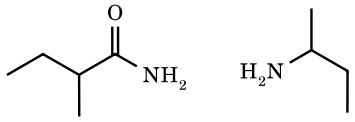
Answer: E: N-(4-methylphenyl)acetamide (aryl migrates); Z: N-methyl-4-methylbenzamide
The migrating group is always the one anti to the leaving group; under acid conditions oximes interconvert and the more stable E isomer predominates, but under non-equilibrating conditions (TsCl/pyridine, PCl_5 at low T) the product reflects the oxime geometry.

NI-19.
State whether each of the following contains a free nitrene intermediate under the stated conditions: (a) CH_3CON_3, 80 °C; (b) CH_3CON_3, hv; (c) PhN_3, hv; (d) TsN=IPh + Cu(I); (e) o-nitrobiphenyl + $P(OEt)_3$.
Answer: (a) no (concerted Curtius); (b) partly yes; (c) yes (singlet then triplet); (d) no (copper–imido, nitrenoid); (e) yes (arylnitrene, or nitrenoid with phosphorus)
Thermal Curtius gives isocyanate cleanly; photolysis yields insertion/addition products indicating real acylnitrene; phenylnitrene is formed from PhN_3 and ring-expands; Cu and Rh give metal-bound nitrene, and Cadogan intermediates are nitrene-like species whose cyclisation is on the nitrene or its deoxygenation precursor (nitroso).

NI-20.
Phenyl azide is heated in benzene and the product is mainly azobenzene together with aniline; when irradiated at −196 °C in a glass, an ESR signal appears. Identify the intermediate.
Answer: triplet phenylnitrene
The ESR signal (zero-field splitting D ≈ 1 cm^{-1}) is the fingerprint of a triplet (S = 1). Dimerisation of two triplet nitrenes gives PhN=NPh; H-abstraction from solvent gives $PhNH^{\bullet}$ and $PhNH_2$.

NI-21.
(Match the column) Reaction – leaving group lost from nitrogen: (A) Hofmann, (B) Curtius, (C) Lossen, (D) Beckmann (P) N_2, (Q) Br^-, (R) $RCOO^-$, (S) H_2O.
Answer: A–Q; B–P; C–R; D–S
Hofmann: N-bromoamide anion → bromide; Curtius: acyl azide → N_2; Lossen: O-acyl hydroxamate → carboxylate; Beckmann: protonated oxime → water.

NI-22.
A compound A ($C_5H_{11}NO$) is a primary amide. On Hofmann degradation it gives an optically active amine B ($C_4H_{11}N$). Identify A and B.

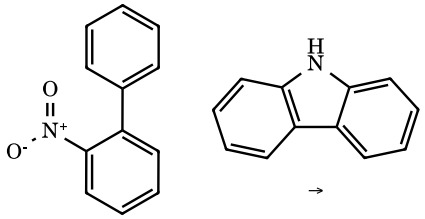


A B

Answer: A = 2-methylbutanamide; B = butan-2-amine
Of the four C_5 primary amides only 2-methylbutanamide gives a chiral amine, since the migrating sec-butyl carbon retains configuration. The others give n-butyl, isobutyl and tert-butylamines, all achiral.

NI-23.
Explain why bromamide anion $RCON^-Br$ rearranges at 60–80 °C but N-chloroamides or N-bromoamides are stable at 0 °C.
Answer: the amide N–H must be removed to give the anion; the neutral N-bromoamide is stable
Rearrangement requires the negatively charged nitrogen whose lone pair forms the new C=N π bond as the migrating group moves; in the neutral N-haloamide that lone pair is delocalised into C=O. Only with base is the nitrogen deprotonated and the isocyanate-forming step possible.

NI-24.
Predict the product: 2-nitrobiphenyl + triethyl phosphite, 160 °C.



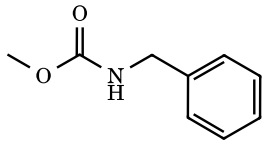
Answer: carbazole
$P(OEt)_3$ deoxygenates the nitro group twice (to the nitroso then nitrene stage; 2 equiv of phosphite give 2 $O{=}P(OEt)_3$); the aryl nitrene then inserts into the adjacent ring C–H bond (electrocyclisation/1,5-insertion) to form the five-membered ring (Cadogan).

NI-25.
N-aminophthalimide is oxidised with $Pb(OAc)_4$ in the presence of cis-2-butene and trans-2-butene separately. Predict the stereochemistry.
Answer: cis → cis-aziridine; trans → trans-aziridine
N-Phthalimidonitrene is a singlet aminonitrene; singlets add in a single concerted syn step, so alkene stereochemistry is retained (Atkinson).

NI-26.
Which is more stable, a triplet or singlet phenylnitrene, and why does phenyl azide photolysis in diethylamine give an azepine rather than N,N-diethyl-N′-phenylhydrazine?
Answer: triplet; azepine arises from singlet-derived ring expansion
Triplet PhN: is the ground state, but the initially formed singlet is converted ($k \approx 10^9$–10^{10} s^{-1}) to benzazirine then didehydroazepine before it can intersystem-cross; the strained cyclic ketenimine reacts with nucleophiles (Et_2NH adds) to give 2-diethylamino-3H-azepine. With ortho substituents or at low T the ISC pathway wins.

NI-27.
Distinguish nitrene, nitrenium ion, nitrenoid and nitrile by (i) electron count, (ii) charge, (iii) typical reactivity.
Answer: see reasoning
Nitrene R–N: (6 e, neutral, electrophilic/insertion/addition); nitrenium R_2N^+ (6 e, +, electrophile; attacks aromatic ring or DNA bases); nitrenoid M=NR or $X–NR^-$ (transfer of NR, with metal/halide still attached); nitrile $R–C \equiv N$ (8 e, stable).

NI-28.
In the Hofmann degradation of the amide $PhCH_2CONH_2$ with Br_2/KOH in methanol (instead of water) the product is a carbamate. Give its structure and explain.



Answer: methyl N-benzylcarbamate
The isocyanate $PhCH_2N{=}C{=}O$ is trapped by the solvent methanol (faster than by hydroxide). This ‘modified Hofmann’ is used to obtain protected amines; the carbamate can be hydrolysed or hydrogenolysed to the amine.

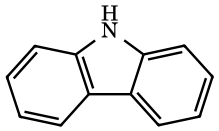
NI-29.
How many different primary amines $C_4H_{11}N$ can be obtained by Hofmann degradation of primary amides $C_5H_{11}NO$?
Answer: 4
The four pentanamide skeletons (pentanamide, 2-methylbutanamide, 3-methylbutanamide, 2,2-dimethylpropanamide) give n-butylamine, sec-butylamine, isobutylamine and tert-butylamine – the four primary $C_4H_{11}N$ amines.

NI-30.
Why do aryl azides decompose at lower temperatures than alkyl azides, and why do aryl nitrenes ring-expand while alkyl nitrenes undergo 1,2-H shifts?
Answer: conjugation with the arene lowers the TS energy; the aryl π system offers cyclisation, the alkyl group offers a migrating H
Conjugation of the azide with the ring stabilises the transition state for N_2 loss (aryl azides lose N_2 at ≈ 100–140 °C, alkyl azides at higher temperatures or by photolysis). The singlet arylnitrene can attack the ortho carbon (benzazirine → didehydroazepine); an alkylnitrene has no π system and the adjacent C–H bond migrates to nitrogen to give an imine.

NI-31.
(S)-2-Phenylpropanoic acid is converted to its acyl azide, heated, and hydrolysed. Give the product and its configuration.
Answer: (S)-1-phenylethylamine
Curtius migration of the 1-phenylethyl group occurs with retention. In the acid the priority order is COOH > Ph > CH_3 > H and in the amine NH_2 > Ph > CH_3 > H, so NH_2 takes the rank of COOH and the descriptor stays S.

NI-32.
Write the balanced equation for the Hofmann degradation of $RCONH_2$ with Br_2/NaOH and state the moles of NaOH consumed per mole of amide.
Answer: $RCONH_2 + Br_2 + 4\ NaOH \rightarrow RNH_2 + Na_2CO_3 + 2\ NaBr + 2\ H_2O$; 4 mol
The Na, O and H balance: Na 4 = 2 + 2; O 1 + 4 = 3 + 2; H 2 + 4 = 2 + 4. NaOH is used to form NaOBr, to deprotonate the N–H, and to convert CO_2 (from the isocyanate) to carbonate.

NI-33.
2-Azidobiphenyl is heated in an inert solvent. Give the main product and the intermediate.

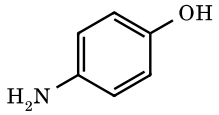


Answer: carbazole via an aryl nitrene (insertion into the adjacent ring C–H)
Loss of N_2 gives the nitrene on one ring, which inserts into the ortho C–H of the other ring to form the five-membered pyrrole ring. The same carbazole arises from 2-nitrobiphenyl and $P(OEt)_3$ (Cadogan).

NI-34.
Ethyl azidoformate is photolysed in cyclohexene. Name the two kinds of product and say how stereochemistry would reveal the spin state.
Answer: N-ethoxycarbonyl-7-azabicyclo[4.1.0]heptane (aziridine) and allylic C–H insertion product
A singlet carbalkoxynitrene adds to C=C in one step (syn, stereospecific) and can insert into allylic C–H. A triplet nitrene adds in two steps via a diradical; with cis- or trans-2-butene it would give aziridines of mixed configuration.

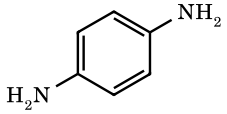
NI-35.
A carboxylic acid is treated with DPPA/Et_3N and then t-BuOH under reflux. Give the product type and the intermediates.
Answer: R–NH–Boc via acyl azide and isocyanate
RCOOH is converted to the acyl azide (via the mixed phosphoryl anhydride), which rearranges on heating to RN=C=O; t-BuOH adds to give the Boc carbamate (Shioiri–Yamada). Retention at R.

NI-36.
N-Phenylhydroxylamine is warmed in dilute H_2SO_4. Give the product and the intermediate.



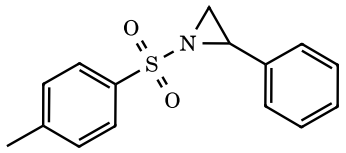
Answer: p-aminophenol via a nitrenium ion (Bamberger rearrangement)
Protonation of OH and loss of water give the phenylnitrenium ion, delocalised onto the para carbon; water attacks there and tautomerisation gives p-aminophenol.

NI-37.
Terephthalamide is treated with excess NaOBr. Give the product.



Answer: p-phenylenediamine
Each $CONH_2$ undergoes Hofmann degradation, losing its carbonyl carbon as carbonate and leaving NH_2 on the ring carbon.

NI-38.
Styrene reacts with PhI=NTs and $Cu(OTf)_2$. Give the product, and say where a nucleophile will open it.

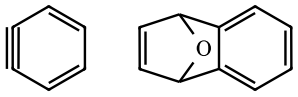


Answer: 2-phenyl-1-tosylaziridine; opening at the benzylic carbon
A copper-nitrenoid transfers NTs to the C=C (Evans aziridination). Nucleophiles open the activated aziridine at the benzylic carbon, which best stabilises the developing positive charge.

Part 12 Benzyne (Arynes): Structure, Generation, Evidence and Reactions

Benzyne (1,2-didehydrobenzene, C_6H_4) is a highly strained, short-lived reaction intermediate in which two adjacent carbons of a benzene ring have lost their substituents and share an additional, in-plane, bond. It is the simplest **aryne** (aromatic ring with a formal triple bond), first proposed by Wittig (1942) and demonstrated by Roberts (1953, 14C-labelled chlorobenzene) and by trapping with furan (Wittig, 1955).

12.1 Structure and bonding



benzyne **furan adduct**

Left: benzyne drawn as cyclohexa-1,3-dien-5-yne (the 'triple bond' is one σ + one π of the aromatic sextet + one extra in-plane bond). Right: the Diels–Alder adduct with furan (1,4-epoxy-1,4-dihydronaphthalene), the standard proof of existence.

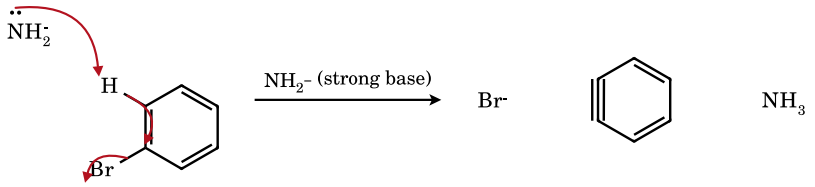
- The **aromatic π sextet** (six p orbitals perpendicular to the ring plane) is intact: benzyne is still aromatic.
- The 'third' bond is formed by the sideways overlap of the two **sp^2 orbitals** that held the leaving groups (H and X); these orbitals lie *in the plane of the ring*, pointing away from the ring centre at ≈ 120°, so they cannot overlap well. The overlap is poor, the extra bond is **very weak and strained** (strain ≈ 50 kcal mol^{-1}) and the LUMO is low: benzyne is a strong electrophile/dienophile.
- **It is not a true alkyne**: the C ≡ C is not linear (ring angles ≈ 127° at the benzyne carbons), the bond is longer (≈ 1.24 Å) than a normal triple bond (1.20 Å) and shorter than an aromatic bond, and the carbons are sp^2 (not sp). It has no net charge, no unpaired electrons (diamagnetic) and a dipole moment; lifetime ≈ 10^{-6}–10^{-3} s; observed in matrix isolation (IR 1846 cm^{-1}, 'C ≡ C' stretch, intermediate between a double and a triple bond).

12.2 Generation of benzyne

Method	Reagents	Equation	Comment
Elimination–addition (strong base)	Ar–X + $NaNH_2$/liquid NH_3; KNH_2; PhLi; LDA	ArX + NH_2^- → C_6H_4 + X^- + NH_3; benzyne + NH_2^- → anilide → aniline	**requires an ortho H** (2,6-disubstituted halides do not react); rate X: Br > I > Cl ⋙ F
From o-dihalides	1-bromo-2-fluorobenzene + Mg (THF); o-dibromobenzene + n-BuLi (–100 °C); Li–Hg	metal–halogen exchange gives o-haloaryl-M, which loses MX	o-FC_6H_4MgBr → C_6H_4 + MgBrF
From benzenediazonium-2-carboxylate	anthranilic acid + isopentyl nitrite (diazotisation), or the zwitterion	$C_6H_4(N_2^+)(COO^-)$ → C_6H_4 + N_2 + CO_2 (Δ)	clean, but the zwitterion is explosive when dry
Kobayashi precursor	2-(trimethylsilyl)phenyl triflate + F^- (CsF, Bu_4NF)	F^- attacks Si; C–Si electrons expel OTf^-	mild, neutral, the modern method
From phthalic anhydride / benzothiadiazole dioxide / 1-aminobenzotriazole (oxidation)	$Pb(OAc)_4$, hν	loss of N_2, CO, CO_2 from a fused five-membered ring	gas-phase and photochemical routes
Dehalogenation of o-halo triflates	ArBr(OTf) + n-BuLi	Li–Br exchange then loss of LiOTf	fast, low temperature

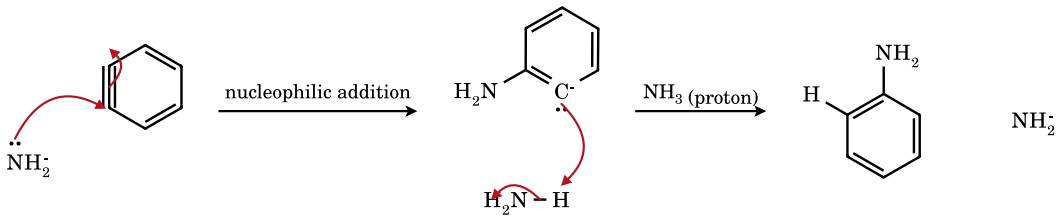
12.3 Mechanisms (arrow-validated)

12.1 Elimination step: bromobenzene + NH_2^- → benzyne



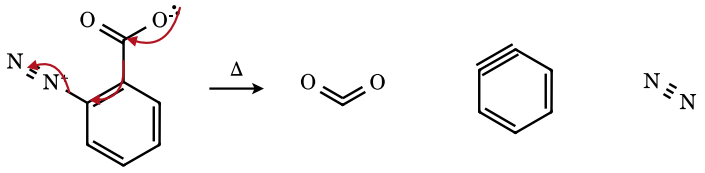
Amide removes the proton ortho to Br (the sp^2 C–H is acidified by the adjacent C–Br dipole), the C–H electrons form the in-plane bond to C-1, and bromide leaves. In terms of mechanism this is E2 (or E1cB with the aryl anion of F). The reaction needs a base strong enough to remove an aromatic H ($pK_a \approx 43$): $NaNH_2$ (NH_3 $pK_a \approx 38$) is effective only because C–X departure is concerted.

12.2 Addition step: benzyne + NH_2^- → aryl anion → aniline



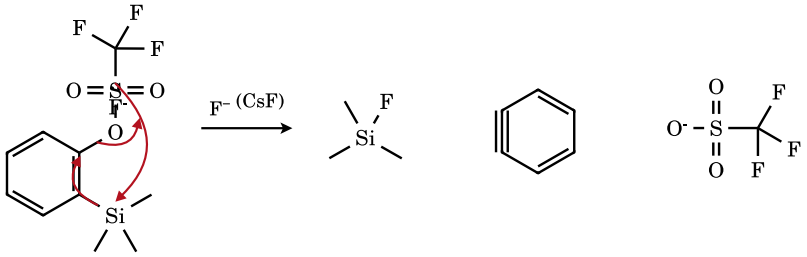
The nucleophile adds to one carbon of the strained C ≡ C and the in-plane π pair moves to the neighbouring carbon, giving an aryl *σ*-carbanion (lone pair in the sp^2 orbital in the plane, not conjugated with the ring), which is protonated by ammonia: aniline. In the Roberts experiment 1-^{14}C-chlorobenzene gave aniline with 50 % of the label at C-1 and 50 % at C-2, because the two benzyne carbons are equivalent: this is cine substitution evidence.

12.3 Benzyne from benzenediazonium-2-carboxylate



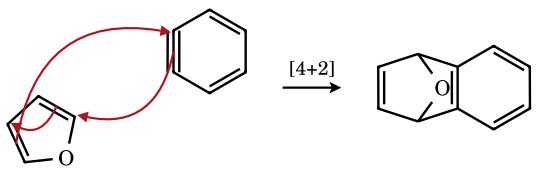
Three simultaneous electron shifts: the carboxylate oxygen forms the second C=O of CO_2, the C–C σ pair becomes the new in-plane bond, and N_2 leaves with the C–N pair. All three leaving groups (CO_2, N_2) are gases, so the reaction is irreversible and clean.

12.4 Kobayashi benzyne: 2-(trimethylsilyl)phenyl triflate + F^-



Fluoride, a hard nucleophile with a huge affinity for silicon (Si–F 135 kcal mol^{-1}), attacks Si; the C–Si bond pair becomes the in-plane bond while triflate (an excellent leaving group) departs. No strong base is needed, so benzyne can be generated in the presence of base-sensitive traps.

12.5 Diels–Alder trapping of benzyne with furan (the proof of benzyne)



The diene (furan) and the dienophile (benzyne) join in one concerted step in which the weak in-plane bond of benzyne is converted into two σ bonds. The adduct, 1,4-dihydronaphthalene-1,4-endoxide, is stable and isolated; it is the classical evidence that benzyne forms when no nucleophile is present. Anthracene (→ triptycene) and cyclopentadiene are also used as traps.

12.4 Evidence for the benzyne mechanism

Observation	Meaning
Roberts (1953): PhCl(1-^{14}C) + KNH_2/NH_3 gave aniline with 14C at C-1 and C-2 in equal amounts	the two benzyne carbons are equivalent: elimination then addition (cine substitution)
o-, m- and p-bromotoluene with $NaNH_2$ give mixtures of toluidines (o-: o- + m-; m-: o- + m- + p-; p-: m- + p-)	different benzynes are formed; the nucleophile may attack either carbon
2,6-dimethylbromobenzene (no ortho H) does not react with $NaNH_2/NH_3$	an ortho H is required for the elimination
Trapping with furan, anthracene, tetraphenylcyclopentadienone, cyclopentadiene, or reaction with a nucleophile present	an intermediate that is both an electrophile and a dienophile
Isotope exchange: PhBr in KNH_2/ND_3 exchanges ortho H with D before reaction	the aryl anion (E1cB) is formed reversibly
Halide reactivity order Br > I > Cl ⩾ F in $NaNH_2/NH_3$	for F the C–F cleavage (leaving group loss) is the slow step; for Cl, Br the C–H proton removal is fast and C–X cleavage is rate-determining at intermediate X; the order reflects both acidity and leaving-group ability

12.5 Regiochemistry of nucleophilic addition to substituted benzynes

The nucleophile adds to that carbon of the benzyne triple bond that gives the **more stable aryl carbanion** (an sp^2 σ-carbanion). The carbanion is not conjugated to the ring π system, so **only the inductive effect** of the substituent matters: an –I group stabilises the carbanion best when it is **closest** to the carbanion carbon.

Benzyne	Substituent effect	Where Nu^- adds; major product
3-methoxybenzyne (from o- or m-bromoanisole)	–I of OMe (σ-withdrawing)	C-1 (meta to OMe) so that the carbanion forms at C-2, next to OMe → **m-anisidine** (3-methoxyaniline)
3-methylbenzyne (from o-bromotoluene)	CH_3 is +I: slight effect; steric	≈ 50 : 50 or slightly favouring meta → **o-toluidine and m-toluidine** (nearly equal)
4-methylbenzyne (from p-bromotoluene or m-bromotoluene)	the methyl is remote from both carbons	m- and p-toluidine, ≈ 1 : 1
3-(trifluoromethyl)benzyne	strong –I (CF_3)	attack at C-1 (meta) gives the carbanion next to CF_3: **meta** product
4-(trifluoromethyl)benzyne	–I of CF_3 is felt more at the nearer carbon (C meta to CF_3 is closer than the para carbon)	attack so that the carbanion forms meta to CF_3 (nearer) – i.e. the nucleophile ends up **para** to CF_3

Counting problems. A halotoluene with $NaNH_2$ gives 2 aminotoluenes (o- or p-halo) or 3 (m-halo). o-Bromotoluene has one ortho H and gives one benzyne (3-methylbenzyne → o- and m-toluidine). m-Bromotoluene has two different ortho H (at C-2 and C-4) and gives two benzynes (3-methyl- and 4-methylbenzyne → o-, m-, p-toluidine: 3 products with m- from both). p-Bromotoluene has two equivalent ortho H and gives one benzyne (4-methylbenzyne → m- and p-toluidine). A substrate with **no** ortho H (2,6-disubstituted aryl halide) cannot form benzyne with $NaNH_2$.

12.6 Reactions of benzyne

Reaction	Example	Product
Nucleophilic addition	Nu^- = NH_2^-, RO^-, R_2N^-, RS^-, enolates, Grignards	ArNu after protonation (aniline, phenyl ethers, aryl ketones by enolate addition)
[4+2] Diels–Alder	furan, anthracene, cyclopentadiene, 1,3-butadiene	bicyclic adducts (1,4-epoxynaphthalene, triptycene)
[2+2] with alkenes	benzyne + enol ether, ketene acetals	benzocyclobutenes
Ene reaction	benzyne + propene	allylbenzene
σ-Insertion	benzyne + C–C(=O), N–Si, Sn–Sn	ortho-disubstituted arenes
Trimerisation (Pd) / dimerisation	3 C_6H_4 → triphenylene; 2 C_6H_4 → biphenylene	in the absence of traps
Reaction with a second molecule of aryl anion	ArH + benzyne → biaryl	diarylamine/ biaryl by-products

12.8 Electronic structure of benzyne in depth

- **Orbital picture.** Benzyne (1,2-didehydrobenzene, C_6H_4) retains the aromatic sextet in the π system perpendicular to the ring (six p orbitals, one on each carbon). The 'triple bond' is a formal description: the two ortho carbons each have an sp^2 orbital that would have held a C–H or C–X bond; these two orbitals point outward in the ring plane and lie nearly parallel, overlapping only sideways and poorly (the ring geometry forces an inter-orbital angle of 120° rather than 180°). The resulting **in-plane 'π' bond is weak** and high in energy, and the corresponding antibonding orbital is low-lying, which makes benzyne a strong electrophile (LUMO energy ≈ −1.0 eV) and a very reactive dienophile.
- **Geometry and spectroscopy.** Computed C ≡ C ≈ 1.24 Å (between a double bond, 1.34, and a triple bond, 1.20), with the ring angles at the 'triple bond' ≈ 127° and a distorted hexagon. Matrix-isolation IR (Chapman, 1973, from phthaloyl peroxide or benzocyclobutenedione photolysis at 8 K) shows a band near 1846 cm^{-1} assigned to the strained C ≡ C stretch, lower than that of a normal alkyne (≈ 2150) because of the weak bond. The ground state is a closed-shell **singlet**; its first triplet is ≈ 37 kcal mol^{-1} above.
- **Resonance forms.** Benzyne is described by the hybrid of the cumulene-like form (four electrons in two in-plane orbitals forming a weak bond) and a zwitterionic/diradical form (C^+/C^-), which explains both its electrophilicity at carbon and its ability to react with nucleophiles in two orientations. The in-plane bond does *not* interact with the aromatic π system (orthogonal), so benzyne retains its aromatic stabilisation.
- **Why alkynes are linear but benzyne is not.** A linear sp-hybridised carbon needs 180°; the six-membered ring cannot accommodate it. The cost is ≈ 50 kcal mol^{-1} of strain, the energy that is released when benzyne reacts; cyclohexyne, cycloheptyne and cyclooctyne are analogous strained cycloalkynes (cyclooctyne is the smallest stable one; strain-promoted azide–alkyne click chemistry uses it).
- **The three isomers.** 1,2-Didehydrobenzene (ortho-benzyne, the usual one), 1,3-didehydrobenzene (meta-benzyne) and 1,4-didehydrobenzene (para-benzyne) are all C_6H_4. Meta- and para-benzyne are singlet diradicals with the two radical centres separated by one or two carbons; **p-benzyne** is formed by the **Bergman cyclisation** of (Z)-hex-3-ene-1,5-diynes at 200 °C, and by the 'enediyne' antitumour antibiotics (calicheamicin, esperamicin, dynemicin, neocarzinostatin) under physiological conditions: the generated p-benzyne abstracts two H atoms from DNA sugar rings (C-4′, C-5′), leading to strand cleavage and cell death. Enyne-allenes cyclise by the analogous Myers–Saito reaction (to α,3-didehydrotoluene).

12.9 Generation of benzyne: complete catalogue

Precursor	Conditions	Mechanism / remarks
Aryl halide (ArX with ortho H)	$NaNH_2/NH_3$(l), −33 °C; KNH_2; LDA; LiTMP; PhLi; KOtBu/DMSO	E2 or E1cB: base removes the ortho H (acidified by the inductive effect of X), X^- leaves. Elimination rate (leaving group in the E1cB step) Br > I > Cl ⩾ F (in NH_3/KNH_2); however, for F the ortho H is the most acidic (strong –I) and ArF reacts with LiTMP/PhLi via ArLi, in which F^- is lost slowly: the C–F bond is strong. Substrates with no ortho H (2,6-disubstituted halides) do not give benzyne
o-Dihaloarenes + Mg or Li	o-BrC_6H_4F + Mg, THF; o-FC_6H_4Li (–78 °C → warm)	Wittig 1942: ArMgX/ArLi with ortho-F eliminates MgXF/LiF at room temperature; in the presence of furan gives the 1,4-epoxy-1,4-dihydronaphthalene adduct, the first trapping evidence
Anthranilic acid (via diazonium)	o-$H_2NC_6H_4COOH$ + isoamyl nitrite (or HONO) → benzenediazonium-2-carboxylate (zwitterion), Δ	loses CO_2 and N_2 (Stiles–Miller 1960); aprotic route; the zwitterion is explosive when dry, so is prepared in situ
1-Aminobenzotriazole	$Pb(OAc)_4$ or NBS	Campbell–Rees: oxidation gives benzyne + 2 N_2
Phthaloyl peroxide, benzothiadiazole dioxide, benzocyclobutenedione	hν or Δ	photolytic routes used for matrix isolation and spectroscopy
2-(Trimethylsilyl)aryl triflates (Kobayashi, 1983)	CsF or TBAF, MeCN or THF, 25 °C	fluoride attacks Si; C–Si electrons form the in-plane bond while triflate leaves: the standard mild method (no strong base); used for natural-product synthesis
o-(Trimethylsilyl)aryl iodonium triflates	$Bu_4N^+F^-$	Kitamura: loses PhI as leaving group
Hexadehydro-Diels–Alder (HDDA)	triyne (1,3-diyne + diynophile), 80–120 °C	Hoye: thermal [4+2] of a diyne with an alkyne gives a benzyne from acyclic precursors without any base or halide (Hoye 2012)
Benzene-1,2-diazonium and aryl diazonium salts + base	ArN_2^+, ortho-H, base	rare
Photolysis of o-iodophenyl... or flash vacuum pyrolysis	FVP of phthalic anhydride (950 °C)	gas-phase benzyne; mass spectrometric detection

12.10 Evidence for benzyne as an intermediate: Roberts, trapping, isotope effects

- **Roberts' ^{14}C-labelling experiment (1953).** Chlorobenzene labelled with ^{14}C at C-1 reacts with KNH_2 in liquid NH_3 to give aniline with 50 % ^{14}C at C-1 and 50 % at C-2. A direct substitution (S_NAr, ipso) would have given 100 % C-1 aniline. The 1 : 1 distribution requires a symmetrical intermediate in which C-1 and C-2 are equivalent: benzyne.
- **Absence of ortho hydrogen.** 2-Bromo-1,3-dimethylbenzene (both ortho positions blocked by CH_3) does not react with KNH_2; 2,6-dimethylhalobenzenes are inert — in contrast, S_NAr reactions are not blocked by methyl groups in the 2,6-positions.
- **Isotope effect and exchange.** Fluorobenzene-2-d and bromobenzene-2-d: H/D exchange at the ortho position occurs faster than substitution in NH_3/KNH_2 for the weakly leaving fluorine, showing a reversible aryl carbanion (E1cB), while for bromobenzene the removal of H is rate-limiting (E2-like, $k_H/k_D \approx 5.5$).
- **Trapping.** Benzyne gives Diels–Alder adducts with furan (1,4-epoxy-1,4-dihydronaphthalene), anthracene (triptycene, Wittig 1956), tetraphenylcyclopentadienone (1,2,3,4-tetraphenylnaphthalene after loss of CO), cyclopentadiene (benzonorbornadiene), 2,5-dimethylfuran and 1,3-diphenylisobenzofuran, even when no nucleophile is present. Dimer biphenylene and trimer triphenylene arise in the absence of traps.
- **Cine substitution.** Substituted halobenzenes give mixtures of isomers in which the nucleophile enters at the carbon bearing the leaving group (*ipso*) or the adjacent carbon (*cine*): o-bromoanisole → m-anisidine only; p-chlorotoluene → p- and m-toluidine (1 : 1); m-chlorotoluene → o-, m-, p-toluidine; o-chlorotoluene → o- and m-toluidine (≈ 1 : 1). The incoming group is never found at positions 'para' to the leaving group unless the substituent pattern allows it.
- **Spectroscopic detection.** Matrix IR (1846 cm^{-1}), NMR of benzyne trapped inside a hemicarcerand (Cram, 1997: the cage prevents dimerisation; ^{13}C shift of the triple-bond carbons 182 ppm), and gas-phase photoionisation mass spectrometry.

12.11 Regiochemistry of nucleophilic addition to substituted benzynes: Roberts and Gilchrist rules

For a benzyne with a substituent Z at C-3 (adjacent to the triple bond) or C-4 (one carbon removed), the nucleophile attacks the position that gives the more stable aryl carbanion. The anion is in the plane of the ring and cannot be stabilised by resonance; its stability is controlled by **inductive/field effects** (and in some cases by coordination of the metal cation). Rules:

Substituent position	Z	Preferred site of Nu^- attack	Product
3-Z-benzyne (Z adjacent)	–I (OMe, OR, F, Cl, CF_3, NR_3^+)	C-1 (remote from Z), so the carbanion at C-2 is adjacent to Z and is stabilised inductively	meta-substituted product predominates (e.g. o-bromoanisole → m-anisidine, > 99 %)
3-Z-benzyne	+I (CH_3)	~1 : 1 (small steric preference for C-1)	mixture of o- and m-
3-Z-benzyne	large steric group (t-Bu, $SiMe_3$)	C-1 (steric)	meta
4-Z-benzyne (Z one carbon removed)	–I (OMe, CF_3)	attack at C-1 gives carbanion at C-2 (meta to Z); attack at C-2 gives carbanion at C-1 (para to Z). The carbanion nearer to Z is favoured, but the effect is small (the distances are similar)	mixture of meta and para (m ≥ p; e.g. p-chloroanisole → ca. 1 : 1)
4-Z-benzyne	alkyl	~1 : 1	m : p ≈ 1 : 1
Distortion model (Garg–Houk, 2010)	—	aryne is distorted by the substituent so that the more linear carbon (bigger bond angle, more electrophilic) is attacked: 3-OMe: C-1 attack; indolynes and 3,4-pyridynes with the substituent-induced distortion	predicts selectivity in heteroarynes: 3,4-pyridyne with 5-bromo-substituent → attack at C-3 selectively

12.12 Reactions of benzyne: a systematic classification

Class	Partner	Product	Notes
Nucleophilic addition–protonation	NH_2^-, RO^-, RS^-, R_2N^-, CN^-, enolates, ArLi, RMgX, amines (neutral), halide, azide	ArNu after protonation of the aryl anion by the solvent (NH_3, ROH)	α-arylation of ketones/nitriles (Caubère: $NaNH_2$/ t-BuONa 'complex base' with ArX and a ketone); intramolecular attack gives rings (indoles, benzofurans)

Diels–Alder [4+2]	furan, cyclopentadiene, anthracene, pyrrole (N-protected), isobenzofuran, cyclohexadiene, α-pyrone, tetracyclone, 2-pyridones	benzo-fused bicyclics, triptycene, naphthalenes (with extrusion of CO_2 or CO)	most reliable trapping; the endo/exo and π-facial selectivity are governed by sterics
[2+2] cycloaddition	enol ethers, ketene acetals, alkenes with e-donors, ketenes	benzocyclobutenes (with ArH loses nothing)	Ar + CH_2=C(OR)$_2$ → 1,1-dialkoxybenzocyclobutene; in the absence of trap, 2 ArH → biphenylene
[3+2] 1,3-dipolar	azides, nitrones, diazo compounds, nitrile oxides, sydnones	benzotriazoles, benzisoxazolines, indazoles	click-type; mild with Kobayashi precursors
Ene reaction	propene, allylbenzene, alkenes with allylic H	allylbenzenes (Ar–C–C=C with H transfer)	benzyne acts as enophile: a concerted six-membered TS; e.g. benzyne + propene → allylbenzene
σ-Bond insertion	C–C (β-ketoester α-acylation), C–N (amides), N–Si, S–S, Sn–Sn, C–Cl	ortho-disubstituted arenes	Stoltz acyl-alkylation of β-ketoesters (aryne inserts into the C–C bond of the enolate, via a four-membered ring)
Dimerisation, trimerisation	benzyne + benzyne; Pd(0)-catalysed	biphenylene; triphenylene (3 arynes, Pd)	the last is a metal-templated [2+2+2]
Radical/ligand	transition-metal arynes (Ni, Zr, Pt complexes)	stable benzyne–metal complexes	e.g. Zr(η^2-C_6H_4)Cp_2(PMe_3)
Multi-component coupling	aryne + isocyanide + carbonyl; aryne + DMF + NaN_3	one-pot ortho difunctionalisation	typical modern application of arynes in diversity synthesis

12.13 Heteroarynes and cycloalkynes

- **3,4-Pyridynes** (from 3-halopyridines + base: 3-bromopyridine + $NaNH_2$ → 3- and 4-aminopyridine mixture; 4-halopyridine similarly) are the reason that meta- and para-halopyridines give mixtures with amide, but 2-halopyridines undergo normal S_NAr. 2,3-Pyridynes also form.
- **Indolynes** (3,4-, 4,5-, 5,6-, 6,7-indolyne) are made from bromoindoles with strong base, or from silyl triflates; Garg used them in total synthesis of indolactam V, tubingensin A, and the distortion model above explains their regioselectivity.
- **Cyclohexyne, cycloheptyne** exist only as intermediates (trapped by Diels–Alder, e.g. cyclohexyne from 1-chloro-cyclohexene + $NaNH_2$ (Wittig, Scardiglia) or from Kobayashi precursors); **cyclooctyne** is the smallest isolable cycloalkyne (bp 90 °C/ 65 mmHg), and strained dibenzocyclooctynes (DIBO, DBCO) enable copper-free click chemistry (Bertozzi, Nobel Prize 2022).
- **1,2-Cyclohexadiene, 1,2,3-cyclohexatriene** (Garg 2012): strained cyclic allenes and cumulenes generated from silyl triflates and trapped in cycloadditions; the logical extension of the aryne concept.

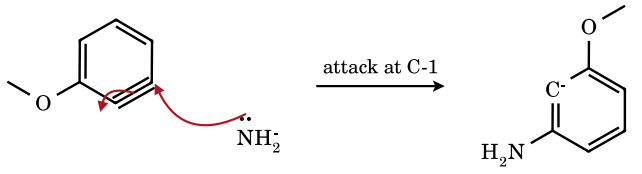
12.14 Three pathways to nucleophilic aromatic substitution compared

	S_NAr (addition–elimination)	**Benzyne (elimination–addition)**	**S_N1 (diazonium)**	**$S_{RN}1$ (radical chain)**
Substrate	ArX with strong EWG (NO_2, CN, COR) ortho/para to X	ArX with ortho-H and no activating group	ArN_2^+	ArX (X = I, Br) + photostimulation; no EWG needed
Reagent	Nu^- or neutral nucleophile	very strong base (NH_2^-, RLi, LDA) at low T	H_2O/Δ (phenol), HX/Cu, KI, H_3PO_2	enolate, thiolate, amide; electron-transfer initiation
Intermediate	Meisenheimer complex (carbanion delocalised into EWG)	benzyne then aryl anion	aryl cation (very unstable), or radical (Cu)	radical anion $ArX^{\bullet-}$ → Ar•
Leaving-group order	F ⩾ Cl ≈ Br > I (rate-limiting addition)	Br > I > Cl ⩾ F (elimination)	N_2	I > Br > Cl
Regiochemistry	ipso only; Nu enters where X was	ipso + cine mixtures (nucleophile at either end of 'triple bond')	ipso	ipso
Key evidence	isolable Meisenheimer salts; second-order kinetics	^{14}C scrambling; trapping; no reaction when ortho-H are blocked	N_2 evolution; salt-free arenes	chain inhibition by radical traps

Typical errors in benzyne questions. (1) Treating chlorobenzene + NaOH (aqueous, 300 °C Dow process) as benzyne: the Dow process includes benzyne-type cine products with some substrates, but plain chlorobenzene + NaOH at 623 K, 300 atm gives phenol (ipso/cine mixtures confirmed by ^{14}C labelling). (2) Forgetting that p-substituted halobenzenes give two products and o-substituted ones can give one or two. (3) Choosing the nitro-substituted halide: 1-chloro-2,4-dinitrobenzene reacts by S_NAr, not benzyne. (4) Drawing benzyne with a linear triple bond. (5) Assuming the NH_2^- adds with the usual Markovnikov-like selectivity. (6) Thinking benzyne has a stable carbanion (aryl anion lone pair is in an sp^2 orbital, orthogonal to the π system, not delocalised). (7) Missing that the ortho-H must be present: 2-bromo-1,3-dimethylbenzene cannot form a benzyne.

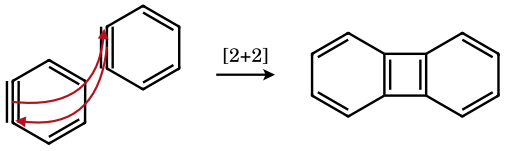
12.15 Additional validated mechanisms

12.6 meta-Selectivity: 3-methoxybenzyne + NH_2^- → m-anisidine (carbanion adjacent to OMe)



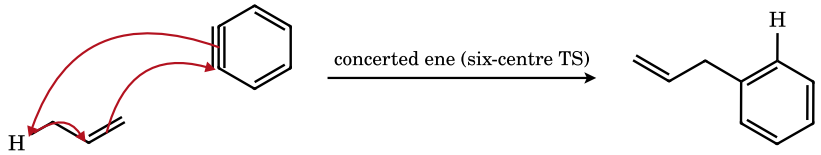
The nucleophile adds to the benzyne carbon farther from the OMe group (C-1), so the developing aryl carbanion lies on C-2, next to the electronegative oxygen where the σ-withdrawing (–I) effect stabilises the localised in-plane lone pair. The aryl anion (an sp^2 lone pair in the ring plane) is then protonated by NH_3 (the solvent) and the product is meta-anisidine only. The same product is formed from 2-bromoanisole (benzyne with OMe at C-3) and from 3-bromoanisole (via deprotonation at C-2, between OMe and Br).

12.7 Dimerisation of benzyne to biphenylene ([2+2])



In the absence of a trap, the strongly electrophilic and strongly dienophilic benzyne reacts with a second molecule: the in-plane π bonds of two molecules combine to form the two C–C σ bonds of a cyclobutadiene-like four-membered ring flanked by two benzene rings (biphenylene). A third molecule gives triphenylene (the trimer, more common with Pd(0) catalysis).

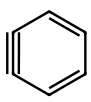
12.8 Aryne ene reaction: benzyne + propene → allylbenzene



The benzyne acts as an enophile: the C=C π bond of propene forms a new C–C σ bond to one aryne carbon, the allylic C–H bond shifts to the other aryne carbon, and the double bond migrates; all three electron pairs move in a single six-membered transition state. The product is allylbenzene (3-phenylprop-1-ene). In competition, benzyne also adds [2+2] to alkenes, the ene reaction being preferred when allylic H are available.

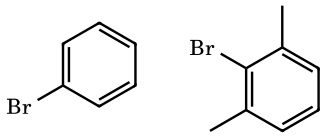
12.16 Problems on benzyne

BZ-1.
Which statements about benzyne are correct? (A) the ‘triple bond’ is a normal sp–sp bond; (B) the extra bond is formed by overlap of two in-plane sp^2 orbitals; (C) the aromatic sextet is retained; (D) it is a strong dienophile.



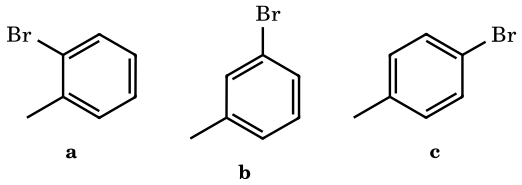
Answer: (B), (C), (D)
The ring carbons stay sp^2; the weak in-plane overlap makes the bond very strained and reactive, so benzyne is a powerful electrophile and dienophile while remaining aromatic.

BZ-2.
Bromobenzene is treated with $NaNH_2$ in liquid NH_3. Name the intermediate and the product. What if the starting material is 2,6-dimethylbromobenzene?



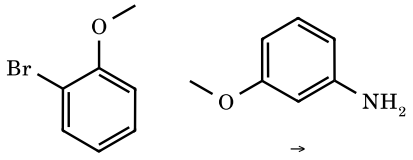
Answer: benzyne → aniline; no reaction for 2,6-dimethylbromobenzene
The mechanism needs an H on the carbon ortho to Br to be removed by the base. 2,6-Dimethylbromobenzene has none, so it cannot form an aryne (it could only react by the very slow S_NAr).

BZ-3.
How many distinct aminotoluenes are formed from (a) o-, (b) m-, (c) p-bromotoluene with $NaNH_2/NH_3$?



Answer: (a) 2; (b) 3; (c) 2
(a) one benzyne (3-methylbenzyne): attack at either carbon gives o- and m-toluidine. (b) two benzynes (2,3 and 3,4 pairs): 3-methylbenzyne → o-, m-; 4-methylbenzyne → m-, p-: o-, m-, p- in total. (c) one benzyne (4-methylbenzyne): m- and p-toluidine.

BZ-4.
o-Bromoanisole + $NaNH_2/NH_3$ gives mainly which product and why?

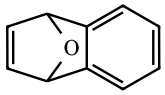


Answer: m-anisidine (3-methoxyaniline)
The benzyne is 3-methoxybenzyne. NH_2^- attacks C-1, the carbon farther from OMe (meta), so that the carbanion forms at C-2, adjacent to the electronegative OMe where its –I effect stabilises the in-plane carbanion.

BZ-5.
Chlorobenzene and the following reagents: (i) $NaNH_2/NH_3$, (ii) NaOH, 350 °C (Dow process), (iii) OH^-, 25 °C. Which proceed through benzyne?
Answer: (i) and (ii)
Strong bases (NH_2^-; HO^- at 350 °C and high pressure) remove ortho H to give benzyne. At 25 °C hydroxide does not react with unactivated PhCl at all. With an o- or p-nitro group the reaction is addition–elimination (S_NAr, Meisenheimer complex), not benzyne.

BZ-6.
Arrange the aryl halides in decreasing reactivity toward KNH_2/NH_3 via benzyne: C_6H_5F, C_6H_5Cl, C_6H_5Br, C_6H_5I.
Answer: Br > I > Cl ⩾ F (Roberts)
The rate depends on two steps – removal of the ortho H and loss of X^-. The H of fluorobenzene is the most acidic (strong –I of F) but F is a poor leaving group; for I the C–I bond is weak but the ortho H is less acidic. Bromine balances both best.

BZ-7.
Benzenediazonium-2-carboxylate is heated in the presence of furan. Give the product.



Answer: 1,4-dihydro-1,4-epoxynaphthalene
Loss of CO_2 and N_2 gives benzyne, trapped by the diene in a [4+2] cycloaddition.

BZ-8.

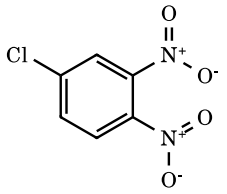
In the benzyne mechanism, what is the hybridisation of the carbons of the strained bond and what is the C ≡ C bond angle?

Answer: sp^2; ≈ 120–127° (not 180°)

A linear sp-hybridised alkyne carbon would need 180°; in a six-membered ring the angle is forced to ≈ 120–127°, so each benzyne carbon is sp^2 with an in-plane orbital.

BZ-9.

A halobenzene with a strongly electron-withdrawing group ortho or para to the halogen reacts with HO^- at room temperature. By which mechanism?



Answer: addition–elimination (S_NAr), not benzyne

The nucleophile adds to the ipso carbon, giving a resonance-stabilised Meisenheimer anion with negative charge on the nitro groups, then chloride is lost. The benzyne path needs a very strong base and has no requirement of an EWG. (Chlorine as a leaving group: F ⩾ Cl ≈ Br ≈ I in S_NAr, the reverse of the benzyne order.)

BZ-10.

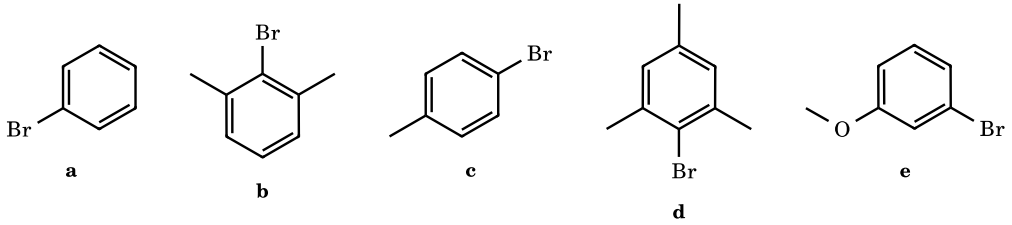
Compare benzyne with the other reactive intermediates: how many π electrons in the ring, and is it charged?

Answer: 6π (aromatic) + 2 electrons in the extra in-plane bond; neutral

Benzyne has all electrons paired, no charge and no unpaired electrons; the unusual reactivity arises from strain and from the low-lying in-plane LUMO.

BZ-11.

Which of the following aryl halides cannot form a benzyne with $NaNH_2/NH_3$: (a) bromobenzene, (b) 2-bromo-1,3-dimethylbenzene, (c) 4-bromotoluene, (d) 1-bromo-2,4,6-trimethylbenzene, (e) 3-bromoanisole?

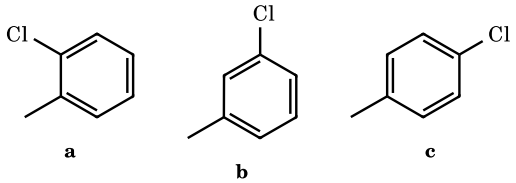


Answer: (b) and (d)

Benzyne formation requires an H on a carbon ortho to the leaving group. Both ortho positions of (b) and (d) carry CH_3, so no elimination; in S_NAr there is no such requirement (but no EWG either), so these halides are essentially inert to amide.

BZ-12.

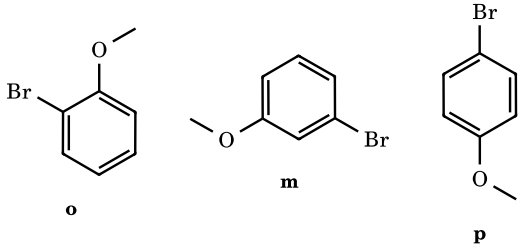
How many isomeric toluidines (aminotoluenes) are formed from (a) o-chlorotoluene, (b) m-chlorotoluene, (c) p-chlorotoluene with KNH_2/NH_3?



Answer: (a) 2 (o- and m-); (b) 3 (o-, m-, p-); (c) 2 (m- and p-)

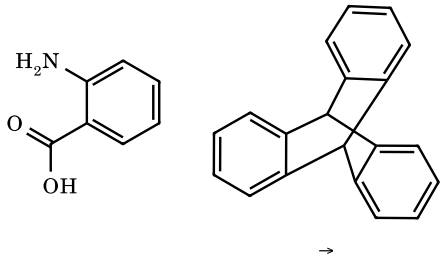
(a) only one ortho H (C-6): 3-methylbenzyne, attack at C-1 (→ o-toluidine) or C-6 (→ m-toluidine). (b) two ortho Hs: C-2 gives 3-methylbenzyne (o- and m-) and C-6 gives 4-methylbenzyne (m- and p-); total three. (c) 4-methylbenzyne: attack at either carbon gives m- or p-toluidine in a ca. 1 : 1 ratio.

BZ-13.
o-Bromoanisole, m-bromoanisole and p-bromoanisole each react with KNH_2/NH_3. Predict the amine(s) formed in each case.



Answer: o → m-anisidine (only); m → m-anisidine (only); p → m- and p-anisidine (≈ 1 : 1)
OMe is –I at the carbanion position: the carbanion forms adjacent to OMe, so Nu attacks the carbon meta to OMe. For the p-isomer both aryne carbons are meta/para to OMe and the inductive differentiation is weak, giving a nearly equal mixture.

BZ-14.
Complete and balance the sequence: anthranilic acid + isoamyl nitrite (aprotic, Δ) + anthracene.

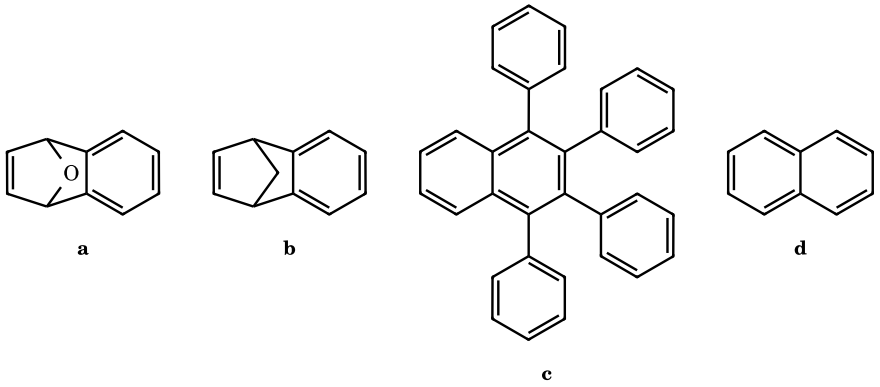


Answer: triptycene (+ CO_2 + N_2 + 2 isoamyl alcohol)
The diazonium carboxylate generates benzyne, which adds across the 9,10-positions of anthracene (Diels–Alder) to give triptycene, a rigid three-bladed paddle-wheel hydrocarbon (Wittig 1956).

BZ-15.
(Assertion–Reason) Assertion: the reaction of chlorobenzene with KNH_2/NH_3 gives a 1 : 1 mixture of aniline-1-^{14}C and aniline-2-^{14}C when chlorobenzene is labelled at C-1. Reason: the nucleophile adds equally to the two carbons of a symmetrical benzyne intermediate.
Answer: both correct; the reason explains the assertion
The symmetrical intermediate makes C-1 and C-2 equivalent, so label ends up equally at both positions. A direct displacement (ipso) would give 100 % at C-1.

BZ-16.
The reaction of 1-chloro-2-nitrobenzene with NaOH (aq) gives o-nitrophenol, while chlorobenzene + NaOH under the same conditions does not react. Identify the mechanism in each case.
Answer: o-chloronitrobenzene: S_NAr (addition–elimination); chlorobenzene: no reaction (needs benzyne conditions, $NaNH_2$, or 623 K/300 atm)
The ortho NO_2 stabilises the Meisenheimer anion by resonance, so addition of HO^- is feasible. Chlorobenzene has no activating group; at very high T the Dow process proceeds in part by benzyne (some cine substitution).

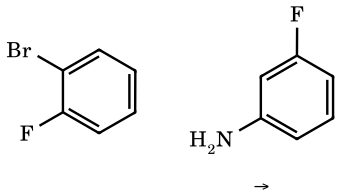
BZ-17.
Give the product of benzyne + (a) furan, (b) cyclopentadiene, (c) tetraphenylcyclopentadienone, (d) 2-pyrone. Which loses a small molecule?



a **b** **c** **d**

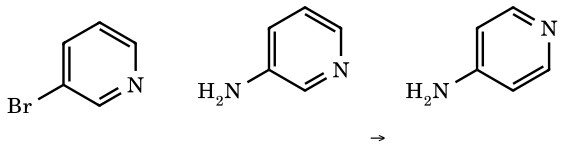
Answer: (a) 1,4-epoxy-1,4-dihydronaphthalene; (b) benzonorbornadiene; (c) 1,2,3,4-tetraphenylnaphthalene (loses CO); (d) naphthalene (loses CO_2)
(c) and (d): the bicyclic adduct has a bridging C=O (ketone) or O–C(=O) that extrudes CO or CO_2 in a retro-Diels–Alder (cheletropic), aromatising the new ring.

BZ-18.
What is the product of o-bromofluorobenzene with $NaNH_2/NH_3(l)$? Explain the regiochemistry.



Answer: m-fluoroaniline
The only ortho hydrogen is at C-6 (next to Br); loss of H and Br^- gives 3-fluorobenzyne. The nucleophile adds to C-6, the carbon remote from F, so that the aryl carbanion is on C-1 adjacent to the electronegative fluorine (–I stabilisation). Protonation gives the meta isomer.

BZ-19.
Account for the following: 2-chloropyridine + KNH_2 gives 2-aminopyridine (Chichibabin-like S_NAr); 3-bromopyridine + KNH_2 gives a 1 : 1 mixture of 3- and 4-aminopyridine.



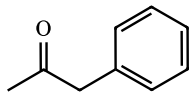
Answer: 3-bromopyridine forms 3,4-pyridyne; 2-halopyridine reacts by S_NAr
In 2-halopyridine the ring nitrogen stabilises the Meisenheimer anion (like an ortho/para nitro group). 3-Halopyridine has no such activation, so the strong base eliminates HBr to 3,4-pyridyne; the amide then adds at C-3 or C-4 in nearly equal amounts (cine substitution).

BZ-20.
A student heats 2,6-dimethylbromobenzene with $NaNH_2/NH_3$ and recovers it unchanged. Another student treats 2,4-dinitrobromobenzene with aqueous ammonia and obtains 2,4-dinitroaniline quickly. Explain both.
Answer: no ortho H: no benzyne; but strongly activated ring: S_NAr
The first lacks a hydrogen ortho to Br (both are CH_3) and has no EWG; the second has activating nitro groups ortho and para to Br and reacts by addition–elimination via a Meisenheimer complex.

BZ-21.
Why does fluorobenzene give benzyne with PhLi or LDA, but the benzyne from fluorobenzene is formed more slowly with KNH_2 than that from bromobenzene?
Answer: C–F bond is strong and F^- is a poor leaving group, although the ortho-H is more acidic
The first step (deprotonation) is easier for F because of its strong –I effect; the aryl-lithium or aryl-potassium is formed readily but the loss of fluoride (strong C–F bond, 125 kcal mol^{-1}) is slow, so for KNH_2 the elimination step is rate-limiting (E1cB) and H/D exchange is faster than substitution. Bromide is a much better leaving group; with bromobenzene proton removal is rate-limiting.

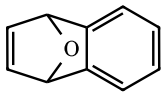
BZ-22.
Compare the reaction of m-chloro-trifluoromethylbenzene vs. o-chloro-trifluoromethylbenzene with KNH_2 (Roberts, 1956). Which product(s) form and why?
Answer: o-chloro: m-CF_3-aniline only; m-chloro: m-CF_3-aniline predominantly
CF_3 is strongly –I. In 3-CF_3-benzyne the nucleophile adds to C-1, remote from CF_3, giving the anion on C-2 adjacent to CF_3. Both substrates therefore lead to 3-(trifluoromethyl)aniline (meta), exactly as for OMe and F.

BZ-23.
The Stobbe-type (Caubère) arylation: benzyne is generated from bromobenzene in the presence of the sodium enolate of acetone. Give the product and name the reaction type.



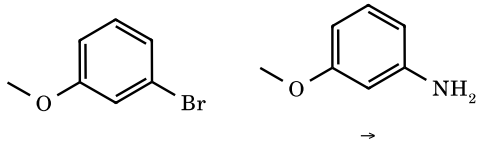
Answer: phenylacetone (1-phenylpropan-2-one)
The enolate carbon adds to benzyne (nucleophilic addition) and the aryl anion is protonated. This is a useful transition-metal-free α-arylation of ketones; with excess base, di-arylation can occur.

BZ-24.
o-Bromofluorobenzene is treated with magnesium in THF at room temperature in the presence of furan. Name the intermediate and the final product.



Answer: o-fluorophenylmagnesium bromide, then benzyne (loss of MgBrF); 1,4-epoxy-1,4-dihydronaphthalene
Mg inserts selectively into the C–Br bond to give the ortho-fluoro Grignard reagent, which eliminates MgBrF at room temperature to give benzyne; furan traps it in a Diels–Alder reaction (Wittig, 1942).

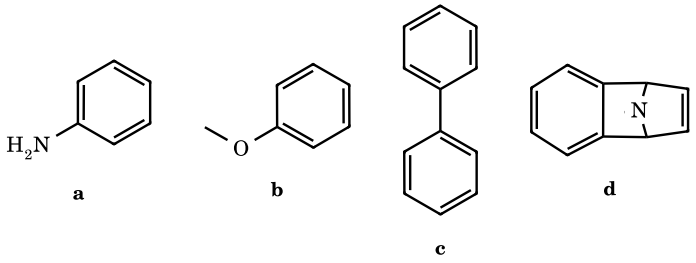
BZ-25.
m-Bromoanisole is treated with KNH_2 in liquid ammonia. How many isomeric anisidines are possible by benzyne pathways, and which is actually obtained?



Answer: 3 possible (o-, m-, p-); m-anisidine is obtained
Two ortho H (C-2 and C-6). Removal of C-2–H (between Br and OMe, the most acidic) gives 3-methoxybenzyne: attack at the carbon remote from OMe gives m-anisidine (attack at the adjacent carbon would give o-anisidine). Removal of C-6–H gives 4-methoxybenzyne with m- and p-anisidine as possible products. Experimentally the more acidic C-2 route and the –I orientation rule give essentially only m-anisidine.

BZ-26.
Why does the 1,4-didehydrobenzene (p-benzyne) formed from enediynes cleave DNA, while ortho-benzyne reacts with nucleophiles?
Answer: p-benzyne is a σ,σ-diradical; ortho-benzyne is a strained electrophilic alkyne
p-Benzyne has two radical centres at para positions; it abstracts two H-atoms (from the sugar C–H bonds of DNA) to give benzene, creating DNA radicals that cleave the strand. ortho-Benzyne has a closed-shell electronic structure with a polarised in-plane bond that accepts nucleophiles and cycloaddition partners.

BZ-27.
Give the products of (a) benzyne + $NH_3(l)/NH_2^-$, (b) benzyne + CH_3OH/CH_3O^-, (c) benzyne + PhLi, (d) benzyne + N-methylpyrrole.



a **b** **c** **d**

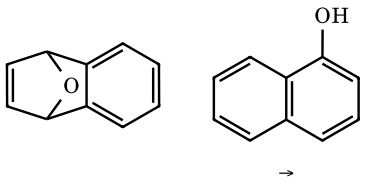
Answer: (a) aniline; (b) anisole; (c) biphenyl; (d) N-methyl-1,4-dihydronaphthalen-1,4-imine (Diels–Alder adduct)
Nucleophilic additions give the aryl anion which is protonated (a, b) or, in (c), gives aryl lithium and then protonation (biphenyl after work-up). The pyrrole acts as the diene.

BZ-28.
Explain why the nitro group in 1-chloro-4-nitrobenzene accelerates S_NAr but 4-chlorotoluene with the same reagents (NH_2^-) reacts via benzyne.
Answer: NO_2 stabilises the Meisenheimer anion; CH_3 cannot, so the strong base removes the ortho H
In 1-chloro-4-nitrobenzene the NO_2 (–M, para) delocalises the negative charge of the addition intermediate. In 4-chlorotoluene there is no stabilisation, so S_NAr is impossible; the base instead deprotonates the ortho H giving 4-methylbenzyne.

BZ-29.
A mixture of 3- and 4-aminopyridine is obtained from 3-bromopyridine and KNH_2, but only 4-aminopyridine from 4-bromopyridine. Explain.
Answer: 3,4-pyridyne from 3-bromopyridine gives both; 4-bromopyridine reacts partly by S_NAr
4-Bromopyridine has N para to the leaving group: strongly activated for S_NAr (the Meisenheimer anion puts charge on N), so ipso substitution dominates; the 3-isomer lacks that activation and goes through 3,4-pyridyne, with attack at either carbon.

BZ-30.
1-Bromo-3-methoxy-2-methylbenzene is treated with $NaNH_2/NH_3$. How many benzyne intermediates and how many isomeric amines are possible?
Answer: 1 benzyne, 2 amines
The only hydrogen ortho to Br is at C-6 (C-2 carries CH_3), so there is one aryne (C-1 ≡ C-6, with CH_3 at C-2 and OMe at C-3). The nucleophile can add at C-1 or C-6, giving two isomeric amines: 3-methoxy-2-methylaniline (Nu at C-1, anion at C-6) and 4-methoxy-3-methylaniline (Nu at C-6, anion at C-1 next to the electron-releasing CH_3, which destabilises it). The alkyl group therefore favours the first, but the effect is small, so a mixture is expected.

BZ-31.
The oxabenzonorbornadiene (furan–benzyne adduct) is treated with dilute HCl. Give the product.



→

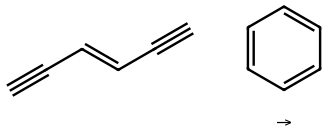
Answer: 1-naphthol
Protonation opens the strained C–O bridge to a benzylic/allylic cation; a 1,2-hydride shift and loss of H^+ aromatise the new ring, giving 1-naphthol.

BZ-32.
o-Dibromobenzene is treated with 1 equiv of n-BuLi in THF at (a) –100 °C and (b) 0 °C. What species forms in each case?
Answer: (a) o-bromophenyllithium; (b) benzyne (LiBr lost)
Bromine–lithium exchange is fast at –100 °C and gives the stable ortho-bromoaryllithium. On warming, elimination of LiBr is thermally easy (the C–Li bond and the ortho C–Br are syn-periplanar in the plane), giving benzyne, which is trapped if a diene is present.

BZ-33.
How many ^1H NMR signals (aromatic and bridgehead types) does triptycene show, and from which reagents is it made?
Answer: 3 signal types; benzyne + anthracene
Each of the three benzo rings has two types of aromatic H (AA′BB′ pattern: H next to the bridgehead and H farther away), and the two bridgehead H are equivalent: 3 signal types. Benzyne from anthranilic acid/isoamyl nitrite adds across C-9/C-10 of anthracene.

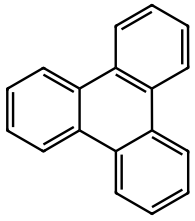
BZ-34.
Account for the observation that 1-chloro-2,4-dinitrobenzene reacts with aqueous NaOH at 25 °C to give the phenol but 4-chlorotoluene needs $NaNH_2/NH_3$ and gives two products.
Answer: S_NAr versus elimination–addition
The nitro groups ortho and para to Cl stabilise the Meisenheimer anion, so addition of HO^- occurs without strong base (ipso substitution, one product). 4-Chlorotoluene has no activation; only a very strong base can remove the ortho H, giving 4-methylbenzyne, which is attacked at either carbon to give m- and p-cresol-type products (here m- and p-toluidine).

BZ-35.
(Z)-Hex-3-ene-1,5-diyne is heated in 1,4-cyclohexadiene. Name the intermediate and the final aromatic product.



Answer: 1,4-didehydrobenzene (p-benzyne) diradical; benzene
Bergman cyclisation forms the σ,σ-diradical, which abstracts two H atoms from 1,4-cyclohexadiene (itself aromatising to benzene). The same chemistry in enediyne antibiotics cleaves DNA.

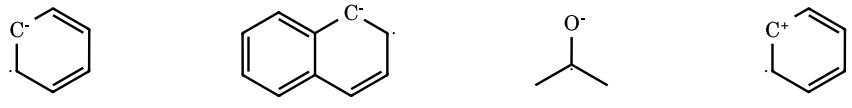
BZ-36.
Three benzyne molecules combine in the presence of Pd(0). Give the product and its molecular formula.



Answer: triphenylene, $C_{18}H_{12}$
A palladium-templated [2+2+2] cyclotrimerisation of three C_6H_4 units gives the fully fused triphenylene ($3 \times C_6H_4 = C_{18}H_{12}$).

Part 13 Radical Ions: Radical Anions and Radical Cations

A **radical ion** has both an unpaired electron and a net charge. A **radical anion** forms when a neutral molecule accepts one electron into its lowest empty π* orbital; a **radical cation** forms when it loses one electron from its highest filled orbital. They are the intermediates of single-electron-transfer (SET) chemistry: dissolving-metal reductions, electrochemistry, radiolysis, mass spectrometry ($M^{\bullet+}$ is the molecular ion) and photoinduced electron transfer. They are normally observed by ESR spectroscopy, the hyperfine splittings giving the spin density at each atom.



benzene radical anion **naphthalene radical anion** **ketyl radical anion** **benzene radical cation**

Radical-ion structures (one resonance contributor; the unpaired electron and the charge are delocalised over the ring or over C and O in the ketyl).

Species	Formation	Typical chemistry	Why it matters
Arene radical anion (naphthalenide, benzene$^{\bullet-}$)	arene + Na or Li in THF / liquid NH_3 (electron into π*)	strong reducing agent; protonated by ROH to a cyclohexadienyl radical, then reduced again to a carbanion and protonated	**Birch reduction**: benzene + $Na/NH_3(l)/ROH \rightarrow$ 1,4-cyclohexadiene (EDG on the ring end up on the double bond carbons; EWG on sp^3 carbons)
Ketyl radical anion $R_2C{\bullet}–O^-$	ketone + Na, Mg, SmI_2	radical combination (dimerisation) or further reduction to the carbanion/alkoxide	**Pinacol coupling** (2 $R_2C{=}O$ + Mg(Hg) → diol), **acyloin condensation** (2 RCOOR + Na → α-hydroxyketone), the benzophenone ketyl (a deep-blue, solvent-drying indicator)
Nitroarene / quinone radical anion	electrolytic or chemical one-electron reduction	semiquinone ($ArO{\bullet}/ArO^-$) are redox intermediates in biology (ubiquinone)	ESR detection; antibiotic and DNA-damage mechanisms

Radical anion in $S_{RN}1$	ArX + e^- → $ArX^{\bullet -}$ → Ar• + X^-	chain: Ar• + Nu^- → $ArNu^{\bullet -}$; $ArNu^{\bullet -}$ + ArX → ArNu + $ArX^{\bullet -}$	substitution without an activating group
Arene radical cation	anodic oxidation, SET to a photoexcited acceptor, Ce^{4+}, mass-spectrometric ionisation	deprotonation at benzylic C–H gives benzyl radicals; nucleophile addition	**mass spectrometry**: $M^{\bullet +}$ fragmentation; Kolbe anodic oxidation; conducting polymers (polyaniline, polypyrrole)
Amine radical cation $R_3N^{\bullet +}$ (aminium)	oxidation of amines (e.g. Wurster's blue from N,N,N′,N′-tetramethyl-p-phenylenediamine)	α-C–H deprotonation gives α-amino radical; the N-centred radical performs 1,5-HAT (Hofmann–Löffler)	stable, coloured radical cations

Birch reduction in four steps (benzene, Na, NH_3(l), EtOH): (1) electron transfer gives the radical anion; (2) protonation by ethanol (a weak acid; NH_3 is too weak) gives a cyclohexadienyl radical; (3) a second electron gives the cyclohexadienyl anion (a pentadienyl carbanion, charge at C-1, C-3, C-5); (4) protonation at the central carbon (C-3, the position of greatest charge density) gives the non-conjugated 1,4-diene. Electron-donating groups (OMe, alkyl) sit on the sp^2 carbons of the product; electron-withdrawing groups (COOH) sit at the sp^3 end because they stabilise the carbanion at their own carbon.

13.2 Dissolving-metal reductions: regiochemistry and scope

- **Birch reduction rules.** Arene + Na (or Li)/NH_3(l) with an alcohol gives a 1,4-cyclohexadiene. Electron-*donating* substituents (alkyl, OR, NR_2) end up on the **sp^2** carbons of the product (the radical anion is protonated, and the carbanion forms, at the positions least destabilised: *ortho*/*meta* to the donor). Electron-*withdrawing* substituents (COOH, $CONH_2$, CN) end up on the **sp^3** carbon (para protonation, carbanion stabilised by the acceptor). Anisole → 1-methoxycyclohexa-1,4-diene (hydrolysis gives the β,γ-unsaturated ketone, then the conjugated enone: a route to cyclohexenones); benzoic acid → cyclohexa-2,5-diene-1-carboxylic acid; toluene → 1-methylcyclohexa-1,4-diene.
- **Why an alcohol is needed:** NH_3 ($pK_a \approx 38$) is too weak an acid to protonate the benzene radical anion at a useful rate; ROH ($\approx$ 16–18) does so. The first protonation is rate-determining. The 1,4-diene (not the conjugated 1,3-diene) forms because the pentadienyl anion is protonated at its central carbon, where the charge density (and the kinetic basicity) is highest.
- **Alkynes → trans-alkenes (Na/NH_3).** The vinyl radical anion (intermediate) is protonated to a vinyl radical that adopts the less hindered trans geometry before a second electron gives the trans-vinyl anion, protonated to the trans-alkene; terminal alkynes are deprotonated first (acetylide) and are not reduced. Lindlar (H_2, Pd/$CaCO_3$, quinoline) gives the cis-alkene by syn addition.
- **Bouveault–Blanc** (ester + Na/EtOH → two alcohols) via ketyl and carbanion stages; replaced by $LiAlH_4$ today. **Acyloin condensation** (diester + Na, toluene, then TMSCl) closes rings from C_5 to C_{30} through a diradical dianion on the metal surface (high dilution is not needed because the sodium surface holds both ends). **McMurry coupling** (2 $R_2C{=}O$ + $TiCl_3$/$LiAlH_4$ or Zn–Cu → $R_2C{=}CR_2$): ketyl radical anions on low-valent titanium couple to a pinacolate and deoxygenate (Ti–O strength); used for tetrasubstituted alkenes and macrocycles. **SmI_2 (Kagan)** (THF/HMPA) reduces ketones, halides, and promotes ketyl–olefin cyclisations (a ketyl adds to an alkene in 5-exo mode).
- **Benkeser reduction** (Li/ethylamine): further reduces arenes to cyclohexenes. **Clemmensen** (Zn(Hg)/HCl) and **Wolff–Kishner** are not Birch-type; Clemmensen proceeds by zinc-carbenoid surface species.
- **Pinacol coupling:** 2 $R_2C{=}O$ + Mg(Hg)/or Al/or SmI_2 → vicinal diol. Intramolecular versions give cis-1,2-diols in rings (1,5- and 1,6-dicarbonyls give 5-/6-membered rings with cis selectivity because the metal chelates both oxygens). The acid-catalysed pinacol–pinacolone rearrangement of the product is a carbocation 1,2-shift (Part 7).
- **Nitroarene reduction steps** (one-electron stages): $ArNO_2$ → $ArNO_2^{\bullet -}$ → ArNO → ArNHOH → $ArNH_2$; with Zn/NH_4Cl stops at hydroxylamine; in alkali, bimolecular products (azoxy, azo, hydrazo) form by coupling of nitroso and hydroxylamine intermediates (Part 14 table).

13.3 Radical cations: mass spectrometry, anodic chemistry, photoredox

- **Electron-impact ionisation (70 eV)** removes the electron of highest energy: n(O), n(N) lone pairs > π > σ. The molecular ion $M^{\bullet +}$ then fragments by pathways that give the most stable cation plus a neutral radical (or a neutral molecule plus a radical cation). Rules: (i) **nitrogen rule**: an odd number of N gives an odd $M^{\bullet +}$ mass; (ii) **α-cleavage** next to O, N or C=O (radical on the heteroatom pairs with an α C–C electron): $CH_3CH_2COCH_3^{\bullet +}$ → CH_3CO^+ (m/z 43, loss of the larger radical is favoured) and $C_2H_5CO^+$ (57); (iii) **McLafferty rearrangement** of carbonyls with a γ-H: six-membered TS 1,5-H transfer to O and β-cleavage gives an enol radical cation and an alkene (e.g. butanal m/z 44; ketone/ester/acid/amide); (iv) **benzylic cleavage** to the tropylium ion (m/z 91) from alkylbenzenes; (v) **loss of small neutral molecules**: H_2O (M –

18) from alcohols, CO (28), CO_2 (44), HCN (27), C_2H_4 (28, retro-Diels–Alder of cyclohexenes), HX; (vi) isotopic patterns: Cl (3 : 1 for M : M+2), Br (1 : 1).

- **Anodic oxidation (electro-organic).** Kolbe (carboxylates, Part 9), Shono oxidation (carbamate/amide → N-acyliminium via α-C–H, then nucleophile trapping), oxidation of enol ethers, aromatic ethers (anisole → quinone ketals in MeOH). The radical cation of an electron-rich arene reacts with nucleophiles at the position of highest spin/charge density (e.g. para to OMe).
- **Chemical oxidants:** CAN (Ce^{4+}), DDQ (hydride/SET abstraction from benzylic C–H of electron-rich arenes, deprotection of PMB ethers: $ArCH_2OR \rightarrow ArCH{=}O^+R \rightarrow ArCHO + ROH$), Fe(III)/Cu(II) in Scholl coupling, $NO^+BF_4^-$, Wurster-type amine radical cations. **Photoredox catalysis** ($[Ru(bpy)_3]^{2+*}$, $Ir(ppy)_3$, organic dyes; Nobel-adjacent field) uses visible light to generate radical anions/cations by SET: reductive quenching gives a radical anion of the substrate, oxidative quenching a radical cation; applications include radical dehalogenation, amine α-functionalisation, alkene hydrofunctionalisation, [2+2] cycloadditions via radical cation chains.
- **Radical-cation chain reactions.** The Bauld cation-radical Diels–Alder: an electron-rich dienophile radical cation (e.g. from trans-anethole with a triarylaminium salt) adds to a diene much faster than the neutral pair; the new radical cation oxidises another neutral dienophile (chain). Rate acceleration > 10^6 and complementary regiochemistry.
- **Amine radical cations:** loss of α-H^+ gives α-aminoalkyl radicals; loss of an electron again gives iminium ions (the basis of photoredox α-functionalisation of tertiary amines and of the biological oxidation of N-methyl amines by cytochrome P450 to formaldehyde + dealkylated amine).

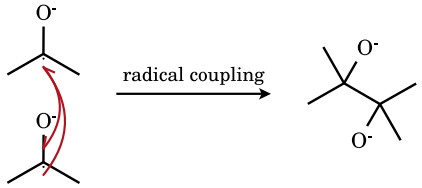
13.4 Radical anions in substitution: $S_{RN}1$ and electron-transfer mechanisms

- **$S_{RN}1$ (Bunnett, 1970):** $ArX + e^- \rightarrow ArX^{\bullet-} \rightarrow Ar\bullet + X^-$ (initiation by hν, Na/NH_3, electrochemistry, or Fe^{2+}); $Ar\bullet + Nu^- \rightarrow ArNu^{\bullet-}$; $ArNu^{\bullet-} + ArX \rightarrow ArNu + ArX^{\bullet-}$ (chain). Features: no EWG needed; leaving-group order I > Br > Cl ≫ F (opposite to S_NAr); nucleophiles are enolates, thiolates, amide, phosphide, acetone enolate; inhibited by radical traps and O_2 and by radical-anion scavengers (p-dinitrobenzene); promoted by light and by electron-transfer initiators; no cine substitution (distinguishes it from benzyne). The nucleophile enters where the leaving group was (ipso).
- **Single-electron transfer (SET) in organometallic reactions:** the formation of Grignard reagents (RX + Mg surface: $RX^{\bullet-} \rightarrow R\bullet + X^-$), the reaction of RLi/RMgX with hindered ketones (benzophenone gives the deep-blue/purple ketyl observed by ESR), the Wurtz reaction (Na + RX), the Ullmann coupling. The SET versus polar-mechanism question is generally settled by radical probes: cyclopropylcarbinyl and 5-hexenyl halides give ring-opened/cyclised products if a free radical forms (Part 9).
- **Radical anions of alkyl halides** have a three-electron σ*(C–X) bond that dissociates in < 10^{-12} s (concerted dissociative ET), giving $R\bullet + X^-$ directly.

> **Typical errors in radical-ion questions.** (1) Placing the Birch product double bonds in conjugation: they are 1,4 (skipped). (2) Putting EWG on sp^2 carbon and EDG on sp^3: it is the opposite. (3) Using NH_3 alone as the proton source without an alcohol. (4) Reducing a terminal alkyne with Na/NH_3: it is deprotonated first. (5) Assuming that the ionisation in MS removes an σ electron: it removes the highest-energy lone-pair/π electron. (6) Forgetting the nitrogen rule. (7) Treating $S_{RN}1$ as S_NAr.

13.5 Additional validated mechanisms

13.1 Pinacol coupling: two ketyl radical anions (from acetone + Mg) combine at carbon



Each ketyl carries the unpaired electron on carbon (the spin is largely on C, the charge on O). Two ketyls pair their electrons to form the new C–C σ bond; the dialkoxide is protonated on work-up to the 1,2-diol (pinacol), or held by Mg^{2+} as a five-membered chelate.

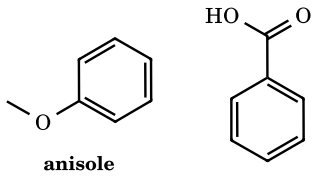
13.2 Mass spectrum of butan-2-one: α-cleavage of the radical cation (m/z 43)



The radical on oxygen pairs with one electron of the adjacent C–C σ bond to form the third bond of the acylium ion $CH_3–C \equiv O^+$ (m/z 43); the other σ electron stays on the departing ethyl radical, which is lost as a neutral. Cleavage that expels the larger alkyl radical is preferred, so m/z 43 is the base peak of butanone (m/z 57 is smaller).

13.1 Problems on radical ions

RI-1.
What is the major product of $Na/NH_3(l)/EtOH$ on anisole, and on benzoic acid?



anisole

benzoic acid

Answer: 1-methoxycyclohexa-1,4-diene; cyclohexa-2,5-diene-1-carboxylic acid
A donor (OMe) stays on the double bond (sp^2 carbon), because the carbanion is least stabilised there, so the protonation occurs at the ortho and meta carbons; an acceptor (COOH) stays on the sp^3 carbon because it stabilises the carbanion on its own carbon, which is protonated.

RI-2.
Mg(Hg), then H_2O, is allowed to react with acetone. Name the intermediate, the product, and the acid-catalysed product.

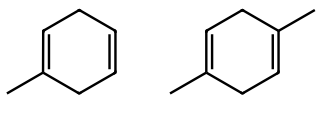


Answer: ketyl radical anion; pinacol; pinacolone
Mg transfers one electron to C=O (ketyl radical anion); two ketyls couple at carbon to the dimeric dialkoxide (pinacol after hydrolysis); H^+ then rearranges pinacol to pinacolone (Part 7.0, 7.10).

RI-3.
Which species is paramagnetic? (a) benzene radical anion, (b) tropylium cation, (c) naphthalene, (d) benzophenone ketyl.
Answer: (a) and (d)
Both have one unpaired electron. The tropylium cation and naphthalene have all electrons paired.

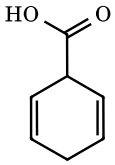
RI-4.
Mass spectrometry: what is $M^{\bullet+}$, and why do its fragments follow carbocation-stability rules?
Answer: the molecular radical cation; fragmentation gives the more stable carbocation + radical
Electron impact removes one electron (usually from a lone pair or π bond) to give $M^{\bullet+}$. Its cleavage into a cation and a neutral radical follows the order of carbocation stability (tertiary > secondary > primary; benzyl/tropylium and oxonium ions are particularly abundant, e.g. m/z 91 for alkylbenzenes).

RI-5.
Predict the Birch product from (a) toluene, (b) p-xylene, (c) benzoic acid, (d) anisole, all with $Na/NH_3(l)/EtOH$.

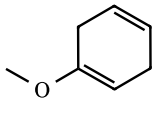


a

b



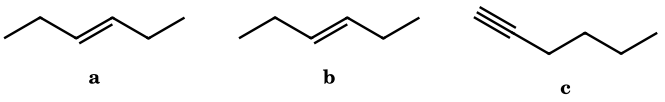
c



d

Answer: (a) 1-methylcyclohexa-1,4-diene; (b) 1,4-dimethylcyclohexa-1,4-diene; (c) cyclohexa-2,5-diene-1-carboxylic acid; (d) 1-methoxycyclohexa-1,4-diene
Donor groups (CH_3, OMe) stay on the double bonds; the acceptor (COOH) sits on the sp^3 carbon, with protonation at the ipso and para carbons. The two products of the second (para-dialkyl) case are the 1,4-diene with both methyls on sp^2 carbons.

RI-6.
Complete: (a) hex-3-yne + $Na/NH_3(l)$; (b) hex-3-yne + H_2, Lindlar; (c) hex-1-yne + $Na/NH_3(l)$ (1 equiv Na).



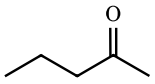
Answer: (a) (E)-hex-3-ene; (b) (Z)-hex-3-ene; (c) the terminal alkyne is deprotonated, giving sodium hex-1-ynide (not reduced)
Dissolving-metal reduction proceeds through a trans-configured vinyl radical/anion; Lindlar hydrogenation is syn. A terminal C–H (pK_a 25) is removed by the amide/NH_2^- formed in solution before reduction.

RI-7.
(Integer) How many of the following form a radical anion on treatment with sodium metal in THF: naphthalene, benzophenone, cyclohexane, anthracene, acetone, tert-butyl alcohol?
Answer: 4 (naphthalene, benzophenone, anthracene, acetone)
Sodium transfers an electron into a low-lying π* orbital: arenes with extended π systems and ketones form radical anions (naphthalenide green, benzophenone ketyl blue/purple, anthracenide green; acetone mainly pinacolises). Cyclohexane has no low-energy acceptor orbital; t-BuOH is deprotonated (H_2 evolution).

RI-8.
A ketyl radical anion from benzophenone is generated with Na in dry THF. (a) What is its colour and what is it used for? (b) What happens on adding water?
Answer: (a) deep blue/purple; indicator of dryness and O_2-free conditions in solvent stills; (b) the colour is discharged and benzhydrol/benzopinacol form
The persistent ketyl is destroyed by traces of water or O_2, so its colour signals that the THF is dry. Protonation gives $Ph_2C^{\bullet}–OH \rightarrow$ benzhydrol by H-atom pickup or benzopinacol by dimerisation.

RI-9.
Arrange the following in decreasing order of intensity of the $M^{\bullet+}$ peak in EI-MS: benzene, hexane, 2-methylpentan-2-ol, ethanol.
Answer: benzene > hexane > ethanol > 2-methylpentan-2-ol
Aromatic molecular ions are the most stable (charge delocalised); linear alkanes give weak $M^{\bullet+}$, primary alcohols weaker still (rapid loss of H_2O and α-cleavage), and tertiary alcohols are almost absent (very fast α-cleavage and dehydration).

RI-10.
In the EI mass spectrum of 2-pentanone ($C_5H_{10}O$), the base peak is m/z 43 and there is a significant peak at m/z 58. Account for both.



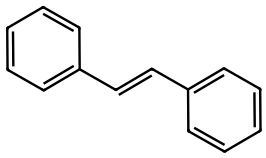
Answer: m/z 43: CH_3CO^+ by α-cleavage; m/z 58: McLafferty rearrangement (loss of ethene)
α-Cleavage on the propyl side expels $^{\bullet}C_3H_7$ to give $CH_3C\equiv O^+$ (43). The γ-H (on C-5) is transferred to O through a six-membered transition state with cleavage of the α,β C–C bond, giving $CH_2{=}C(OH)CH_3^{\bullet+}$ (58) and neutral C_2H_4.

RI-11.
A chloro compound shows $M^{\bullet+}$ peaks at m/z 78 and 80 in a 3 : 1 ratio, and an odd/even pattern that fits C_3H_7Cl. Identify the isotope pattern and give one fragmentation.
Answer: one chlorine ($^{35}Cl : {}^{37}Cl \approx 3 : 1$); loss of Cl$^{\bullet}$ gives $C_3H_7^+$ (m/z 43)
The 3 : 1 intensity ratio of M and M+2 is the signature of one Cl atom (Br gives 1 : 1). C_3H_7Cl has M = 78 (^{35}Cl) and 80 (^{37}Cl); C–Cl cleavage gives the propyl cation, m/z 43.

RI-12.
Why does $S_{RN}1$ show leaving-group order I > Br > Cl ⩾ F, opposite to S_NAr, and why is there no cine substitution?
Answer: the rate-limiting step is C–X cleavage of the radical anion, not nucleophilic addition; no aryne is involved
In S_NAr the addition (needing a polarised C–X, i.e. F) is rate-limiting. In $S_{RN}1$ the fragmentation of $ArX^{\bullet-}$ depends on C–X bond strength, favouring I and Br. The aryl radical couples with Nu^- at the carbon that carried X, so the product is ipso (no cine).

RI-13.
Acetone enolate reacts with iodobenzene under photostimulation in liquid ammonia to give phenylacetone; no product forms in the dark or in the presence of di-tert-butyl nitroxide. What is the mechanism?
Answer: $S_{RN}1$ radical chain
Light supplies the initial electron transfer; phenyl radical (from $PhI^{\bullet-}$) couples with the enolate to give the radical anion of phenylacetone, which transfers an electron to PhI to propagate the chain. Radical scavengers (nitroxide) terminate the chain; the dark reaction does not initiate.

RI-14.
Give the products of (a) acetophenone + SmI_2/THF (ketyl coupling) and (b) benzaldehyde + $TiCl_3$/Zn–Cu (McMurry).



b

Answer: (a) 2,3-diphenylbutane-2,3-diol (pinacol); (b) (E)-stilbene
(a) ketyl radical anions couple at carbon. (b) Low-valent titanium couples the ketyls to a titanium pinacolate and then deoxygenates it (Ti–O bond formation), giving mainly trans-stilbene.

Part 14 Reaction Intermediates Compared, and Integrated Problems

14.1 The six intermediates at a glance

	Carbocation	Carbanion	Free radical	Carbene	Nitrene	Benzyne
Species	R_3C^+	R_3C^-	$R_3C\bullet$	R_2C:	R–N:	C_6H_4
Valence electrons on the key atom	6 (sextet)	8 (octet)	7	6	6	ring carbons octet; extra in-plane bond
Charge	+	–	0	0	0	0
Hybridisation / shape	sp^2, planar, 120°; empty p	sp^3, pyramidal (planar if delocalised)	sp^2, planar or shallow pyramid	singlet: sp^2 bent; triplet: bent or linear	singlet/triplet; one σ bond	sp^2 carbons, in-plane overlap, strained
Magnetism	diamagnetic	diamagnetic	paramagnetic (1 e^-)	singlet dia; triplet para (2 e^-)	singlet dia; triplet para	diamagnetic
Character	electrophile, Lewis acid	nucleophile, base	weakly electrophilic or nucleophilic	electrophile (ambiphilic)	electrophile (ambiphilic)	strong electrophile, dienophile
Formed by	heterolysis (loss of LG), H^+/E^+ addition to π bond	deprotonation, metal insertion, decarboxylation	homolysis (Δ, hν, peroxides)	α-elimination, diazo compounds	azides, N-halo/N-sulfonyloxy amides	ArX + $NaNH_2$, anthranilate, Kobayashi
Stability order	3° > 2° > 1° > CH_3; benzyl, allyl; aromatic ($C_7H_7^+$); +M groups	$sp > sp^2 > sp^3$; CH_3 > 1° > 2° > 3°; –M, –I; aromatic ($C_5H_5^-$)	benzyl ≈ allyl > 3° > 2° > 1° > CH_3 > vinyl	X_2C: with π-donor X (F > Cl > Br > I); NHC	—	—
Rearrangement	1,2-H, 1,2-alkyl, 1,2-aryl; ring expansion/contraction (to a more stable cation)	rare (1,2-Wittig, Favorskii-type)	1,2-aryl (neophyl), vinyl, acyl; **no H/alkyl shift**	1,2-H, Wolff 1,2-shift	acyl → isocyanate; aryl → azepine	none
Stereochemistry at the centre	racemisation (planar)	racemisation (rapid inversion)	racemisation	singlet: syn, stereospecific; triplet: loses stereochemistry	same as carbene	—
Typical reaction	S_N1, E1, Friedel–Crafts, hydration, pinacol	Grignard, aldol, Claisen, E1cB, Michael	halogenation, anti-Markovnikov HBr, polymerisation, Kolbe, Wurtz	cyclopropanation, Reimer–Tiemann, carbylamine, Wolff	Hofmann, Curtius, aziridination	nucleophilic aromatic via aryne, Diels–Alder

14.2 Decision guide: which intermediate does this reaction go through?

Clue in the question	Intermediate
Alkyl halide + Lewis acid or $AgNO_3$; alcohol + conc. H_2SO_4; alkene + H^+; ArN_2^+ on warming	carbocation (and look for a 1,2-shift)

Aldehyde/ketone + base; Grignard; acetylide; soda lime; E1cB; Michael	carbanion
Light, peroxides, high temperature, NBS, AIBN, Kolbe, HBr + peroxide, Cl_2/hν	free radical
$CHCl_3$ + KOH; CH_2N_2; CH_2I_2/Zn(Cu); Reimer–Tiemann; carbylamine; Wolff	carbene (or carbenoid)
$RCONH_2$ + Br_2/NaOH; $RCON_3$ + Δ; azide photolysis; one-carbon-less amine	nitrene / nitrenoid (isocyanate)
Aryl halide + $NaNH_2/NH_3$; no EWG on ring; mixture of meta and ortho/para products; no reaction if both ortho positions blocked	benzyne

14.3 Integrated problems

IP-1.
Identify the intermediate in each: (a) n-propyl chloride + $AlCl_3$ (in benzene gives cumene), (b) toluene + NBS/peroxide, (c) chlorobenzene + $NaNH_2$, (d) CH_3COONa + soda lime, (e) benzamide + Br_2/KOH, (f) $CHCl_3$ + KOH + aniline.
Answer: (a) carbocation (after a hydride shift); (b) benzylic radical; (c) benzyne; (d) carbanion CH_3^-; (e) isocyanate via acyl nitrenoid; (f) dichlorocarbene (→ phenyl isocyanide)
Each reagent combination is the textbook signature of the intermediate. (a) The primary cation is not formed; ionisation occurs with a concerted hydride shift to the isopropyl cation (Part 2.E3).

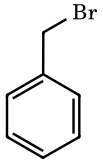
IP-2.
Which of the six intermediates have an incomplete octet on the key atom?
Answer: carbocation (6), carbene (6), nitrene (6) and free radical (7)
The carbanion has 8 electrons; benzyne has a complete octet at each ring carbon (the extra bond is an in-plane partial bond).

IP-3.
Arrange by the number of unpaired electrons (decreasing): triplet carbene, free radical, carbocation, singlet carbene, triplet nitrene.
Answer: triplet carbene = triplet nitrene (2) > free radical (1) > carbocation = singlet carbene (0)
Spin multiplicity 2S + 1: triplet = 3, doublet = 2, singlet = 1.

IP-4.
How many of these species can undergo a 1,2-hydride shift? $CH_3CH_2CH_2^+$, $CH_3CH_2CH_2\bullet$, $CH_3CH_2CH_2^-$, CH_3CH_2CH:, $CH_3CH_2CH_2N$:.
Answer: 3 (cation, carbene, nitrene)
A 1,2-H shift is fast in the cation (to the secondary ion), in the alkylcarbene (CH_3CH_2CH: → propene) and in the alkylnitrene (→ imine). The radical does not undergo 1,2-H shifts and the carbanion does so only under forcing conditions.

IP-5.
Phenol is treated with $CHCl_3$/NaOH. Name every intermediate in order.
Answer: CCl_3^- (carbanion) → $:CCl_2$ (carbene) → o-dienone carbanion → o-(dichloromethyl)phenoxide → $ArCH(OH)_2$ → ArCHO
Hydroxide deprotonates $CHCl_3$ (carbanion) which expels Cl^- (carbene); phenoxide attacks the carbene (Part 10.7). Hydrolysis of the benzal chloride gives the aldehyde. So: two carbanions, one carbene.

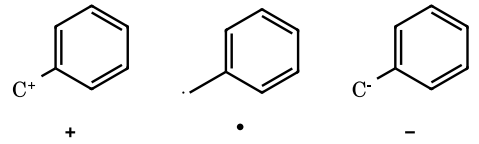
IP-6.
A compound C_7H_7Br gives a pale yellow precipitate with alcoholic $AgNO_3$ immediately and, on reaction with $NaNH_2$, forms no benzyne. Identify it.



Answer: benzyl bromide
Alcoholic $AgNO_3$ precipitates AgBr at once through the resonance-stabilised benzyl cation; the bromine is on sp^3 carbon so there is no aryne pathway. The isomeric bromotoluenes (aryl bromides) give nothing with $AgNO_3$ but do react with $NaNH_2$ via benzyne.

IP-7.
Match: (A) Wurtz reaction, (B) Hofmann bromamide, (C) Kolbe electrolysis, (D) Reimer–Tiemann; with (P) carbene, (Q) free radical, (R) nitrene-type, (S) carbanion.
Answer: A–S,Q; B–R; C–Q; D–P,S
Wurtz: organosodium (carbanion) and radical character; Hofmann: isocyanate formed by a nitrene-type shift; Kolbe: acyloxy and alkyl radicals; Reimer–Tiemann: CCl_3^- and $:CCl_2$.

IP-8.
Toluene can be converted into the benzyl cation, benzyl radical and benzyl anion. Give a reagent for each and say what stabilises all three.

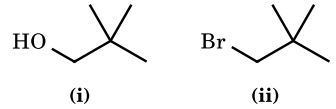


Answer: cation: hydride abstraction (Ph_3C^+); radical: NBS or Cl_2/hν; anion: n-BuLi/KNH_2
In all three the benzylic p orbital overlaps with the six π electrons of the ring, so the empty orbital (cation), half-filled orbital (radical) or filled orbital (anion) is delocalised onto the ortho and para carbons. Resonance stabilises cations, radicals and anions alike, whereas inductive/hyperconjugative effects stabilise cations and radicals but destabilise anions.

IP-9.
Compare the products and stereochemistry: (a) 1-phenylethyl chloride (R) + $SbCl_5$, (b) the radical bromination of the corresponding alkane, (c) the Hofmann degradation of (R)-2-phenylpropanamide.
Answer: (a) racemic substitution product (planar cation); (b) racemic bromide (planar radical); (c) retention: (R)-1-phenylethylamine
Carbocation and radical centres are planar, so stereochemical information at that carbon is lost; the 1,2-shift to nitrogen (Hofmann, Curtius) is intramolecular and goes with retention at the migrating carbon.

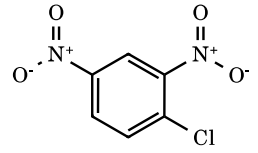
IP-10.
In which reaction are a carbanion and a carbene formed one after the other?
Answer: $CHCl_3$ + KOH (α-elimination)
HO^- removes the proton to give CCl_3^- (carbanion), which expels Cl^- from the same carbon to give $:CCl_2$ (carbene).

IP-11.
Predict the major product and the intermediate: (i) neopentyl alcohol + HBr; (ii) neopentyl radical (from neopentyl bromide, Bu_3SnH).



Answer: (i) 2-bromo-2-methylbutane via the t-amyl cation (methyl shift); (ii) neopentane (no rearrangement)
The same carbon skeleton gives a rearranged product through a cation but not through a radical: radicals do not shift methyl groups. This is the cleanest experimental distinction between cationic and radical pathways.

IP-12.
1-Chloro-2,4-dinitrobenzene reacts with NH_3 at room temperature to give 2,4-dinitroaniline only, whereas chlorobenzene needs $NaNH_2$ in liquid NH_3 and may give regioisomeric products when substituted. Explain.



Answer: S_NAr (addition–elimination) vs benzyne (elimination–addition)
The two nitro groups make the ring electron-poor and stabilise the Meisenheimer anion, so the weak nucleophile NH_3 adds directly at the carbon bearing Cl: a single (ipso) product. Chlorobenzene has no such activation; only a base strong enough to remove an ortho H produces benzyne, and the nucleophile then adds at either carbon.

Answers Part 3 — Leaving-Group Problems: Answers and Explanations

LG1.
Answer: D > C > B > A
Rate-limiting heterolysis of C–X. Bond strength (C–I 57 ⩽ C–F 110 kcal mol^{-1}), polarisability and conjugate-acid strength (HI −10, HBr −9, HCl −7, HF +3.2) all point the same way. Solvation of F^- by hydrogen bonding is the strongest of the four but cannot make up for the C–F bond: t-BuF is usually considered inert to neutral solvolysis. The three heavier halides differ only by modest factors (a few-fold between I and Br; tens between Br and Cl) because the TS is highly ionic and the solvent levels the differences.

LG2.

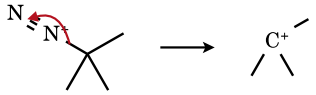
Answer: A > B > C > D

Leaving-group quality follows conjugate-acid strength: CF_3SO_3H (≈ −14) ⩾ TsOH (≈ −2.8) ⩾ CH_3CO_2H (4.76) ⩾ H_2O (15.7). Triflates ionise ≈ 10^4–10^8 times faster than tosylates depending on substrate; acetates are practically inert under these conditions, and the alcohol **D** does not ionise at all (HO^- is not a leaving group without protonation). A secondary substrate is chosen so that all three active derivatives react by borderline S_N1/S_N2 with solvent assistance; the order is the same for either.

LG3.

Answer: A > B > C > D

N_2 is a vastly better leaving group than anything else (a very stable, non-basic molecule; ionisation of an alkyl diazonium is practically barrierless). H_2O leaves from the protonated alcohol readily at 25 °C in acid (conjugate acid H_3O^+, pK_a −1.7). NH_3 is a much stronger base (NH_4^+ pK_a 9.25), so the t-butylammonium ion is stable at room temperature and needs heat or very strong acid to lose NH_3. Hydroxide in **D** is the worst: C–OH ionisation does not occur without protonation (D is therefore last, essentially inert).



LG4.

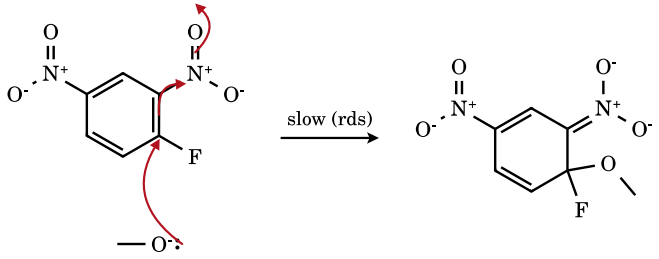
Answer: C > D > B > A

Carboxylate leaving groups follow the pK_a of the parent acid: CF_3CO_2H 0.5 < p-$O_2NC_6H_4CO_2H$ 3.44 < $C_6H_5CO_2H$ 4.20 < CH_3CO_2H 4.76. The same trend is the basis of the classical use of p-nitrobenzoates as 'better' leaving groups than acetates in mechanistic work on tertiary benzylic systems. Resonance donation within the ester (which strengthens the C–O bond) is similar for all four and cancels.

LG5.

Answer: A > B > C

In SNAr the rate-determining step is *addition* of the nucleophile to give the anionic Meisenheimer complex; C–X cleavage comes afterwards (fast). Fluorine, the most electronegative halogen, lowers the energy of the first TS by induction and polarises C–X; the C–F bond strength is irrelevant to the rate. Hence the 'element effect' F ⩾ Cl ≳ Br > I, the reverse of S_N2 – a decisive demonstration that leaving-group ability is mechanism-dependent. (Cl and Br are close; iodine is slowest.)



LG6.

Answer: D > C > B > A

The poorer the leaving group, the less C–LG bond cleavage has occurred in the E2 transition state; the TS becomes carbanion-like (E1cB-like) and the base removes the more acidic, more accessible hydrogen from the CH_3 group, giving the Hofmann alkene. Literature trend for 2-pentyl X with ethoxide: I < Br (≈ 30 %) ≲ Cl (≈ 33 %) < F (≈ 70 %) < NMe_3^+ (≈ 95 % 1-pentene); I and Br differ only slightly. Here LG ability controls *selectivity*, not only rate.

LG7.

Answer: A > B > C > D > E

Y (recalled values): H_2O 3.49 > HCO_2H 2.05 > CF_3CH_2OH 1.15 > CH_3OH −1.09 > C_2H_5OH −2.03. Ionising power combines hydrogen-bond donation to the departing chloride (high for water, HCO_2H, fluorinated alcohols) with dielectric screening and low nucleophilicity. TFE is a poor nucleophile yet an excellent ionising medium, so S_N1 reactions of secondary substrates are favoured there; a leaving group's 'ability' is therefore also a property of the solvent.

LG8.

Answer: A > C > B

E2 needs the C–H and C–Br bonds anti-periplanar, which on a chair means *both axial*. In the cis isomer (A) the Br is axial (the locked t-Bu is equatorial), so reaction is fast. In the trans isomer (B) Br is equatorial, no anti-periplanar H exists, the ring would have to flip into the diaxial-t-Bu chair, and E2 is slow (substitution/E1 intrude). Bromocyclohexane (C) populates the axial conformer to a modest extent (≈ 30 %), so its rate lies between. The axial/equatorial labels refer to chairs, which these flat drawings cannot show; wedge/dash give the relative (cis/trans) configuration.

LG9.

Answer: A > B > C > D

The rate follows both the leaving-group stability (Cl^-, $pK_a(HCl)$ –7; $CH_3CO_2^-$ 4.76; EtO^- 16; NH_2^- 38) and the ability of the substituent to donate into C=O (N ⩾ O > Cl). The tetrahedral intermediate expels the best leaving group; amides, with strong n(N) → π*(C=O) donation and a very poor LG, are the least reactive. Contrast with S_N1/S_N2 at sp^3 carbon, where resonance donation is irrelevant.

LG10.

Answer: A > B > C

Trimethyloxonium (Meerwein) salts are among the strongest methylating agents; trimethylsulfonium is a weak methylating agent (it needs warming), and tetramethylammonium is inert. The leaving groups are Me_2O, Me_2S and Me_3N. Basicity (pK_{aH} of Me_3N 9.8 ⩾ Me_2O ≈ –3.8, Me_2S ≈ –6 to –7) explains why ammonium is last, but *not* why oxonium beats sulfonium: sulfur is the weaker base. Here the intrinsic C–X^+ bond strength and the charge density at the cationic centre dominate – a clear failure of a pure pK_a rule. (Experimental trend: O^+ ⩾ S^+ ⩾ N^+.)

LG11.

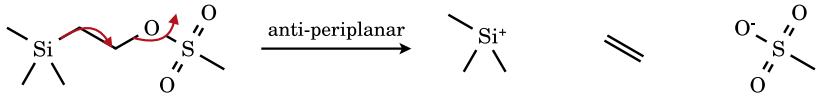
Answer: A > B > C > D

Ag^+ coordinates the halogen lone pair and 'pulls' the bromide off (halophilic assistance); the substrate then ionises. The ease of forming the cation is 3° ⩾ 2° > 1° (t-Bu^+ ⩾ i-Pr^+ ⩾ n-Bu^+); the n-butyl bromide reacts only slowly (a partial S_N2-like displacement by solvent) and bromobenzene (an sp^2 aryl cation would be needed) gives no precipitate. Same leaving group, different cation – the LG is only as good as the cation it can generate.

LG12.

Answer: A > B > C

β-Silicon effect: a C–Si σ bond is an excellent hyperconjugative donor (σ_{C-Si} is higher in energy than $\sigma_{C-C}/\sigma_{C-H}$) and stabilises a β cation by σ→p donation (a bridged, silicon-stabilised cation). The overlap is best when C–Si and C–OTs are **anti-periplanar**, which the trans isomer reaches in the diaxial chair (A: very large acceleration, ≈ 10^{12} in rigid models). The cis isomer (B) has them gauche (synclinal) only: still accelerated (≈ 10^4 in rigid models) but far less. Cyclohexyl tosylate (C) has no silicon. (Orders of magnitude quoted from Lambert's work, recalled.)



LG13.

Answer: A > B > C > D

The aryl ring attacks the carbon carrying OTs from the rear, forming a bridged phenonium (spiro-cyclohexadienyl) ion; the rate rises with the electron-donating power of the para substituent (σ^+: OMe –0.78, Me –0.31, H 0, NO_2 +0.79). The p-nitro compound cannot use the ring and reacts by the slower, unassisted S_N2-like route (k_s). The 'leaving group' is identical (TsO^-) in all four; the substituent changes how effectively it is displaced.

LG14.

Answer: A > B > C > D

The leaving group is the protonated oxygen (an alcohol or ether oxygen that becomes neutral on C–O cleavage), and the driving force is the release of ring strain: oxirane (≈ 27 kcal mol^{-1}) and oxetane (≈ 25) are strained; oxolane (THF, ≈ 6) is slightly strained; oxane (THP, ≈ 0.5) is not. Epoxides react far faster than oxetanes despite similar strain energies because the TS gains from poorer C–O–C orbital overlap (bent bonds) in the three-membered ring; THF is cleaved only under more forcing conditions, THP hardly at all.

LG15.

Answer: A > B > C > D

The sulfonate anion is stabilised by electron-withdrawing substituents (lower pK_a of $ArSO_3H$), so the leaving-group ability follows the Hammett σ constants: NO_2 (σ_p +0.78) > Br (+0.23) > CH_3 (–0.17) > OCH_3 (–0.27); ρ for the leaving group is positive and modest. Brosylates and nosylates are therefore used when tosylates are too unreactive; mesylates are comparable to tosylates.

Answers Part 7.1 — Counting Problems: Answers and Explanations

R1.
Answer: 5 (a, d, e, h, j)

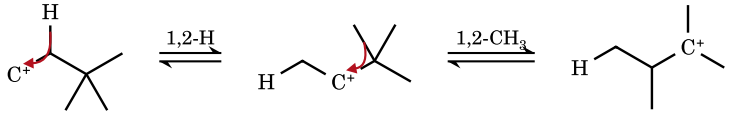
Item	?	Reason
a n-butyl	**Yes**	primary; 1,2-H shift → 2-butyl (secondary)
b isopropyl	**No**	H or CH_3 shift would give a primary cation, or a degenerate secondary one
c tert-butyl	**No**	any shift gives a primary cation
d 3,3-dimethylbut-2-yl	**Yes**	1,2-CH_3 shift → tertiary 2,3-dimethylbut-2-yl
e neopentyl	**Yes**	1,2-CH_3 shift → tertiary tert-amyl
f cyclohexyl	**No**	H shift gives another secondary ring cation (degenerate); the ring does not strain-relieve
g 1-methylcyclohexyl	**No**	tertiary; shifts lead to less stable ions
h cyclobutylmethyl	**Yes**	ring expansion → cyclopentyl (secondary, strain relief), or H shift → 1-methylcyclobutyl (tertiary)
i benzyl	**No**	resonance-stabilised; a shift would give an unstabilised ion
j 2-methylcyclohexyl	**Yes**	1,2-H shift from the ring C–H bearing CH_3 → tertiary 1-methylcyclohexyl

R2.
Answer: 4 (a, b, d, e)

Item	?	Reason
a cyclobutylmethyl	**Yes**	ring C–C bond migrates to the CH_2^+: 4 → 5 ring, primary → secondary
b 1-cyclobutylethyl	**Yes**	ring bond migrates to the exocyclic cationic carbon: 2-methylcyclopentyl (secondary, ring strain relieved)
c cyclopentyl	**No**	the cation is in the ring; no exocyclic carbon to take a ring bond
d (1-methylcyclobutyl)methyl	**Yes**	ring bond migrates → 1-methylcyclopentyl, a tertiary cation with relief of strain (much better than the methyl-shift alternative)
e (1-methylcyclopentyl)methyl	**Yes**	ring bond migrates → 1-methylcyclohexyl (tertiary, strain-free); the competing methyl shift gives 1-ethylcyclopentyl (also tertiary), so ring expansion is the thermodynamic choice
f 1-methylcyclohexyl	**No**	tertiary cation on the ring: expansion impossible
g cyclobutyl	**No**	cation is in the ring (it can ring-contract to cyclopropylcarbinyl, the reverse)
h neopentyl	**No**	no ring: methyl shift is a rearrangement but not a ring expansion

R3.
Answer: (i) 2 (ii) 1
Shifts available: **H from C-3** → 2-methylbut-2-yl, tertiary (more stable); **CH_3 from C-3** → regenerates a 3-methylbutan-2-yl cation (degenerate, not a new ion); **H from the CH_3 on C-2** → 3-methylbutyl, primary (less stable). Distinct new cations: tertiary and primary = 2; more stable: only the tertiary = 1. (Hydride shift is the productive path, reached through a three-centre transition state with σ(C–H) → p overlap.)

R4.
Answer: 2
Step 1: 1,2-hydride shift from the CH_2 next to the quaternary carbon → 3,3-dimethylbut-2-yl (secondary). Step 2: 1,2-methyl shift from the quaternary carbon → 2,3-dimethylbut-2-yl (tertiary, the most stable ion in this manifold). A direct primary → tertiary shift is impossible because the tertiary centre is two bonds away. (The same cation is the one formed from 3,3-dimethylbutan-2-ol; see R5.)



R5.
Answer: 3
Unrearranged secondary cation (3,3-dimethylbut-2-yl): loses H^+ only from CH_3 (the other neighbour is quaternary) → **3,3-dimethylbut-1-ene** (minor). After the 1,2-CH_3 shift the tertiary 2,3-dimethylbut-2-yl cation loses H^+ from the CH (→ **2,3-dimethylbut-2-ene**, tetrasubstituted, major) or from a CH_3 (→ **2,3-dimethylbut-1-ene**). Total 3; the Zaitsev (2,3-dimethylbut-2-ene) dominates.

R6.
Answer: 4 (a, d, e, h)

Item	?	Reason
a n-propylamine	**Yes**	$R–N_2^+$ loses N_2 (with 1,2-H shift) → n-propyl/isopropyl cation; mixture of propanols, propene, chloropropanes
b aniline	**No**	$ArN_2^+ Cl^-$ is stable at 0–5 °C (resonance, sp^2 C–N); the aryl cation forms only on warming
c dimethylamine	**No**	secondary amine → N-nitrosodimethylamine, $Me_2N–N=O$
d tert-butylamine	**Yes**	t-Bu–N_2^+ → tert-butyl cation
e benzylamine	**Yes**	$PhCH_2N_2^+$ → benzyl cation
f trimethylamine	**No**	tertiary amine: only the ammonium nitrite salt; no diazonium
g N-methylaniline	**No**	secondary arylamine → N-nitroso-N-methylaniline
h cyclohexylamine	**Yes**	c-$C_6H_{11}N_2^+$ → cyclohexyl cation (and cyclohexene, cyclohexanol, ring-contracted products)

R7.
Answer: 4 (c, d, f, g)

Item	?	Reason
a butan-1-ol	**No**	primary, unbranched: S_N2 → 1-bromobutane
b butan-2-ol	**No**	secondary cation (S_N1/S_N2 mixture); a shift would give a primary or an identical secondary ion
c 3-methylbutan-2-ol	**Yes**	H shift → tertiary → 2-bromo-2-methylbutane
d neopentyl alcohol	**Yes**	concerted CH_3 shift → 2-bromo-2-methylbutane
e tert-butyl alcohol	**No**	tertiary cation
f 3,3-dimethylbutan-2-ol	**Yes**	CH_3 shift → tertiary → 2-bromo-2,3-dimethylbutane
g cyclobutylmethanol	**Yes**	ring expansion / rearrangement → bromocyclopentane and 1-bromo-1-methylcyclobutane type products
h cyclohexanol	**No**	secondary ring cation; H shift is degenerate

R8.
Answer: 4 (a, b, e, g)

Item	?	Reason
a 1-chloropropane	**Yes**	H shift in the R–Cl···$AlCl_3$ complex → isopropylbenzene (cumene)
b 1-chloro-2-methylpropane	**Yes**	H shift → tert-butyl → tert-butylbenzene
c chloroethane	**No**	no more stable cation reachable (ethyl is the only possibility)
d tert-butyl chloride	**No**	tertiary cation, no shift
e 1-chloro-2,2-dimethylpropane	**Yes**	CH_3 shift → tert-amyl → (2-methylbutan-2-yl)benzene
f benzyl chloride	**No**	benzylic cation; gives diphenylmethane without rearrangement
g 1-chlorobutane	**Yes**	H shift → sec-butyl → sec-butylbenzene as the major component (n-butylbenzene is also formed; the ratio depends on temperature, roughly 2 : 1 in favour of sec-butyl at 0 °C – recalled value)
h 2-chlorobutane	**No**	secondary → secondary: the same sec-butyl product